THE COMPLETE

Nursing School BUNDLE®

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SCAN ME!

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NOTES

Every
ACCOMPLISHMENT
starts with the
decision to **TRY.**

HEAD-TO-TOE ASSESSMENT

BROUGHT TO YOU BY NURSE IN THE MAKING®

HEAD-TO-TOE ASSESSMENT

1. INSPECT
2. PALPATE
3. PERCUSS
4. AUSCULTATE

Introduction

- Knock
- Introduce yourself
- Wash hands
- Provide privacy
- Verify client ID and DOB
- Explain what you are doing (using non-medical language)

Orientation

- What is your name?
- Do you know where you are?
- Do you know what month it is?
- Who is the current U.S. president?
- What are you doing here?
- **A&O X4** = Oriented to Person, Place, Time, and Situation

"Normal" Vital Signs

- **PULSE:** 60-100 bpm
- **BLOOD PRESSURE:** 120/80 mmHg
- **O_2 SATURATION:** 95-100%
- **TEMPERATURE:** 97.8-99.1°F
- **RESPIRATIONS:** 12-20 breaths per min

PULSE SCALE

0	PULSE IS ABSENT
1+	DIMINISHED
2+	NORMAL
3+	FULL
4+	BOUNDING, STRONG

Assessing the strength of the pulse

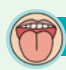
Head & Face

HEAD

- Inspect head/scalp/hair
- Palpate head/scalp/hair

FACE

- Inspect
- Check for symmetry
  - To assess **CRANIAL NERVE 7**, check....



VII: FACIAL

- Raise eyebrows
- Smile
- Frown
- Show teeth
- Puff out cheeks
- Tightly close eyes

EYES

- Inspects external eye structures
- Inspect color of conjunctiva and sclera
- **PERRLA**
  - **P**upils **E**qual, **R**ound, **R**eactive to **L**ight, & **A**ccommodation

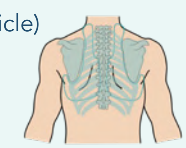
ASSESS THE DEPTH OF THE RESPIRATIONS

EFFORT	note if it's **LABORED** or **UNLABORED**
RHYTHM	note if it's **REGULAR** or **IRREGULAR**

Neck, Chest (Lungs) & Heart

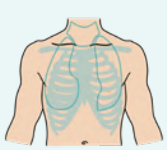
NECK

- Inspect and palpate
- Palpate carotid pulse
- Check skin turgor (under clavicle)



POSTERIOR CHEST

- Inspect
- Auscultate lung sounds in posterior and lateral chest
  - Note any crackles or diminished breath sounds



ANTERIOR CHEST

- Inspect:
  - Use of accessory muscles
  - AP to transverse diameter
  - Sternum configuration
- Palpate: symmetric expansion
- Auscultate lung sounds ➜ anterior and lateral
  - Note any crackles or diminished breath sounds

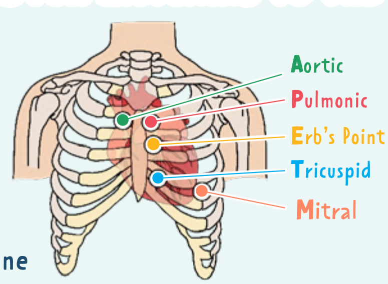
HEART

- Auscultate heart sounds (**A, P, E, T, M**) with diaphragm and bell
  - Note any murmurs, whooshing, bruits, or muffled heart sounds

5 AREAS FOR LISTENING TO THE HEART

MNEMONIC

All
People
Enjoy
Time
Magazine



HEAD-TO-TOE ASSESSMENT

Peripherals

UPPER EXTREMITIES

- Inspect and palpate
- Note any texture, lesions, temperature, moisture, tenderness, & swelling
- Palpate radial pulses bilaterally

0	PULSE IS ABSENT
1+	DIMINISHED
2+	NORMAL
3+	FULL
4+	BOUNDING, STRONG

SHOULDER

- Inspect, palpate, and assess

ELBOWS

- Inspect, palpate, and assess

HANDS AND FINGERS

- Inspect hands/fingers/nails
- Palpate hands and finger joints
- Check muscle strength of hands bilaterally
  - Does each hand grip evenly?

Spine

- Have the client stand up (if able)
- Inspect the skin on the back
- Inspect: spinal curvature (cervical/thoracic/lumbar)
- Palpate spine
- Note any lesions, lumps, or abnormalities

If we were to percuss + palpate before listening (auscultating), we would alter the bowel sounds. This would lead to inaccurate results.

Lower Extremities (hips, knees, ankles)

LOWER EXTREMITIES

- Inspect:
  - Overall skin coloration
  - Lesions
  - Hair distribution
  - Varicosities
  - Edema
- Palpate: Check for edema (pitting or non-pitting)
- Check capillary refill bilaterally

CAPILLARY REFILL TIME (CRT)
Time taken for capillary bed to regain its color after pressure has been applied

NORMAL <2-3 SECONDS

HIPS

- Inspect and palpate

KNEES

- Inspect and palpate

ANKLES

- Inspect and palpate
- Posterior pulse
- Dorsal pedis pulse bilaterally
  - Check strength bilaterally
    - Dorsiflexion flexion against resistance

0	PULSE IS ABSENT
1+	DIMINISHED
2+	NORMAL
3+	FULL
4+	BOUNDING, STRONG

Abdomen

- Inspect:
  - Skin color
  - Contour
  - Scars
  - Aortic pulsations
- Auscultate bowel sounds: all 4 quadrants (start in RLQ and go clockwise)
- Light palpation: all 4 quadrants

Assess in different order:
1. INSPECT
2. AUSCULTATE
3. PERCUSS
4. PALPATE

ABSENT: Must listen for at least 5 minutes to chart absent bowel sounds

HYPOACTIVE: One bowel sound every 3-5 minutes

NORMOACTIVE: Gurgles 5-30 times per minute

HYPERACTIVE: Can sometimes be heard without a stethoscope. Constant bowel sounds (> 30 sounds per minute)

OVERALL

- Positions and drapes client appropriately during exam (gave client privacy)
- Gave client feedback/instructions
- Exhibits professional manner during exam, treated client with respect and dignity
- Organized: exam followed a logical sequence (order of exam "made sense")

NOTES

It's a beautiful thing when a **CAREER** and a **PASSION** come together.

DOSAGE CALCULATION

ABBREVIATIONS

TIMES OF MEDICATIONS

ac	before meals
pc	after meals
daily	every day
bid	two times a day
tid	three times a day
qid	four times a day
qh	every hour
ad lib	as desired
stat	immediately
q2h	every 2 hours
q4h	every 4 hours
q6h	every 6 hours
prn	as needed
hs	at bedtime

EXAMPLE

QUESTION: A patient is receiving 1 mg tid.
How many mg will they receive in one day?

Remember: tid = 3X a day
ANSWER: If they are receiving 1 mg for 3X a day, that's 1 mg x 3 = 3 mg per day

ROUTES OF ADMINISTRATION

PO	by mouth
IM	intramuscularly
PR	per rectum
SubQ	subcutaneously
SL	sublingual
ID	intradermal
GT	gastrostomy tube
IV	intravenous
IVP	intravenous push
IVPB	intravenous piggyback
NG	nasogastric tube

DRUG PREPARATION

tab, tabs	tablet
cap, caps	capsule
gtt	drop
EC	enteric coated
CR	controlled release
susp	suspension
el, elix	elixir
sup, supp	suppository
SR	sustained release

METRIC

g (gm, Gm)	gram
mg	milligram
mcg	microgram
kg (Kg)	kilogram
L	liter
mL	milliliter
mEq	milliequivalent

APOTHECARY & HOUSEHOLD

gtt	drop
min, m, mx	minim
tsp	teaspoon
pt	pint
gal	gallon
dr	dram
oz	ounce
T, tbs, tbsp	tablespoon
qt	quart

CONVERSIONS

BASED ON VOLUME

1 mg = 1,000 mcg
1 g = 1,000 mg
1 oz = 30 mL
8 oz = 1 cup
1 tsp = 5 mL
1 dram = 5 mL
1 tbsp = 15 mL
1 tbsp = 3 tsp
1 L = 1,000 mL

THE METRIC SYSTEM

LARGE unit to **SMALL** unit ➔ move decimal to the **RIGHT**

SMALL unit to **LARGE** unit ➔ move decimal to the **LEFT**

MOVING TO A LARGER UNIT?
Move the decimal place to the Left
(Ex: mcg → mg)

MEMORY TRICK

Larger unit think Left

EXAMPLE

1500 mcg = _____mg

A **mg** is larger than a **mcg**
Therefore you move decimal
3 places to the Left

1500. mcg = 1.500 mg (1.5 mg)

BASED ON WEIGHT

1 kg = 2.2 lbs
1 lb = 16 oz

lb ➔ kg
DIVIDE by **2.2**

EXAMPLE

120 **lbs** = _____ **kg**

120 **lbs** / **2.2** = 54.545 **kg**

kg ➔ lb
MULTIPLY by **2.2**

EXAMPLE

45.6 **kg** = ______ **lbs**

45.6 **kg** x **2.2** = 100.32 **lbs**

DOSAGE CALC RULES

Medication error kills, prevention is crucial!

1 Show ALL your work.

2 Leading zeros must be placed before any decimal point.

The decimal point may be missed without the zero

EXAMPLE **.2** mg should be **0.2** mg

WHY? **.2** could appear to be **2**

(0.2 mg of morphine is VERY different than 2 mg of morphine!)

3 No trailing zeros.

EXAMPLE **0.7** mL *NOT* **0.70** mL

1 mg *NOT* **1.0** mg

WHY? **1.0** could appear to be **10**!

4 Do not round until you have the final anwser!

HOW TO ROUND YOUR FINAL ANSWER

If the number in the thousands place is **5 OR GREATER** → The # in the hundredth place is rounded up

EXAMPLES **1.995** mg is rounded to **2** mg
1.985 mg is rounded to **1.99** mg

If the number in the thousands place is **4 OR LESS** → The # is dropped

EXAMPLES **0.992** mg is rounded to **0.99** mg

DECIMAL REFERENCE GUIDE

34.732

tens
ones
tenths
hundredths
thousandths

5 Most nursing schools, if not all, do not give partial credit.

This means every step must be done correctly!

FORMULA METHOD

For Volume-Related Dosage Orders

$$\frac{D}{H} \times V = A$$

D = Desired

EXAMPLE: "The physician orders **120 mg**..."

! Some medications like heparin and insulin are prescribed in **units/hour**

H = Dosage of medication available

EXAMPLE: "The medication is supplied as **100 mg**/5 mL"

V = Volume the medication is available in

EXAMPLE: "The medication is supplied as 100 mg/**5 mL**"

A = Amount of Medication required for administration

YOUR ANSWER

! You should assume that all questions are asked "**per dose**" unless the question gives a timeframe (example: "how many tablets will you give in 24 hours?")

EXAMPLE 1

Ordered: Drug C 150 mg
Available: Drug C 300 mg/tab
How many tablets should be given?

$$\frac{D}{H} \times V = A$$

What's our desired? Drug C 150mg PO
What do we have? Drug C 300mg/tab
What's our quantity/volume? tablets

150 ~~mg~~ ÷ 300 ~~mg~~ x 1 tab = 0.5 tabs
150 ÷ 300 = 0.5 x 1 = 0.5 tabs

FINAL ANSWER: 0.5 tabs

EXAMPLE 2

Ordered: Drug C 10,000 units SubQ
Available: Drug C 5,000 units/mL
How many mL should be given?

$$\frac{D}{H} \times V = A$$

What's our desired? Drug C 10,000 SubQ
What do we have? Drug C 5,000 units
What's our quantity/volume? 1 mL

10,000 ~~units~~ ÷ 5,000 ~~units~~ x 1 mL = 2 mL
10,000 ÷ 5,000 = 2 x 1 = 2 mL

FINAL ANSWER: 2 mL

IV FLOW RATES

mL / hour

$$\frac{\text{mL of solution}}{\text{total hours}} = \text{mL/hr}$$

What if the question is given in **MINUTES?**

Since there are 60 minutes in one hour, use this formula:

$$\frac{\text{mL of solution}}{\text{min}} \times \underset{(minutes)}{60} = \text{mL/hr}$$

! If the question is asking for flow rate and you're given units of mL, you need to write the answers in **mL/hr!**

mL/hr is always rounded to the nearest **WHOLE** number!

EXAMPLE #1

ORDERED: 1000 mL D5W to infuse over 3 hours. What will the flow rate be?

$$\frac{1000 \text{ mL}}{3 \text{ hr}} = 333.333 \text{ mL/hr}$$

ANSWER: 333 mL/hr
(rounded to the nearest whole number)

EXAMPLE #2

ORDERED: Infuse 3 grams of Penicillin in 50 mL normal saline over 30 minutes.

$$\frac{50 \text{ mL}}{30 \cancel{\text{min}}} \times 60 \cancel{\text{min}} = 100 \text{ mL/hr}$$

ANSWER: 100 mL/hr

gtt / min

$$\frac{\text{mL of solution}}{\text{total minutes}} \times \text{drop factor} = \text{gtt/min}$$

What if the question is given in **HOURS?**

Since there are 60 minutes in one hour, use this formula:

Convert hours to minutes!

EXAMPLES:
1 hour = 60 minutes
2.5 hours = 150 minutes

! If a drop factor is included, the question is asking for flow rate in gtt/min.

You need to write the answers in **gtt/minute**!

REMEMBER OUR ABBREVIATIONS:
gtt means "drop"!

EXAMPLE #1

ORDERED: 1000 mL of Lactated Ringer's to infuse at 50 mL/hr. Drop factor for tubing is a 5 gtt/mL. (*Convert: 1 hour = 60 min*)

$$\frac{50 \cancel{\text{mL}}}{60 \text{ min}} \times 5 \text{ gtt}/\cancel{\text{mL}} = 4 \text{ gtt/min}$$

50 ÷ 60 = 0.833 x 5 = 4.166
Round to the nearest whole number → 4

FINAL ANSWER: 4 gtt/min

! **REMEMBER RULE #4**
Don't round till the end!

EXAMPLE #2

ORDERED: 100 mL of Metronidazole to infuse over 45 minutes. The tubing you are using has a drop factor of 10 gtt/mL.

$$\frac{100 \cancel{\text{mL}}}{45 \text{ min}} \times 10 \text{ gtt}/\cancel{\text{mL}} = 22 \text{ gtt/min}$$

100 ÷ 45 = 2.222 x 10 = 22.222
Round to the nearest whole number → 22

FINAL ANSWER: 22 gtt/min

! **REMEMBER RULE #4**
Don't round till the end!

PRACTICE QUESTIONS

Do all 10 questions without looking at the correct answers on the following pages. Don't forget to show all your work. After you are done, walk through each question…even the questions you got correct!

1 **ORDERED:** Rosuvastatin 3000 mcg PO ac
AVAILABLE: Rosuvastatin 2 mg tablet (scored)
How many tabs will you administer in 24 hours?

2 **ORDERED:** Tylenol supp 2 g PR q6h
AVAILABLE: Tylenol supp 700 mg
How many supp will you administer?
Round to nearest tenth.

3 **ORDERED:** Potassium chloride 0.525 mEq/lb PO dissolved in 6 oz of juice at 0930
AVAILABLE: Potassium cholride 12 mEq/mL
How many mL of potassium chloride will you add to the juice for a 66.75 kg patient? Round to nearest tenth.

4 **1000 mL D5W to infuse over 4 hours.**

5 **150 mL Cipro 250 mcg to infuse over 45 minutes.**

6 **250 mL normal saline over 5 hours.**
Tubing drop factor of 10 gtt/mL.

7 **Humulin R 200 units in 100 mL of normal saline to infuse at 4 units/hr.**

8 **Dopamine 600 mg in 200 mL of normal saline to infuse at 10mcg/kg/min. Pt weight = 190 lbs.**

9 **2.5 L normal saline to infuse over 48 hours.**

10 **ORDERED:** Morphine 100 mg IM q12h prn pain
AVAILABLE: Morphine 150 mg/2.6 mL
How many mL will you administer?
Round to nearest hundredth.

COMPREHENSIVE REVIEW

1

ORDERED: Rosuvastatin 3000 mcg PO ac
AVAILABLE: Rosuvastatin 2 mg tablet (scored)

How many tabs will you administer in 24 hours?

STEP 1: CONVERT DATA

mcg → mg

3000 mcg = 3 mg

REMEMBER **SMALL TO BIG:** move the decimal point 3 to the left
unit is getting Larger think Left

STEP 2: READY TO USE DATA

ORDERED: 3 mg
AVAILABLE: 2 mg
VOLUME: 1 tab
ADMINISTERED AC: before each meal
QUESTION IS ASKING: dosage in 24 hours

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{3\ \cancel{mg}}{2\ \cancel{mg}} = 1.5$$

1.5 x 1 tab = 1.5

1.5 x 3 = 4.5 tabs per day

ROUND: *No rounding necessary*

DON'T FORGET TO CHECK TIMES OF MEDICATION!

The medication is ordered to be given AC, which means before each meal. Since there are 3 meals in a day (24 hours), the answer must be multiplied by 3.

FINAL ANSWER: 4.5 tabs

2

ORDERED: Tylenol supp 2 g PR q6h
AVAILABLE: Tylenol supp 700 mg

How many supp will you administer? Round to nearest tenth.

STEP 1: CONVERT DATA

g → mg

2g = 2000 mg

REMEMBER **BIG TO SMALL:** move the decimal point 3 to the right

STEP 2: READY TO USE DATA

ORDERED: 2000 mg
AVAILABLE: 700 mg
VOLUME: 1 supp

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{2000\ \cancel{mg}}{700\ \cancel{mg}} = 2.857$$

REMEMBER RULE #4
Don't round till the end!

2.857 x 1 supp = 2.857 supp

ROUND: *Nearest tenth*

2.857 supp → 2.9 supp

FINAL ANSWER: 2.9 supp

COMPREHENSIVE REVIEW

3

ORDERED: Potassium chloride 0.525 mEq/lb PO dissolved in 6 oz of juice at 0930
AVAILABLE: Potassium chloride 12 mEq/mL
How many mL of potassium chloride will you add to the juice for a 66.75 kg patient? Round to nearest tenth.

STEP 1: CONVERT DATA

kg → lb

66.75 ~~kg~~ x 2.2 (lb/~~kg~~) = 146.85 lb

In this case, ordered amount depends on patient weight

mEq/lb → mEq

(0.525 mEq/~~lb~~ x 146.85 ~~lb~~ = 77.096 mEq)

STEP 2: READY TO USE DATA

ORDERED: 77.096 mEq
AVAILABLE: 12 mEq
VOLUME: 1 mL

STEP 3: IRRELEVANT DATA

Dissolved in 12 oz of juice at 0930

Question asked for "per dose" because no timeframe was given

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{77.096\ \cancel{mEq}}{12\ \cancel{mEq}} = 6.424$$

6.424 X 1 mL = 6.424 mL

REMEMBER RULE #4
Don't round till the end!

ROUND: *Nearest tenth*

6.424 mL → 6.4 mL

FINAL ANSWER: 6.4 mL

4

1000 mL D5W to infuse over 4 hours.

STEP 1: CONVERT DATA

N/A

STEP 2: READY TO USE DATA

1000 mL
4 hr

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{\text{mL of solution}}{\text{total hours}} = \text{mL/hr}$$

SHOW YOUR WORK

$$\frac{1000\text{ mL}}{4\text{ hr}} = 250\text{ mL/hr}$$

mL/hr is always rounded to the nearest **WHOLE** number!

ROUND: *No rounding necessary*

FINAL ANSWER: 250 mL/hr

COMPREHENSIVE REVIEW

5

150 mL Cipro 250 mcg to infuse over 45 minutes.

If the question is asking for flow rate ("to infuse") and you're given mL of solution, you need to write the answer in **mL/hours!**

STEP 1: CONVERT DATA

N/A

STEP 2: READY TO USE DATA

ML OF SOLUTION: 150 mL
TOTAL HOURS: 45 min

STEP 3: IRRELEVANT DATA

Cipro 250 mcg

IMPORTANT: don't let this information lead you to use the wrong formula. In this example, we're asked for a flow rate which requires mL of solution and total time.

STEP 4: FORMULA USED

$$\frac{\text{mL of solution}}{\text{total minutes}} \times 60 = \text{mL/hr}$$

SHOW YOUR WORK

$$\frac{150 \text{ mL}}{45 \text{ min}} = 3.333 \times 60 = 200 \text{ mL/hr}$$

REMEMBER RULE #4 Don't round till the end!

mL/hr is always rounded to the nearest **WHOLE** number!

ROUND: *No rounding necessary*

FINAL ANSWER: 200mL/hr

6

250 mL normal saline over 5 hours. Tubing drop factor of 10 gtt/mL.

STEP 1: CONVERT DATA

hr → min

1 hour = 60 minutes

$$5 \cancel{\text{hr}} \times \frac{60 \text{ min}}{1 \cancel{\text{hr}}} = 300 \text{ min}$$

STEP 2: READY TO USE DATA

ML OF SOLUTION: 250 mL
TOTAL MINUTES: 300 min
DROP FACTOR: 10 gtt/mL

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{\text{mL of IV solution}}{\text{time in minutes}} \times \text{drop factor} = \text{gtt/min}$$

SHOW YOUR WORK

$$\frac{250 \text{ mL}}{300 \text{ min}} = 0.8333 \text{ mL/min}$$

REMEMBER RULE #4 Don't round till the end!

$$0.8333 \cancel{\text{mL}}\text{/min} \times 10 \text{ gtt/}\cancel{\text{mL}} = 8.3333 \text{ gtt/min}$$

ROUND: *gtt/mL is always rounded to the nearest whole number!*

8.3333 gtt/min → 8 gtt/min

The question may not specify to round the final answer to a whole number; you are expected to know this with gtt/min units.

FINAL ANSWER: 8 gtt/min

COMPREHENSIVE REVIEW

7

Humulin R 200 units in 100 mL of normal saline to infuse at 4 units/hr.

STEP 1: CONVERT DATA

N/A

STEP 2: READY TO USE DATA

DESIRED: 4 units/hr
AVAILABLE: 200 units
VOLUME: 100 mL

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{4 \cancel{\text{units}}/\text{hr}}{200 \cancel{\text{units}}} = 0.02\ /\text{hr}$$

0.02 /hr x 100 mL = 2 mL/hr

ROUND: *No rounding necessary*

mL/hr is always rounded to the nearest **WHOLE** number!

FINAL ANSWER: 2 mL/hr

8

Dopamine 600 mg in 200 mL of normal saline to infuse at 10 mcg/kg/min.
Pt weight = 190 lbs.

If the question is asking for flow rate ("to infuse") and you're given mL of solution, you need to write the answer in **mL/hours!**

STEP 1: CONVERT DATA

mcg → mg

10 mcg = 0.010 mg

lb → kg

190 lb / 2.2 = 86.363 kg

REMEMBER

SMALL TO BIG:
move the decimal point 3 to the left
unit is getting Larger think Left

$$\frac{\text{mg/kg}}{\text{min}} \rightarrow \frac{\text{mg}}{\text{min}}$$

In this case, ordered amount depends on patient weight

0.010 mg/$\cancel{\text{kg}}$/min x 86.363 $\cancel{\text{kg}}$ = 0.863 mg/min

STEP 2: READY TO USE DATA

DESIRED: 0.863 mg/min
AVAILABLE: 600 mg
VOLUME: 200 mL

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{0.863 \cancel{\text{mg}}/\text{min}}{600 \cancel{\text{mg}}} = 0.00143\ /\text{min}$$

0.00143 /min x 200 mL = 0.2878 mL/min

0.2878 mL/$\cancel{\text{min}}$ x 60 $\cancel{\text{min}}$ = 17.2727 mL/hr

WAIT!

This is in mL/min ... we need units of mL/hr!

ROUND: *mL/hr is always rounded to nearest whole number!*

17.2727 mL/hr → 17 mL/hr

FINAL ANSWER: 17 mL/hr

9

2.5 L normal saline to infuse over 48 hours.

! If the question is asking for flow rate ("to infuse") and you're given mL of solution, you need to write the answer in **mL/hours!**

STEP 1: CONVERT DATA

L → mL

REMEMBER **BIG TO SMALL**: move the decimal point 3 to the right

2.5 L = 2500 mL

STEP 2: READY TO USE DATA

ML OF SOLUTION: 2500 mL
TOTAL HOURS: 48 hr

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{\text{mL of solution}}{\text{total hours}} = \text{mL/hr}$$

SHOW YOUR WORK

$$\frac{2500 \text{ mL}}{48 \text{ hours}} = 52.0833 \text{ mL/hr}$$

ROUND: *mL/hr is always rounded to nearest whole number!*

52.0833 mL/hr → 52 mL/hr

FINAL ANSWER: 52 mL/hr

10

ORDERED: Morphine 100 mg IM q12h prn pain
AVAILABLE: Morphine 150 mg/2.6 mL
How many mL will you administer?
Round to nearest hundredth.

STEP 1: CONVERT DATA

N/A

STEP 2: READY TO USE DATA

ORDERED: 100 mg
AVAILABLE: 150 mg
VOLUME: 2.6 mL

STEP 3: IRRELEVANT DATA

IM q12h prn pain

! Question asked for "per dose" because no timeframe was given

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{100 \cancel{\text{mg}}}{150 \cancel{\text{mg}}} = 0.6666$$

0.6666 x 2.6 mL = 1.7333 mL

ROUND: *nearest hundredth*

1.7333 mL → 1.73 mL

FINAL ANSWER: 1.73 mL

LAB VALUE CHEAT SHEET

WITH MEMORY TRICKS

BROUGHT TO YOU BY NURSE IN THE MAKING®

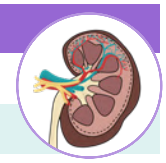
LAB VALUE CHEAT SHEET

VITAL SIGNS

BLOOD PRESSURE	SYSTOLIC	120 mmHg
	DIASTOLIC	80 mmHg
HEART RATE	60 – 100 bpm	
RESPIRATIONS	12 – 20 breaths/min	
TEMPERATURE	97.8 – 99°F (36.5 - 37.2°C)	
OXYGEN	95 – 100%	
OXYGEN IN COPD PT.	as low as 88%	

COPD pts are expected to have low O_2 levels

RENAL



CALCIUM	9 - 11 mg/dL
MAGNESIUM	1.5 - 2.5 mg/dL
PHOSPHORUS	2.5 - 4.5 mg/dL
SPECIFIC GRAVITY	1.010 - 1.030
GFR	90 - 120 mL/min/1.73 m^2
BUN	7 - 20 mg/dL
CREATININE	0.6 – 1.2 mg/dL

PANCREAS

AMYLASE	30 - 110 U/L
LIPASE	0 - 150 U/L

COMPLETE BLOOD COUNT (CBC)

WBCs	4,500 - 11,000 /µL
RBCs	4.5 - 5.5 X10^6 /µL
PLTs	150,000 - 450,000 /µL
HEMOGLOBIN (HGB)	FEMALE: 12 - 16 g/dL
	MALE: 13 - 18 g/dL
HEMATOCRIT (HCT)	FEMALE: 36% - 48%
	MALE: 39% - 54%

BASIC METABOLIC PANEL (BMP)

SODIUM	135 – 145 mEq/L
POTASSIUM	3.5 – 5.0 mEq/L
CHLORIDE	95 - 105 mEq/L
CALCIUM	9 - 11 mg/dL
BUN	7 - 20 mg/dL
CREATININE	0.6 – 1.2 mg/dL
ALBUMIN	3.4 - 5.4 g/dL
TOTAL PROTEIN	6.2 - 8.2 g/dL

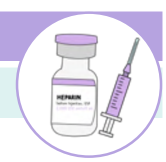
HBA1C

NON-DIABETIC	4 - 5.6%
PRE-DIABETIC	5.7 - 6.4%
DIABETIC	> 6.5%
Goal for diabetic:	< 7%

LIVER FUNCTION TEST (LFT)

ALT	7 - 56 U/L
AST	5 - 40 U/L
ALP	40 - 120 U/L
BILIRUBIN	0.1 - 1.2 mg/dL

COAGs



PT	10 - 13 sec
PTT	25 - 35 sec
aPTT	NOT ON HEPARIN: 30-40 secs
	ON HEPARIN: 47-70 secs
INR	NOT ON WARFARIN: < 1 sec
	ON WARFARIN: 2 - 3 sec

BMI

UNDERWEIGHT	<18.5
HEALTHY WEIGHT	18.5 - 24.9
OVERWEIGHT	25.0 - 29.9
OBESITY	> 30.0

ABGs

PH	7.35 - 7.45
$PaCO_2$	35 - 45 mmHg
PaO_2	80 - 100 mmHg
HCO_3	22 - 26 mEq/L

LIPID PANEL

TOTAL CHOLESTEROL	<200 mg/dL
TRIGLYCERIDE	<150 mg/dL
LDL	<100 mg/dL
HDL	>60 mg/dL

LDL bad cholesterol - we want LOW levels

HDL Happy cholesterol - we want HIGH levels

OTHER

MAP (mean arterial pressure)	70 - 100 mmHg
ICP (intracranial pressure)	5 - 15 mmHg
GLASGOW COMA SCALE	BEST = 15

MILD: 13-15 MODERATE: 9-12 SEVERE: 3-8

LAB VALUE MEMORY TRICKS

ELECTROLYTES

SODIUM: 135 - 145

*Commit to memory!

POTASSIUM: 3.5 - 5

BANANAS:
There are about 3-5 in every bunch & you want them half ripe (½)

So, think 3.5 - 5.0

PHOSPHORUS: 2.5 - 4.5

PHOR: 4
US: 2 (me + you = 2)

*don't forget the .5

CALCIUM: 9 - 11

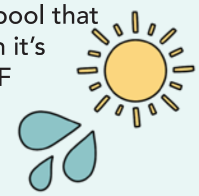
CALL 911

MAGNESIUM: 1.5 - 2.5

MAGnifying glass you see 1.5 - 2.5 bigger than normal

CHLORIDE: 95 -105

Think of a chlorinated pool that you want to go in when it's SUPER HOT: 95 - 105 °F



COMPLETE BLOOD COUNT (CBC)

- Hemoglobin (Hgb)
  Female: 12 - 16 g/dL
  Male: 13 - 18 g/dL
- Hematocrit (HCT)
  Female: 36% - 48%
  Male: 39% - 54%

To remember HCT, multiply Hgb by 3

12 X 3 = 36
16 X 3 = 48 *(Female)*

13 X 3 = 39
18 X 3 = 54 *(Male)*

BASAL METABOLIC PANEL (BMP)

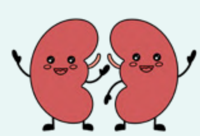
BUN: 7 - 20 mg/dL

Think hamburger **BUN**s...
Hamburgers can cost anywhere from $7 - $20 dollars

CREATININE: 0.6 – 1.2 mg/dL

This is the same value as **LITHIUM'**s therapeutic range (0.6 - 1.2 mmol/L)

Lithium is excreted almost solely by the kidneys...
And creatinine is a value that tests how well your kidneys filter



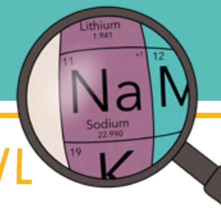
NOTES

It doesn't get
EASIER,
you just get
STRONGER!

ELECTROLYTE IMBALANCES

BROUGHT TO YOU BY NURSE IN THE MAKING

SODIUM (Na+) IMBALANCE



SODIUM is a major ELECTROLYTE found in ECF.
Essential for acid-base, fluid balance, active & passive transport mechanism, irritability & CONDUCTION of nerve-muscle tissue

135 - 145 mEq/L

> 145 mEq/L = HYPERNATREMIA

SIGNS & SYMPTOMS

BIG & BLOATED

MNEMONIC **"FRIED SALT"**

- **F** Flushed skin
- **R** Restless, anxious, confused, irritable
- **I** Increased BP & fluid retention
- **E** Edema (pitting)
- **D** Decreased urine output
- **S** Skin flushed & dry
- **A** Agitation
- **L** Low-grade fever
- **T** Thirst (dry mucous membranes)

RISK FACTORS

- Increased sodium intake
  - Excess oral sodium ingestion
  - Excess administration of IV fluids w/ sodium
  - Hypertonic IV fluids

- LOSS OF FLUIDS!
  - Fever
  - Watery diarrhea
  - Diabetes insipidus
  - Excessive diaphoresis
  - Infection

HEMOCONCENTRATION = INCREASED SODIUM!

- Decreased sodium excretion
  - Kidney problems

MANAGEMENT

- If due to fluid loss:
  - Administer IV infusions
- If the cause is inadequate renal excretion of sodium:
  - Give diuretics that promote sodium loss
- Restrict sodium & fluid intake as prescribed

< 135 mEq/L = HYPONATREMIA

SIGNS & SYMPTOMS

HYPOVOLEMIC HYPONATREMIA: ↓ of fluid & sodium

HYPERVOLEMIC HYPONATREMIA: ↑ body water that is greater than Na+

MNEMONIC **"SALT LOSS"**

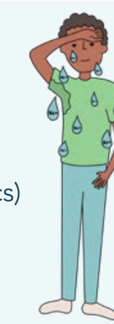
- **S** Stupor/coma
- **A** Anorexia (nausea/vomiting)
- **L** Lethargy (weakness/fatigue
- **T** Tachycardia (thready pulse)
- **L** Limp muscles (muscle weakness)
- **O** Orthostatic hypotension
- **S** Seizures/headache
- **S** Stomach cramping (hyperactive bowels)

RISK FACTORS

- Increased sodium excretion

4 D'S

  - **D**iaphoresis (ex: high fever)
  - **D**iarrhea & vomiting
  - **D**rains (NGT suction)
  - **D**iuretics (thiazide & loop diuretics)



- SIADH
- Adrenal insufficiency (adrenal crisis)
- Inadequate sodium intake
  - Fasting, NPO, Low-salt diet
- Kidney disease
- Heart failure

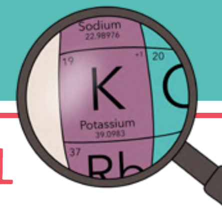
MANAGEMENT

MNEMONIC **"ADD SALT"**

- **A** **ADMINISTER** IV sodium chloride infusions (only if due to hypovolemia)
- **D** **DIURETICS** (If due to hypervolemia) Hyponatremia → high fluids & low salt = hemodilution
- **D** **DAILY WEIGHTS** Where sodium goes, water FLOWS
- **S** **SAFETY** (orthostatic hypotension AKA risk for falls)
- **A** **AIRWAY PROTECTION** (NPO) Don't give food to a lethargic, confused client (Increased Risk For Aspiration)
- **L** **LIMIT WATER INTAKE** Hypervolemic hyponatremia (high fluid & low salt)
- **T** **TEACH** about foods high in sodium (canned food, packaged/processed meats, etc.)

POTASSIUM & SODIUM = OPPOSITES **EXAMPLE: ↑ NA = ↓ K+**

POTASSIUM (K) IMBALANCE



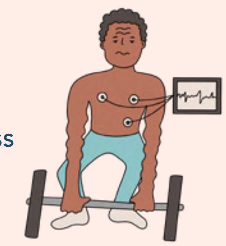
POTASSIUM imbalance plays a vital role in cell METABOLISM, and TRANSITION of nerve impulses, the functioning of cardiac, lung, muscle tissues, & acid-base balance.

3.5 - 5 mEq/L

> 5 meq/L = HYPERKALEMIA

SIGNS & SYMPTOMS

- **Muscles contract for TOO long = Tight & Contracted**



MNEMONIC "MURDER"

- **M** Muscle cramps & weakness
- **U** Urine abnormalities
- **R** Respiratory distress
- **D** Decreased cardiac contractility (↓HR, ↓BP)
- **E** ECG changes
  - Tall peaked T waves
  - Flat P waves
  - Widened QRS complexes
  - Prolonged PR intervals
- **R** Reflexes (↑ DTR)

RISK FACTORS

- Medication
  - Potassium-sparing diuretics (Spironolactone)
  - Ace inhibitors
  - NSAIDs
- Excessive potassium intake
  (*Example:* rapid infusion of potassium-containing IV solutions)
- Kidney disease or those on Dialysis
  - Decreased potassium excretion
- Adrenal insufficiency (Addison's disease)
- Tissue damage
- Acidosis
- Hyperuricemia
- Hypercatabolism

MANAGEMENT

- Monitor EKG
- Discontinue IV & PO potassium
- Initiate a potassium-restricted diet
- Potassium-excreting diuretics
- Prepare the client for dialysis
- Prepare for administration:
  - IV calcium gluconate & IV sodium bicarb
- Avoid the use of salt substitutes or other potassium-containing substances

< 3.5 meq/L = HYPOKALEMIA

SIGNS & SYMPTOMS

- Not enough contraction = Weak
- Thready, weak, irregular pulse
- Orthostatic hypotension
- Shallow respirations
- Anxiety, lethargy, confusion, coma
- Paresthesias
- Hyporeflexia
- Hypoactive bowel sounds (constipation)
- Nausea, vomiting, abdominal distention
- ECG changes
  - ST depression
  - Shallow or inverted T wave
  - Prominent U wave

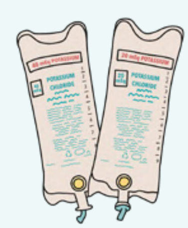
RISK FACTORS

- Actual total body potassium loss
- Inadequate potassium intake
  - Fasting, NPO
- Movement of potassium from the extracellular fluid to the intracellular fluid
  - Alkalosis
  - Hyperinsulinism
- Dilution of serum potassium
  - Water intoxication
  - IV therapy with potassium-deficient solutions

Potassium imbalance can cause **CARDIAC DYSRHYTHMIAS** that can be **LIFE-THREATENING!**

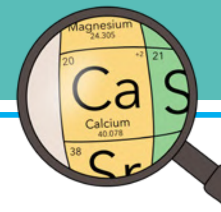
MANAGEMENT

- Oral potassium supplements
- Liquid potassium chloride
- Potassium-retaining diuretic
- Potassium is **NEVER** administered by IV push, IM, or subQ routes
  - IV potassium is always diluted & administered using an infusion device!



POTASSIUM & SODIUM = OPPOSITES — **EXAMPLE: ↑ NA = ↓ K+**

CALCIUM (Ca+) IMBALANCE



CALCIUM is found in the body's cells, bones, and teeth. Needed for proper functioning of the CARDIOVASCULAR, NEUROMUSCULAR, ENDOCRINE systems, blood clotting & teeth formation

9 - 11 mg/dL

SIGNS & SYMPTOMS

> 11 mg/dL = HYPERCALCEMIA

MNEMONIC "BACKME"

- **B** Bone pain
- **A** Arrhythmias
- **C** Cardiac arrest (bounding pulses)
- **K** Kidney stones
- **M** Muscle weakness ↓ (DTR)
- **E** Excessive urination

< 9 mg/dL = HYPOCALCEMIA

MNEMONIC "CATS GO NUMB"

- **C** Convulsions
- **A** Arrhythmias
- **T** Tetany
- **S** Spasms & stridor
- **GO NUMB** Numbness in fingers, face, limbs

POSITIVE TROUSSEAU'S: Carpal spasm caused by inflating a blood pressure cuff

CHVOSTEK'S SIGNS: Contraction of facial muscles w/ light tap over the facial nerve.

MEMORY TRICK: Think "C" for Cheesy smile

RISK FACTORS

Hypercalcemia

- Increased calcium absorption
- Decreased calcium excretion
- Kidney disease
- Thiazide diuretics
- Increased bone resorption of calcium
  - Hyperparathyroidism / Hyperthyroidism
  - Malignancy (bone destruction from metastatic tumors)
- Hemoconcentration

Hypocalcemia

- Inhibition of calcium absorption from the GI tract
- Increased calcium excretion
  - Kidney disease, diuretic phase
  - Diarrhea & steatorrhea
  - Wound drainage
- Conditions that decrease the ionized fraction of calcium

A client with a calcium imbalance is at risk for a **PATHOLOGICAL FRACTURE.** Move the client carefully and slowly

MANAGEMENT

Hypercalcemia

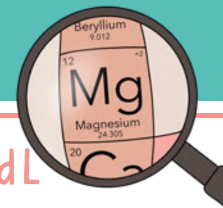
- D/C IV or PO calcium
- D/C Thiazide diuretics
- Administer phosphorus, calcitonin, bisphosphonates, & prostaglandin synthesis inhibitors (NSAIDs)
- Avoid foods high in calcium

Hypocalcemia

- Adm. calcium PO or IV
  - For IV, warm before & adm. slowly
- Adm. aluminum hydroxide & Vit D
- Initiate seizure precautions
- 10% calcium (acute calcium deficit)
- Consume foods high in calcium

CALCIUM & PHOSPHATE = INVERSE EXAMPLE: ↑ CA+ = ↓ PO_4

MAGNESIUM (Mg) IMBALANCE



Most of the MAGNESIUM found in the body is found in the BONES.
Regulates BP, blood sugar, muscle contraction & nerve function.

1.5 - 2.5 mg/dL

MEMORY TRICK: Magnesium is a **SEDATIVE!**

> 2.5 mg/dL = HYPERMAGNESEMIA

SIGNS & SYMPTOMS

LOW (↓) everything, AKA SEDATED

- ↓ energy (drowsiness / coma)
- ↓ HR (bradycardia)
- ↓ BP (hypotension)
- ↓ RR (bradypnea)
- ↓ Respirations (shallow)
- ↓ Bowel sounds
- ↓ DTRs (deep tendon reflex)

RISK FACTORS

- Increased magnesium intake
  - Magnesium-containing antacids (TUMS) & laxatives
  - Excessive adm. of magnesium IV
- Renal insufficiency
  - ↓ renal excretion of Mg = ↑ Mg in the blood
- DKA (Diabetic Ketoacidosis)

MANAGEMENT

- Diuretics
- IV adm. calcium chloride or calcium gluconate
- Restrict dietary intake of Mg containing foods
- Avoid the use of laxatives & antacids containing magnesium
- Hemodialysis

< 1.5 mg/dL = HYPOMAGNESEMIA

SIGNS & SYMPTOMS

HIGH (↑) everything, AKA NOT SEDATED

- ↑ HR (tachycardia)
- ↑ BP (hypertension)
- ↑ deep tendon reflex (hyperreflexia)
- Shallow respirations
- Twitches, paresthesias
- Tetany & seizures
- Irritability & confusion

REMEMBER: Also seen in hypocalcemia. Ca & Mg rise and fall together!

POSITIVE TROUSSEAU'S: Carpal spasm caused by inflating a blood pressure cuff

CHVOSTEK'S SIGNS: Contraction of facial muscles w/ light tap over the facial nerve.

MEMORY TRICK: Think "C" for Cheesy smile

RISK FACTORS

- Insufficient magnesium intake
  - Malnutrition/vomiting/diarrhea
  - Malabsorption syndrome
  - Celiac & Crohn's disease
- Increased magnesium excretion
  - Diuretics or chronic alcoholism
- Intracellular movement of magnesium
  - Hyperglycemia & Insulin adm.
  - Sepsis

MANAGEMENT

- Magnesium sulfate IV or PO
- Seizure precautions
- Instruct the client to increase magnesium-containing foods

MAGNESIUM & CALCIUM = SAME EXAMPLE: ↑ MG = ↑ CA+

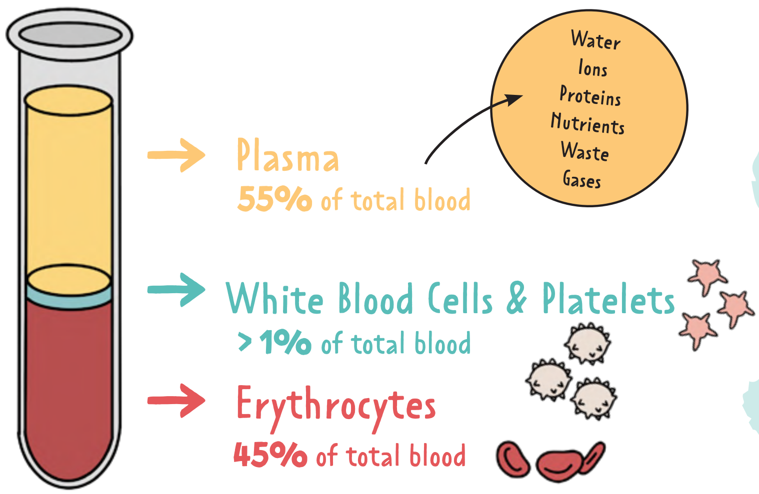
NOTES

You are
CLOSER THAN
you were
YESTERDAY.

FUNDAMENTALS

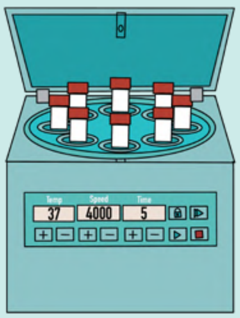
BLOOD TYPES

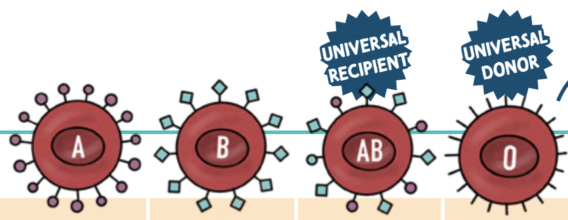
Before a blood transfusion happens, a patient's blood should be sent to the lab to be typed & cross-matched.
If a patient receives blood that is not a compatible type, it can lead to a transfusion reaction and potentially death.



CENTRIFUGE

A device that uses force to separate components of fluids. It separates fluids of different densities. This is how labs separate blood.





MEMORY TRICK: O think universal dOnOr

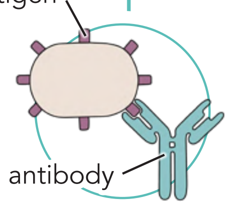
	A	B	AB	O
ANTIGEN	A	B	A&B	NONE
ANTIBODIES	B	A	NONE	A&B
RECIPIENT	A, O	B, O	ALL	O
DONOR	A, AB	B, AB	AB	ALL

A person who can receive blood of any type

Compatible with any blood type

ANTIGENS
- Proteins that elicit immune responses
- Identifies the cell

PLASMA ANTIBODIES
- Protects body from "invaders" (think ANTI)
- Opposite of the type of antigen that is found on the RBC



RH FACTOR

Rhesus (Rh) factor is an inherited protein found on the surface of red blood cells.

If your blood has the protein, you're Rh **positive**.

If your blood lacks the protein, you're Rh **negative**.

+	–
HAS Rh on surface.	**DOES NOT HAVE** Rh on surface.
CAN RECEIVE + –	**CAN RECEIVE** –

DONOR BLOOD TYPES

RECIPIENT BLOOD TYPES

	O-	O+	A-	A+	B-	B+	AB-	AB+
O-	♥							
O+	♥	♥						
A-	♥		♥					
A+	♥	♥	♥	♥				
B-	♥				♥			
B+	♥	♥			♥	♥		
AB-	♥		♥		♥		♥	
AB+	♥	♥	♥	♥	♥	♥	♥	♥

Always check with your hospital's protocol about blood product administration

ABBREVIATIONS

AAA Abdominal Aortic Aneurysm
Abd Abdomen
Ac Before Meals
ACLS Advanced Cardiac Life Support
AD Admitting Diagnosis
A&D Admission and Discharge
Ad lib As Desired
ALL Acute Lymphocytic Leukemia
ADL Activities of Daily Living
Adm. Admission
Amb Ambulation
AKA Above-the-Knee Amputation
AV Atrioventricular
AP or **A.P.** Appendectomy
Bid Twice a Day
BLS Basic Life Support
BM Bowel Movement
BP Blood Pressure
BKA Below-the-Knee Amputation
BUN Blood Urea Nitrogen
BPH Benign Prostatic Hyperplasia
BX Biopsy
CABG Coronary Artery Bypass Graft
C/O Complaining Of
CAD Coronary Artery Disease
CBC Complete Blood Count
CCU Cardiac Care Unit / Coronary Care Unit
C&S Culture & Sensitivity
CF Cystic Fibrosis
CHF Congestive Heart Failure
CKD Chronic Kidney Disease
CPR Cardiopulmonary Resuscitation
COPD Chronic Obstructive Pulmonary Disease
CVA Cerebrovascular Accident (stroke)
CVC Central Venous Catheter
D/C Discontinue or Discharge
D&C Dilatation and Curettage
DI Diabetes Insipidus
DIC Disseminated Intravascular Coagulation
DKA Diabetic Ketoacidosis
DM Diabetes Mellitus
DVT Deep Vein Thrombosis
DX Diagnosis
ECG or **EKG** Electrocardiogram
ED Emergency Department
EENT Eye, Ears, Nose and Throat
ETT Endotracheal Tube
FBS Fasting Blood Sugar
Fx Fracture
Gtt or **G.T.T.** Glucose Tolerance Test
HOB Head of Bed
HS Bedtime
Hx History
ICU Intensive Care Unit
LMP Last Menstrual Period
LOC Level of Consciousness
LES Lower Esophageal Sphincter
LP Lumbar Puncture
I&O Intake and Output
MAP Mean Arterial Pressure
MRI Magnetic Resonance Imaging
MVA Motor Vehicle Accident
NGT Nasogastric Tube
NPO Nothing by Mouth
NKA No Known Allergies
O_2 Oxygen
OB Obstetrics
OOB Out of Bed
OR Operating Room
OA Osteoarthritis
Ortho Orthopedics
OT Occupational Therapist
Pc After Meals
Prn or **p.r.n.** As Needed
Pre op Before Surgery
PFT Pulmonary Function Test
PLT Platelets
PTCA Percutaneous Transluminal Coronary Angioplasty
PRBC Packed Red Blood Cells
PVC Premature Ventricular Contraction
Rom/R.O.M. Range of Motion
RBC Red Blood Cell
RT Respiratory Therapist
RA Rheumatoid Arthritis
SOB Shortness of Breath
SBAR Situation, Background, Assessment, Recommendation
SSE or **S.S.E.** Soap Suds Enema
Stat At Once, Immediately
SLE Systemic Lupus Erythematosus
STD Sexually Transmitted Disease
SIADH Syndrome of Inappropriate Antidiuretic Hormone Secretion
Tid Three Times a Day
T&S Type and Screen
TPN Total Parenteral Nutrition
TIA Transient Ischemic Attack
TB Tuberculosis
TURP Transurethral Resection of the Prostate
UA Urinalysis
UTI Urinary Tract Infection
US Ultrasound
VS Vital Signs
WBC White Blood Count
WNL Within Normal Limits

DO NOT USE	POTENTIAL PROBLEM	INSTEAD, WRITE:
U	Mistaken for "0" (zero) or "cc"	unit
IU	Mistaken for IV (intravenous) or the number 10 (ten)	"international unit"
Q.D., QD, q.d., qd, Q.O.D.,QOD, q.o.d, qod	Mistaken for each other	"daily" or "every other day"
Trailing zero (X.0 mg) Lack of leading zero (.X mg)	Decimal point is missed	"X mg" "0.X mg"
MS, MSO4, MgSO4	Can mean morphine sulfate or magnesium sulfate	"morphine sulfate" "magnesium sulfate"
@	Mistaken for the number "2" (two)	"at"
cc	Mistaken for U (units) when poorly written	"mL" or "milliliters"

THE NURSING PROCESS

MNEMONIC

"A DELICIOUS PIE"

SUBJECTIVE DATA

What the client tells the nurse

OBJECTIVE DATA

Data the nurse obtains through their assessment & observation

ASSESS

- Gather information
- Verify the information collected is clear & accurate

DIAGNOSE

- **Interpret the information collected**
- Identify & prioritize the problem through a nursing diagnosis (be sure it's NANDA approved)

PLAN

- **Set goals to solve the problem**
- Prioritize the outcomes of care

IMPLEMENT

- Reaching those goals through performing the nursing actions
- "Implementing" the goals set above in the planning stage

EVALUATE

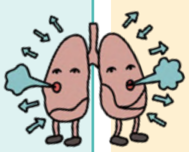
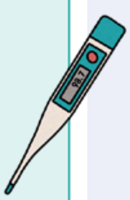
- Determine the outcome of goals
- Evaluate client's compliance
- Document client's response to pain
- Modify & assess for needed changes

Set SMART Goals

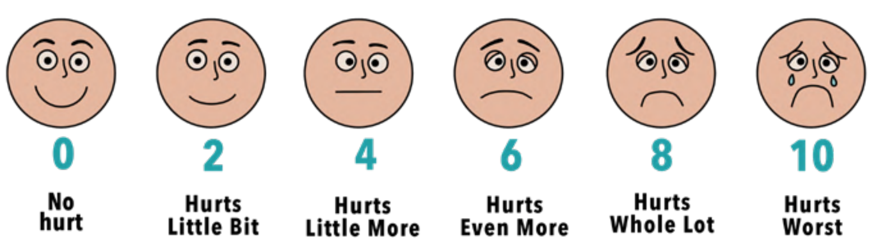
MNEMONIC

Specific
Measurable
Achievable
Relevant
Time frame

VITAL SIGNS

 **BLOOD PRESSURE (BP)**	**SYSTOLIC** 120 mmHg **DIASTOLIC** 80 mmHg	Hypotension = low blood pressure Hypertension = high blood pressure
 **HEART RATE (HR)**	60 – 100 bpm	Bradycardia = <60 bpm Tachycardia = >100 bpm
 **RESPIRATION RATE (RR)**	12 – 20 breaths/min	Bradypnea = <12 breaths/min Tachypnea = >20 breaths/min
 **TEMPERATURE (T)**	97.8 – 99°F (36.5 - 37.2°C)	Hypothermia = <95 °F (<35 °C) Hyperthermia = >104 °F (>40 °C)
 **OXYGEN (O_2)**	95 – 100%	Low oxygen levels = hypoxemia
PAIN	Pain is subjective data given to you by the patient	Can be measured in various ways: The numerical scale, Wong-Baker Faces®, or verbal rating scale

Wong-Baker FACES® Pain Rating Scale



PRIORITY QUESTIONS

! You know you are being asked a **PRIORITY QUESTION** when the question asks:

- What is the **most important** ?
- What is the **initial response** ?
- Which action should the nurse take **first** ?

When you see these questions, you should immediately think of **MASLOW'S HIERARCHY OF NEEDS** and **ABCs**!

ABCs

A Airway　**B** Breathing　**C** Circulation

#1 Patent Airway

Patent means "open"; the airway is clear!

ASK YOURSELF:

Can they successfully breathe oxygen in and breathe CO_2 out?

#2 Breathing

Gas exchange taking place inside the lungs

ASK YOURSELF:

Can gas exchange successfully happen in their lungs?

#3 Circulation

Can they circulate blood through their body and are their organs being perfused?

ASK YOURSELF:

Is there a reason that the blood isn't pumping/circulating in the body?

(Example: The heart is working to pump the blood to the vital organs)

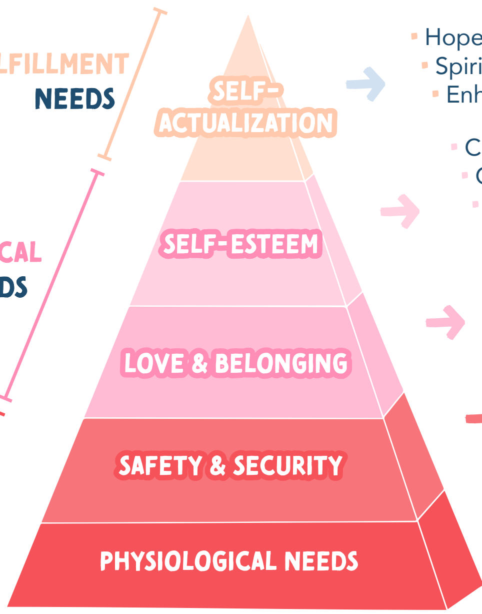
MASLOW'S HIERARCHY OF BASIC NEEDS

This shows the **5** levels of human needs, **PHYSIOLOGICAL NEEDS** being the most important (oxygen, fluids, nutrition, shelter). **ABCs** fall into Maslow's **PHYSIOLOGICAL** need!

SELF-FULFILLMENT NEEDS

PSYCHOLOGICAL NEEDS

BASIC NEEDS



Self-actualization:
- Hope
- Spiritual well-being
- Enhanced growth

Self-esteem:
- Control
- Competence
- Positive regard
- Acceptance/worthiness

Love & belonging:
- Maintain support systems
- Protect from isolation

Safety & security:
- Protection from injury
- Promote feeling of security
- Trust in nurse-client relationship

Physiological needs:
- Airway
- Respiratory effort
- Heart rate, rhythm, and strength of contraction
- Nutrition
- Elimination

NCLEX TIP

Pain is considered "**psychological**" meaning it does not take priority. *Pain rarely kills people

NURSING ETHICS & LAW

ETHICAL PRINCIPLES

AUTONOMY

Respect for an individual's right to make their own decisions

NONMALEFICENCE

Obligation to do & cause no harm to others

BENEFICENCE

Duty to do good to others

JUSTICE

Distribution of benefits & services fairly

VERACITY

Obligation to tell the truth

FIDELITY

Following through with a promise

HIPAA

THE HEALTH INSURANCE PORTABILITY & ACCOUNTABILITY ACT

- Clients records are private & they have the right to ensure the medical information is not shared without permission
- All health care professionals must inform the client how their health information is used
- The client has the right to obtain a copy of their personal health information

PATIENT RIGHTS

THE RIGHT TO...

- Privacy
- Considerate & respectful care
- Be informed
- Know the names & roles of the persons who are involved in care
- Consent or refuse treatment
- Have an advance directive
- Obtain their own medical records & results

CONSENT

TYPES OF CONSENT:

- Admission agreement
- Immunization consent
- Blood transfusion consent
- Surgical consent
- Research consent
- Special consents

- Treatment can not be done without a client's consent
- In the case of an emergency when a client cannot give consent, then consent is implied through emergency laws
- Minors (under 18), consent must be obtained from a parent or legal guardian

!

Before signing the consent, the client must be informed of the following: risks & benefits of surgery, treatments, procedures, & plan of care in layman's terms so the client understands clearly what is being done.

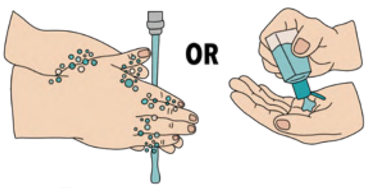
INFECTION CONTROL

PPE → PERSONAL PROTECTIVE EQUIPMENT

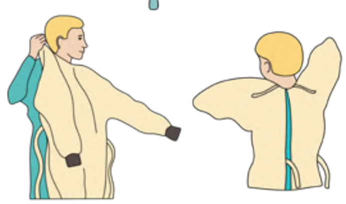
DONNING
Putting on PPE

- Put on PPE before entering the client's room
- Do not touch your face while wearing PPE
- Minimize contact with items in the client's room

1 HAND HYGIENE



2 GOWN



3 MASK / RESPIRATOR

4 GOGGLES / FACE SHIELD

5 GLOVES

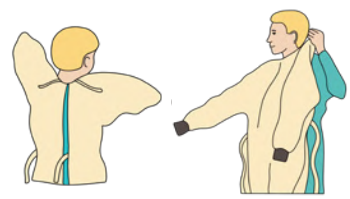
DOFFING
Removing PPE

- Remove PPE at the client's doorway or outside the room
- If hands become soiled while removing PPE, stop & perform hand hygiene
- After hand hygiene, continue with PPE removal

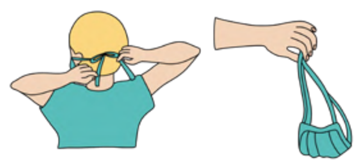
1 REMOVE GLOVES

2 REMOVE PROTECTIVE EYEWEAR

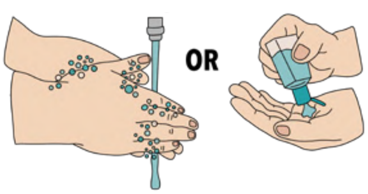
3 REMOVE GOWN



4 REMOVE & DISCARD RESPIRATOR



5 PERFORM HAND HYGIENE



HOSPITAL-ASSOCIATED INFECTIONS (HAIs)

CAUTI........ Catheter-associated urinary tract infection
SSI .... Surgical site infection
CLABSI...... Central line-associated blood infection
VAP .... Ventilator-associated pneumonia

Meticulous hand hygiene practices and use of chlorhexidine washes helps in preventing HAIs

INFECTION CONTROL

CHAIN of INFECTION

CAUSATIVE AGENT
- Bacteria
- Virus
- Fungus
- Prion
- Parasite

RESERVOIR
- Human
- Animal
- Surfaces
- Food
- Soil
- Insects

PORTAL OF EXIT
- Skin (wound)
- Mouth (vomit, saliva)
- Blood (cuts on the skin)
- Respiratory tract

MODE OF TRANSMISSION
- Contact
- Droplet
- Airborne
- Vector borne

PORTAL OF ENTRY
- How it gets to the host
- Same as portal of exit

SUSCEPTIBLE HOST
Leaves the host more susceptible to infections

STAGES OF INFECTION

INCUBATION
Interval between the pathogen entering the body & the presentation of the first symptom

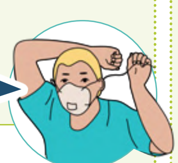
PRODROMAL STAGE
Onset of general symptoms to more distant symptoms; the pathogen is multiplying

ILLNESS STAGE
Symptoms specific to the infection appear

CONVALESCENCE
Acute symptoms disappear and total recovery could take days to months

TRANSMISSION BASED PRECAUTIONS

AIRBORNE
- Single room under negative pressure
- Door remains closed
- Health care workers wear a respiratory mask (N95 or higher level)



MNEMONIC Think "MTV"

Measles
Tuberculosis
Varicella (Chickenpox) & Disseminated herpes-zoster (Shingles)

*Airborne precaution is no longer needed when all lesions have crusted over

DROPLET
- Private room or a client whose body cultures contain the same organism
- Wear a surgical mask
- Place a mask on the client whenever they leave the room

- **Adenovirus**
- **Diphtheria (pharyngeal)**
- **Epiglottitis**
- **Influenza (flu)**
- **Meningitis**
- **Mumps**
- **Parvovirus B19**
- **Pertussis**
- **Pneumonia**
- **Rubella**
- **Scarlet fever**
- **Streptococcal pharyngitis**

CONTACT
- Private room or cohort client
- Use gloves & a gown whenever entering the client's room

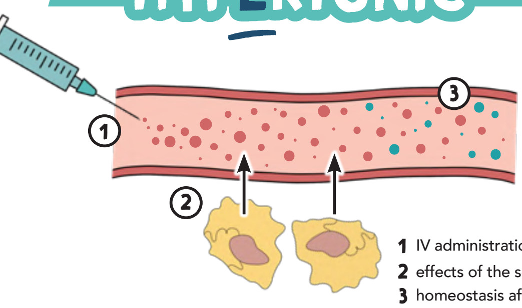
- **Colonization or infection with a multidrug-resistant organism**
- **Enteric infections** (Clostridium difficile)
- **Respiratory infections** (RSV, Influenza)
- **Wound & skin infections** (cutaneous diphtheria, herpes simplex, impetigo, pediculosis, scabies, staphylococci, & varicella-zoster)
- **Eye infections** (conjunctivitis)

When in contact with C. Diff, patient's hands must be washed with *soap & water* when performing hand hygiene

IV THERAPY: TYPES OF IV SOLUTIONS

HYPERTONIC

MEMORY TRICK "ENTER THE VESSEL FROM THE CELLS"



1 IV administration
2 effects of the solution
3 homeostasis after

EXAMPLES:
5% saline
3% saline
5% dextrose in 0.9% saline (D5NS)
5% dextrose in 0.45% saline (D5 ½ NS)
5% dextrose in LR (D5LR)
10% dextrose in water (D10W)

USED FOR:
Cerebral edema
Hyponatremia
(low levels of sodium)
Metabolic alkalosis
Maintenance fluid
Hypovolemia

MONITOR FOR:
Fluid Volume Overload

MORE **SALT** in the solution,
LESS **WATER** in the solution. The vessel becomes MORE concentrated than the cell.
Water then **LEAVES** the cell.
Therefore, the cells will **SHRINK**.

MEMORY TRICK
HYPERtonic think
HIGH numbers

*The only exception to this memory trick is
5% DEXTROSE IN WATER (D5W)

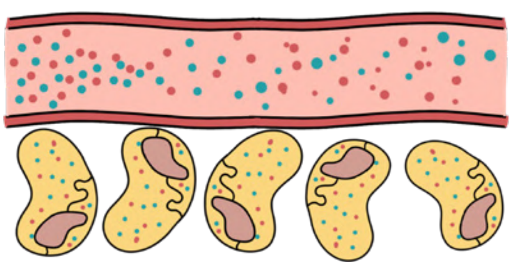
→ **5% DEXTROSE IN WATER (D5W)** starts as **ISOTONIC** and then changes to **HYPOTONIC** when the dextrose is metabolized.

ISOTONIC

ISO means EQUAL

MEMORY TRICK "STAYS WHERE I PUT IT"

EXPANDS intravascular fluid volume & replaces fluid loss



Same osmolality as body fluids
(Equal water & particle ratio)

EXAMPLES:
0.9% sodium chloride (NS) (normal saline)
5% dextrose in water (D5W)*
Lactated Ringers (LR)

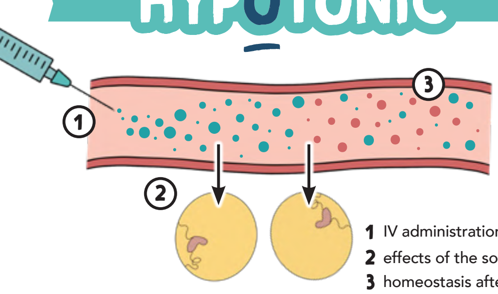
Use with **BLOOD PRODUCTS**

USED FOR:
Blood loss
(hemorrhage, burns, surgery)
Dehydration
(vomiting & diarrhea)
Fluid maintenance

NORMAL SALINE
is the only solution compatible to use with blood or blood products

HYPOTONIC

MEMORY TRICK "GO OUT OF THE VESSEL" + INTO THE CELL



1 IV administration
2 effects of the solution
3 homeostasis after

EXAMPLES:
0.45% saline (1/2 NS)
0.33% saline (1/3 NS)
0.225 saline (1/4 NS)
5% dextrose in water (D5W)*

USED FOR:
Diabetic ketoacidosis (DKA)
Helps kidneys excrete excess fluids
Hypernatremia
(high levels of sodium)

In DKA, there is so much glucose in the cells, they need water!

DO NOT GIVE WITH:
↑ ICP
Burns
Trauma

LESS **SALT** in the solution, MORE **WATER** in the solution.
The vessel becomes LESS concentrated than the cell.
Water then **ENTERS** the cell. Therefore, the cells will **SWELL**.

IV THERAPY: BASICS

Fluid in our body is found in **2** places:

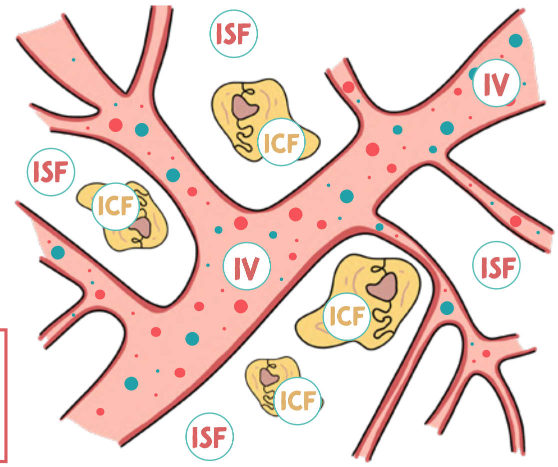
INTRACELLULAR & **EXTRACELLULAR**

(ICF)
fluid INSIDE the cell
(Millions of these cells in our body)

(ECF)
fluid OUTSIDE the cell

INTERSTITIAL FLUID (ISF)
fluid that surrounds the cell
AKA fluid in the tissues

INTRAVASCULAR (IV)
plasma/fluid in
the blood vessels



THE CELLS & HOMEOSTASIS

The cells love to have everything equal (homeostasis).
But when fluids/solutes shift, **DIFFUSION/OSMOSIS**
occurs to get back to homeostasis again.

DIFFUSION

the movement of a **SOLUTE** from a
HIGHER
concentration
to a
LOWER
concentration
(until there is equal concentration)

TIP
Sodium is a solute!

OSMOSIS

the movement of **WATER** through a semipermeable membrane from a
LOWER
solute concentration
to a
HIGHER
solute concentration
(until there is equal concentration)

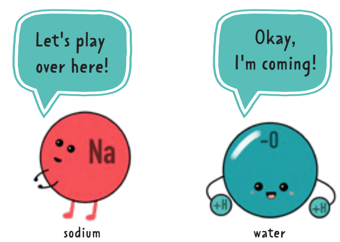
said another way...
from a
HIGHER
water concentration
to a
LOWER
water concentration
(until there is equal concentration)

SODIUM & WATER

MEMORY TRICK

WHERE SODIUM GOES WATER FLOWS!

Sodium is the cool kid,
so water wants to be his friend.



EXAMPLE: *If sodium shifts into the cell (intracellular space) water will follow and leave the extracellular space (the vessel)*

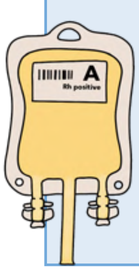
COLLOIDS & CRYSTALLOIDS

COLLOIDS

PLASMA EXPANDERS!

Large molecules

Colloids have **LARGE** molecules making it more efficient at increasing fluid volume in the blood.



EXAMPLES:
- Albumin
- Fresh frozen plasma (FFP)

USED FOR:
- Shock
- Pancreatitis
- Burns
- Excessive bleeding

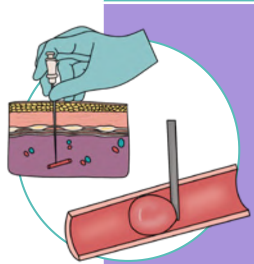
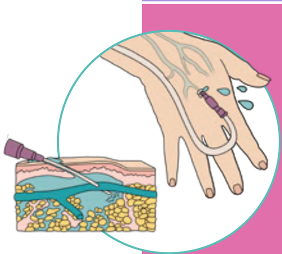
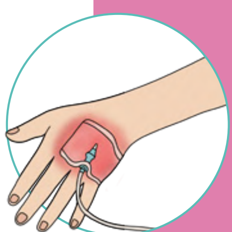
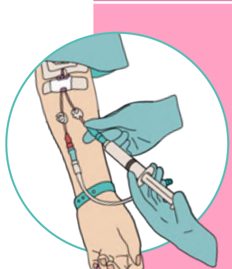
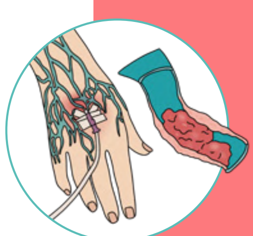
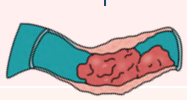
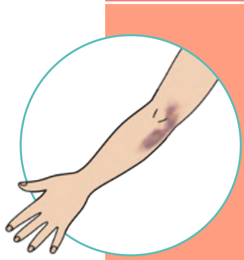
CRYSTALLOIDS

Small molecules

Crystalloids have **SMALL** molecules.
They are less expensive than colloids and provide immediate fluid resuscitation.

EXAMPLES:
- Hypertonic solution
- Isotonic solution
- Hypotonic solution

IV THERAPY: COMPLICATIONS

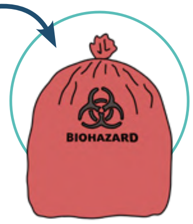
	PATHOLOGY	SYMPTOMS	TREATMENT
 AIR EMBOLISM	Air enters the vein through the IV tubing	▪ Tachycardia ▪ Chest pain ▪ Hypotension ▪ ↓ LOC ▪ Cyanosis	▪ Clamp the tubing ▪ Turn client on the left side & place in Trendelenburg position ▪ Notify the HCP
 INFILTRATION	IV fluid leaks into surrounding tissue	AT THE SITE: ▪ Pain ▪ Swelling ▪ Coolness ▪ Numbness ▪ No blood return	▪ Remove the IV ▪ Elevate the extremity ▪ Apply a warm or cool compress ▪ Do not rub the area
 INFECTION	Entry of microorganism into the body via IV	▪ Tachycardia ▪ Redness ▪ Swelling ▪ Chills & fever ▪ Malaise ▪ Nausea & vomiting	▪ Remove the IV ▪ Obtain cultures ▪ Possible antibiotic administration
 CIRCULATORY OVERLOAD	Administration of fluids too rapidly (Fluid Volume Overload)	▪ ↑ blood pressure ▪ Distended neck veins ▪ Dyspnea ▪ Wet cough & crackles	▪ ↓ flow rate (keep-vein-open rate) ▪ Elevate the head of the bed ▪ Keep the client warm ▪ Notify the HCP
 PHLEBITIS	Inflammation of the vein Can lead to a **clot** (thrombophlebitis) 	AT THE SITE: ▪ Heat ▪ Redness ▪ Tenderness ▪ ↓ flow of IV	▪ Remove the IV ▪ Notify the HCP ▪ Restart the IV on the opposite side
 HEMATOMA	Collection of blood in the tissues	AT THE SITE: ▪ Blood ▪ Hard & painful lump ▪ Ecchymosis	▪ **ELEVATE** the extremity ▪ Apply pressure & ice

BLOOD TRANSFUSIONS

ADMINISTRATION OF BLOOD TRANSFUSION

1. Insert an IV line using a 16g*, 18g, or 20g IV needle
   *commonly used for trauma patients
2. Run it with normal saline 0.9% (keep-vein-open-rate)
   Blood is transfused with a special Y-tubing with an inline-filter
3. Begin the transfusion slowly
   - A. The first 15 min are the **MOST CRITICAL,** the RN must stay at bedside
   - B. Vital signs are monitored every 30 min - 1 hr
   - C. After 15 min, the flow can be increased (unless a transfusion reaction has occurred)
4. Dispose the bag into a red biohazard bag
5. Document the patient's tolerance to the administration of the blood product

If you use too small of a needle (i.e. 24 gauge needle) when administering blood products, it can cause the blood to **LYSIS**.



FACTS ABOUT BLOOD TRANSFUSIONS

- Administered by the RN
- Only normal saline (NS) can be used in conjunction with blood
- Type & screen and a cross match are good for **72 HOURS**

Blood must be hung (started) within **30 MINUTES** from the time the blood is picked up from the blood bank

All blood must be transfused within **4 HOURS** of the time the blood was hung (started)

STOP the transfusion if you suspect a **TRANSFUSION REACTION**

TRANSFUSION REACTION

A transfusion reaction is an adverse reaction that happens as a result of receiving blood transfusions

IMMEDIATE TRANSFUSION REACTION

Chills, diaphoresis, aches, chest pain, rash, hives, itching, swelling, dyspnea, cough, wheezing, or rapid, thready pulse

CIRCULATORY OVERLOAD

Infusion of blood too rapid for the client to tolerate
Cough, dyspnea, chest pain, headache, hypertension, tachycardia, bounding pulse, distended neck vein, wheezing

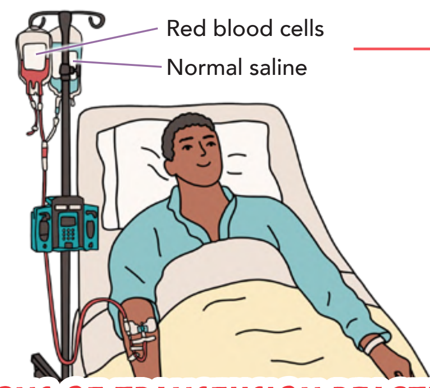
SEPTICEMIA

Blood that is contaminated with microorganisms
Rapid onset of chills, high fever, vomiting, diarrhea, hypotension & shock

IRON OVERLOAD

Complication that occurs in client's who receive multiple blood transfusions
Vomiting, diarrhea, hypotension, altered hematological values

Always check with your hospital's protocol about IV and blood product administration



SIGNS OF TRANSFUSION REACTIONS

- Fast heart rate
- Itching/urticaria/skin rash
- Wheezing/dyspnea/tachypnea
- Anxiety
- Flushing/fever
- Back pain

NURSING ACTIONS:

1. **STOP** the transfusion
2. Change the IV tubing down to the IV site
3. Keep the IV open w/ normal saline
4. Notify the HCP & blood bank
5. Do not leave the patient alone (monitor the patient's vital signs & continue to assess)

MEDICATION ADMINISTRATION

6 RIGHTS OF MED ADMIN

- RIGHT **CLIENT**
- RIGHT **TIME**
- RIGHT **DOSE**
- RIGHT **MED**
- RIGHT **ROUTE**
- RIGHT **DOCUMENTATION**

TYPES OF ORDERS

ROUTINE
Given on a regular schedule with or without a termination date.

SINGLE "ONE-TIME"
Used for a single case.
Not a routine medication.

STAT
Only for administration once and given immediately.

PRN
"As needed" must have an indication for use such as pain, nausea & vomiting.

COMMON MEDICATION ERRORS

Medication error kills, prevention is crucial!

- ✘ Wrong medication
- ✘ Incorrect dose
- ✘ Wrong...
  - Client
  - Route
  - Time
- ✘ Administer a medication the client is allergic to
- ✘ Incorrect D/C of Medication
- ✘ Inaccurate prescribing

SCOPE OF PRACTICE

RN

- Post-op assessment
- Initial client teaching
- Starting blood products
- Sterile procedures
- IVs & IV medications
- Discharge education
- Clinical assessment

NOTE:

When a registered nurse delegates tasks to others, responsibility is transferred but accountability for patient care is not transferred.
The RN is still responsible!

LPN/LVN

- Stable client
- Monitor RN's findings & gather data
- Specific assessments
- Reinforce teaching
- Routine procedures (catheterization, ostomy care, wound care)
- Monitors IVFs & blood products
- Administer injections & narcotics (not IVs meds & 1st IV bag)
- Tube potency & enteral feedings
- Sterile procedures

SPECIFIC ASSESSMENTS

Lung sounds, bowel sounds, & neurovascular checks

UAP

- Routine, stable vital signs
- Documenting input and output
- Can get blood from the blood bank
- Activities of daily living (ADLs)

ADLS

- **Feeding** (not with aspiration risk)
- **Positioning**
- **Ambulation**
- **Cleaning**
- **Linen change**
- **Hygiene care**

RN = Registered Nurse **LPN** = Licensed Practical Nurse **LVN** = Licensed Vocational Nurse **UAP** = Unlicensed Assistive Personnel

PHARMACOKINETICS

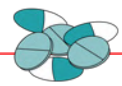
PHARMACOKINETICS: The study of how drugs are moved throughout the body

MEMORY TRICK: "ADME" some medications

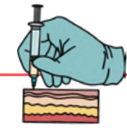
A — ABSORPTION

Medication going from the location of administration to the bloodstream

ORAL
Takes the longest to absorb



SUBQ & IM
Depends on the site of blood perfusion.
More blood perfusion = rapid absorption



IV
Quickest absorption time

D — DISTRIBUTION

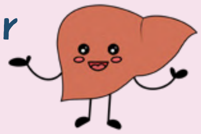
Transportation by bodily fluids of the medication to where it needs to go

INFLUENCING FACTORS:
- Circulation
- Permeability of the cell membrane
- Plasma protein binding

M — METABOLISM

How is the medication going to be broken down?

MOST COMMON SITE: Liver

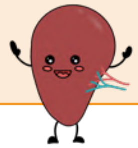


INFLUENCING FACTORS:
- Age (Infants & elderly have a limited med-metabolizing capacity)
- Medication type
- First-pass effect
  A drug given orally gets metabolized and its effects are greatly reduced before it reaches the systemic circulation. It's generally related to the liver or gut. It may need to be administered via parenteral route (subQ, IM, or IV) because this route bypasses the liver and gut.
- Nutritional status

E — EXCRETION

How is the medication going to be eliminated from the body?

MOST COMMONLY DONE BY: Kidneys



INFLUENCING FACTORS:
- Kidney dysfunction
  Leads to an increase in the duration and intensity of a medication response

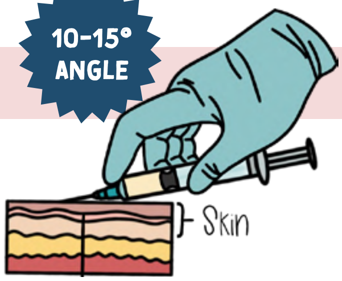
If the kidneys aren't working/excreting waste, the medication will stay in the body which leads to toxic levels

PARENTERAL ADMINISTRATION

Any route of administration that does not involve drug absorption through the GI tract

SLOWEST ABSORPTION → QUICKEST ABSORPTION

10-15° ANGLE



INTRADERMAL (ID)

Should form a "BLEB"

USES:
- TB testing
- Allergy sensitivities

NEEDLE SIZE:
25 - 27 gauge

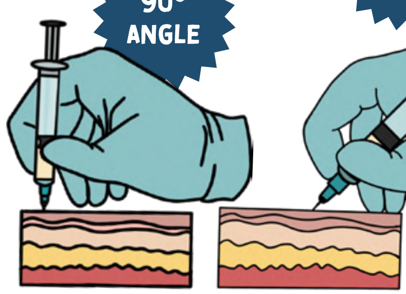
USUAL SITE:
Inner forearm

Normal to overweight clients

90° ANGLE

Thin clients

45° ANGLE



SUBCUTANEOUS (SUBQ)

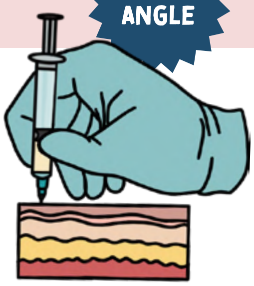
USES:
non-irritating, water-soluble medication (insulin & heparin)

NEEDLE SIZE:
23 - 25 gauge

USUAL SITE:
- Abdomen
- Posterior upper arm
- Thigh

Giving a malnourished/thin client a medication at a 90° angle could lead to accidental intramuscular injury!

90° ANGLE



INTRAMUSCULAR (IM)

Use the Z TRACK METHOD

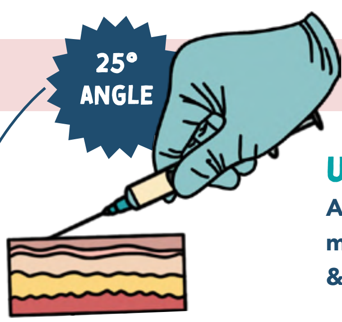
Do not inject more than 3 mL (2 mL for the deltoid)
Divide larger volumes into two syringes & use two different sites

USES:
Irritating, solutions in oils, and aqueous suspensions

NEEDLE SIZE:
22 - 25 gauge

USUAL SITE:
- Deltoid
- Vastus lateralis
- Ventrogluteal

25° ANGLE



INTRAVENOUS (IV)

25° angle used when starting an IV

USES:
Administering medications, fluids, & blood products

NEEDLE SIZE:
16 gauge:
clients who have trauma
18 gauge:
surgery & blood administration
22 - 24 gauge:
children, older adults,
& clients who have medical issues or are stable post-op

USUAL SITE:
- Hand
- Wrist
- Cubital fossa
- Foot
- Scalp

The smaller the gauge number, the larger the IV bore

GAUGES & IV USES

16 G	18 G	20 G*	22 G	24 G
Trauma, surgery, rapid fluid administration (bolus)	Administering blood, rapid infusions (bolus), CT scans with IV dye	Medications, routine therapies, IV fluids	IV fluids, medications	Pediatric patients, elderly patients, very fragile/small veins

LARGEST ← SMALLEST

*Some hospitals allow blood to be administered with 20 G

Always check with your hospital's protocol about IV and blood product administration

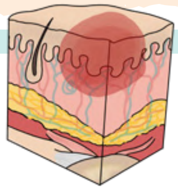
PRESSURE INJURIES (ULCERS)

"Decubitus Ulcer" "Bed Sores"

WHAT IS A PRESSURE INJURY?

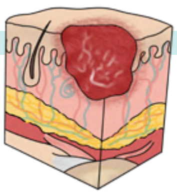
The break down of skin integrity due to unrelieved pressure

Most commonly seen in bedridden and/or incontinent patients



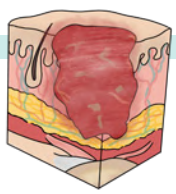
TYPE 1

- Skin is intact (unbroken)
- Nonblanchable redness
- Swollen tissue
- Darker skin ➔ may appear blue / purple



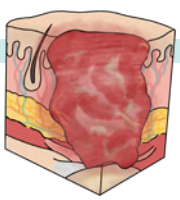
TYPE 2

- Skin is NOT intact
- Partial thickness loss
- No fatty tissue is visible
- Superficial ulcer



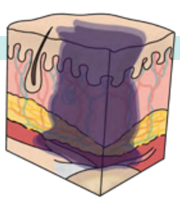
TYPE 3

- Skin is NOT intact
- Full thickness SKIN loss
  - ➔ Damage to or necrosis of subQ tissue
  - ➔ No bone, muscle, or tendon exposed
- Ulcer extend down to the underlying fascia, but not through it
- Deep crater with or without tunneling



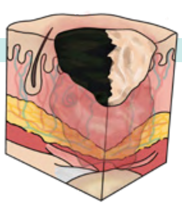
TYPE 4

- Skin is NOT intact
- Full thickness TISSUE loss
  - ➔ Destruction of tissue
  - ➔ Bone, muscle, or tendon exposed
- Deep pockets of infection & tunneling



DEEP TISSUE INJURY (DTI)

- Skin is intact (unbroken)
- Tissue beneath the surface is damaged
- Appears purple or dark red



UNSTAGEABLE

Stage cannot be determined due to eschar or slough covering the visibility of the wound

RISK FACTORS

MNEMONIC "AVOIDS PRESS"

- **A** Aging skin
- **V** Vascular disorders
- **O** Obesity
- **I** Immobility & incontinence
- **D** Diabetes
- **S** Skin friction
- **P** Poor nutrition
- **R** Reduced RBCs (anemia)
- **E** Edema
- **S** Sensory deficits
- **S** Sedation

BRADEN SCALE

Asses your client's skin **EVERY** shift for pressure injuries using the Braden Scale!

Looks at **6** categories

- **SENSORY PERCEPTION**
- **MOISTURE**
- **ACTIVITY**
- **MOBILITY**
- **NUTRITION**
- **FRICTION & SHEAR**

Interpretation

LOW RISK: 22 – 23
LESS RISK: 19 – 21
HIGH RISK: <18

NURSING CONSIDERATIONS

As a nurse, it's important to **PREVENT** pressure injuries while in the hospital!

RELIEVE PRESSURE

- Apply pressure relieving devices (overlays, specialty beds, air cushions, foam-padded seat cushions, etc.)
- Do not use donut-type devices or synthetic sheepskins

PROPER NUTRITION

- ↑ protein intake
- Adequate hydration
- Possible enteral nutrition

protein promotes wound healing

SKIN HYGIENE

- Clean skin with mild soap
- Clean incontinent patients
- Do not scrub or rub bony prominences
- Barrier for incontinence
- Moisturizer for hydration

REPOSITIONING

Turn/reposition patient every 2 hours while in the bed

- LIFT, don't PULL
  - Pulling could cause shearing & friction from force

MONITOR:

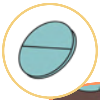
- Size & color of the wound
- Braden Scale (tool for anticipating the risk of pressure ulcers)

NONPARENTERAL ADMINISTRATION

Absorbed into the system through the digestive tract

ORAL OR ENTERAL

- CONTRAINDICATIONS: vomiting, aspiration precautions/absence of a gag reflex, decreased LOC, difficulty swallowing
- Have client sit at 90 angle to help with swallowing
- NEVER crush enteric-coated or time-release medications
- Break or cut scored tablets only!



TRANSDERMAL

- Place the patch on a dry and clean area of skin (free of hair)
- Rotate the sites of the patch to prevent skin irritation
- Always take off the old patch before placing a new one on

INHALATION

- Rinse mouth after the use of steroids
- 20 - 30 seconds between puffs
- 2 - 5 minutes between different medications
- Use a spacer if possible to prevent thrush

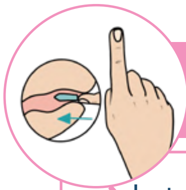
SUBLINGUAL & BUCCAL

SUBLINGUAL: Under the tongue
BUCCAL: Between the cheek & the gum

Do not swallow!

Keep the medication under the tongue (sublingual) or in between the cheek and gum (buccal) until it has completely absorbed

SUPPOSITORIES



RECTAL

- Lateral or sims' position
- Use lubrication
- Insert beyond the internal sphincter
- Leave it in for 5 minutes

VAGINAL

- Supine with knees bent & feet flat on the bed, close to hips
- Insert the suppository along the posterior wall of the vagina (3 - 4 inches deep)
- Stay supine for at least 5 minutes

INSTALLATION (DROPS, OINTMENTS, SPRAYS)

EYES

- If there is dried section use a moisten sterile gauze and wipe from inner to outer canthus to prevent bacterial from entering the eye
- Have the client tilt their head back slightly
- Pull lower eye lid down gently to expose the conjunctival sac
- Hold the dropper 1 - 2 cm above the conjunctiva sac & drop medication directly into the sac
- Close eye lid & apply gentle pressure on the nasolacrimal duct for 30 - 60 seconds

EARS

- Have client tilt their head
- Warm the solution before adm. to prevent vertigo & dizziness
- **ADULTS**: pull ear UPward & outward
- **< 3 YEARS OF AGE**: pull ear DOWN & back

ADULT – UP
CHILD – DOWN

NOSE

- Have client lie supine
- Do not blow nose for 5 min after drop instillation

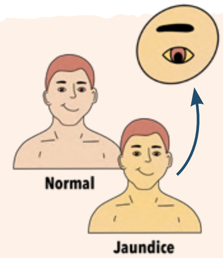
INTEGUMENTARY (SKIN) OVERVIEW

INSPECTION OF THE SKIN

Color changes are more difficult to notice in clients with darker skin

	DESCRIPTION	INDICATION	LOCATIONS
PALLOR	Loss of color	Lack of blood flow, anemia, shock	Face, conjunctiva, nail beds, palm, lips, mucous membranes
ERYTHEMA	Redness can be blanchable or non-blanchable	Inflammation, localized vasodilation, sun exposure, rash, hyperthermia	Skin (areas of trauma or pressure)
JAUNDICE	Yellow to orange	Liver-dysfunction	Skin, sclera, mucous membranes
CYANOSIS	Bluish	Hypoxia (not enough oxygen) or impaired venous return	Lips, mucous membranes, nail beds, skin

The best way to assess for **JAUNDICE** is to press gently on the forehead or nose. If the skin looks yellow where you applied pressure, it indicates jaundice.



PERIPHERAL CYANOSIS

Cyanosis of the peripherals (fingertips, palms, toes)

Rarely a life-threatening medical emergency

CENTRAL CYANOSIS

Cyanosis around the mouth, tongue or mucous membranes

Medical emergency!

EDEMA

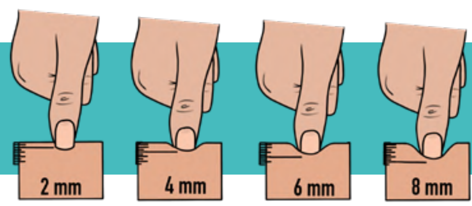
EDEMA is accumulation of excess fluid in the body's tissues that causes swelling of the skin

EDEMA CAN BE:
- → Non-pitting
- → Pitting

WEEPING EDEMA
Areas that have pitting edema can leak fluid out directly from the skin

PITTING is when you press the edematous area for a few seconds and it dimples or pits

GRADING PITTING EDEMA



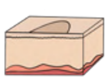
2 mm	4 mm	6 mm	8 mm
+1 = TRACE	+2 = MILD	+3 = MODERATE	+4 = SEVERE

TYPES OF WOUND DRAINAGE

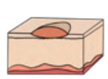
SEROUS	Clear, watery plasma.
SANGUINEOUS	Bright red blood. *(Indicates active bleeding)*
SEROSANGUINEOUS	Pale, pink, watery. Mixture of clear and red fluid.
PURULENT	Thick, yellowish-green. Foul odor. *(May indicate infection)*

PRIMARY LESION

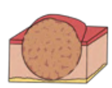
Develops as a result of a disease process



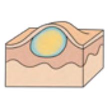
MACULE
Flat discoloration of the skin <1 cm
Example: freckles



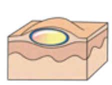
PAPULE
Solid, slightly elevated lesion <1cm
Example: moles



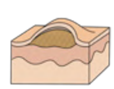
NODULE
Solid & elevated lesion >1cm
Example: lipomas



PUSTULE
Enclosed pus-filled cavity
Example: acne



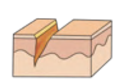
WHEAL
Superficial, raised lesion
Example: allergic reactions



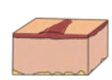
VESICLE
Elevated cavity containing clear fluid
Example: chickenpox, shingles

SECONDARY LESION

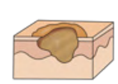
Results from a primary lesion or due to a client's actions (scratching, picking)



FISSURE
Linear crack/tear with abrupt edge
Example: anal fissures, athletes foot



SCAR
Normal tissue is lost & replaced with connective tissue causing a scar
Example: healed area after surgery/injury



EROSION
Scooped-out, shallow depression
Example: severe pressure injuries

SCALE
Compact, flaky skin (silvery or white)
Example: psoriasis

HYPOVOLEMIA VS. HYPERVOLEMIA

SCAN FOR HYPO VS. HYPERVOLEMIA VIDEO

HYPOVOLEMIA

"LOW" "VOLUME" "IN THE BLOOD"

ALSO CALLED: Dehydration • Fluid volume deficit (FVD) • Hypovolemic shock

Causes

- Loss of fluid from ANYWHERE
  - Thoracentesis
  - Paracentesis
  - Hemorrhage
  - NG tube
  - Trauma
  - GI losses
    - Vomiting
    - Diarrhea
- Third spacing
  - Burns
  - Ascites
- Polyuria (peeing a lot)
  - Diabetes
  - Diuretics
  - Diabetes insipidus

THIRD SPACING shifts the fluids from the **INTRAVASCULAR SPACE** (the vein) into the **INTERSTITIAL SPACE** (third space).

This causes a drop in the circulating blood volume

Signs & Symptoms

- Flat neck veins
- ↑ HR (weak & thready)
- ↑ Respirations
- ↑ Urine specific gravity
- ↓ BP

LESS VOLUME = LESS PRESSURE

- ↓ CVP
- ↓ Weight
- ↓ Skin turgor
- ↓ Urine output
- Dry mucous membranes
- Thirst

Labs

MEMORY TRICK

CONCENTRATED (DEHYDRATED) MAKES THE # GO UP

- ↑ Urine specific gravity
- ↑ Hematocrit (%)
- ↑ Serum sodium
- ↑ BUN

Nursing Considerations / Treatment

- Fluid replacement
  - Fluids (PO or IV)

monitor for fluid volume overload

- Safety precautions
  - Risk for fall due to orthostatic hypotension
- Daily I&O + weights

HYPERVOLEMIA

"HIGH" "VOLUME" "IN THE BLOOD"

ALSO CALLED: Over-hydration • Fluid volume excess

Causes

- Heart failure
- Kidney dysfunction
  - Can't filter the blood = back up of fluids
- Cirrhosis
- ↑ Sodium intake

MEMORY TRICK

WHERE SODIUM GOES WATER FLOWS!

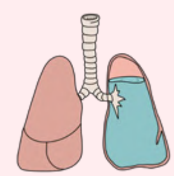
Sodium is the cool kid, so water wants to be his friend.

Signs & Symptoms

- Distended neck vein (JVD)
- ↑ HR (bounding)
- ↑ BP
- ↑ Weight
- ↑ CVP

MORE VOLUME = MORE PRESSURE

- Wet lung sounds
  - Crackles / dyspnea
  - Due to back flow of fluid from the heart
- Edema
- Polyuria
  - Kidneys are trying to get rid of the excess fluid



Labs

MEMORY TRICK

DILUTED (OVER-HYDRATED) MAKES THE # GO DOWN

- ↓ Urine specific gravity
- ↓ Hematocrit (%)
- ↓ Serum sodium
- ↓ BUN

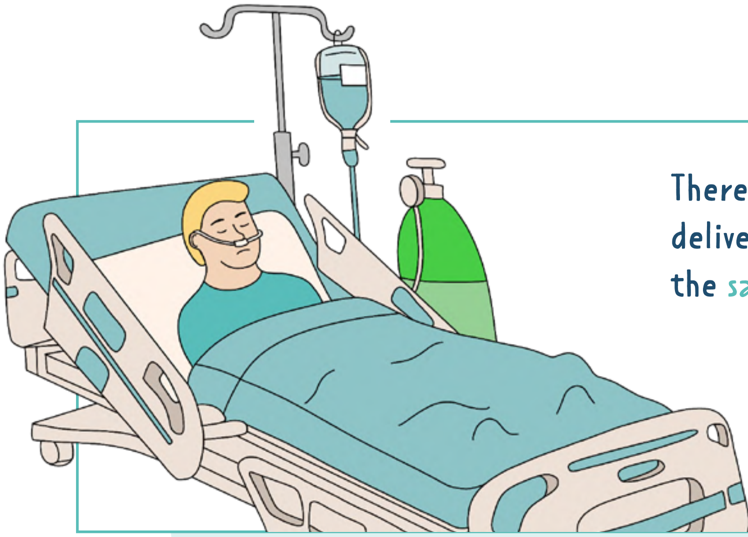
Nursing Considerations / Treatment

- Low sodium diet
- Daily I&O + weights
- Diuretics
- High-Fowler's or Semi-Fowler's position
  - Easier to breathe

MEMORY TRICK

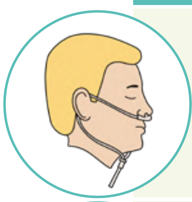
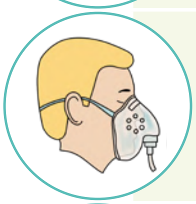
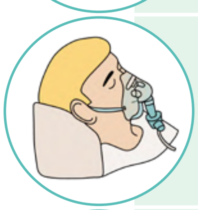
WHERE SODIUM GOES WATER FLOWS!

OXYGEN DELIVERY SYSTEM



There are many types of oxygen delivery systems, but they all have the same goal:

They are used to **ADMINISTER, REGULATE,** and **SUPPLEMENT OXYGEN.**

MASK TYPE	FLOW RATE	FiO_2	DESCRIPTION
 Nasal cannula	2 - 6 L/min	24 - 44%	**Low-flow device** Used for non-acute situations
 Simple masks	6 - 10 L/min	40 - 60%	**Low-flow device** Used for non-acute situations
 Non-rebreather mask	10 - 15 L/min	80 - 90%	**Low-flow device** Used for acutely ill patients
 High-flow oxygen therapy	Up to 60 L/min	21 - 100%	**High-flow oxygen** Often a high flow nasal cannula
 Venturi mask	2 - 15 L/min	24 - 50%	**High-flow device** Best for patients with chronic lung disease
 Face tent	at least 10 L/min	24 - 100%	**High-flow device** Great for those who don't tolerate masks well

Can be given as humidified air to ↓ nasal irritation/dryness

Most precise O_2 delivery without intubation

MEMORY TRICK

Venturi mask think **V**ery accurate O_2

MENTAL HEALTH

BROUGHT TO YOU BY NURSE IN THE MAKING

THERAPEUTIC COMMUNICATION TECHNIQUES

Client-centered type of communication to build and help relationships with clients, families, and all relationships

✓ DO

- Allow client to control the discussion
- Give recognition/validation
- Active listening!
- Use open-ended questions

Don't be a **LOSER,** be an active listener!

- **L**ean forward toward the client
- **O**pen posture
- **S**it squarely facing the client
- **E**stablish eye contact
- **R**elax & listen

EXAMPLES

"Is there something you would like to talk about?"

"Tell me more about that"

"So you are saying you haven't been sleeping well?"

"Tell me more about ______"

X DON'T

- Ask "why"
- Ask too many questions
- Give advice
- Give false reassurance
- Change the conversation topic
- Give approval or disapproval
- Use close-ended questions/statements

EXAMPLES

"Don't worry!"

"I think you should _____"

"Don't be silly"

"That's great!"

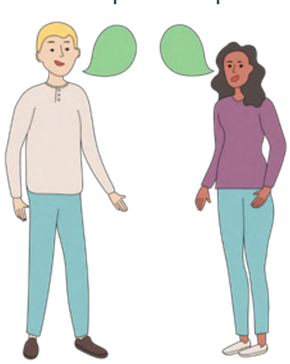
THERAPEUTIC COMMUNICATION CAN BE BOTH...

VERBAL COMMUNICATIONS

&

NON-VERBAL COMMUNICATIONS

Words a person speaks



You may say all the "right" things but deliver it poorly

- **Facial expressions**
- **Eye contact**
- **Posture**
- **Movement**
- **Appearance**
- **Body language**
- **Vocal cues**
  (yawning, tone of voice, pitch of voice)

PERSONALITY DISORDERS

CLUSTER A
Odd or Eccentric

PARANOID
- Suspicious of others
- Thinks everyone wants to harm them

SCHIZOID
- Indifferent
- Seclusive
- Detached
- Doesn't care for close relationships

SCHIZOTYPAL
- Indifferent
- Seclusive
- Detached
- Doesn't care for close relationships

CLUSTER B
Dramatic or Emotional

ANTISOCIAL
- No care for others
- Aggressive
- Manipulative
- Doesn't follow the rules

BORDERLINE
- Unstable
- Manipulative to self & others
- Fear of neglect

HISTRIONIC
- Seeks attention
- Center of attention by being seductive & flirtatious

NARCISSISTIC
- Egocentric AKA narcissus
- Needs consistent applause

CLUSTER C
Anxious or Insecure

AVOIDANT
- Anxious in social settings
- Avoids social interactions but desires close relationships
- Fear of abandonment

DEPENDENT
- Extreme dependency on someone
- Searches urgently to find a new relationship when the other fails

OBSESSIVE COMPULSIVE
- Perfectionist
- Control issue
- Rigid

NURSING CARE
- Safety is a priority

Clients with a personality disorder are at a ↑ risk for violence & self-harm

- Develop a therapeutic relationship
- Respect the client's needs while still setting limits and consistency
- Give the client choices to improve their feeling of control

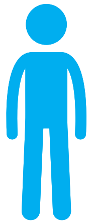
TREATMENT


Medications such as:
- Antidepressants
- Anxiolytics
- Antipsychotics
- Mood stabilizers

Therapies such as:
- Psycho
- Group
- Cognitive
- Behavioral

For more information about psychiatric medications, see the Pharmacology Bundle

EATING DISORDERS



ANOREXIA NERVOSA

- ↓ Weight (BMI <18.5)
- ↓ Blood pressure
- ↓ Heart rate

from dehydration & electrolyte imbalance

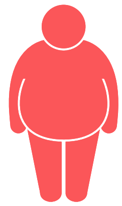
- ↓ Sexual development
- ↓ Subcutaneous tissue = Hypothermia
- ↓ Period regularity
- Amenorrhea *(period may stop)*
- Refuses to eat
- Lanugo *(thin hair to keep the body warm)*
- Typically does not purge
- Restricts self from eating
- Fear of gaining weight
- Constipation *(from dehydration)*

TREATMENT

- ↑ **WEIGHT SLOWLY** (2 -3 lbs a week)
- **MONITOR EXERCISE**

REFEEDING SYNDROME

Potential complications when fluids, electrolytes, and carbohydrates are introduced too quickly to a malnourished client. Treatment should be done **slowly** to avoid this syndrome.

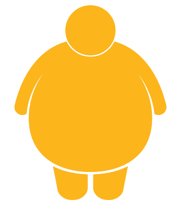


BULIMIA NERVOSA

- Binge eating followed by purging
- Normal weight to overweight (BMI 18.5 - 30)
- Teeth erosion
- Bad breath
- May use laxatives and/or diuretics

TREATMENT

Monitor client during and after meals for acts of purging



BINGE EATING

- Binge eating not followed by purging
- Tend to be overweight
- Binging causes:
  - Depression
  - Hatred
  - Shame

TREATMENT FOR ALL EATING DISORDERS

Teach coping skills

Maintain trust

Have the client be a part of the decision making & the plan of care!

Therapy
group, individual or family

BIPOLAR DISORDER

MOOD SWINGS:

Depression to mania with periods of normalcy

SWINGS FROM

MANIC PHASE

- Periods of **HIGH** mood
- Irritable & hyper
- May require hospitalization

SIGNS & SYMPTOMS

Restless

Flight of ideas
Conversation is all over the place with rapid speech

Grandiosity

Hyper mood
Leads to exhaustion

Poor judgment

Manipulative behavior

↓ Sleep

Delusions

Hallucinations

Impulsive
Examples: maxing out credit cards, engaging in risky behavior

Elevated activity
Leads to malnutrition & dehydration

DEPRESSIVE PHASE

- Periods of **LOW** mood

SIGNS & SYMPTOMS

Sad

Low energy levels

Sleep disturbances:
too much or too little sleep

For clients with mania, the nurse should offer energy & protein-dense foods that are easily consumed on the go (finger foods!)

HAMBURGERS • SANDWICHES
FRUIT JUICES • GRANOLA BARS • SHAKES

TREATMENT

NURSING CONSIDERATIONS FOR THE ACUTE PHASE

- **Provide a safe environment**
  Remove harmful objects from the room
- **Set limits on manipulative behavior**
- **Provide finger foods & fluids**
- **Re-channel energy for physical activity**
- **↓ Stimuli**
  - Turn off or turn down the TV & music
  - Keep away from other clients if they are bothersome

PHARMACOLOGY

- **Lithium carbonate**
- **Anticonvulsants**
- **Antidepressants**
- **Antipsychotics**
- **Antianxiety**

See pharmacology section for more details

SCHIZOPHRENIA SPECTRUM DISORDER OVERVIEW

PHASES

1	PRE-MORBID	Normal functioning. Symptoms have not become apparent yet.
2	PRODROMAL	More tempered form of the disorder. Can be months to years for the disorder to become obvious.
3	SCHIZOPHRENIA	Positive symptoms are noticeable and apparent.
4	RESIDUAL	Periods of remission. Negative symptoms may remain, but S&S of the acute stage (positive symptoms) are gone.

POSSIBLE CAUSES

(not fully known)

- ↑ in the neurotransmitter **DOPAMINE**
- **ILLICIT SUBSTANCE** (LSD & Marijuana)
- **ENVIRONMENTAL** (malnutrition, toxins, viruses during pregnancy)
- **GENETICS** (family history)

SIGNS & SYMPTOMS

POSITIVE

Delusions

Anxiety/agitation

Hallucinations

Auditory **most common*

Jumbled speech

Disorganized behavior

NEGATIVE

Flattened/bland effect

Lack of energy

Reduced speech

Avolition
Lack of motivation

Anhedonia
Not capable of feeling joy or pleasure

Lack of social interaction

TREATMENT

- **Medication**
  - Antipsychotic medications
  - Antidepressants
  - Mood stabilizers (lithium)
  - Benzodiazepines

  **For more information about psychiatric medications, see the Pharmacology Bundle*
- **Therapy**
- **Exercise**

NURSING CONSIDERATIONS

- ☞ Try to establish trust with the client
- ☞ Encourage compliance with the medications
- ☞ Promote self-care
- ☞ Encourage group activities
- ☞ Offer therapeutic communication

HOW TO ADDRESS HALLUCINATIONS?

- Don't address the hallucinations
- Be compassionate
- Bring the conversation back to reality
- Do not argue with the client
- Provide safety for the client & the staff!

EXAMPLE:
"I don't see spiders on the wall but I see you are scared"

TYPES OF DEPRESSION

MAJOR DEPRESSIVE DISORDER (MDD)

Has at least 5 of these symptoms every day for at least 2 weeks:

- Depressed mood
- Too much or too little sleep
- Indecisiveness
- Thoughts of death (suicide)
- ↓ ability to think/concentrate
- Not able to feel pleasure
- ↑ or ↓ motor activity
- Weight fluctuations *(5% change within a month)*

FACTS

- MDD impairs the client's normal functioning
- MDD is not the same depression seen in bipolar disorder
- MDD is not a mood swing, it's constant

TREATMENT PHASES FOR MDD

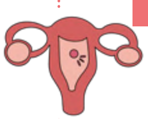
ACUTE: 6 - 12 weeks
Hospitalization & medications may be prescribed
GOALS:
- ↓ Depressive symptoms
- ↑ Functionality

CONTINUATION: 4 - 9 months
Medication is continued
GOALS:
- Prevent relapse

MAINTENANCE: 1+ year
Medication may be continued or be phased out
GOALS:
- Prevent relapse & further depressive episodes

Treatment for the client will reflect what phase they are in!

PREMENSTRUAL DYSPHORIC DISORDER (PMDD)



Depression that occurs during the luteal phase of the menstrual cycle.

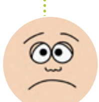
SYMPTOMS

- Emotional
- ↑ Eating
- ↓ Energy
- ↓ Concentration

SUBSTANCE INDUCED DEPRESSIVE DISORDER

Depression associated with withdrawal or the use of alcohol and drugs.

PERSISTENT DEPRESSIVE DISORDER (DYSTHYMIA)



A more mild form of depression compared to MDD, although it can turn into MDD later in life.

SEASONAL AFFECTIVE DISORDER (SAD)

Depression that occurs seasonally.
Often occurs during the winter months when there is less sunshine.

TREATMENT: Light therapy

POSTPARTUM

Depression that happens after a woman goes through childbirth. The woman may feel disconnected from the world. She may have a fear of harming her newborn.

TREATMENT

ANTIDEPRESSANTS

- SSRIs
- SNRIs
- TCAs
- MAOIs

NON-PHARMACOLOGICAL THERAPIES

- Light therapy
- St. John's wort

ELECTROCONVULSIVE THERAPY (ECT)

Used for clients who are unresponsive to other treatments.
Transmits a brief electrical stimulation to the patient's brain.

THE PROCEDURE
- The client is asleep under anesthesia
- The client will not remember and is unaware of the procedure
- Muscle relaxants may be given to ↓ seizure activity & ↓ risk for injury
- Client may have memory loss, confusion, & headache post-procedure

**For more information about antidepressants, see the psychiatric section in the Pharmacology Bundle*

NURSING CONSIDERATIONS

- Safety is a priority. Those struggling with depression have a higher suicide risk.

  INITIATE SUICIDE PRECAUTIONS:
  - *Remove sharp things*
  - *Keep medications out of reach*
  - *Remove objects that may be used for strangulation (wires)*
- Help the client identify coping methods & teach alternatives if needed
- Provide local resources such as churches, local programs, community resources, etc.
- **ENCOURAGE:**
  - Physical activity
  - Self-care
  - Supportive relationships
    Individual therapy, support groups, & peer support

DIFFERENT TYPES OF ANXIETY DISORDERS

NORMAL → WORST

	MILD	MODERATE	SEVERE	PANIC
LEVELS OF ANXIETY	Normal/healthy amount of anxiety. Allows one to have sharp focus & problem solve.	Thinking ability is impaired. Sharp focus & problem-solving can still happen just at a lower level.	Focus & problem solving are not possible. Feelings of doom may be felt.	Most extreme anxiety. Unstable & not in touch with reality.
SYMPTOMS	Nail-biting Tapping Foot jitters	GI upset Headache Voice is shaky	Dizziness Headache Nausea Sleeplessness Hyperventilation	Pacing Yelling Running Hallucinations

ANXIETY DISORDERS

Separation Anxiety Disorder	Experiences extreme fear of anxiety when separated from someone they are emotionally connected to. This is a normal part of infancy, but not a normal part of adulthood.
Specific Phobia	Irrational fear of a particular object or situation. **SOME EXAMPLES:** ▪ Monophobia - *Fear of being alone* ▪ Zoophobia - *Fear of animals* ▪ Acrophobia - *Fear of heights*
Social Anxiety Disorder (Social Phobia)	Fear of social situations or presenting in front of groups. They fear embarrassment. They may have symptoms (real or fake) to escape the situation.
Panic Disorder	Reoccurring panic attacks that last 15 - 30 minutes with physical manifestations.
Agoraphobia	Extreme fear of certain places where the client feels unsafe or defenseless. May even be too fearful of places to maintain employment. AGORA MEANS "open space"
Generalized Anxiety Disorder (GAD)	Uncontrolled extreme worry for at least **6** months that causes impairment of functionality.

OBSESSIVE COMPULSIVE DISORDERS

Obsessive Compulsive Disorder (OCD)	**OBSESSION:** Recurrent thoughts **COMPULSION:** Recurrent acts or behaviors This obsessiveness is usually because it decreases stress & helps deal with anxiety.
Hoarding Disorder	Compulsive desire to save items even if they have no value to the person. It may even lead to unsafe living environments.
Body Dysmorphic Disorder	Preoccupied with perceived flaws or imperfections in physical appearance that the client thinks they have.

SOMATIC SYMPTOM & RELATED DISORDERS (Somatoform Disorders)

SOMATIC SYMPTOM DISORDER

Somatization is psychological stress that presents through physical symptoms that can not be explained by any pathology or diagnosis.

NURSING CONSIDERATIONS

- SAFETY is a priority — Asses for symptoms or thoughts of self-harm or suicide
- Understand the somatic symptoms are real to the client even though they are not real
- Help the client verbalize their feelings while limiting the amount of time talking about their somatic symptoms
- Assess coping mechanism & educate on alternative ways of coping

MANIFESTATIONS

- Consumed by physical manifestations to the point it disrupts daily life
- Seeks medical help from multiple places
- Remission & exacerbations
- Over-medicates with analgesic and antianxiety medications
- ↑ Stress = ↑ somatic symptoms

PHQ-15: PATIENT HEALTH QUESTIONNAIRE 15

An assessment tool used to identify 15 of the most common somatic symptoms

CONVERSION DISORDER

Sudden onset of neurological manifestations & physical symptoms without a known neurological diagnosis. It can be related to a psychological conflict/need beyond their conscious control.

NURSING CONSIDERATIONS

- Ensure SAFETY
- Gain trust & rapport with the client
- Assess coping mechanism & educate on alternative ways of coping
- Assess stress management methods
- Encourage therapy such as:
  - Individual therapies
  - Group therapies
  - Support groups

MANIFESTATIONS

MOTOR

Paralysis pseudoseizures

PSEUDOCYESIS:
Signs & symptoms of pregnancy without the presence of a fetus AKA false pregnancy. This may be present in a client who desires to become pregnant

SENSORY

Blindness
Deafness
Sensations (burning/tingling)
Inability to smell/speak

MEDICATIONS

The client may be prescribed **antidepressants** or **anxiolytics**

POST TRAUMATIC STRESS DISORDER (PTSD)

Mental health condition where exposure to a traumatic event has occurred.

NURSING CONSIDERATIONS

- Teach relaxation techniques
- Teach ways to ↓ anxiety
- Support groups

MANIFESTATIONS

Lasting longer than 1 month:

- Anxiety
- Detachment
- Nightmares of the event

MEDICATIONS

Antidepressants may be prescribed

*For more information about antidepressants, see the psychiatric section in the Pharmacology Bundle

NEUROCOGNITIVE DISORDERS

DEMENTIA & ALZHEIMER'S ARE NOT THE SAME
Dementia is a general term that refers to a group of symptoms, not a specific disease. Dementia may advance to a major neurocognitive disorder such as Alzheimer's disease.

DELIRIUM

ALZHEIMER'S

ONSET

Delirium:
SHORT TERM / SUDDEN CHANGE
Impairment (hours – days)

Alzheimer's:
CONTINUOUS
Decline of function (months – years)

RISK FACTORS

Delirium:
- Hospitalization
- ICU delirium
- Polypharmacy
- Old age
- Stroke
- Surgery
- Restraints
- Secondary to a medical condition
(infection, electrolyte imbalance, substance abuse, etc.)

There is always an underlying cause... something is causing the delirium!

Alzheimer's:
- **Genetics**
  Family history (immediate family)
- **Head injury**
  Traumatic brain injuries (TBI) & head trauma
- **Advanced age**
  >65 have the highest risk
- **Cardiovascular disease & lifestyle factors**
  Inactivity, unhealthy diet, high cholesterol, obesity, & diabetes

MANIFESTATIONS

Delirium:
Delirium is a medical emergency and requires prompt diagnosis & treatment

- Disorganization
  - *Most common to time & place*
  - *Happens mostly at night*
- ↓ Memory
- Anxiety & agitation
- Delusional thinking
- Ranges from lethargic to hypervigilance

Alzheimer's:
STAGES OF ALZHEIMER'S DISEASE

MILD
EARLY STAGE
Not noticeable to others
- Memory lapse
- Misplacing things
- Difficulty focusing
- Can still accomplish own ADLs

MODERATE
MIDDLE STAGE
Noticeable to others
- Forgetfulness
- Short term memory loss
- ★ **Personality changes**
- Gets lost & wanders often
- Unable to do some ADLs & self-care (may be incontinent)

SEVERE
LATE STAGE
Requires full assistance
- Needs assistant with all ADLs
- Losing physical skills (walking, sitting, swallowing)
- May result in death or coma

INTERVENTIONS

Delirium:
- Safety: prevent physical harm
- Avoid restrains when possible
- Remember physical needs
  (hygiene, food, water, sleep, etc.)
- May be prescribed anti-anxiety/antipsychotic medications

Alzheimer's:
Caring for a client with Alzheimer's is very complex!

- Help families in planning for extended care
- Monitor nutrition, weight, & fluids status
- Maintain a quiet environment to ↓ stimuli
- **Cholinesterase inhibitor** may be prescribed to improve quality of life but does NOT cure the disease.

COMMUNICATION
- Speak slowly
- Give one direction at a time
- Don't ask complex or open-ended questions
- Ask simple, direct questions
- Face the client directly when speaking

USES
Used in early & moderate stages of dementia & Alzheimer's disease. May also be used for Parkinson's disease.

GENERIC	TRADE NAME
donepezil	Aricept
galantamine	Razadyne
rivastigmine	Exelon

CURE?

Delirium: Reversible if prompt treatment is initiated

Alzheimer's: ~~CURE~~ Irreversible

MOTHER BABY

BROUGHT TO YOU BY

ABBREVIATIONS

IUP/IUFD ...... Intrauterine pregnancy / intrauterine fetal demise
SAB .... Spontaneous abortion
TAB .... Therapeutic abortion
LMP .... Last menstrual period
ROM .... Rupture of membranes
SROM .... Spontaneous rupture of membranes
AROM .... Artificial rupture of membranes
PROM .... Prolonged rupture of membranes (>24 hours)
PPROM .... Preterm premature rupture of membranes
SVD .... Spontaneous vaginal delivery
FHR .... Fetal heart rate
EFM .... Electronic fetal monitoring
US .... Ultrasound transducer (detects FHR)
FSE .... Fetal scalp electrode (precise reading of FHR)
IUPC .... Intrauterine pressure catheter (strength of contractions)
LTV .... Long term variability
SVE .... Sterile vaginal exam
MLE .... Midline episiotomy

NST .... Non-stress test
CST .... Contraction stress test
BPP .... Biophysical profile
VBAC .... Vaginal birth after cesarean
AFI .... Amniotic fluid index
BUFA .... Baby up for adoption
NPNC .... No prenatal care
PTL .... Preterm labor
BOA .... Born on arrival
BTL .... Bilateral tubal ligation
D&C / D&E ... Dilation & curettage / dilation & evacuation
LPNC .... Late prenatal care
TIUP .... Term intrauterine pregnancy
VMI / VFI ...... Viable male infant / viable female infant
EDB .... Estimated date of birth
EDC .... Estimated date of confinement
EDD .... Estimated date of delivery

PREGNANCY DURATION

40 WEEKS GESTATIONAL AGE

The number of completed weeks counting from the 1st day of the last normal menstrual cycle (LMP).

38 WEEKS FETAL AGE

This refers to the age of the developing baby, counting from the estimated date of conception. The fetal age is usually 2 weeks less than the gestational age.

TRIMESTERS

FIRST TRIMESTER →	0 – 13 WEEKS
SECOND TRIMESTER →	14 – 26 WEEKS
THIRD TRIMESTER →	27 – 40 WEEKS

PRENATAL TERMS

Gravida / Gravidity

A woman who is pregnant / the number of pregnancies

NULLIGRAVIDA	PRIMIGRAVIDA	MULTIGRAVIDA
Never been pregnant	Pregnant for the first time	A woman who has had 2+ pregnancies

Parity

The number of pregnancies that have reach viability (20 weeks of gestation) whether the fetus was born alive or not

NULLIPARA	PRIMIPARA	MULTIPARA
0 Zero pregnancies beyond viability (20 weeks)	1 One pregnancy that has reached viability (20 weeks)	2+ Two or more pregnancies that have reached viability (20 weeks)

PRETERM

Pregnancies that have reached 20 weeks but ended before 37 weeks

TERM

Pregnancies that have lasted between week 37 and week 42

EARLY TERM: 37 – 38 6/7
FULL TERM: 39 – 40 6/7
LATE TERM: 41 – 41 6/7

POSTDATE/POSTTERM

A pregnancy that goes beyond 42 weeks

GTPAL

An acronym used to assess pregnancy outcomes

SCAN FOR GTPAL VIDEO

G — GRAVIDITY → The number of pregnancies
- Includes the present pregnancy
- Includes miscarriages / abortions
- Twins / triplets count as one

T — TERM BIRTHS → The number born at term
- > 37th week of gestation
- Includes alive or stillborn
- Twins / triplets count as one

P — PRE-TERM BIRTHS → The number of pregnancies delivered beginning with the 20^{th} – $36\,\frac{6}{7}^{th}$ weeks of gestation
- Includes alive or stillborn
- Twins / triplets count as one

A — ABORTIONS / MISCARRIAGES → The number of pregnancies delivered before 20 weeks gestation
- Counts with gravidity
- Twins / triplets count as one

L — LIVING CHILDREN → The number of current living children
- Twin / triplets count individually

PRACTICE QUESTION 1

You are admitting a client to the mother-baby unit. Two hours ago she delivered a boy on her due date. She gives her obstetric history as follows: she has a three-year-old daughter who was delivered a week past her due date and last year she had a miscarriage at 8 weeks gestation. How would you note this history using the GTPAL system?

A. 2-2-1-0-2
B. 3-2-1-0-1
C. 3-2-1-0-2
D. 3-2-0-1-2

PRACTICE QUESTION 2

A prenatal client's obstetric history indicates that she has been pregnant 3 times previously and that all her children from previous pregnancies are living. One was born at 39 weeks gestation, twins were born at 34 weeks gestation, & another child was born at 38 weeks gestation. She is currently 38 weeks pregnant. What is her gravidity & parity using the GTPAL system?

A. 4-1-3-0-4
B. 4-1-2-0-3
C. 4-2-1-0-4
D. 4-2-2-0-4

ANSWER KEY
Q#1 is (D) 3-2-0-1-2
Q#2 is (C) 4-2-1-0-4

PREGNANCY SIGNS & SYMPTOMS

PRESUMPTIVE

SUBJECTIVE

NOT a definite diagnosis for pregnancy!

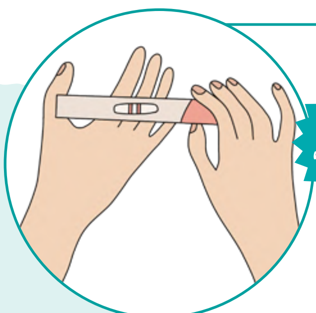
Think "mom"

These are changes felt by the woman and are subjective. Can be associated with other things.

- **P** Period absent (amenorrhea)
- **R** Really tired
- **E** Enlarged breasts
- **S** Sore breasts
- **U** Urination increased (urinary frequency)
- **M** Movement perceived (quickening)
- **E** Emesis & nausea

Why is quickening not a positive sign?

Quickening can be difficult to distinguish from peristalsis or gas so it can not be a positive sign.



PROBABLE

OBJECTIVE

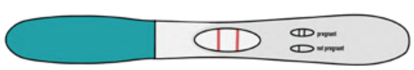
Think "doctor"

Pregnancy signs that the nurse or doctor can observe

- **P** Positive (+) pregnancy test (high levels of the hormone: hCG)
- **R** Returning of the fetus when uterus is pushed w/ fingers (ballottement)
- **O** Objective
- **B** Braxton hicks contractions
- **A** A softened cervix (Goodell's sign)
- **B** Bluish color of the vulva, vagina, or cervix (Chadwick's sign)
- **L** Lower uterine segment soft (Hegar's sign)
- **E** Enlarged uterus

Why is a positive pregnancy test not a positive sign?

High levels of hCG can be associated with other conditions such as certain medications or hydatidiform mole (molar pregnancy).



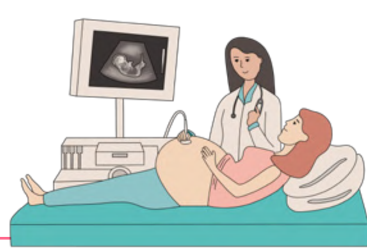
POSITIVE

OBJECTIVE

Definite diagnosis for pregnancy!

Think "Baby"

Can only be attributed to a fetus



- **F** Fetal movement palpated by a doctor or nurse
- **E** Electronic device detects heart tones
- **T** The delivery of the baby
- **U** Ultrasound detects baby
- **S** Seeing visible movements

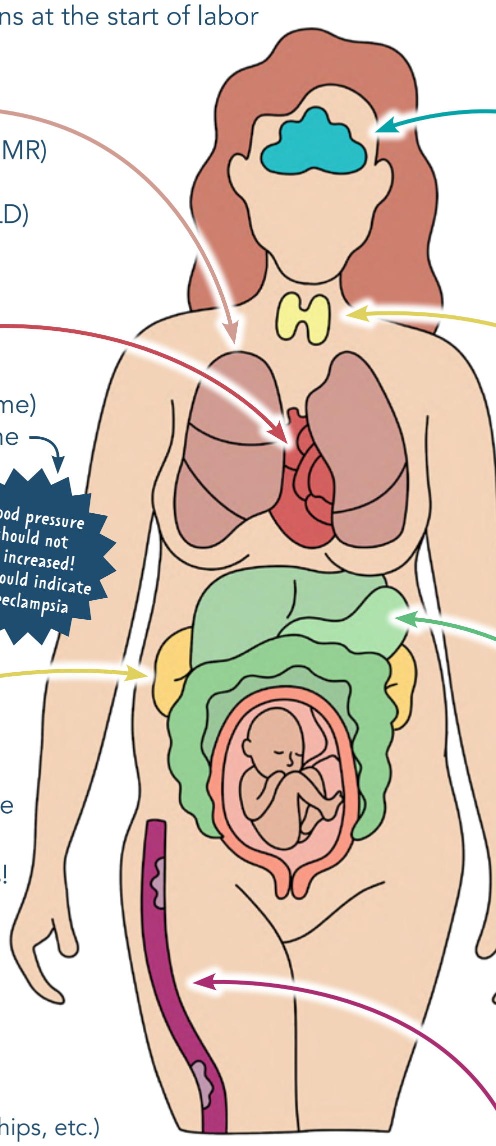
PREGNANCY PHYSIOLOGY

HORMONES

Prolactin: Allows for breast milk production
Estrogen: Growth of fetal organs & maternal tissues
Progesterone & Relaxin: Relaxes smooth muscles
hCG: Produced by placenta, prevents menstruation
Oxytocin: Stimulates contractions at the start of labor

MUSCULOSKELETAL

- **Lordosis**: center of gravity shifts forward leading to inward curve of spine
- Low back pain
- Carpal tunnel syndrome
- Calf cramps



RESPIRATORY

- ↑ Basal metabolic rate (BMR)
- ↑ O_2 needs
- Respiratory alkalosis (MILD)

PITUITARY

- ↓ FSH/LH due to ↑ Progesterone
- ↑ Prolactin
- ↑ Oxytocin

CARDIOVASCULAR

- ↑ Cardiac output (↑ Heart rate + ↑ stroke volume)
- Blood pressure stays the same or a slight decrease
- ↑ in plasma volume
- ❤ Enlarges *(May develop systolic murmurs)*

Blood pressure should not be increased! This could indicate preeclampsia

THYROID

- ↑ Thyroxine
- May have moderate enlargement of the thyroid gland (goiter)
- ↑ Metabolism & ↑ appetite

RENAL

- ↑ GFR from ↑ plasma volume
- Smooth muscle relaxation of the uterus = ↑ risk of UTIs!
- ↑ Urgency, frequency & nocturia
- EDEMA!

GASTROINTESTINAL

- **Pyrosis**
  ↑ Progesterone = LOS to relax = ↑ heartburn
- **Constipation & hemorrhoids**
  ↑ Progesterone = ↓ gut motility
- **Pica**
  Non-food cravings such as ice, clay, and laundry starch

SKIN

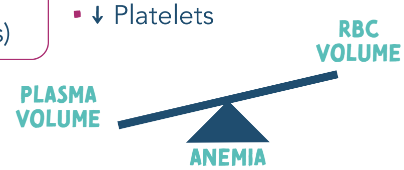
- **Striae**
  Stretch marks (abdomen, breasts, hips, etc.)
- **Chloasma**
  - Mask of pregnancy
  - Brownish hyperpigmentation of the skin
- **Linea Nigra**
  "Pregnancy line" dark line that develops across your belly during pregnancy
- **Montgomery glands / Tubercles**
  Small rough / nodular / pimple-like appearance of the areola (nipple)

HEMATOLOGICAL

FIBRINOGEN
- **Non-pregnant levels:** 200-400 mg/dL
- **Pregnant levels:** up to 600 mg/dL

Pregnant women are **HYPERCOAGULABLE** (increased risk for DVTs)

- ↑ White blood cells
- ↓ Platelets



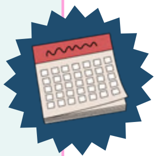
ANEMIA

Plasma volume is greater than the amount of red blood cell (RBC) = hemodilution = **physiological anemia**

NAEGELE'S RULE

Used for estimating the expected date of delivery (EDD) based on LMP (last menstrual period)

DATE OF LAST MENSTRUAL PERIOD — 3 CALENDAR MONTHS + 7 DAYS + 1 YEAR



REMEMBER: How many days are in each month?

"30 days hath September, April, June & November. All the rest have 31, except February alone (28 days)"

EXAMPLE

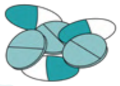
1st day of last period:	September 2, 2015
Minus 3 calendar months:	June 2, 2015
Plus 7 days:	June 9, 2015
Plus 1 year:	June 9, 2016 → EDD

FACTS ABOUT NAEGELE'S RULE

- Bases calculation on a woman who has a 28-day cycle (most women vary)
- The typical gestation period is 280 days (40 weeks)
- First-time mothers usually have a slightly longer gestation period

WHAT TO AVOID DURING PREGNANCY

TERATOGENIC DRUGS



MNEMONIC **"TERA-TOWAS"**

- **T** Thalidomide
- **E** Epileptic medications (valproic acid, phenytoin)
- **R** Retinoid (vit A)
- **A** Ace inhibitors, ARBs
- **T** Third element (lithium)
- **O** Oral contraceptives
- **W** Warfarin (coumadin)
- **A** Alcohol
- **S** Sulfonamides & sulfones

TORCH INFECTIONS

TORCH infections are a group of infections that cause fetal abnormalities. Pregnant women should avoid these infections!

MNEMONIC **"TORCH"**

- **T** Toxoplasmosis
- Parvo **O** Virus-B19 (fifth disease)
- **R** Rubella
- **C** Cytomegalovirus
- **H** Herpes simplex virus

STAGES OF LABOR

STAGE 1 — Cervix DILATES from 0-10 cm

LONGEST STAGE

Latent (early)

- **Cervix dilates**: 1- 3 cm
- **Intensity**: Mild
- **Contractions**: 15 - 30 min

Active

- **Cervix dilates**: 4 - 7 cm
- **Intensity**: Moderate
- **Contractions**: 3 -5 min (30-60 sec in duration)

Transition

- **Cervix dilates**: 8 - 10 cm
- **Intensity**: Strong
- **Contractions**: Every 2-3 min (60-90 sec in duration)

INTERVENTIONS

- Promote comfort
  - Warm shower, massage, or epidural
- Offer fluids & ice chips
- Provide a quiet environment
- Encourage voiding every 1 - 2 hours
- Encourage participation in care & keep informed
- Instruct partner in **effleurage** (light stroking of the abdomen)
- Encourage effective breathing patterns & rest between contractions

MNEMONIC

Labor
Actively
Transitioning

STAGE 2 — The BABY is delivered

→ Starts when cervix is fully dilated & effaced
→ Ends after the baby is delivered

PUSHING!

INTERVENTIONS

- Provide ice chips & ointment for dry lips
- Provide praise & encouragement to the mother
- Monitor uterine contractions & mothers vital signs
- Maintain privacy & encourage rest between contractions
- Encourage effective breathing patterns & rest between contractions
- Monitor for signs of birth (perineal bulging or visualization of fetal head)

STAGE 3 — The PLACENTA is delivered

>30 min = Retained placenta

The PLACENTA is expelled (5 - 30 min after birth)

SIGNS OF A PLACENTA DELIVERY

- Lengthening umbilical cord
- Gush of blood
- Uterus changes from oval to globular shape

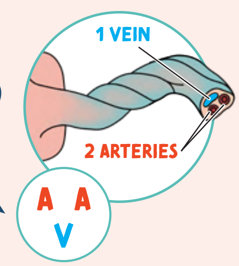
DELIVERY MECHANICS

"Shiny Schultz"
Side of *baby* delivered 1st

"Dirty Duncan"
Side of *mother* delivered 1st

INTERVENTIONS

- Assessing mothers vital signs
- Uterine status (fundal rubs every 15 minutes)
- Provide warmth to the mother
- Promote parental-neonatal attachment
- Examine placenta & verify it's intact
  - Should have 2 arteries & 1 vein



MEMORY TRICK looks like a smiley face!

2 "**A**" for **ARTERIES**
1 "**V**" for **VEIN**

STAGE 4 — Recovery!

RECOVERY: first 1-4 hours after delivery of the placenta

- Assessing the fundus
- Continue to monitor vital signs & temperature for infection
- Administer IV fluids
- Monitor lochia discharge (lochia may be moderate in amount & red).
- Monitor for respiratory depression, vomiting, & aspiration if general anesthesia was used
- *Great time to watch for complications such as bleeding (postpartum hemorrhage)*

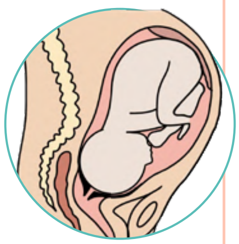
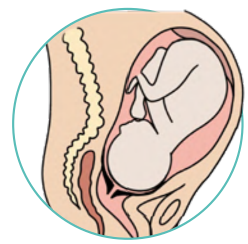
✓
- FIRM
- Midline

✗
- Soft
- Boggy
- Displaced

TRUE VS. FALSE LABOR

	FALSE LABOR	TRUE LABOR

CONTRACTIONS	• Irregular
	• Stops with walking/position change
	• Felt in the back or the abdomen above the umbilicus
	• Often stops with comfort measures
		• Occur regularly
		  ▪ Stronger
		  ▪ Longer
		  ▪ Closer together
		• More intense with walking
		• Felt in lower back → radiating to the **lower** portion of the abdomen
		• Continue despite the use of comfort measures

CERVIX	
	• May be soft
	• NO significant change in....
	  ▪ Effacement
	  ▪ Dilation
	• No bloody show
	• In posterior position (baby's head facing mom's front of belly)
		
		• Progressive change
		  ▪ Softening
		  ▪ Effacement
		  ▪ Dilation signaled by the appearance of bloody show
		  ▪ Moves to an increasingly anterior position (baby's head facing mom's back)

FETUS	• Presenting part is usually not engaged in the pelvis	• Presenting parts become engaged in the pelvis • Increased ease of breathing (more room to breathe) • Presenting part presses downward & compresses the bladder = urinary frequency

SIGNS OF LABOR

LABOR

Moving the fetus, placenta, & the membranes out of the uterus through the birth canal

Signs of Preceding Labor

- Lightening
- Increased vaginal discharge (bloody show)
- Return of urinary frequency
- Cervical ripening
- Rupture of membranes "water breaking"
- Persistent backache
- Stronger Braxton Hicks contractions
- Days preceding labor
  - Surge of energy
  - Weight loss (1- 3.5 pounds) from a fluid shift

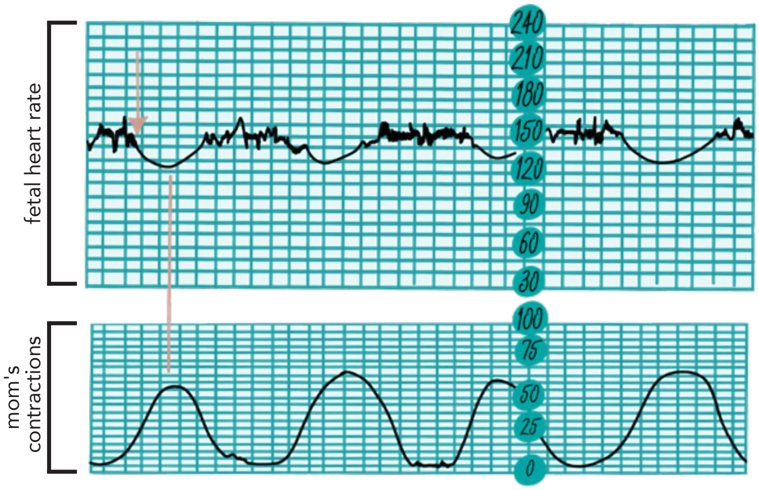
FETAL HEART TONES

EARLY DECELERATIONS

MEMORY TRICK "Mirror" image of mom's contractions (they don't technically come early)

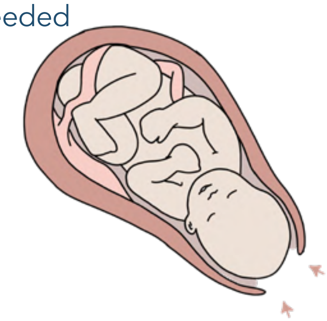
NORMAL FETAL HEART RATE:
120 - 160 BPM

NORMAL! ✓



CAUSE:
- From head compression

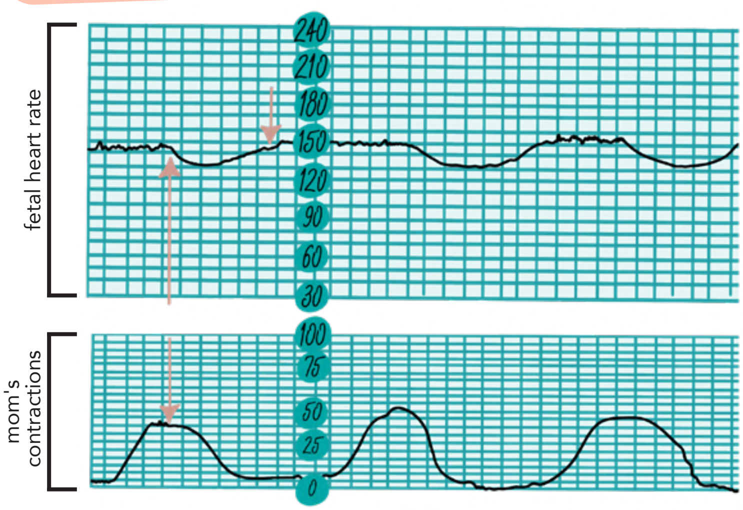
INTERVENTION:
- Continue to monitor
- No intervention needed



LATE DECELERATIONS

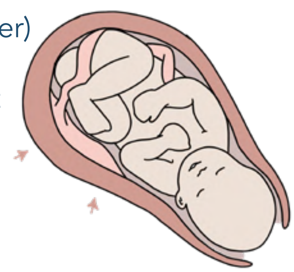
MEMORY TRICK Literally comes late after mom's contraction

NON-REASSURING ✗



CAUSE:
- Uteroplacental insufficiency

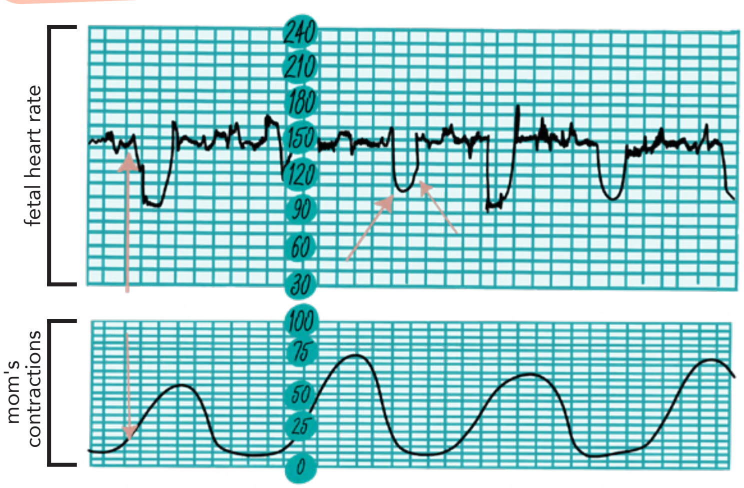
INTERVENTION:
- D/C oxytocin
- Position change
- Oxygen (non-rebreather)
- Hydration (IV fluids)
- Elevate legs to correct the hypotension



VARIABLE DECELERATIONS

MEMORY TRICK *Variable: Looks "V" shaped

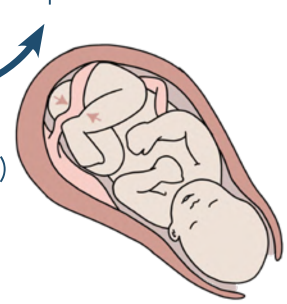
NON-REASSURING ✗



CAUSE:
- Cord compression

INTERVENTION:
- D/C Oxytocin
- Amnioinfusion
- Position change
- Breathing techniques
- Oxygen (non-rebreather)

Side-lying or knee chest will relieve pressure on cord



VEAL CHOP

A tool to help interpret fetal strips

V Variable Decelerations	→	**C** Cord Compression
E Early Decelerations	→	**H** Head Compression
A Accelerations	→	**O** OK (normal fetal oxygenation)
L Late Decelerations	→	**P** Placental Insufficiency

ASSESSMENT OF UTERINE CONTRACTIONS

DURATION	**BEGINNING** of the contraction to the **END** of that same contraction	▪ Lasts 45 - 80 seconds ▪ Should not exceed 90 seconds *Only measured through external monitoring*	
FREQUENCY	Number of contractions from the **BEGINNING** of one contraction to the **BEGINNING** of the next	▪ 2 - 5 contractions every 20 minutes ▪ Should not be more FREQUENT then every 2 minutes *Only measured through external monitoring*	
INTENSITY	Strength of a contraction at its **PEAK**	▪ 25 - 50 mm Hg ▪ Should not exceed 80 mm HG *Can be palpated*	**MILD** - *nose* **MODERATE** - *chin* **STRONG** - *forehead*
RESTING TONE	**TENSION** in the uterine muscle between contractions (relaxation of the uterus = fetal oxygenation between contractions)	▪ Average: 10 mm HG ▪ Should not exceed 20 mm HG *Can be palpated*	**SOFT** = *good* **FIRM** = *not resting enough*

PREECLAMPSIA OVERVIEW

Overview of Hypertensive disorders during pregnancy

1ST TRIMESTER | 20 WEEKS | **2ND TRIMESTER** | **3RD TRIMESTER**

CHRONIC HTN: Before pregnancy or before 20 weeks!

PREECLAMPSIA: HTN after 20 weeks gestation with systemic features

GESTATIONAL HTN: HTN after 20 weeks without systemic features

WHAT IS HYPERTENSION?

SYSTOLIC > 140
OR
DIASTOLIC > 90

Hypertension may be abbreviated "HTN"

SIGNS & SYMPTOMS

"PRE" eclampsia

- **P**roteinuria
- **R**ising BP
- **E**dema

Triad Signs

- Severe headache
- RUQ or epigastric pain
- Visual disturbances
- ↓ Urine output
- Hyperreflexia
- Rapid weight gain

PATHOLOGY

PLACENTA is the root cause

Pathology is not completely known

- Defective spiral artery remodeling
- Systemic vasoconstriction & endothelial dysfunction

RISK FACTORS

- HX of preeclampsia in previous pregnancies
- Family history of preeclampsia
- 1st pregnancy
- Obesity
- Very young (<18) or very old (>35)
- Medical conditions (Chronic HTN, renal disease, diabetes, autoimmune disease)

>35 = AMA advanced maternal age

HELLP SYNDROME

LIFE-THREATENING COMPLICATION

Variant of preeclampsia

- **H** – **H**emolysis
- **EL** – **E**levated **L**iver enzymes
- **LP** – **L**ow **P**latelet count

ECLAMPSIA

(seizures activity or a coma)

IMMEDIATE CARE:

- Side-lying
- Padded side rails with pillows/blankets
- O_2
- Suction if needed
- Do not restrain
- Do not leave

MAGNESIUM SULFATE

RX **given to prevent seizures during & after labor.**

***Remember**: magnesium acts like a depressant*

THERAPEUTIC RANGE: 4 – 7 mg/dL

TOXICITY!

- RR <12
- ↓ DTRs
- UOP <30 mL/hr
- EKG changes

**Mag is excreted in urine ↓UOP → ↑mag levels*

ANTIDOTE: CALCIUM GLUCONATE

*because magnesium sulfate can cause respiratory depression

LABOR & BIRTH PROCESSES

5 P's 5 factors that affect the process of labor & birth

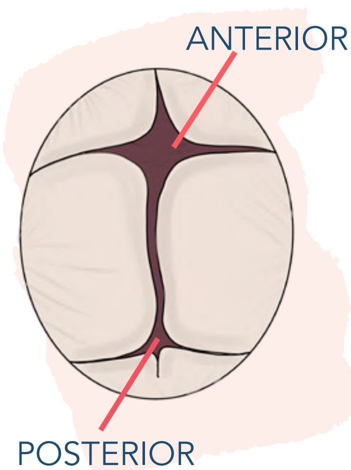
Passenger	Passageway	Position	Powers	Psychology
FETUS & PLACENTA	THE BIRTH CANAL	POSITION OF THE MOTHER	CONTRACTIONS	EMOTIONAL RESPONSE

Passenger — FETUS & PLACENTA

SIZE OF THE FETAL HEAD

FONTANELS
- Space between the bones of the skull allows for molding
- Anterior (larger)
  - Diamond-shaped
  - Ossifies in 12-18 months
- Posterior
  - Triangle shaped
  - Closes 8 - 12 weeks

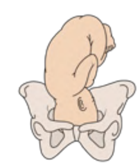
MOLDING
- Change in the shape of the fetal skull to "mold" & fit through the birth canal



FETAL PRESENTATION

Refers to the part of the fetus that enters the pelvic inlet first through the birth canal during labor

1 CEPHALIC MOST COMMON
- Head first
- Presenting part: Occipital (back of head/skull)



2 BREECH
- Buttocks, feet, or both first
- Presenting part: Sacrum

3 SHOULDER
- Shoulders first
- Presenting part: Scapula

FETAL LIE

Relation of the long axis (spine) of the fetus to the long axis (spine) of the mother

LONGITUDINAL OR VERTICAL
- The long axis of the fetus is parallel with the long axis of the mother
- Longitudinal: cephalic or breech

TRANSVERSE, HORIZONTAL, OR OBLIQUE
- Long axis of the fetus is at a right angle to the long axis of the mother
- Transverse: vaginal birth **CANNOT** occur in this position
- Oblique: usually converts to a longitudinal or transverse lie during labor

CONTINUED →

LABOR & BIRTH PROCESSES

Passenger CONTINUED

FETAL ATTITUDE

GENERAL FLEXION
- Back of the fetus is rounded so that the chin is flexed on the chest, thighs are flexed on the abdomen, legs are flexed at the knees

BIPARIETAL DIAMETER
- 9.25 cm at term, the largest transverse diameter and an important indicator of fetal head size

SUBOCCIPITOBREGMATIC DIAMETER
- *Most critical* & smallest of the anteroposterior diameters

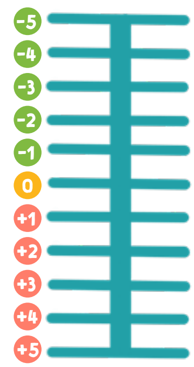
FETAL POSITION

FETAL STATION

Head, foot, butt (closest to exit of uterus)

- Where the baby's **PRESENTING PART** is located in the pelvis
- Measured in centimeters (cm)
  - Find the ischial **spine** = **ZERO**
  - **Above the ischial spine is (-)**
  - **Below the ischial spine is (+)**
    - **+4** / **+5** = Birth is about to happen

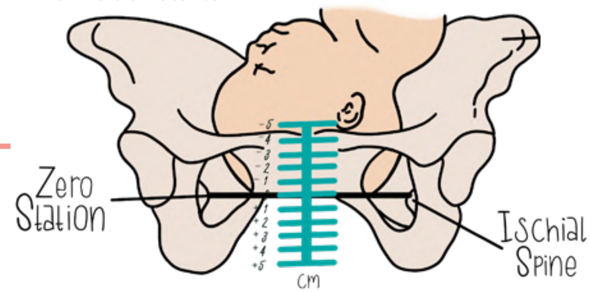
MEMORY TRICK: I'm (+) that I'm getting this baby out



ENGAGEMENT
- Fetal station **ZERO** = baby is "**engaged**"
- Presenting parts have entered down into the pelvis inlet & is at the ischial spine line **(0)**

When does this happen?
- **FIRST-TIME MOMS:** 38 weeks
- **ALREADY HAD BABIES:** can happen when labor starts

LIGHTENING
When the baby "drops" into the mother's pelvis



Passageway

THE BIRTH CANAL: Rigid bony pelvis, soft tissue of cervix, pelvic floor, vagina & introitus

TYPES OF PELVIS

GYNECOID
- Classic female type
- Most common

ANDROID
- Resembling the male pelvis

ANTHROPOID
- Oval-shaped
- Wider anteroposterior diameter

PLATYPELLOID
- The flat pelvis
- Least common

SOFT TISSUE

LOWER UTERINE SEGMENT
- Stretchy

CERVIX
- Effaces (thins) & dilates (opens)
- After fetus descends into the vagina, the cervix is drawn upward and over the first portion

PELVIC FLOOR MUSCLES
- Helps the fetus rotate anteriorly

VAGINA

INTROITUS
- External opening of the vagina

LABOR & BIRTH PROCESSES

Position

POSITION OF THE MOTHER DURING BIRTH

UPRIGHT POSITION
Sitting on a birthing stool or cushion

LITHOTOMY POSITION MOST COMMON
Supine position with buttocks on the table

"ALL FOURS" POSITION
On all fours: putting your weight on your hands & feet

LATERAL POSITION
Laying on a side

Frequent changes in position helps with:

- Relieving fatigue
- Increasing comfort
- Improving circulation

Powers

CONTRACTIONS: PRIMARY & SECONDARY

PRIMARY POWERS

INVOLUNTARY uterine contractions
Signals the beginning of labor

DILATION

- Dilation of the cervix is the gradual enlargement or widening of the cervical opening & canal once labor has begun
- Pressure from amniotic fluid can also apply force to dilate

MEASURED IN CM

0 - 10 cm
closed — full dilation

EFFACEMENT

- Shortening & thinning of the cervix during the first stage of labor
- Cervix normally:
  - **2 -3 CM** long
  - **1 CM** thick
- The cervix is "pulled back / thinned out" by a shortening of the uterine muscles

Degree of **EFFACEMENT** is **EXPRESSED** in **%** (0-100%)

SECONDARY POWERS

VOLUNTARY bearing-down efforts by the women once the cervix has dilated

- Does not affect cervical dilation but helps with expulsion of infant once the cervix is fully dilated
- When the presenting part reaches the pelvic floor, the contractions change in character & become expulsive.
- Laboring women start to feel an involuntary urge to push & she uses secondary powers to aid in the expulsion of the fetus

FERGUSON REFLEX

- When the stretch receptors release oxytocin, it triggers the maternal urge to bear down

Psychology

EMOTIONAL RESPONSE

Anxiety can increase pain perception & the need for more medications (analgesia & anesthesia)

THINGS TO CONSIDER:

SOCIAL SUPPORT — PAST EXPERIENCE — KNOWLEDGE

NEWBORN ASSESSMENT

APGAR

7 - 10 supportive care
4 - 6 moderate depression
< 4 aggressive resuscitation

SCORE	0 POINTS	1 POINT	2 POINTS
A **ACTIVITY** (Muscle tone)	Absent	Flexed arms & legs	Active
P **PULSE**	0	< 100	> 100
G **GRIMACE** (Reflex irritability)	Floppy	Minimal response to stimulation	Prompt response to stimulation
A **APPEARANCE** (Skin color)	Blue / pale all over	Pink body Blue extremities (acrocyanosis)	Pink
R **RESPIRATION** (Effort)	No breathing	Slow & irregular	Vigorous cry

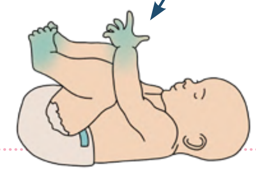
INITIAL GOALS

1ST PRIORITY = AIRWAY
Suction with bulb syringe / deep suction
*Newborns are obligatory nose breathers

2ND PRIORITY = WARMTH
Dry with a blanket or place in warmer

CIRCULATORY SYSTEM

- Blood flow from umbilical vessels & placenta stop at birth
- **Acrocyanosis**: Blueness of hands & feet
  (normal during the first 24 hours of life)
- Closure of:
  - Ductus arteriosus
  - Foramen ovale
  - Ductus venosus
- Transient murmurs are normal



VITAL SIGNS

BLOOD PRESSURE (BP) (Not done routinely)	SYSTOLIC / DIASTOLIC	**60 - 80** mmHg / **40 - 50** mmHg
HEART RATE (HR)	**110 - 160** bpm can be **180** if *crying* can be **100** if *sleeping*	
RESPIRATORY RATE (RR)	**30 - 60** breaths/min	
TEMPERATURE (T) (Axillary)	**97.7 – 99.5°F (36.5 - 37.5°C)**	
MAP	Equal to the # of weeks gestation or higher	

Take apical pulse for 1 full min

SIGNS OF RESPIRATORY DISTRESS
- Retractions
- Nasal flaring
- Grunting

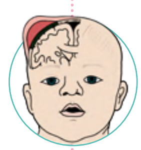
Breathing pattern is **IRREGULAR.**
Newborns are **ABDOMINAL** breathers.

To count breaths, place your hand on their abdomen.
Count for a full minute!

HEAD

CAPUT SUCCEDANEUM:
- Edema (collection of fluid)
- Crosses the suture lines

MEMORY TRICK: Like a baseball **CAP**



CEPHALOHEMATOMA:
- Birth trauma (collection of blood)
- Does not cross the suture lines

MOLDING:
Abnormal head shape that results from pressure (*normal*)

FONTANELLES:
Bulging = increase ICP or hydrocephalus
Sunken = dehydration

Fontanelles may be **BULGING** when the newborn cries, vomits, or is lying down. This is normal.

GENERAL CHARACTERISTICS

Length & Weight

EXPECTED LENGTH	**44 - 55** cm **17 - 22** in
EXPECTED WEIGHT	**2,500 - 4,000** g **5** lb, **8** oz - **8** lb, **14** oz

Head & Chest Circumference

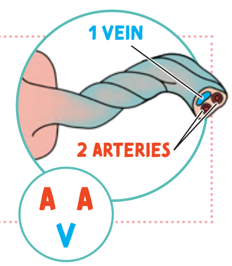
HEAD CIRCUMFERENCE	**32 - 39** cm **14 - 15** in **measure above eyebrows*
CHEST CIRCUMFERENCE	**30 - 36** cm **12 - 14** in **measure above nipple line*

UMBILICAL CORD

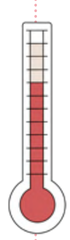
Should have
2 ARTERIES & **1 VEIN**

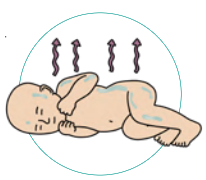
Should be dry, no odor & no drainage

MEMORY TRICK: looks like a smiley face!

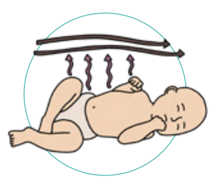


↓ TEMP → HEAT LOSS DUE TO:

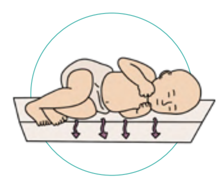




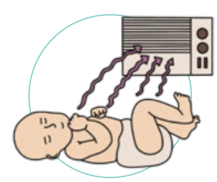
EVAPORATION:
Moisture from skin & lungs



CONVECTION:
Body heat to cooler air



CONDUCTION:
Body heat to a cooler surface in direct contact



RADIATION:
Body heat to a cooler object nearby

NEWBORN REFLEXES

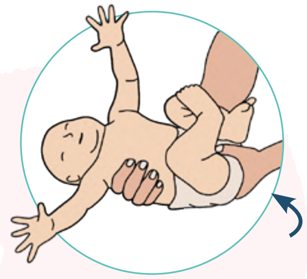
BABINSKI REFLEX

When the bottom of the foot is stroked from the heel upward. The big toe dorsiflexes (bends back) and the other toes spread out.

MEMORY TRICK
Babinski =
Big toe fans out

ROOTING REFLEX

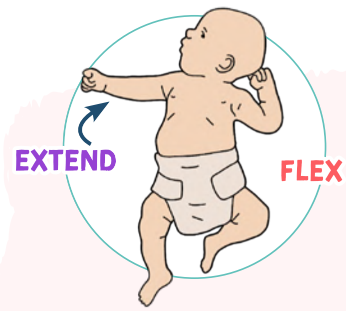
When the baby's mouth is stroked, the baby will turn its head and open the mouth. This helps the baby find the food source when feeding.



MORO REFLEX

"Startle Reflex"

Can be triggered by a sudden loud noise or unexpected movement. The infant will extend the arms with palms up and then move the arms back to the body

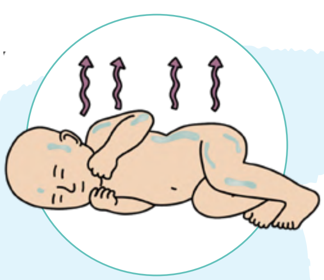


TONIC NECK REFLEX

"Fencing"

When an infant is lying on its back, and quickly turns their head to one side. The leg and arm on that side will **EXTEND**, while the leg and are on the opposite side will **FLEX**.

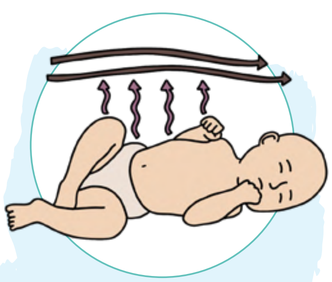
TYPES OF HEAT LOSS & PREVENTION



EVAPORATION

Moisture from skin & lungs

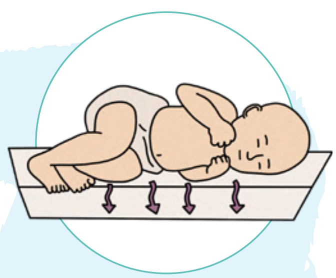
PREVENTION:
Dry infant immediately after birth



CONVECTION

Body heat to cooler air

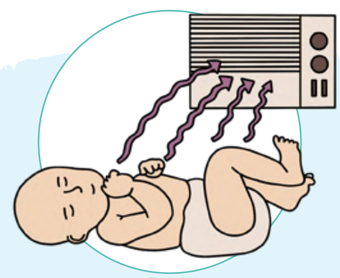
PREVENTION:
Keep bed away from open windows



CONDUCTION

Body heat to a cooler surface in direct contact

PREVENTION:
Warm stethoscope & other instruments before use



RADIATION

Body heat to a cooler object nearby

PREVENTION:
Keeping infant away from any cold objects nearby

POSTPARTUM ASSESSMENT

MNEMONIC "BUBBLES"

B BREASTS

- May be sore after breastfeeding
- Breastfeed every 2 - 3 hours (15 - 20 minutes each breast)
- Position newborn "tummy to mummy"
- Latch should be completely around the areola

MASTITIS

Infection & inflammation of breast tissue

- Continue breastfeeding
- Warm compress
- Hydration
- Rest
- Analgesics
- Wash hands!

U UTERUS

UTERINE ATONY

RISK FACTORS

- Retained placenta
- Chorioamnionitis (infection)
- Uterine fatigue
- Full bladder

SYMPTOMS

- Enlarged
- Soft
- Boggy
- Not midline
- Poorly contracted uterus

INTERVENTIONS

- Fundal massage
- Assist to void or use in-and-out catheter

B BOWELS

Constipation is common after birth. Increasing **FLUIDS & FIBER** may help!

HEMORRHOIDS

- May see blood in the stool
- Should begin to shrink following birth

INTERVENTIONS

- Tucks / witch hazel
- Ice pack
- Squeeze bottle
- Sitz Bath

B BLADDER

- Postpartum urinary retention is common
  - In-and-out catheterization may be needed
  - Bladder distention can cause a displaced & boggy uterus!

L LOCHIA

⚠ SIGNS OF INFECTION ⚠

- Foul smelling or purulent lochia
- Fever (>100.4 F)
- Abdominal tenderness
- Tachycardia

MEMORY TRICK "Really Sore After"

- **Rubra** — bright red, 1 - 3 days
- **Serosa** — pinkish/brown, 4 - 10 days
- **Alba** — whitish-yellow, 10 - 14 days
  **Can last up to 6 weeks*

E EMOTIONAL STATUS

- Postpartum depression (PPD) is common for women following childbirth →
  - Crying
  - Irritable
  - Sleep disturbances
  - Anxiety
  - Feelings of guilt
- As the nurse ask about feelings of...
  depression ▪ *hopelessness* ▪ *self-harm* ▪ *harm to the newborn*

S SECTION (c-section incisions) / Episiotomy

- Promote proper wound healing
- Report to the health care provider: *pain* ▪ *inflammation* ▪ *surrounding skin is warm to touch*

POSTPARTUM HEMORRHAGE

POSTPARTUM HEMORRHAGE is defined as:

VAGINAL BIRTH: loss of >500 ml of blood

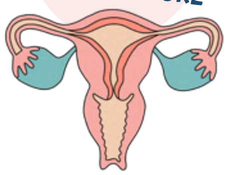
CESAREAN BIRTH: loss of >1,000 ml of blood

A change in hematocrit by 10%

PATHOLOGY

The uterus is like a **BASKET WEAVE OF MUSCLE FIBERS** that crimps off vessels **protecting mom from hemorrhage.**

The uterus is often called the **LIVING LIGATURE**



If the uterus is not doing this crimping off, it causes bleeding!

SIGNS & SYMPTOMS

- Hypotonia of the uterus
- Atony / boggy uterus
- Deviated to the right
- Uncontrolled bleeding

#1 cause of uterine atony is **A FULL BLADDER**

RISK FACTORS

- Multiple gestations
- Polyhydramnios
- Macrosomic fetus (> 8 lbs)
- Multifetal gestation

overdistended uterus

DRUGS

MEMORY TRICK

"**O**h **M**y **H**emorrhage"

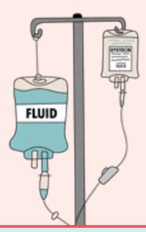
This is a way to remember the order in which the drugs are used

#1 Oxytocin

"**Pitocin**"

ACTION

Stimulates contraction of the uterine smooth muscle



#2 Methergine

"**Methylergonovine**"

ACTION

Vasoconstriction

CONTRAINDICATIONS

Contraindicated in people with hypertension

**Remember vasoconstriction causes blood pressure to rise*

#3 Hemabate

ACTION

Hemabate is a prostaglandin! Hemabate helps control blood pressure and muscle contractions (uterine contractions).

CONTRAINDICATIONS

Contraindicated in people with asthma

Another medication that can be used: Misoprostol

given rectally

ACTION

Stimulates contraction of the uterine smooth muscle

PEDIATRICS

BROUGHT TO YOU BY

PEDIATRIC MILESTONES

INFANCY

BIRTH – 1 year

FONTANELLE CLOSURE

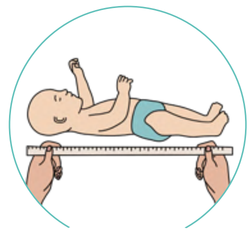
ANTERIOR (larger)
→ Diamond-shaped
→ Closes in 12-18 *months*

POSTERIOR
→ Triangle shaped
→ Closes 8 - 12 *weeks*

WEIGHT

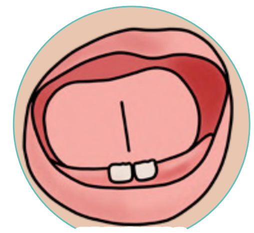
6 MONTHS:
Should **double** from birth weight

12 MONTHS:
Should **triple** from birth weight



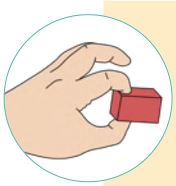
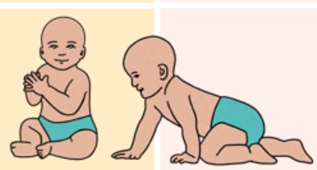
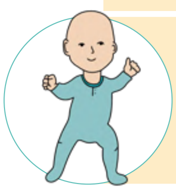
LENGTH

Should be growing **½ - 1 INCH** every month



TEETH

First teeth to show are the **LOWER CENTRAL INCISORS** (usually show around 10 months of age)

		MOTOR SKILLS	LANGUAGE
2 MONTHS		▪ Raises head & chest ▪ Head control improving ▪ Moves head side to side ▪ Should be **SMILING**	Makes verbal noise (coos)
4 MONTHS		▪ Begins to **PLAY** ▪ Rolls from prone to supine ▪ Holds & reaches for toys ▪ Head leads body when pulled to sit	Babbling (copies noises)
6 MONTHS		▪ Can sit up w/support ▪ **STRANGER ANXIETY** begins ▪ Tripod sit 	Babbles (nonspecific)
8-9 MONTHS		▪ Sits without support ▪ Crawling ▪ Stands with pulling & holds onto object ▪ **PINCER GRASP** ▪ **OBJECT PERMANENCE** 	
10-12 MONTHS		▪ Walking ▪ Separation anxiety	Simple words like "dada"

MEMORY TRICK

Rolls on the floor rhymes with four!

Realizing that objects that are out of sight still exist

PEDIATRIC MILESTONES

TODDLER 1-3 years

	15 MONTHS	18 MONTHS	24 MONTHS	30 MONTHS
GROSS MOTOR	• Walks independently 	• Climbs stairs • Pulls toys	• Kicks a ball • Able to stand on tiptoes • Climbs on & off furniture MEMORY TRICK Think Terrible Two's!	
FINE MOTOR	• Feeds self finger foods • Uses index finger to point • Full pincer grasp developed	• Uses their hands a lot for: *reaching, grabbing, releasing, stacking blocks* • Turns book pages • Removes shoes and socks • Stacks four cubes	• Builds tower of 6-7 cubes • Right/left-handed • Scribbles, paints, & imitates strokes • Turns doorknobs • Puts round pegs into holes	
RECEPTIVE LANGUAGE	• Understands 100-150 words • Follows commands without gestures • Looks at adults when communicating	• Understands "no" • Understands 200 words • Says: "what's this?"	• Points to named body parts/pictures in books • Listens to simple stories • Says: "my" & "mine"	• Follows a series of 2 independent commands
EXPRESSIVE LANGUAGE	• Repeats words • Babbles sentences	• Vocab: 15-20 words • Uses names of familiar objects	• Vocab: 40-50 words • Sentences of 2-3 words (ex. "want cookie") • Use descriptive words: hungry, hot, cold	• Vocab: 150-300 words
SIGNS OF DELAY	• Persistent tiptoe walking • Does not develop a mature walking pattern	• Not walking • Not speaking 15 words • Does not understand the function of common household items	• Does not: *use two-word sentences, imitate actions, or follow basic instructions* • Cannot push a toy with wheels	

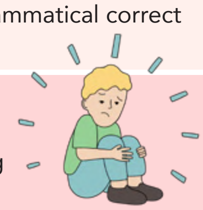
PEDIATRIC MILESTONES

PRESCHOOL 3-6 years

	3 YEARS	4 YEARS	5 YEARS

GROSS MOTOR	▪ Climbs well and runs easily
	▪ Pedals tricycle
	▪ Walks up & down stairs with alternating feet
	▪ Bends over without falling
		▪ Throws ball overhead
		▪ Kicks ball forward
		▪ Can bounce a ball back
		▪ Hops on one foot
		▪ Alternating feet going up & down steps
			▪ May be able to:
			  ▪ Skip
			  ▪ Swim
			  ▪ Skate
			  ▪ Climb
			  ▪ Swing

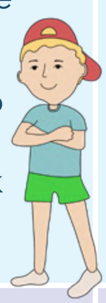
FINE MOTOR	▪ Undresses self ▪ Copies circles ▪ Tower of 9-10 ▪ Holds a pencil ▪ Screws and unscrews lids ▪ Turns book pages one at a time	▪ Uses scissors ▪ Copies capital letter ▪ Draws circles, squares, & traces a cross or diamond ▪ Draws a person with 2-4 body parts ▪ Laces shoes	▪ Can draw a person and some letters ▪ May dress/undress themselves ▪ Can use a fork, spoon, & knife ▪ Mostly cares for own toileting needs
COMMUNICATION	▪ Understands most sentences ▪ Understands physical relation (in, on, under) ▪ Follows a 3-part command ▪ Half of the conversation understood by outside family ▪ Says: "why?" ▪ 3 or 4-word sentences ▪ Talks about past ▪ Vocab: 1,000 words ▪ Says their name, age, & gender ▪ Uses pronouns and plurals	▪ Speaks in complete sentences ▪ Tells a story ▪ 75% of speech understood by outside observers ▪ Stays on topic in conversation ▪ Knows the name of familiar animals ▪ Knows at least one color ▪ Uses language to engage in make-believe ▪ Can count a few numbers ▪ Vocab: 1,500 words	▪ Most of the child's speech can be understood ▪ Explains how an item is used ▪ Participates in long & detailed conversations ▪ Talks about past, future, and imaginary events ▪ Answers questions that use "why" and "when" ▪ Can count to 10 ▪ Says name & address ▪ Recalls part of a story ▪ Speech should be completely intelligible, even if the child has articulation difficulties ▪ Speech is generally grammatical correct ▪ Vocab: 2,000 words
SIGNS OF DELAY	▪ Difficulty with stairs ▪ Falls a lot while walking ▪ Can't build a 4+ block tower ▪ Extreme difficulty separating from parents ▪ No make-believe play ▪ Can't copy a circle ▪ No short paragraphs ▪ Doesn't understand simple instructions ▪ Unclear speech & drooling ▪ Little interest in other kids	▪ Can't jump in place or ride a tricycle ▪ Can't stack 4 blocks ▪ Can't throw a ball overhead ▪ Does not grasp crayon with thumb and fingers ▪ Difficulty with scribbling ▪ Can't copy a circle ▪ Doesn't say 3+ word sentences ▪ Can't use the words "me" & "you" ▪ Ignores other children or doesn't show interest in interactive games ▪ Still clings or cries if parents leave	▪ Sad often ▪ Little interest in playing with other kids ▪ Unable to separate from their parents ▪ Is extremely aggressive, fearful, passive, or timid. ▪ Easy distracted (can't concentrate for 5 minutes) ▪ Can not do ADLs by themselves (brush teeth, undress, wash & dry hands, etc.) ▪ Rarely engages in fantasy play



PEDIATRIC MILESTONES

PHYSIOLOGICAL CHANGES

	EARLY ADOLESCENCE (10-13 years)	MIDDLE ADOLESCENCE (14-16 years)	LATE ADOLESCENCE (17-20 years)
MALE	▪ Pubic hair spread laterally, begins to curl, pigmentation increases ▪ Growth & enlargement of testes & lengthening of the penis ▪ Lengthy look due to extremities growing faster than the trunk	▪ Pubic hair becomes more coarse in texture & takes on adult distribution ▪ Testes, scrotum, & penis continue to grow ▪ The skin around the scrotum darkens ▪ Glands penis develops ▪ May experience breast enlargement ▪ Voice changes	▪ Mature pubic hair distribution & coarseness ▪ Breast enlargement disappears ▪ Adult size & shape of testes, scrotum, and penis ▪ Scrotum skin darkening
FEMALE	▪ First menstrual period (average age is 12 years) ▪ Breasts bud and areola continue to enlarge (no separation of the breasts) ▪ Pubic hair begins to curl & spread over the mons pubis	▪ Pubic hair becomes coarse in texture ▪ Amount of hair increases ▪ Areola & papilla separate from the contour of the breasts to form a secondary mound	▪ Mature pubic hair distribution and coarseness



PEDIATRIC CPR (<12 MONTHS)

Cardiac arrest in infants usually stems from **RESPIRATORY ETIOLOGY**

PEDIATRIC VITAL SIGNS

AGE	RESPIRATIONS	PULSE	SYSTOLIC BP
NEWBORN	30 - 50	120 - 160	60 - 80
6 MO - 1 YR	30 - 40	120 - 140	70 - 80
2 - 4 YR	20 - 30	100 - 110	80 - 95
5 - 8 YR	14 - 20	90 - 100	90 - 100
8 - 12 YR	12 - 20	80 - 100	100 - 110
> 12 YR	12 -20	60 - 90	100 - 120

BREATHS/MIN

BEATS/MIN

ORDER OF EVENTS

1 PULSE

- Check pulse no longer than 10 seconds

  INFANT: Check **BRACHIAL** pulse

  CHILD: Check **CAROTID** pulse

2 CALL FOR HELP

- Active the emergency response system / shout for nearby **HELP**
- Delegate someone else to call 911 / get the AED

3 CHEST COMPRESSIONS

- 2 minutes of CPR before retrieving the AED
  - Rate of 100 - 120 compression/min
  - Using either 2 fingers or 2 thumbs on the sternum
  - Depth: **INFANT:** Equal to one-third of chest's anterior-posterior diameter

    CHILD: 2 inches
  - Allow for recoil between compressions

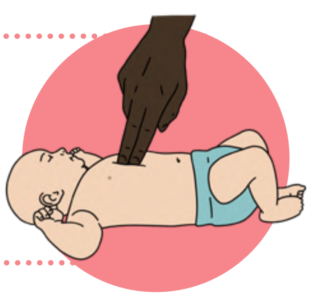
SINGLE RESCUER

30:2 compression-to-breath ratio

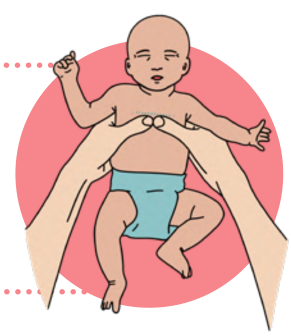
TWO RESCUERS

15:2 compression-to-breath ratio

2-FINGER COMPRESSION TECHNIQUE



2-THUMB ENCIRCLING HAND TECHNIQUE



4 CONTINUE UNTIL SIGNS OF HELP ARRIVE OR AED BECOMES AVAILABLE

PIAGET'S STAGES OF COGNITIVE DEVELOPMENT

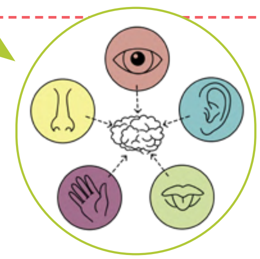
MNEMONIC

Saying **P**iaget's **C**ognitive Stages is **F**un

SENSORIMOTOR STAGE

0 – 2 YEARS

- Development through our 5 senses
- Development through motor response
- **OBJECT PERMANENCE** is developed
- Egocentric
  - ➥ Can only see the world from one's own point of view



Realizing that objects that are out of sight still exist

PREOPERATIONAL STAGE

2 – 7 YEARS

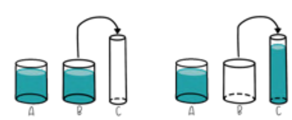
- Symbolic thinking
- Imagination
- Abstract thinking is still difficult
- Asks a lot of questions (intuition)

- Magical thinking
- **ANIMISM** - thinks objects are alive
- Plays pretend

CONCRETE OPERATIONAL STAGE

7 – 11 YEARS

- Develop concrete cognitive operations
  - ➥ Sorting blocks in a certain order
- **CONSERVATION** is developed
- Conductive reasoning (Mathematical advancements)



CONSERVATION
Understanding that something stays the same in volume even though its shape changes.

FORMAL OPERATIONAL STAGE

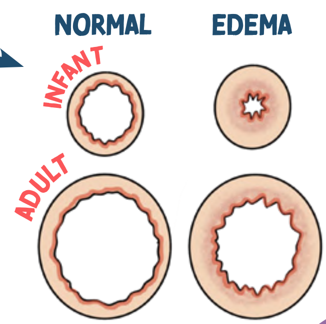
> 11 YEARS

- More rational, logical, organized, moral, and consistent thinking
- **HYPOTHETICAL THINKING** - Can think outside the present
- Abstract concepts
  - ➥ Love, hate, failures, successes
- Deductive reasoning

VARIATIONS IN PEDIATRIC ANATOMY & PHYSIOLOGY

RESPIRATORY

- Narrow airways
- Newborns have ↓ alveoli than an adult
  - Thousands of alveoli grow each day for the first few months of life!
- Floppy airways from less cartilage
- Obligatory nose breathers
- ↑ metabolic rate
- ↑ O_2 requirements

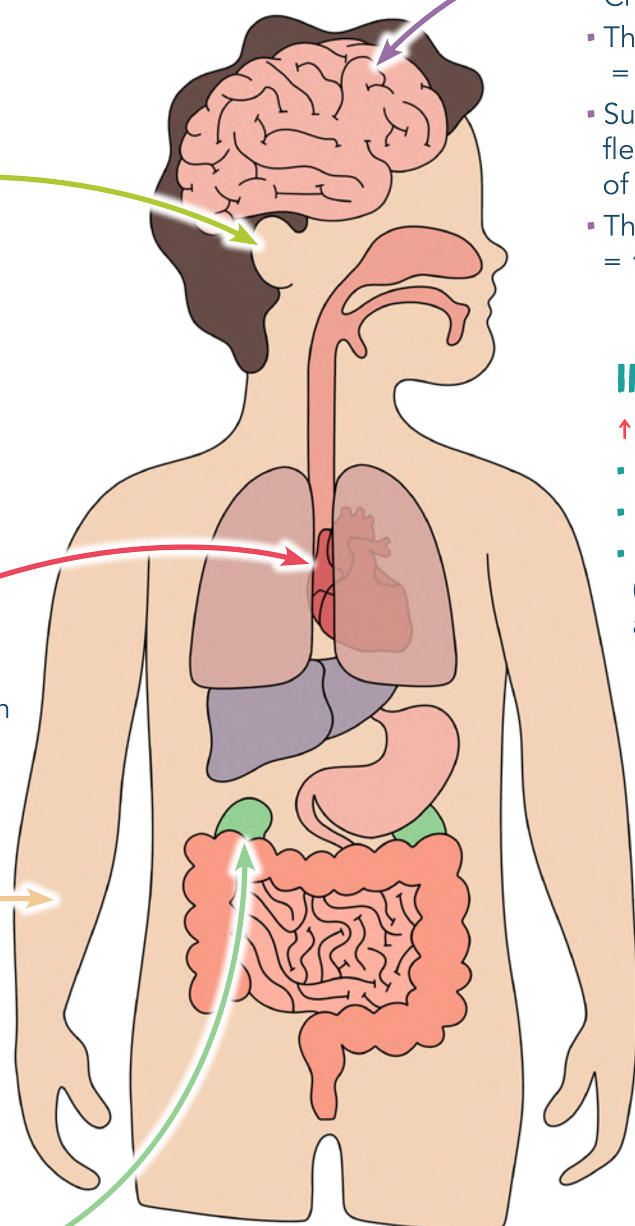


HEAD SIZE

- Head is the fastest growing part of an infant (large in proportion to the body!)
- Head & neck muscles are not well developed

BRAIN & SPINAL CORD

- Cranial bones not completely fused
- The brain is highly vascular = ↑ risk for hemorrhage
- Sutures & fontanels makes the skull flexible and allows for growth of the brain
- The spine is very mobile = ↑ risk for cervical spin injury



EARS

↑ RISK FOR EAR INFECTION

- Eustachian tubes are short, wide, & flat
  = making drainage difficult
  = harbors microorganisms

IMMUNE SYSTEM

↑ RISK FOR INFECTION

- Immature immune systems
- ↓ inflammatory response
- Limited exposure to disease (losing immunity from maternal antibodies)

CARDIOVASCULAR

- The transition from fetal circulation → normal circulation at birth
- Infants hearts are thinner and less compliant

NERVOUS SYSTEM

- Myelinization is incomplete at birth
- Myelinization happens in **CEPHALOCAUDAL DIRECTION** (head to tail)

SKIN

- Epidermis is thinner
- Blood vessels are closer to the surface - loses heat very easily!

CEPHALOCAUDAL DIRECTION

(HEAD TO TAIL)

Head control before walking!

KIDNEYS

- Kidneys are larger in relation to abdomen = less protection
- GFR is slower
- ↓ ability to concentrate urine & reabsorb = ↑ risk for dehydration

PROXIMODISTAL

(INWARD OUTWARD)

SUDDEN INFANT DEATH SYNDROME (SIDS)

Sudden death of a previously healthy infant younger than 1 year of age

RISK FACTORS

- AGE: 1 - 6 months (↑ risk)
- Preterm
- Sleep position
- Sibling death
- Nicotine exposure
- Socioeconomic status
- Lack of prenatal care
- Genetic
- Bedding (can be smothered)
- Room temp (cooler is better)

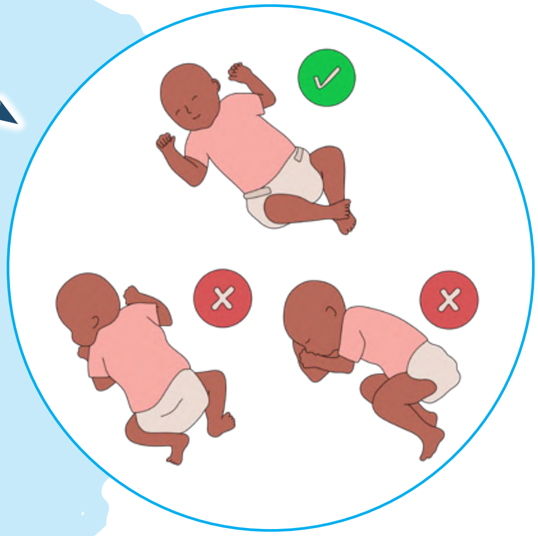
THERE ARE NO SIGNS OR SYMPTOMS!

Sudden death

Leading cause of death in infants

EDUCATION / PREVENTION

- Sleep in **SUPINE POSITION**
- Bedding
    - **Firm** mattress
    - No toys, blankets, pillows, or stuffed animals
- Avoid over bundling or overdressing the infant
- Avoid smoking
- **No co-bedding**
  (Infant should sleep separate from the parents)
- Normal room temp
- Encourage pacifier use



MEMORY TRICK

ABCs OF SAFE SLEEPING

A **A**lone

B On their **B**ack

C In a **C**rib

NEURAL TUBE DEFECTS

NORMAL SPINE

The neural tube closes:
3rd - 4th week of gestation

SPINA BIFIDA

is a general term for a birth defect typically diagnosed during pregnancy where the spinal column fails to close.

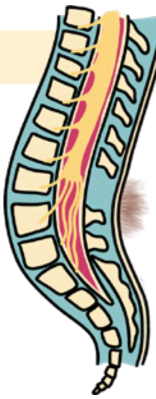
Spina bifida means **"SPLIT SPINE"**

CAUSES

NOT KNOWN...

BUT MANY FACTORS HINDER NORMAL CNS DEVELOPMENT

- Drugs
- Malnutrition
- Chemicals
- Genetics
- Folic acid deficiency (Vitamin B9)
- Diabetes
- Obesity



SPINA BIFIDA OCCULTA

MILDEST FORM

Defect of the vertebral body **WITHOUT** protrusion of the spinal cord or meninges.

Typically asymptomatic

May have dimpling, abnormal patches of hair, or discoloration near the spine.

Does not need immediate medical care if asymptomatic.

If symptoms are present, the client may get an MRI.

MENINGOCELE

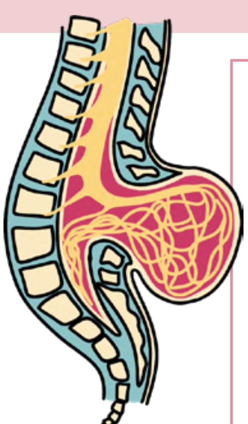
Sac protruding from the spinal area.

Most are covered with skin.

Meninges herniate through a defect in the vertebrae.

Usually minor or no neurological deficits.

Surgical correction of the lesion



MYELOMENINGOCELE

MOST SEVERE FORM

Protrusion of the meninges, cerebrospinal fluid, and spine.

Skin may be exposed as well.

The spinal cord often ends at the point of the defect.

=

Absent motor & sensory function beyond that point.

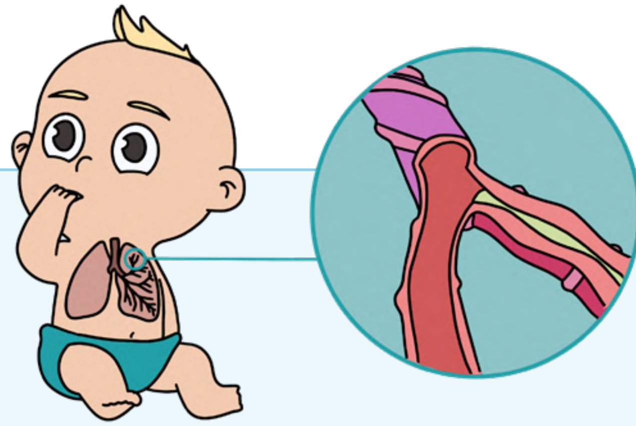
- Multiple surgical procedures
- Paralysis
- Bladder / bowel incontinence
- Neurogenic bladder
- Meningitis (infection)
- Hypoxia
- Hemorrhage
- Freq. catheterization causes...
  - ➡ Latex allergy
  - ➡ UTIs / pyelonephritis
  - ➡ Renal damage

BRONCHIOLITIS (RSV)

PATHO

BRONCHIOLITIS

- BRONCHI → small airways in the lungs
- ITIS → inflammation



- Viral illness usually caused by **Respiratory syncytial virus (RSV)**
- Very contagious
- Starts as an upper respiratory infection & moves into the chest

SIGNS & SYMPTOMS

INITIAL

- Upper respiratory symptoms
  - Nasal congestion
  - Runny nose
  - Cough
  - Sneezing
- Fever

CONTINUED

- Lower respiratory tract symptoms
  - Tachypnea
  - Cough
  - Wheezing

EMERGENT

- Grunting
- Nasal flaring
- Cyanosis
- Hypoxia
- Respiratory failure
- Apneic episodes

TREATMENT

- Self-limited illness & supportive care
- Airway maintenance
  - Oxygen
  - Suctioning
    Saline nose drops & then suction the nares with a bulb syringe to remove the secretions before feeding or at bedtime
  - Position the child at a 30 - 40° angle

- Hydration
  Increase fluid intake (oral or IV) (risk for dehydration)
- Hospitalization
  Only necessary if the child has severe symptoms
- Use contact & standard precautions during care

Most children can be managed at home

REYES SYNDROME

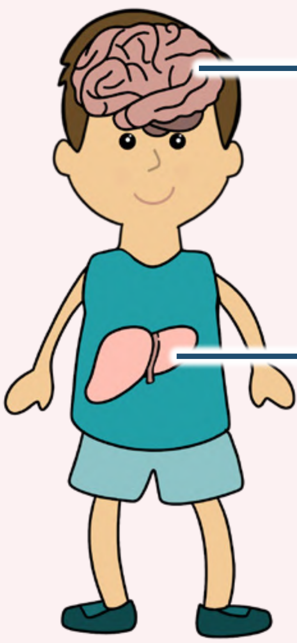
Rare disease affecting young children recovering from a viral illness (flu or chicken pox)

CAUSE

EXACT CAUSE UNKNOWN

Triggered due to the intake of salicylates
or salicylate-containing products
such as **aspirin** *to treat a viral illness*
(Flu / Chickenpox)

SIGNS & SYMPTOMS



ENCEPHALOPATHY / CEREBRAL EDEMA

ACUTE FATTY LIVER FAILURE

LABS
- ↑ LIVER ENZYMES
- ↑ AST
- ↑ ALT

MNEMONIC **"CHILDS"**

- **C** Confusion (changes in mental status)
- **H** Hyperreflexia
- **I** Irritability
- **L** Lethargy
- **D** Diarrhea & vomiting
- **S** Seizures

TREATMENT

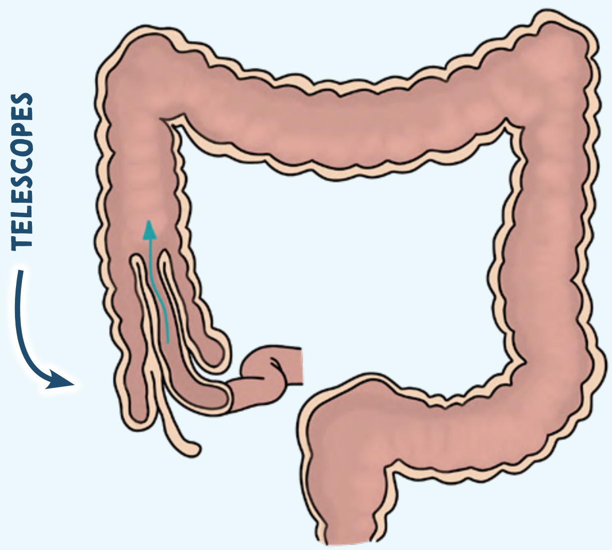
- Early recognition & treatment
- Education on prevention!
- Monitor fluid status
- Swelling of the brain occurs
  - Maintaining cerebral perfusion
  - Managing & preventing increased ICP
  - Seizure precautions

Educate on
products that contain
SALICYLATES:

Aspirin
Alka-Seltzer
Pepto-Bismol
Kaopectate

INTUSSUSCEPTION

PATHO



ILEUM TELESCOPES INTO THE CECUM
↓
OBSTRUCTION = PAIN
↓
COMPRESSION OF BLOOD VESSELS
↓
BLOOD FLOW DECREASES
↓
BOWEL ISCHEMIA
↓
RECTAL BLEEDING (CURRANT JELLY STOOLS!)

SIGNS & SYMPTOMS

- Intermittent pain / cramping
- Child draws up their legs toward the abdomen in severe pain while crying
- Vomiting & diarrhea
- Currant-jelly stools (bloody)
- Lethargy
- Sausage-shaped mass in the upper mid-abdomen

THIS IS BECAUSE TELESCOPING IS INTERMITTENT

CAUSES

- **NOT COMPLETELY KNOWN**
- May be due to a virus that causes swelling
- Condition child is born with
  - Diverticulum
  - Polyps

TREATMENT

- May spontaneously be reduced (Passage of normal, brown stools)
- IV fluids
- Antibiotics
- Decompression via NG tube
- Provide comfort & emotional support to the parents
- Monitor for signs of perforation & shock
- May need air or barium enema
  - Provide education to child & family about pre-op & post-op

DIAGNOSTIC / TREATMENT

AIR or **BARIUM ENEMA** works to diagnose & also helps reduce the intussusception

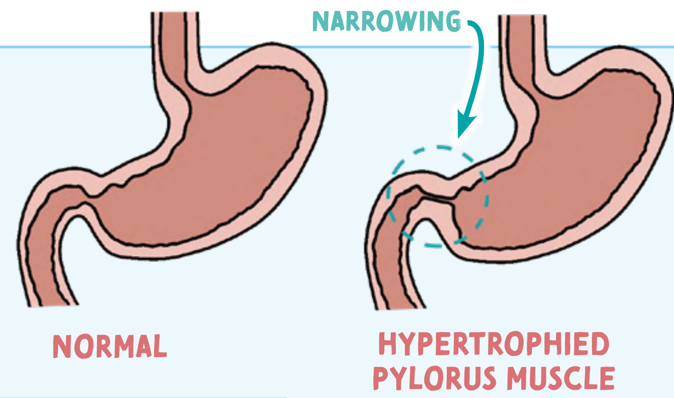
HYPERTROPHIC PYLORIC STENOSIS

PATHO

A hypertrophied pyloric muscle causes narrowing of the pyloric canal

↓

Thickness creates a narrow stomach outlet



HYPERTROPHIC	PYLORIC	STENOSIS
↓	↓	↓
Increase in size	Pylorus	Narrowing

Opening from the stomach into the small intestines

SIGNS & SYMPTOMS

- Projectile vomiting
- Non-bilious emesis
- Olive-shape mass palpable in the right upper quadrant
- Infants will be hungry constantly despite regular feedings
- Weight loss
- DEHYDRATION!

↑ Hematocrit from hemoconcentration
↑ BUN

Stomach contains acid which becomes depleted when vomiting which leads to **METABOLIC ALKALOSIS**

↑PH & ↑HCO3

TREATMENT

- Monitor ...
  - I&O's
  - Vomiting episodes & stools
  - Signs of dehydration & electrolyte imbalances
- Obtain daily weights
- Provide comfort & emotional support to the parents
- Educate about surgery

PYLOROMYOTOMY

Cut the muscle of the pylorus

↓

Relieving the gastric outlet obstruction

EPIGLOTTITIS

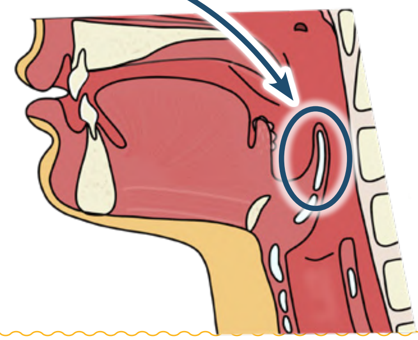
PATHO

Inflammation of the **EPIGLOTTIS** leading to an **UPPER AIRWAY OBSTRUCTION**

WHAT IS THE EPIGLOTTIS?

Piece of cartilage at the back of the tongue

FUNCTION:
Closes the entry to the trachea during swallowing....
AKA prevents aspiration



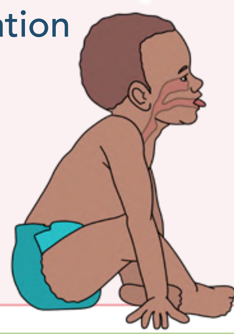
CAUSES

- Most common cause: **HAEMOPHILUS INFLUENZA TYPE B**
- Streptococcus pneumonia

PEDS incident falling due to Hib vaccination

SIGNS & SYMPTOMS

- Tachycardia
- Sore throat
- High fever
- Anxious / apprehensive / agitation
- Difficulty speaking
- Nasal flaring
- Stridor (Frog-like croak on inspiration)



- Drooling / dysphagia
- **Tripod position**
- Sitting forward with the neck extended to breath - mouth open
- Retractions (chest)
- Nasal flaring
- Absent cough!

NURSING MANAGEMENT

- Never leave the client
- Asses oxygen status
- IV access
- May need emergency intubation
- Calm environment
  - Stay with parents
  - Don't restrain the child
  - Help to avoid crying
  - Most comfortable position (usually tripod position)
- Do not place them in supine position. It becomes harder to breathe.

Do not visualize the throat with a tongue blade. Take oral temperature or take throat culture.

WHY?
It can cause **REFLEX LARYNGOSPASMS** (cutting off the airway)

- NPO
- Medications
  - Antibiotics
  - Antipyretics
  - Corticosteroids *(decrease inflammation)*
  - IV Fluids

LARYNGOTRACHEOBRONCHITIS "CROUP"

PATHO

Inflammation of the **LARYNX, TRACHEA, & BRONCHI** occur as a result of viral infection

Most commonly caused by the **PARAINFLUENZA VIRUS**

LARYNGO ↓ Larynx

TRACHEO ↓ Trachea

BRONCHI ↓ Bronchi

ITIS ↓ Inflammation

SIGNS & SYMPTOMS

- Inflammation & edema obstructs the airway

MEMORY TRICK

3 S's *Symptoms occur at night*

- **S**tridor
- **S**ubglottic swelling (causes hoarseness in the voice)
- **S**eal-bark cough

	CROUP	EPIGLOTTITIS
ONSET	Sudden (at night)	Rapid (within hours)
FEVER	Fluctuating	High
COUGH	Yes	No
DYSPHAGIA	No	Yes
CAUSE	Viral	Bacterial
EMERGENCY	Not typically	Yes

TREATMENT

HOME CARE

Self-limiting
(Usually resolves on its own)

- Corticosteroids (↓ inflammation)
- Racemic epinephrine
- Humidified air (steamy bathroom or mist humidifier)
- Encourage rest & fluid intake
- Calm environment for the child

SEEK HELP

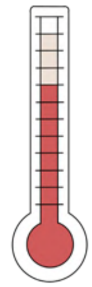
When the child is indicating respiratory distress

- Child is confused/restless
- Blue lips/nails
- ↑ respiration rate (breathing faster, but less air is going in)
- Retractions
- Nasal flaring
- Drooling/can't swallow

FEVER MANAGEMENT

NORMAL TEMP

97.5°F to **98.6°F**
36.4°C to **37.0°C**



FEVER

> 100.4°F (38.0°C)

SIGNS & SYMPTOMS

- Flushed skin
- Diaphoresis (sweating)
- Chills
- Restlessness
- Lethargy

TREATMENT

- Administer antipyretics (ibuprofen) → Do not administer aspirin (risk for Reye's Syndrome)
- Monitor for S&S of dehydration & electrolyte imbalances → Provide adequate fluids!
- Sponge bath → Tepid water for 20-30 min. Squeeze over back & body
- Remove excess clothing & coverings to ↓ the temp
- Cool compress on the forehead

FEBRILE SEIZURE

WHAT IS IT?

Seizures associated with a **FEVER**

Not related to:

- intracranial infection
- metabolic imbalance
- viral illness

Usually **DOES NOT** have long term complications such as epilepsy or intellectual disability

SIGNS & SYMPTOMS

- Rapid ↑ in core temperature
- Child may be drowsy during postictal period

RISK FACTORS

- 6 months - 5 years
- Rapidly developed fever
- HIGH fever
- Family history of febrile seizures
- Certain vaccines
    - DTP & MMR

TREATMENT

- NOT anticonvulsants therapy
- Rectal Diazepam
- Educate the parents to seek help if...
    - Last > 5 min
    - Repeated seizures

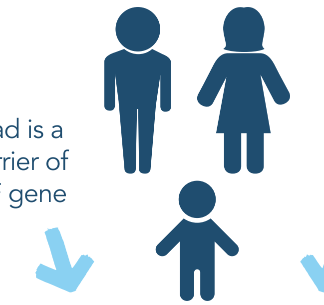
CYSTIC FIBROSIS (CF)

PATHO

- Multisystem disorder of the **EXOCRINE GLANDS** with increased production of thick mucus
- Gene mutation (CFTR): prevents exocrine glands from properly functioning
- **EXOCRINE GLANDS**: Produce & transfer secretions (mucus, tears, sweat, & enzymes) via ducts
- ↑ viscosity of mucus = ↑ resistance to ciliary action = slowing the flow rate of mucus, leading to **mucus plugging**

CF is an **AUTOSOMAL RECESSIVE** genetic disorder

Dad is a carrier of CF gene



Mom is a carrier of CF gene

2 mutated CF genes = **CYSTIC FIBROSIS**

DIAGNOSIS

- Ambry test
- Positive sweat sodium chloride test
- Genetic screen

TREATMENT

- **Treatment of the mucus**
  - Chest physiotherapy (PT)
  - Postural drainage
  - Huff coughing
  - Nebulizers
    - Bronchodilators, mucolytics, anti-inflammatory drugs
- **Treat & prevent infection**
  - Wear a mask, hand washing, up-to-date on vaccines, avoid those who are sick.

- **Nutrition**
- **Prevent GI blockage**
  - Fluids & stool softeners

CHEST PT

- Drains airways of thick mucus to be coughed up
  - Stimulates cough
  - Helps loosen mucus
  - Results in deep breathing
  - Builds up strength and endurance of respiratory muscles
  - Improves cardiovascular fitness

- Done multiple times a day between 1-2 hour increments
  - NOT done right before or after meals!
- Causes vibrations & percussions to break apart the mucus (vests, manual vibration)

- ↑ protein, ↑ fat, ↑ calorie
  - Fat soluble vitamin supplementation A, K, E, D
- Possible supplemental oral feeding or enteral feeding
- **Pancreatic enzymes**:
  - Pancrelipase or Pancreatin
  - Can swallow a capsules or sprinkle enzymes on foods that are acidic such as apple sauce!

MEMORY TRICK

All Kids Eat Donuts

MANIFESTATIONS OF CF

NOSE & SINUSES

- Sinusitis
- Nasal polyps (snoring, stuffiness)

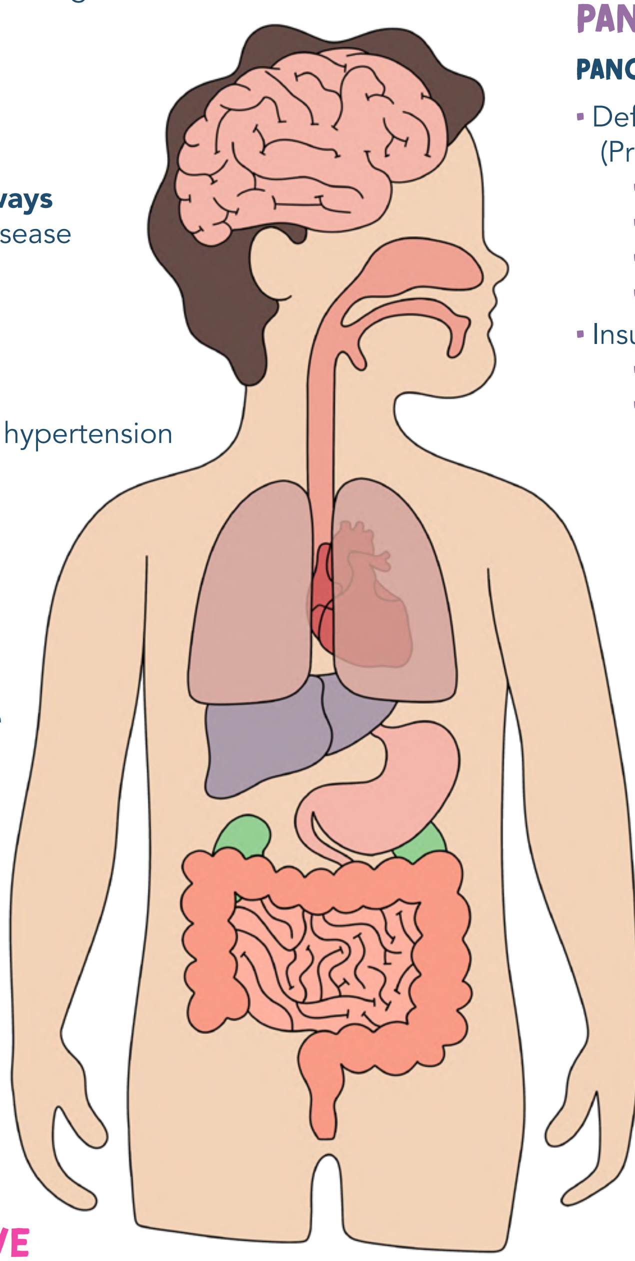
RESPIRATORY

- **INFECTION:** Thick mucus creates a great environment for bacterial growth
  - Pseudomonas
  - Staph. aureus
- Pneumonia
- Bronchitis
- **Thick mucus** = **blocked airways**
  - Obstructive pulmonary disease (Emphysema)
    - Clubbing
    - Barrel-shape chest
- Pneumothorax
- Strain on lungs = pulmonary hypertension

PANCREAS

PANCREAS SECRETES THICK MUCUS

- Deficient in pancreatic enzymes: (Protease, Amylase, Lipase)
  - Weight loss
  - Inadequate protein absorption
  - Deficiency of protein
  - Failure to thrive
- Insulin deficiency
  - Hyperglycemia
  - CF-related diabetes



CARDIOVASCULAR

- Pulmonary hypertension puts strain on the heart
  - **Right-sided heart failure**

LIVER

- Bile duct blocked from **THICK** mucus
  - Gallstones
  - Biliary cirrhosis

INTEGUMENTARY

- Sweat glands produce ↑ chloride = salty skin
- Salty sweat & salty tears which leads to
  - Dehydration
  - Electrolyte imbalance

STOMACH & INTESTINES

- Fecal impaction
- Rectal prolapse
- Bowel obstruction
- Intussusception
- Back up of stool in intestine
  - Constipation
  - Vomiting
  - Abdominal distention
  - Cramping
  - Anorexia
  - RLQ pain
- Meconium ileus in infants
- Steatorrhea
  - Frothy (bulky), fatty, foul-smelling stools

REPRODUCTIVE

BOYS

- Thick mucus blocks the vas deferens = Infertility

BOTH HAVE DELAYED PUBERTY

GIRLS

- Thick cervical mucus blocks sperm from penetrating = Infertility

FETAL CIRCULATION IN UTERO

FORAMEN OVALE

Blood is *SHUNTED* from the right atrium to the left atrium by the **FORAMEN OVALE**

Blood bypasses the lungs...why?

It's already oxygenated blood from the placenta (mom)

How can blood be shunted from the right atrium to the left atrium?

PRESSURE DIFFERENCE!

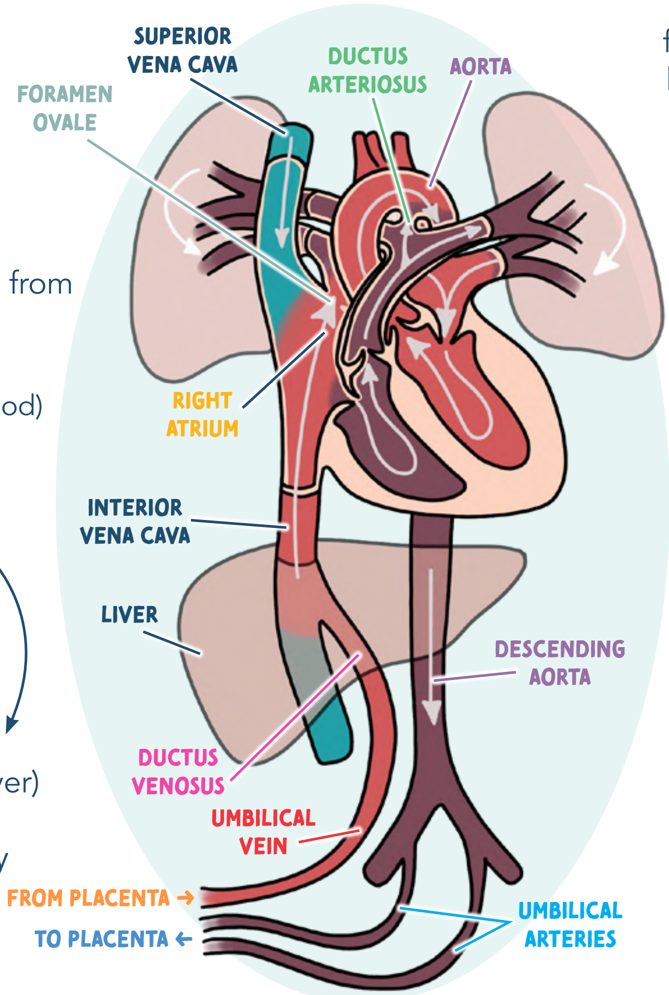
Blood flows from
high resistance
to
low resistance

Lungs: High resistance from all the fluid. So the blood does not want to go in the lungs!

RIGHT ATRIUM

Blood goes from the interior vena cava to the right atrium as well as some deoxygenated blood coming from the **SUPERIOR VENA CAVA**.

So the blood is now MIXED (oxygen-rich & oxygen-poor blood)



DUCTUS ARTERIOSUS

Blood is *SHUNTED* from the pulmonary artery into the aorta by the ductus arteriosus

Liver not fully functioning yet

DUCTUS VENOSUS

Umbilical vein is carrying oxygenated blood from the placenta. It passes the **LIVER** (Some blood will go to the liver) but most will be *SHUNTED* to the **INFERIOR VENA CAVA** by the **DUCTUS VENOSUS**

AORTA

Mixed blood is now in the aorta and being pushed out to oxygenate the fetus

BLOOD GOES BACK TO THE PLACENTA TO GET OXYGENATED AGAIN!

THE PLACENTA IS THE "LIFELINE" BETWEEN MOTHER & BABY

The Placenta is like "temporary lungs" for the fetus while in utero

2 UMBILICAL ARTERIES

1 UMBILICAL VEIN

MEMORY TRICK **A** think **AWAY**

Takes deoxygenated blood + waste **AWAY** from the baby back to the placenta

Gives oxygen rich blood **TO** the baby

START

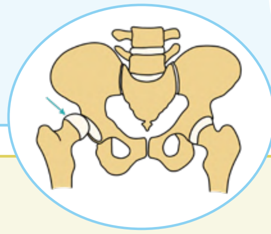
SHUNTS TO KNOW

- Ductus venosus
- Foramen ovale
- Ductus arteriosus

DEVELOPMENT DYSPLASIA OF THE HIPS (DDH)

PATHO

- Abnormal development of the hip joint
- A baby's bones are not ossified yet so they have the ability to dislocate & relocate easily



DISLOCATION No contact between femoral head & acetabulum

SUBLUXATION Partial dislocation (acetabulum is not completely in contact with the hip joint)

DYSPLASIA Hip joint doesn't have the proper shape to fit together correctly

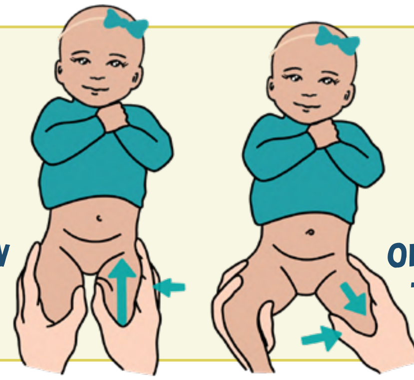
DIAGNOSIS

- Ultrasound for in utero
- X-ray for those older than 6 months
- Barlow test & Ortolani

Listen for any noises during the exam. There should be no "clunks" heard or felt.

If "clunks" are felt or heard = a positive sign for DDH

BARLOW TEST



ORTOLANI TEST

COMPLICATIONS

- Avascular necrosis of the femoral head
- ↓ ROM
- Leg-length discrepancy
- Early osteoarthritis
- Femoral nerve palsy

RISK FACTORS

- **FEMALE** → more lax ligaments from maternal hormones
- Breech positioning
- Oligohydramnios

TREATMENT

Early detection & treatment are crucial. The bones are not ossified in early infancy, so you want to manipulate them to grow properly. If DDH is not treated early the bones will ossify and develop incorrectly.

> 6 MONTHS

- **Pavlik harness**: Stabilizes the hip by preventing hip extension

4 MONTHS - 2 YEARS

- **Closed reduction**:
  - Requires general anesthesia where the hips will be placed back into the acetabulum by the surgeon
  - Spica cast is worn after surgery to maintain reduction
  - After spica cast the child will wear a brace until acetabulum is fully normal

> 2 YEARS OR NO IMPROVEMENTS WITH SURGERY OR HARNESS

- Open surgical reduction followed by casting

INSTRUCTIONS FOR PAVLIK HARNESS

- Must wear the harness at all times!
- Do not adjust the straps or remove harness until instructed by the HCP
- Change the diaper while the baby is in the harness
- Check for redness, irritation or breakdown 2-3 times per day
- Place baby on their back to sleep
- Place long knee socks and undershirt to prevent rubbing of the harness

SCARLET FEVER

PATHO

- Complication of **group A streptococcal infection** AKA **Strep throat**
- Not all children who have strep will develop scarlet fever
- **TRANSMISSION:** Droplets & respiratory tract secretions.
  Transmission happens in close contact such as schools & daycares.

MEMORY TRICK

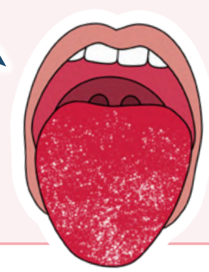
Scarlet Fever think Strep!

SIGNS & SYMPTOMS

- Onset: ABRUPT!
- **RED RASH!**
  Sandpaper-like rash
- Pharyngitis
- Fever, body aches, chills
- Strawberry tongue
- Tender cervical nodes
- Tonsils are red
- Exudate may be present

Begins on the **NECK & CHEST** and spreads outwards to **THE EXTREMITIES!**

Rash is usually not seen on the palms & soles of the feet



MEMORY TRICK

S's of SCARLET FEVER:
Strawberry tongue
Sandpaper rash

COMPLICATIONS

- Rheumatic fever
- Glomerulonephritis
- Abscesses of the throat
- Pneumonia

Early diagnosis & treatment are very important to prevent complications!

TREATMENT

Most children can be cared for at home

Take antibiotics as directed.... Finish the medication even if the child appears to be better!

- Antibiotics (Penicillin V)
  - Erythromycin for those allergic to Penicillin
- Fluids & soft foods
- Provide comfort
- Cool mist humidifier

Soups, teas, Popsicles, slushies

MED-SURG

BROUGHT TO YOU BY NURSE IN THE MAKING

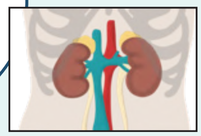
KIDNEY OVERVIEW

FUNCTIONS of the kidneys

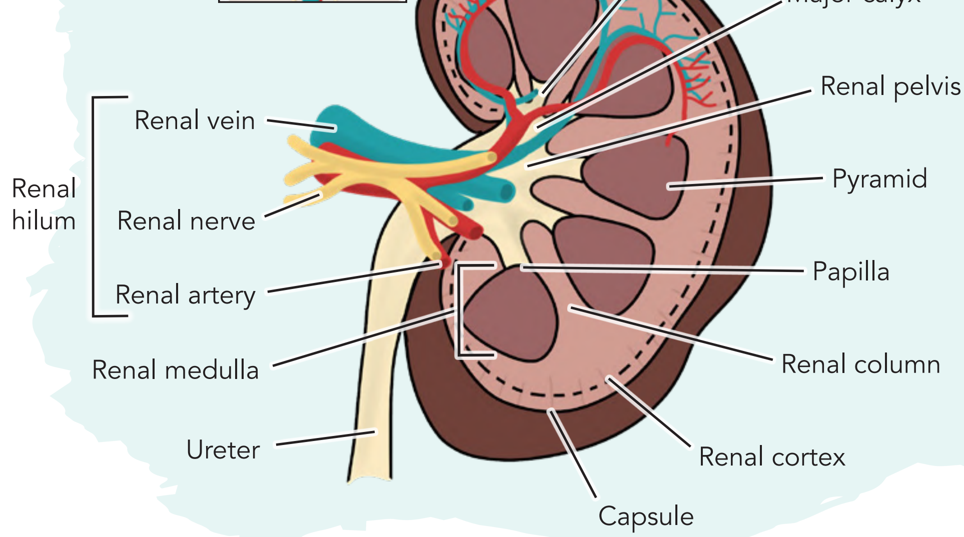
MNEMONIC "A WET BED"

- **A**cid-base balance
- **W**ater balance
- **E**lectrolyte balance
- **T**oxin removal
- **B**lood pressure control
- **E**rythropoietin
- vitamin **D** metabolism

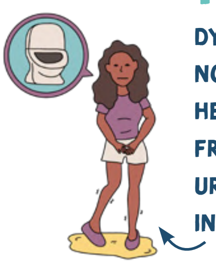
FUN FACT: The right kidney sits lower than the left due to the location of the liver



ANATOMY of the kidney



TERMS TO KNOW



DYSURIA	Pain while urinating
NOCTURIA	Excessive urination at night
HEMATURIA	Bloody urine
FREQUENCY	Voiding more than every 3 hours
URGENCY	Strong desire to void
INCONTINENCE	Involuntary voiding
ENURESIS	Involuntary voiding during sleep
PROTEINURIA	Abnormal amounts of protein in the urine
OLIGURIA	Urine output: <400 mL/day
ANURIA	Urine output: <50 mL/day
MICTURITION	Voiding

URINE FORMATION

1 GLOMERULAR FILTRATION → **2 TUBULAR REABSORPTION** → **3 TUBULAR SECRETION** → **4 URINE EXCRETION**

1. GLOMERULAR FILTRATION

Blood flows into the kidneys: **120 mL/min**

Filters water, electrolytes, & small molecules **into the glomerulus** (Large molecules stay in the bloodstream)

2. TUBULAR REABSORPTION

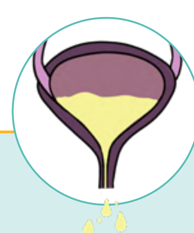
Fluid moves **from renal tubules into the capillaries**. They reabsorb fluid into the venous circulation.

3. TUBULAR SECRETION

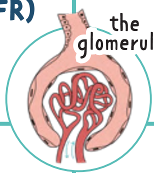
Fluid moves **from the capillaries into the renal tubules** to get eliminated/excreted.

4. URINE EXCRETION

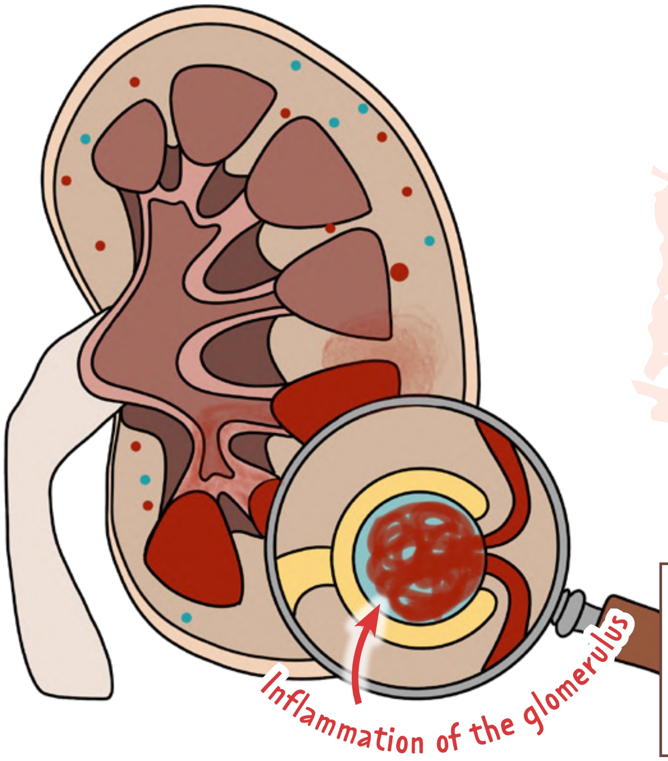
Adults should void 1-2 L/day
No less than 30mL/hr



LAB VALUES RELATED TO THE KIDNEYS

	DESCRIPTION	EXPECTED RANGE	↓	↑
GLOMERULAR FILTRATION RATE (GFR) 	Rate of blood flow through the kidneys	**90 - 120** mL/min	**POSSIBLE CAUSES:** Kidney dysfunction *(such as chronic kidney disease)*	-
CREATININE *Creatinine is a better indicator of kidney function than BUN*	End product of muscle metabolism; solely filtered from the blood via glomerulus	**0.6 - 1.2** mg/dL MEMORY TRICK **RHYME:** Creatinine over 1.3 = think bad kidney	▪ Muscle mass is low ▪ Hyperthyroidism ▪ Starvation ▪ Liver disease	▪ Acute or chronic kidney disease ▪ Congestive heart failure ▪ Dehydration ▪ Certain drugs ANY TIME THE GFR RATE DECREASES
BLOOD UREA NITROGEN (BUN)	Normal waste product resulting from the breakdown of proteins. ↑ levels can indicate a kidney problem & be toxic in the body	**7 - 20** mg/dL MEMORY TRICK Think hamburger BUNs... Hamburgers can cost anywhere from $7 - $20	▪ Liver damage ▪ Malabsorption ▪ Poor diet ▪ Low nitrogen diet	Can be due to PRErenal failure, POSTrenal failure, or INTRArenal failure *See "ACUTE KIDNEY INJURY (AKI)" page*
URINE SPECIFIC GRAVITY	Measures the kidney's ability to excrete or conserve water	**1.010 - 1.030**	▪ Too much fluid intake ▪ Diabetes Insipidus DILUTED URINE MAKES THE #'S GO DOWN	▪ Dehydration ▪ Syndrome of inappropriate antidiuretic hormone secretion (SIADH) CONCENTRATED URINE MAKES THE #'S GO UP
URINE OUTPUT	The amount of urine a person excretes from their bladder via the urethra	Urine output: **AT LEAST 30** mL/hr Average adult: **1500** mL/day 	▪ Shock ▪ Hypotension ▪ Trauma ▪ Infection ▪ Chronic kidney injury	▪ Diabetes mellitus ▪ Diabetes insipidus ▪ Too much diuretics
NORMAL FINDINGS	Free from glucose, ketones, blood, protein, bilirubin, nitrates or leukocyte esterase in the urine			

ACUTE GLOMERULONEPHRITIS (POSTSTREPTOCOCCAL)



PATHOLOGY

1. Untreated strep
2. Immune system response by creating **antigen-antibody complexes** (14 days after infection)
3. These antibodies get "lodged" in the glomeruli
4. Inflammation & scarring
5. ↓ GFR

It's not the strep that causes the inflammation of the kidneys.

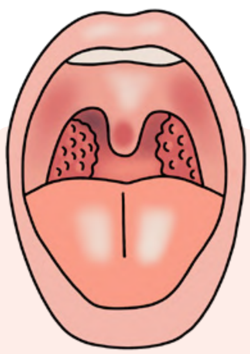
It's the **antigen-antibody complexes** that form due to the strep that causes the inflammation & damage to the glomeruli

SIGNS & SYMPTOMS

- Hematuria → Blood in the urine
- Azotemia → Excessive nitrogenous waste in the blood
- Malaise
- Headache
- Proteinuria (mild)
- Hypoalbuminemia
- ↓ GFR = Oliguria

Tea colored urine (cola color)

- Edema
  - Swelling in the face/eyes
- ↑ Blood pressure
- Retaining sodium
- ↑ Urine specific gravity
- ↑ BUN & creatinine
- (+) ASO (Antistreptolysin) Titer



MAIN CAUSE:

Recent group A beta-hemolytic streptococcal infection

INTERVENTIONS

- Fix the cause! (strep)
- Diet modifications
  - Fluid restriction
  - Sodium restriction
  - ↓ Protein
  - Provide a lot of carbohydrates

Carbohydrates provide energy & stop the breakdown of protein

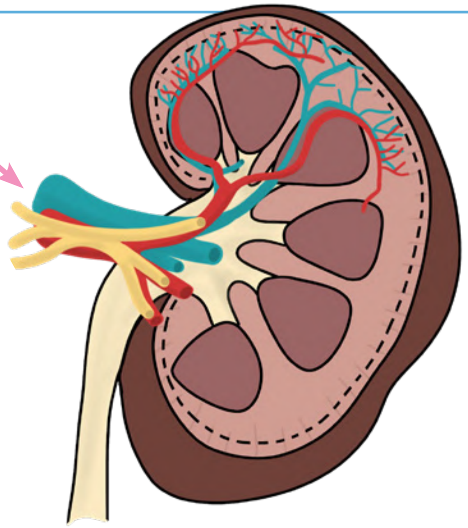
- Monitor
  - Daily intake & output
  - Daily weight
- Bed rest
- Monitor blood pressure
  - Antihypertensives
  - Diuretics

A weight gain of 1 kg is equal to 1,000 mL of retained fluid

ACUTE KIDNEY INJURY (AKI)

WHAT IS IT?

Sudden renal damage! Causes a build-up of waste, fluid, and electrolyte imbalance.
It can be reversible. Formerly called *Acute Renal Failure*.



PRERENAL FAILURE

Damage **BEFORE** the kidneys

↓ VOLUME/PERFUSION TO THE KIDNEYS

- **Cardiac damage**
  - ↓ or impaired cardiac output
  - *Example:* MI
- **Vasodilation**
- **Hemorrhage** (hypovolemia)
- **Burns**
- **GI losses** (vomiting/diarrhea)

INTRARENAL FAILURE

Damage **IN** the kidneys

PROLONGED ISCHEMIA

- **Myoglobinuria**
- **Hemoglobinuria**
- **Rhabdomyolysis**
- **Nephrotoxic drugs**
  - *Examples:* NSAIDs, antibiotics (aminoglycosides), chemo drugs, contrast dyes
- **Infections**
  - *Examples:* Glomerulonephritis

POSTRENAL FAILURE

Damage **AFTER** the kidneys

OBSTRUCTION/BLOCKAGE IN THE URINARY TRACT

- **Renal calculi (stones)**
- **Blood clots**
- **Benign prostatic hyperplasia (BPH)**
- **Tumors**
- **Neuro damage (stroke)**

MEMORY TRICK "OH OH DARN RENAL"

	OH	OH	DARN	RENAL
	ONSET/INITIATION	**OLIGURIA**	**DIURETIC**	**RECOVERY**
PHASES	Triggering event *(Prerenal, intrarenal or postrenal failure)*	↓ Urine output < 400 mL/24 hrs Glomerulus decreases the ability to filter blood (↓ GFR)	Cause of AKI is corrected Gradual ↑ in urinary output	↑ in kidney function May take up to 6 - 12 months
TREATMENT	Correct & identify the underlying cause to prevent long term damage to nephrons!	**DIET MODIFICATIONS:** • Low protein diet • Limit fluid intake • Strict I&O + daily weights **MONITOR** EKG & labs • *Watch for HYPERkalemia > 5.0* • *↑ BUN & creatinine* • Dialysis may be needed until kidney function returns	Large amount of diluted urine with electrolytes **MONITOR** the patient for dehydration & hypokalemia	Some patients may never recover and may develop chronic kidney disease (CKD)

NEPHROTIC SYNDROME

PATHOLOGY

Inflammatory response in the glomerulus

↓

Damage to membrane

↓

Loss of protein (albumin)

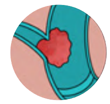
Albumin regulates oncotic pressure

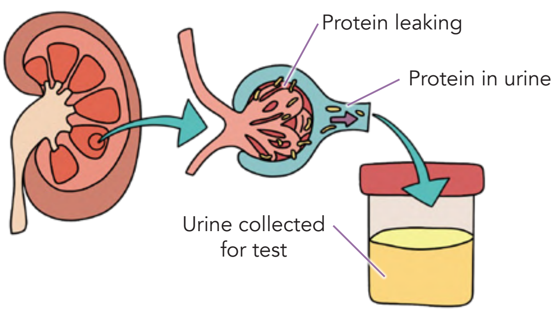
↓

HYPOALBUMINEMIA

Low albumin levels

- Causes synthesis of cholesterol & triglycerides → Hyperlipidemia
- Fluid shift → Generalized edema
- Albumin is a protein which prevents clot formation → Possible blood clots (thrombosis)
- Can lose protein that helps fight infections (immunoglobulins) → Risk for infection





CAUSES

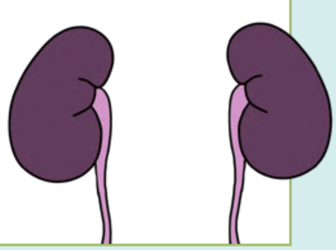
- Bacteria or viral infection
- Cancer
- Genetic predispositions
- Systemic disease (lupus or diabetes)
- NSAIDs

SIGNS & SYMPTOMS

- **Hypoalbuminemia**
  - Edema
  - Fatigue & loss of appetite
  - Hyperlipidemia
- **Proteinuria (> 3 g/day)**
  - Large amounts of protein in the urine

INTERVENTIONS

- **Monitor fluid status**
  - Daily weights & I&O's
  - Swelling & abdominal girth
- **Diet modifications**
  - ↓ Cholesterol & saturated fats
  - ↓ Na+ intake
  - Moderate protein intake
- **Medications**
  - Diuretics
  - Statins (lipid-lowering drugs)
  - Prednisone to ↓ inflammation
  - Antineoplastic agent
  - Immunosuppressant
- **Monitor signs of...**
  - Infection
  - Blood clots



CHRONIC KIDNEY DISEASE (CKD)

PATHOLOGY

- Progressive & irreversible loss of kidney function
- Occurs over a long period of time

CAUSES

- Untreated acute kidney injury (AKI)
- Diabetes mellitus
- Hypertension
- Family history
- Recurrent infections
- Autoimmune disorders

STAGES

Stages are based on the GFR rate
AS CKD WORSENS... GFR DECREASES ↓

STAGE	GFR
STAGE 1	> 90
STAGE 2	60 - 89
STAGE 3	A: 45 - 59 B: 30 - 44
STAGE 4	15 - 29
STAGE 5	< 15 END STAGE RENAL DISEASE

TREATMENT

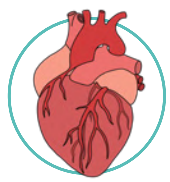
- Dialysis
- Kidney transplant

SIGNS & SYMPTOMS

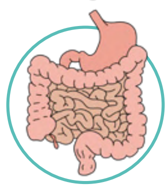
In the end stages of CKD,
ALMOST EVERY BODY SYSTEM is negativity affected

- ↓ Urinary output (UOP)
  - Oliguria = <400 mL/day
  - Anuria = <100 mL/day
- Proteinuria & hematuria

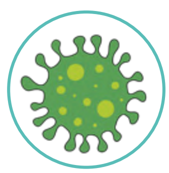
- Lethargy
- Altered LOC/confusion
- Seizures



- Hypertension
- Fluid volume excess (hypervolemia)
- Heart failure

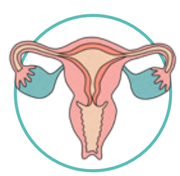


- Anorexia
- Nausea/vomiting
- Uremic fetor (ammonia breath)
- Metallic taste

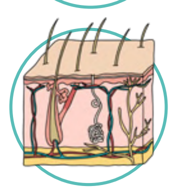


- Impaired immune & inflammatory response

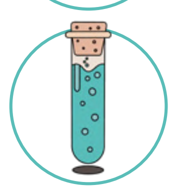
- Anemia (↓ erythropoietin [EPO])
- ↑ Risk for bleeding
- Prolonged bleeding time



- Amenorrhea
- Erectile dysfunction
- ↓ Libido



- Uremic frost
- Pruritus



LABS

- ↑ BUN
- ↑ Creatinine
- ↑ K+
- ↑ Magnesium
- ↓ Calcium
- ↑ Phosphate

TYPES OF DIALYSIS: HEMODIALYSIS

DIALYSIS is a way to remove waste products from the blood in those with kidney dysfunction. In a healthy body, the kidneys are able to filter waste products. But if the kidneys are not functioning properly and are injured, they need help removing excess waste from the blood. Otherwise, waste accumulates and becomes toxic/harmful to the body.

MOST COMMON METHOD OF DIALYSIS

HEMODIALYSIS

uses a dialyzer (an artificial kidney) to remove excess fluids and toxins.

THE PROCESS

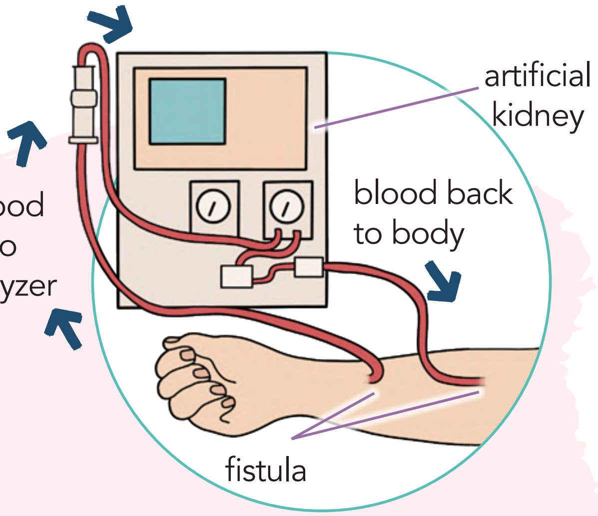
OUTSIDE THE BODY

THE DIALYZER
(Artificial kidney)

↓
Brings blood to the dialyzer
↓
Filters out toxins/waste products
↓
Brings clean blood back to the body

3X a week
(3 - 5 hours per treatment)

Typically done in the hospital or in a dialysis clinic



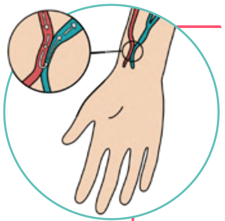
COMPLICATIONS

- Hypotension
- Disequilibrium syndrome
- Hemorrhage
- Air embolus
- Electrolyte imbalances

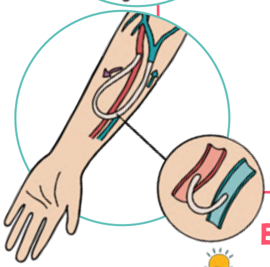
ACCESS

VASCULAR ACCESS

BOTH REQUIRE SURGERY



FISTULA
Joining an artery to a vein



GRAFT
Inserting synthetic graft material between an artery and a vein

Increased risk for infection due to the synthetic material insertion

EVALUATION OF PATENCY:

MEMORY TRICK

✔ **Feel the thrill** (palpating the fistula)
✔ **Hear the bruit** (heard during auscultation)

PATIENT EDUCATION

On the arm that has vascular access, you need to **AVOID:**

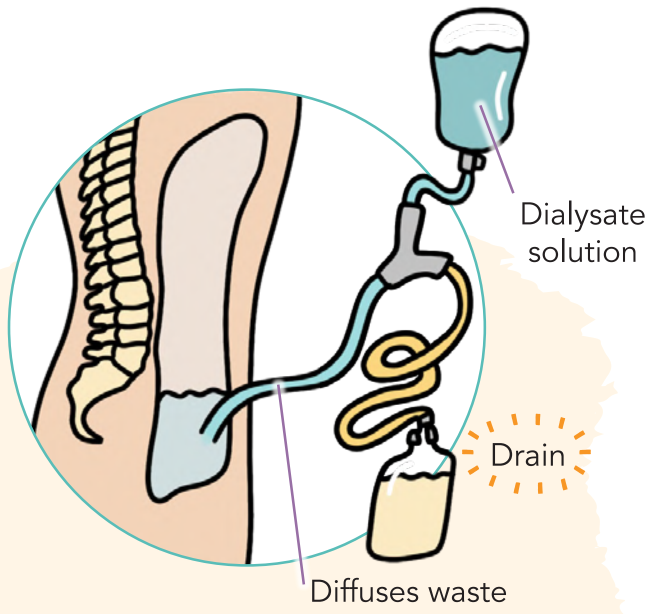
✘ Compression
✘ Blood draws
✘ Blood pressure readings
✘ Tight clothing
✘ Carrying bags
✘ Sleeping on that arm

TYPES OF DIALYSIS: PERITONEAL DIALYSIS

DIALYSIS is a way to remove waste products from the blood in those with kidney dysfunction. In a healthy body, the kidneys are able to filter waste products. But if the kidneys are not functioning properly and are injured, they need help removing excess waste from the blood. Otherwise, waste accumulates and becomes toxic/harmful to the body.

PERITONEAL DIALYSIS

Uses a peritoneum to remove excess fluids and toxins



THE PROCESS

INSIDE THE BODY

Warm the solution!

↓

Dialysate is infused into the peritoneal cavity by gravity

↓

Close the clamp on the infusion line

↓

Dialysate dwells for a set amount of time (dwell time)

↓

The drainage tube is unclamped

↓

Fluid drains from the peritoneal cavity by gravity

↓

A new container of dialysate is infused as soon as drainage is complete

↓

REPEAT!

7X a week
(multiple exchanges per day)

Typically done at home

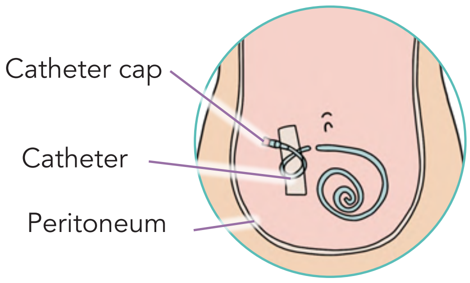
COMPLICATIONS

This procedure is commonly done at home and has an **increased risk for infection** in the peritoneum.

PERITONITIS (INFECTION)

- Cloudy or bloody drainage
- Fever
- Abdominal pain
- Malaise

ACCESS



PERITONEAL CATHETER
performed at the bedside or in the operating room

PATIENT EDUCATION

How to **AVOID** infections:

✔ Good hand hygiene before and after dialysis

✔ Clean site of catheter daily

✔ Keep supplies in a clean, dry place

URINARY TRACT INFECTION

PATHO

Infection within the urinary system caused by either a **BACTERIA**, **VIRAL**, or **FUNGUS**.

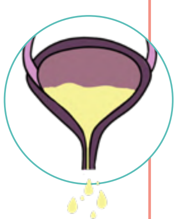
BACTERIA IS THE MOST COMMON specifically E. Coli

CAUSES

- Most common in women (shorter urethra & urethra is close to the rectum)
- Overuse of antibiotics
- Indwelling catheters
- Hormone changes (pregnancy changes)
- Diabetes
- Lifestyle
  - Baths, scented tampons, perfumes, etc.

EDUCATION

- Take entire antibiotics course
- Wipe from front to back
- Void after intercourse
- Avoid caffeine & ETOH
- Void frequently
- Avoid bubble baths, perfumes, or sprays!
- Wear non-tight cotton underwear



NURSING CONSIDERATIONS

- Maintain fluid status
  - 2 - 3 L per day
  - Remove the catheter ASAP (per HCP order)
- Medications
  - Antibiotics
  - Analgesia (control pain)
  - Phenazopyridine (Pyridium)

"flushing" out the urinary tract

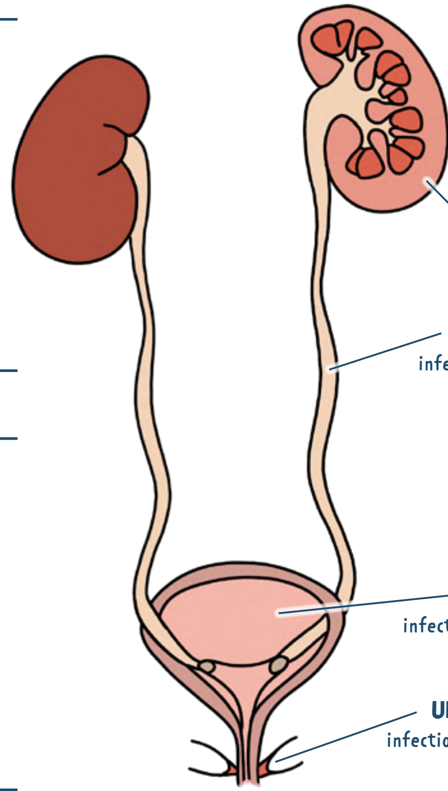
Take urine culture BEFORE giving first dose of antibiotics

Analgesic to ↓ pain
May turn urine orange

UPPER URINARY TRACT

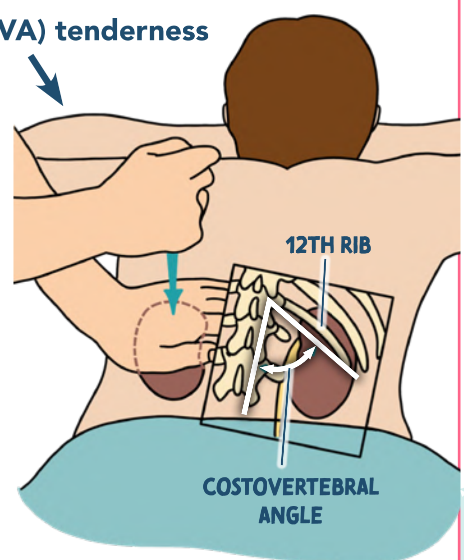
LOWER URINARY TRACT

UTIs typically start in the **lower tract** & move upwards making it to the **upper tract**



SIGNS & SYMPTOMS

- Smelly urine
- Chills & fever
- **Costovertebral angle (CVA) tenderness**
- Nausea & vomiting
- Headache/malaise
- Painful urination (dysuria)
- Burning on urination
- Frequency & urgency
- Nocturia
- Incontinence
- Hematuria
- Fever
- WBCs in the urine



Elderly patients may show atypical symptoms:
- Confusion
- Lethargy
- New incontinence

RENAL CALCULI

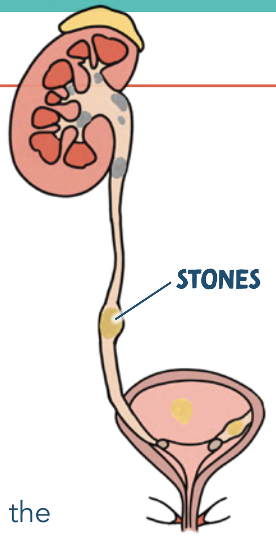
PATHO

Stones (calculi) found in the urinary tract & kidney!

NEPHROLITHIASIS: stones in the **kidneys**

URETEROLITHIASIS: stones in the **ureter**

- Stones can be very large *or* very small
- They can be found inside the **KIDNEYS**, **URETERS**, or the **BLADDER**



TREATMENT

Most commonly, the stone will pass on it's own!

- **Medications to control the *PAIN***
  - NSAIDs
  - Opioid analgesics

  ↓ **Pain & inflammation** *(makes the stone easier to pass)*
- **Strain the urine**
  - keep any stones & send them to the lab to evaluate the type of stone

- **Get them moving or frequently turning them!**

  Push stone forward & out! Decreases risk of infection
- **↑ Fluids!**
- **Diet:**
  - Limit protein, Na+ foods, & calcium
- **Procedures:**

  NONINVASIVE **Extracorporeal Shock Wave Lithotripsy (ESWL)**
  Sends shock waves to break up the stone!

  INVASIVE! **Percutaneous Nephrolithotomy**
  Stone removed by an incision made on the back where the kidneys are located.

SIGNS & SYMPTOMS

- **PAIN!**
- Discomfort
- Hematuria → (RBCs)
- Pyuria → (WBCs)
- Nausea & vomiting

DIAGNOSIS

- KUB: X-ray of kidneys, ureters, bladder
- IVP: intravenous pyelogram
- Ultrasound or CT scan
- Urine test

What is URIC ACID? Uric acid is a waste products of the breakdown of **purines**

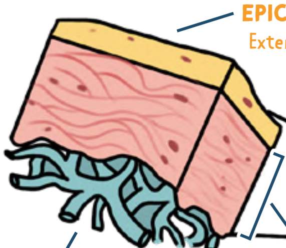



	CALCIUM (MOST COMMON!)	URIC ACID	STRUVITE	CYSTINE
STONE TYPE	Forms due to ↑ amounts of calcium & oxalate in the urine	Too much uric acid in the urine (acidic urine)	Persistent alkaline environment that is ammonia-rich urine Due to a **bacteria**	
CAUSES	• Hypercalcemia • Hypercalciuria • Hyperparathyroidism • ↑ intake of Na+ • Dehydration • GI disorders • ↑ intake of calcium supplements with vitamin D	• Gout • Foods high in purine or animal proteins • Dehydration • Metabolic issues (diabetes)	• Chronic urinary tract infections (UTIs) • Foreign bodies • Neurogenic bladder	Rare, genetic, inherited disorder that affects renal absorption of cystine

CARDIAC OVERVIEW

LAYERS OF THE HEART

There are three layers of the heart:
EPICARDIUM, **MYOCARDIUM**, and **ENDOCARDIUM**



EPICARDIUM
Exterior layer

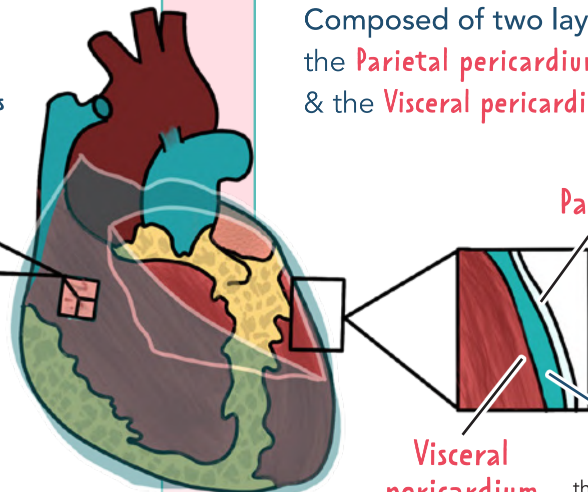
MEMORY TRICK
Epi means "upon"

ENDOCARDIUM
Thin inner layer
Lines the inside of the heart & valves

MEMORY TRICK
Endo means "within"

MYOCARDIUM
Middle layer
Responsible for pumping action

MEMORY TRICK
Myo means "muscle"



PERICARDIUM

Thin sac that encases the heart.

Composed of two layers:
the Parietal pericardium
& the Visceral pericardium.

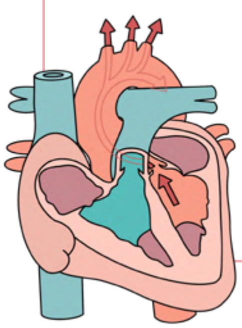
Parietal pericardium

Visceral pericardium
Adheres to the epicardium

Fluid
lubricates the surface of the heart & reduces friction

CARDIAC TERMS

CARDIAC OUTPUT



Total volume of blood ejected (pumped) by the heart per minute.

It's the amount of blood reaching the tissues.

FORMULA

HR x **SV** = **CO**
Heart Rate · *Stroke Volume* · *Cardiac Output*

HR = The # of times the heart contracts each minute (normal **60 - 100** bpm)

SV = Amount of blood ejected from the left ventricle with each heartbeat

NORMAL:
4 - 8 L/min

INTERPRETATION

↓ **CO** = **LESS** volume
(↓ perfusion to the vital organs)

↑ **CO** = **MORE** volume
(could be due to hypervolemia, etc.)

STROKE VOLUME

Amount of blood pumped out of the ventricle with each beat or contraction

CONTRACTILITY

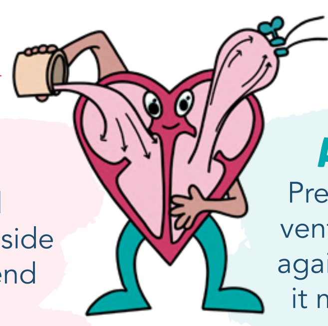
Force / strength of contraction of the heart muscle

EJECTION FRACTION (EF)

% of blood expelled from the left ventricle with every contraction

NORMAL EF: 50 - 70%

EXAMPLE:
If the EF is 55%, the heart is pumping out 55% of what's inside of the left ventricle



PRELOAD

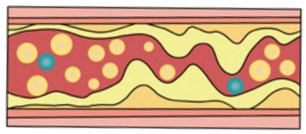
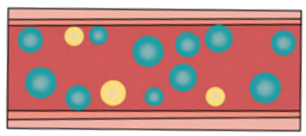
Amount of blood returned to the right side of the heart at the end of diastole

AFTERLOAD

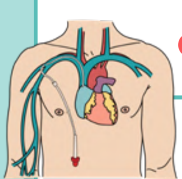
Pressure that the left ventricle has to pump against (the resistance it must overcome to circulate blood)

Clinically measured by systolic blood pressure!

LAB VALUES RELATED TO THE CARDIAC SYSTEM

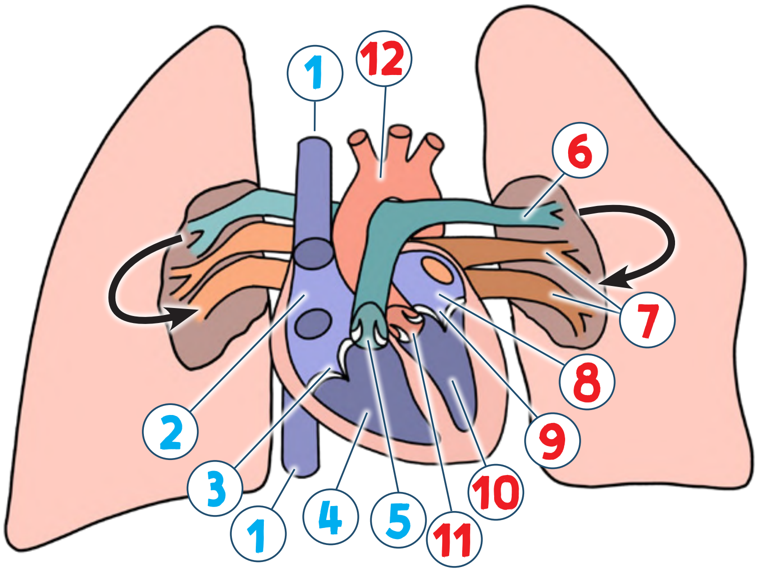
	EXPECTED RANGE	DESCRIPTION	↓	↑
TOTAL CHOLESTEROL	< 200 mg/dL	Measurement of the total amount of cholesterol in your blood	Indicates a lower risk for cardiovascular disease	Increases the risk for heart disease and stroke
TRIGLYCERIDES	< 150 mg/dL	Most common type of fat in the body. Takes the food you eat and stores it as excess energy		
LOW DENSITY LIPOPROTEINS (LDL)	< 100 mg/dL	 LDL bad MEMORY TRICK: LDL think: we want Low levels "bad fat"		
HIGH DENSITY LIPOPROTEINS (HDL)	F > 40 mg/dL M > 55 mg/dL	 HDL good MEMORY TRICK: HDL think: we want High levels, because it's a Happy cholesterol	Increases the risk for heart disease and stroke	Indicates a lower risk for cardiovascular disease
D-DIMER	< 0.5 mcg/mL	D-dimers are fragments of fibrin that are in the blood when a clot dissolves or is broken down. **D-dimer helps to determine if a clot is present somewhere in the body**	 Normal/low Levels • Blood clot is ruled out	 Elevated/high levels (positive result) **POSSIBLE CAUSES:** • Blood clot may be present in the body • Disseminated intravascular coagulation (DIC)
BNP	< 100 pg/mL	BNP is a peptide released when the ventricle is filled with too much fluid and stretches	Helps to indicate heart failure is not present	Congestive heart failure (HF)

HEMODYNAMIC PARAMETERS



CARDIAC OUTPUT (CO)	4 - 8 L/min	Total volume pumped per minute
CARDIAC INDEX (CI)	2.5 - 4.0 L/min/m^2	Cardiac output per body surface area $CI = \frac{CO}{\text{surface area}}$
CENTRAL VENOUS PRESSURE (CVP)	2 - 8 mmHg	Pressure in the superior vena cava. Shows how much pressure from the blood is returned to the right atrium from the superior vena cava
MEAN ARTERIAL PRESSURE (MAP)	70 - 100 mmHg *At least 60 mmHg is required to adequately perfuse the vital organs*	Average pressure in the systemic circulation *(your body)* through the cardiac cycle
SYSTEMIC VASCULAR RESISTANCE (SVR)	800 - 1200 dynes/sec/cm	The **resistance** it takes to push blood through the circulatory system to create blood flow

FLOW OF BLOOD THROUGH THE HEART



RIGHT
Deoxygenated Blood

1. Superior Vena Cava / Inferior Vena Cava
2. Right Atrium (RA)
3. Tricuspid Valve (TV)
4. Right Ventricle (RV)
5. Pulmonary Valve (PV)
6. Pulmonary Artery*

carries **DEOXYGENATED** blood to the **LUNGS**

LEFT
Oxygenated Blood

7. Pulmonary Vein*
8. Left Atrium
9. Bicuspid/Mitral Valve
10. Left Ventricle
11. Aortic Valve
12. Aorta

carries **OXYGENATED** blood to the **TISSUES/BODY**

OVERVIEW OF BLOOD VESSELS

ARTERIES	VEINS
Carry oxygenated blood to tissues	Carry deoxygenated blood back to the heart

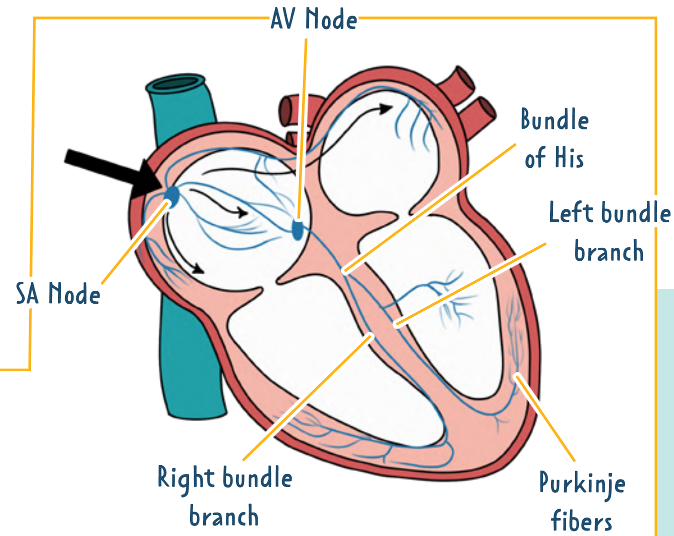
MEMORY TRICK: **A**rteries think **A**way from the heart

* EXCEPTIONS

The only exception to this is the **PULMONARY ARTERY** and **PULMONARY VEIN**

PULMONARY ARTERY	PULMONARY VEIN
brings deoxygenated blood from the heart to the lungs	carries oxygenated blood from the lungs to the heart

ELECTRICAL CONDUCTION OF THE HEART



CARDIAC CONDUCTION SYSTEM:

Generates & transmits **ELECTRICAL IMPULSES** which stimulates contractions of the atria and then the ventricles.

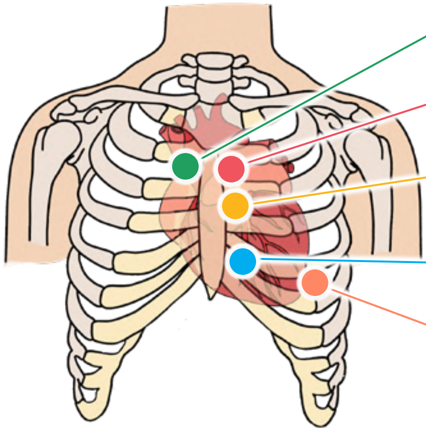
STEPS IN THE HEART'S CONDUCTION SYSTEM

MNEMONIC

SEND	**SA** node (**S**ino**A**trial node)	Primary pacemaker of the heart. Creates electrical impulses of **60 - 100 bpm** (NOTE: This is a normal heart rate)
A	**AV** node (**A**trio**V**entricular)	Secondary pacemaker of the heart "backup pacemaker." If the SA node malfunctions, the AV node takes over at a rate of **40 - 60 bpm**
BIG	**B**undle of His	
BOUNDING	**B**undle branches (right & left)	
PULSE	**P**urkinje fibers	If the SA & the AV nodes fail, the Purkinje fibers can fire at a rate of **30 - 40 bpm**

AUSCULTATING HEART SOUNDS

5 AREAS FOR LISTENING TO THE HEART

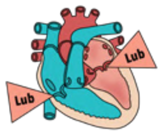


MNEMONIC **A**ll **P**eople **E**njoy **T**ime **M**agazine

- **Aortic** Right 2nd intercostal space
- **Pulmonic** Left 2nd intercostal Space
- **ERB's Point** (S1, S2) Left 3rd intercostal space
- **Tricuspid** Lower left sternal border 4th intercostal
- **Mitral** Left 5th intercostal, medial to midclavicular line

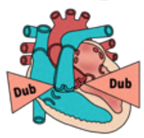
MEMORY TRICK Think **M** for **M**idclavicular

NORMAL



S1 LUB — Tricuspid & mitral valve closure

↓



S2 DUB — Aortic & pulmonic valve closure

CLOSING OF THE VALVES
Valve opening does not normally produce a sound

ABNORMAL

S3 **EARLY DIASTOLE** in rapid ventricle filling

↓

S4 **LATE DIASTOLE** & high atrial pressure (forcing blood into a stiff ventricle)

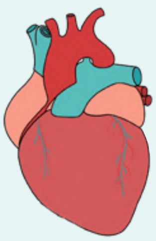
ABNORMAL VENTRICULAR FILLING
Extra ❤ sounds

CONTRACTED

SYSTOLIC

Ventricle pump / ejection = **LUB (S1)**

LUB (S1) ⇄ DUB (S2)



RELAXED

DIASTOLIC

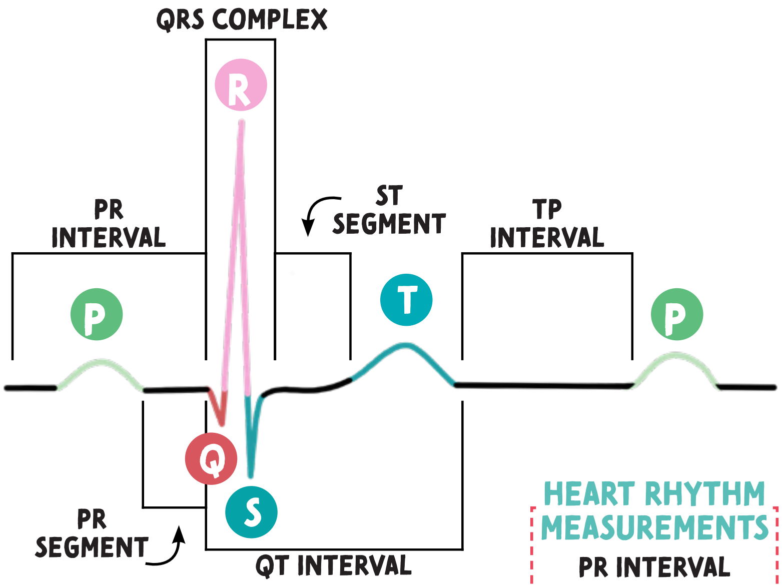
Ventricle relax / filling = **DUB (S2)**

MEMORY TRICK

"COZY RED"

CO (*co*ntract) **ZY** (*sy*stole)
RE (*re*lax) **D** (*d*iastole)

EKG WAVEFORMS



P WAVE	Atrial contraction (**DE**polarization)
PR SEGMENT	Movement of electrical activity from atria to ventricles
QRS COMPLEX	Ventricle contraction (**DE**polarization)
ST SEGMENT	Time between ventricular depolarization & repolarization
T WAVE	Ventricle relaxing (**RE**polarization)
TP INTERVAL	Ventricle relaxing & filling

MEMORY TRICK
DEpolarization think...
DEcompressing

MEMORY TRICK
REpolarization think...
RElaxing
REpolarizing
REfilling with blood

BASIC RHYTHMS

NORMAL SINUS	60 - 100 bpm
SINUS TACHYCARDIA	> 100 bpm
SINUS BRADYCARDIA	< 60 bpm

HEART RHYTHM MEASUREMENTS

PR INTERVAL
0.12 - 0.20

QRS COMPLEX
0.06 - 0.12

QT INTERVAL
< 0.40 seconds

PR INTERVAL

Movement of electrical activity from atria to ventricles

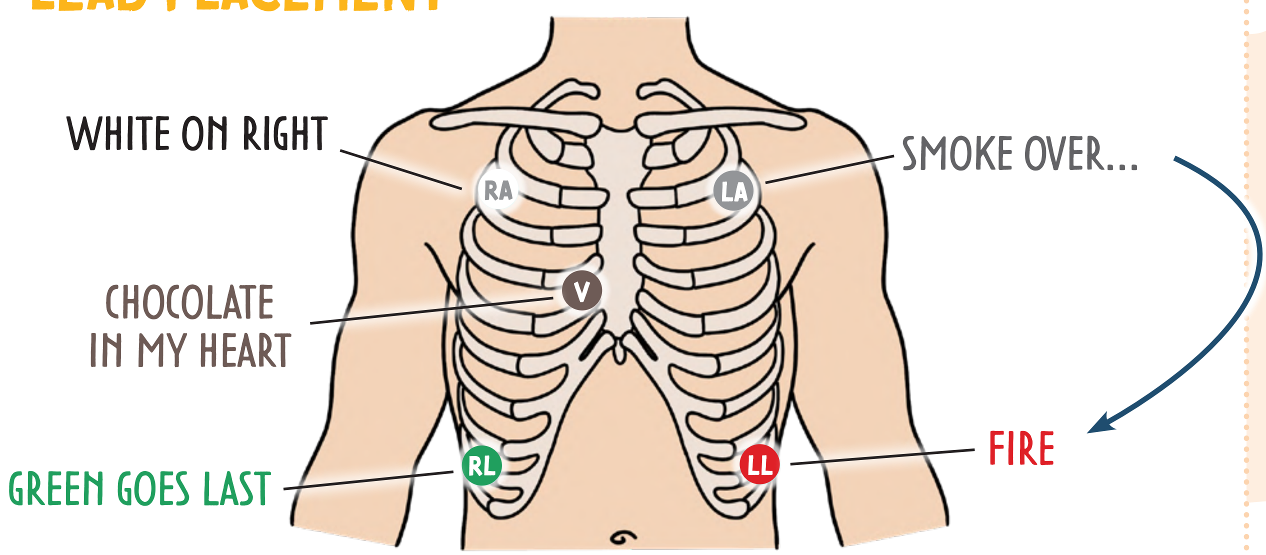
ST SEGMENT

Time between ventricular de-polarization and repolarization *(ventricular contraction)*

QT INTERVAL

Time it takes for ventricles to depolarize and repolarize (to contract and relax)

5-LEAD PLACEMENT



6 STEPS TO INTERPRETING EKGs

#1 P-WAVE

Identify & examine the P-waves

- Should be present & upright
- Comes before QRS complex
- One P-wave for every QRS complex

#2 PR INTERVAL

Measure PR interval

NORMAL PR INTERVAL:
0.12 - 0.20 seconds

#3 QRS COMPLEX

Is every P-wave followed by a QRS complex?

NORMAL QRS COMPLEX:
0.06 - 0.12 seconds

- Should not be widened or shortened – this may indicate a problem!

Widened is often seen in PVCs, Electrolyte imbalances & drug toxicity!

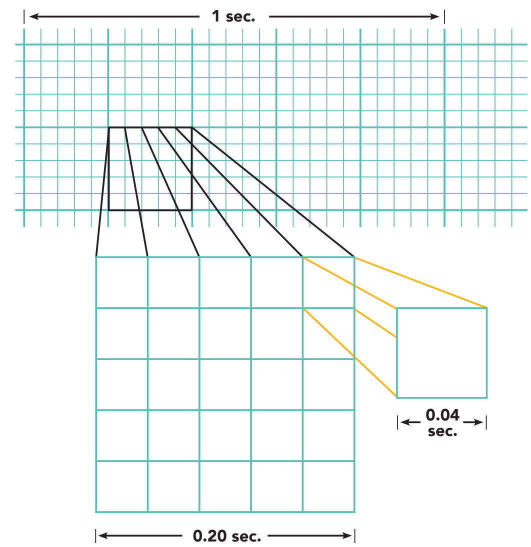
#4 R-R

Are the R-R intervals consistent?

- Regular or irregular?

BASIC RHYTHMS

SINUS TACHYCARDIA	> 100 bpm
NORMAL SINUS	60 - 100 bpm
SINUS BRADYCARDIA	< 60 bpm



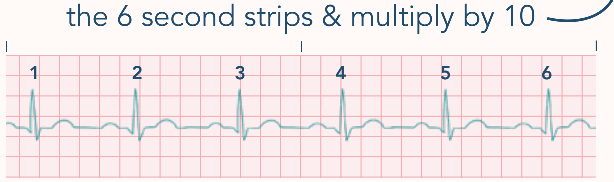
1 large box = 0.20 seconds
5 large boxes = 1 second
1 small box = 0.04 seconds

#5 DETERMINE THE HEART RATE

6 SECOND METHOD

Count the number of R's in between the 6 second strips & multiply by 10

Be sure and check that the strip is 6 seconds! Count the boxes



6 R's X 10 = 60 beats per minutes

BIG BOX METHOD

300 divided by the number of big boxes between 2 R's

300 / 5 = 60 BPM

#6 IDENTIFY THE EKG FINDING!

EKGs

NORMAL SINUS RHYTHM

RATE	60 - 100 bpm
RHYTHM	Regular
P-WAVE	Upright & uniform before each QRS
PR INTERVAL	Normal
QRS COMPLEX	Normal

SINUS BRADY

KEY The sinus node creates an impulse at a ***slower***-than-normal rate

RATE	< 60 bpm
RHYTHM	Regular
P-WAVE	Upright & uniform before each QRS
PR INTERVAL	Normal
QRS COMPLEX	Normal

CAUSES

- Lower metabolic needs
  - Sleep
  - Athletic training
  - Hypothyroidism
- Vagal stimulation
- Medications
  - Calcium channel blockers, beta blockers, Amiodarone

THIS IS NORMAL: Athletes have a low RESTING heart rate. This is because the heart is strong and pumps more efficiently with each heartbeat

TREATMENT

- Correct the underlying cause!
- ↑ the heart rate to normal

SINUS TACHY

KEY The sinus node creates an impulse at a ***faster***-than-normal rate

RATE	> 100 bpm
RHYTHM	Regular
P WAVE	Upright & uniform before each QRS
PR INTERVAL	Normal
QRS COMPLEX	Normal

CAUSES

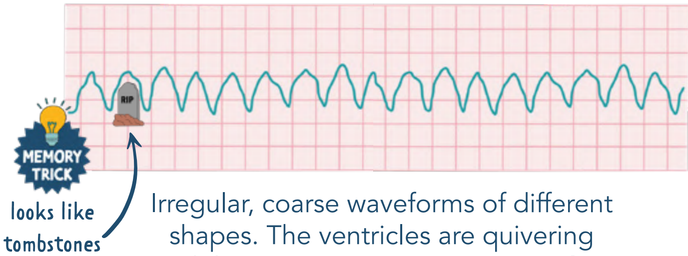
- Physiologic or psychological stress
  - Blood loss, fever, exercise, dehydration, infection, sepsis
- Certain medications
  - **Stimulants**: caffeine, nicotine
  - **Illicit drugs**: cocaine, amphetamines
  - **Stimulate sympathetic response**: epinephrine
  - **Beta-2 agonists**
- Heart failure
- Cardiac tamponade
- Hyperthyroidism

TREATMENT

- Identify the underlying cause!
- ↓ the heart rate to normal

EKGs

VENTRICULAR TACHYCARDIA (VT)



Irregular, coarse waveforms of different shapes. The ventricles are quivering and there is **no contraction or cardiac output** which may be **fatal**!

RATE	100 - 250 bpm
RHYTHM	Regular
P-WAVE	Not visible
PR INTERVAL	None
QRS COMPLEX	Wide (like tombstones) > 0.12 seconds

CAUSES

- Myocardial ischemia / infarction
- Electrolyte imbalances
- Digoxin toxicity
- Stimulants: caffeine & methamphetamine

MANIFESTATIONS

- Patient is usually awake (unlike V-fib)
- Chest pain
- Lethargy
- Anxiety
- Syncope
- Palpitations

No Cardiac Output
=
LOW OXYGEN

TREATMENT

STABLE CLIENT WITH A PULSE

- Oxygen
- Antiarrhythmics
  (ex. Amiodarone...stabilizes the rhythm)
- Synchronized Cardioversion
  - Synchronized administration of shock (delivery in sync with the QRS wave).
  - Cardioversion is NOT defibrillation! (defibrillation is only given with deadly rhythms!)

UNSTABLE CLIENTS WITHOUT A PULSE

Also called **PULSELESS V-TACH**

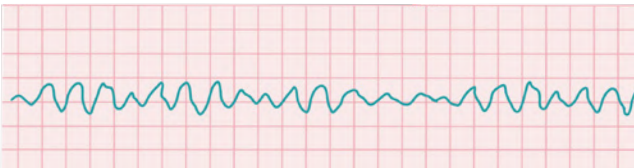
- CPR
- Follow ACLS protocol for defibrillation
- Possible intubation
- Drug therapy
  - Epinephrine, vasopressin, amiodarone

SHOCK!

UNTREATED VT can lead to → **VENTRICULAR FIBRILLATION** → **DEATH** ☠

EKGs

VENTRICULAR FIBRILLATION (V-FIB)



Rapid, disorganized pattern of electrical activity in the ventricle in which electrical impulses arise from many different foci!

RATE	Unknown
RHYTHM	Chaotic & irregular
P-WAVE	Not visible
PR INTERVAL	Not visible
QRS COMPLEX	Not visible

CAUSES

- Cardiac injury
- Medication toxicity
- Electrolyte imbalances
- Untreated ventricular tachycardia

MANIFESTATIONS

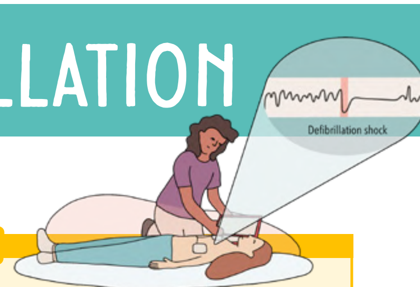
- Loss of consciousness
- May not have a pulse or blood pressure
- Respirations may stop
- Cardiac arrest

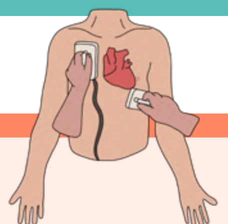
No Cardiac Output = No blood or oxygen to the body

TREATMENT

- CPR
- Oxygen
- Defib — MEMORY TRICK "Defib the Vfib"
  (follow ACLS protocol for defibrillation)
- Possible intubation
- Drug Therapy
  - Epinephrine (causes vasoconstriction)
  - Antiarrhythmics: Amiodarone, lidocaine
  - Possibly magnesium

CARDIOVERSION VS. DEFIBRILLATION





VS

CARDIOVERSION

- **SYNCHRONIZED SHOCK**
  Synced shock delivered only during the R wave of the QRS complex
  If the shock is accidentally delivered during the T-wave, it can cause **R-ON-T PHENOMENON**
- **LOWER** amount of joules (energy) used
- Not done with CPR
- Stable patients
  (must have a QRS complex)

Synchronizer switch must be turned on!

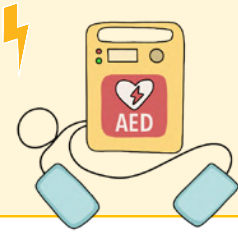
EXAMPLE:

- **A-fib**
  Patients are sedated for this outpatient procedure. It does not require a hospital stay.

MEMORY TRICK Think: **C**ardioversions are **C**arefully planned

DEFIBRILLATION

- **ASYNCHRONOUS**
  Done with an automated external defibrillator (AED)
- **HIGHER** amount of joules (energy) used
- Resume CPR after shock
- Unstable patients



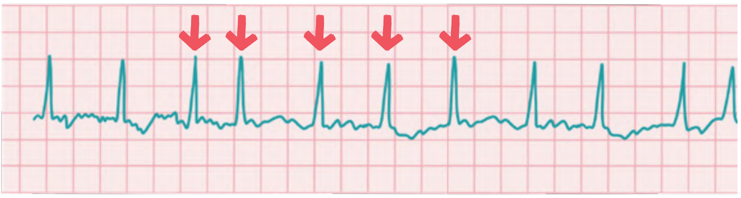
EXAMPLE:

- **Pulseless ventricular tachycardia (VT)**
  or
- **Ventricular fibrillation (VF)**

EKGs

ATRIAL FIBRILLATION (A-FIB)

IRREGULAR R-R INTERVALS



Uncoordinated electrical activity in the atria that causes rapid & disorganized "fibbing" of the muscles in the atrium.

RATE	Usually over 100 bpm
RHYTHM	Irregular
P-WAVE	None. They are irregular (fibrillary waves)
PR INTERVAL	Visible
QRS COMPLEX	Narrow & irregularly irregular

CAUSES

- Open heart surgery
- Heart failure
- COPD
- Hypertension
- Ischemic heart disease

MANIFESTATIONS

THE ATRIA IS QUIVERING!

- Most commonly asymptomatic
- Fatigue
- Malaise
- Dizziness
- Shortness of breath
- Tachycardia
- Anxiety
- Palpitations

All due to Low O_2

TREATMENT

STABLE PT.

- Oxygen
- Drug therapy!
  - Beta blockers
  - Calcium channel blockers
  - Digoxin
  - Amiodarone
  - Anticoagulant therapy to prevent clots

RISK FOR CLOTS

The atria quiver causes pooling of blood in the heart which increases the risk for clots = increased risk for MI, PE, CVA, & DVTs!

UNSTABLE PT.

- Oxygen
- Cardioversion
  - Synchronized administration of shock (delivery in sync with the QRS wave).
  - Cardioversion is NOT defibrillation!

DEFIBRILLATION

Defibrillation is only given with deadly rhythms!

EKGs

PREMATURE VENTRICULAR CONTRACTIONS (PVCs)

Early or premature conduction of a QRS complex

RATE	Depends on the underlying rhythm
RHYTHM	Regular but interrupted due to early P-waves
P-WAVE	Visible but depends on timing of PVC (may be hidden)
PR INTERVAL	Slower than normal but still 0.12 - 0.20 seconds
QRS COMPLEX	Sharp, bizarre, and abnormal during the PVC

CAUSES

- Heart failure
- Cardiomyopathy
- Electrolyte imbalance
- Myocardial ischemia / infarction
- Drug toxicity
- Caffeine, tobacco, alcohol
- Stress or pain
- ↑ workload on the heart
  - Exercise
  - Fever
  - Hypervolemia
  - Heart failure
  - Tachycardia

BIGEMINY: every **other** beat
TRIGEMINY: every **3rd** beat
QUADRIGEMINY: every **4th** beat

R-ON-T PHENOMENON: PVC arises spontaneously from the repolarization gradient (T-wave) may precipitate V-fib

TREATMENT

Treatment based on underlying cause

- May not be harmful if the client has a healthy heart
- Oxygen
- ↓ caffeine intake
- Correct the electrolyte imbalances
- D/C or adjust the drug causing toxicity
- ↓ stress or pain

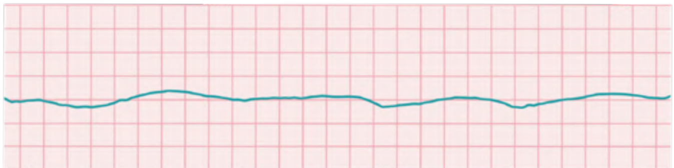
MANIFESTATIONS

- May be asymptomatic
- Feels like your heart...
  - skipped a beat
  - is pounding
- Chest pain

CHEST PAIN

Notify the healthcare provider if the client complains of chest pain, if the PVCs increase in frequency or if the PVCs occur on the T-wave (R-on-T phenomenon).

ASYSTOLE



~~RATE~~
~~RHYTHM~~
~~P-WAVE~~
~~PR INTERVAL~~
~~QRS COMPLEX~~

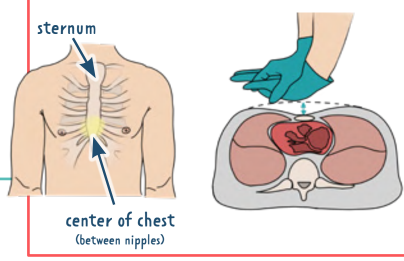
FLATLINE

CAUSES

- Myocardial ischemia/infarction
- Heart failure
- Electrolyte imbalances (common: hypo/hyperkalemia)
- Severe acidosis
- Cardiac tamponade
- Cocaine overdose

TREATMENT

HIGH QUALITY CPR

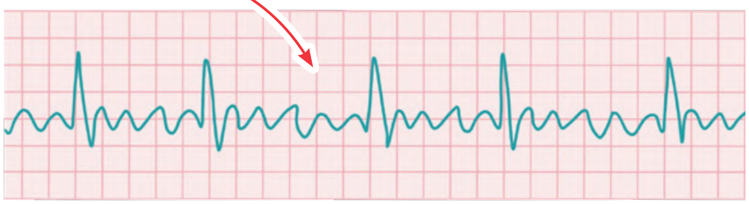


- Heel of hand on center of the chest
- Arms straight
- Shoulders aligned over hands
- Compress at 2-2.4 inches at a rate of 100-120/min
- 30 compressions to 2 rescue breaths
- Minimal interruptions

EKGs

ATRIAL FLUTTER

SAWTOOTH



Similar to A-fib, but the heart's electrical signals spread through the atria. The heart's upper chambers (atria) beat too quickly but at a regular rhythm.

RATE	75-150 bpm
RHYTHM	Usually regular
P-WAVE	"**Sawtooth**" P-wave configuration shaped flutter waves
PR INTERVAL	Unable to measure
QRS COMPLEX	Usually normal & upright

CAUSES

- Coronary artery disease (CAD)
- Hypertension
- Heart failure
- Valvular disease
- Hyperthyroidism
- Chronic lung disease
- Pulmonary embolism
- Cardiomyopathy

MANIFESTATIONS

- May be asymptomatic
- Fatigue / syncope
- Chest pain
- Shortness of breath
- Low blood pressure
- Palpitations
- Dizziness

TREATMENT

STABLE PT.

- Drug therapy!
  - Calcium channel blockers
  - Antiarrhythmics
  - Anticoagulants

RISK FOR CLOTS

Atrial flutter causes pooling of blood in the atria = risk for clots

UNSTABLE PT.

- Cardioversion
  - Synchronized administration of shock (delivery in sync with the QRS wave).
  - Cardioversion is NOT defibrillation!

DEFIBRILLATION

Defibrillation is only given with deadly rhythms!

HEART FAILURE

Can also be referred to as ***congestive heart failure***

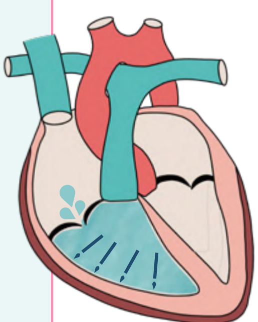
PATHOLOGY

Cardiac disorder that **impairs the ability of the ventricles to fill or eject properly.** The heart muscle can't pump enough blood to meet the body's needs.

DIAGNOSTIC

- ↑ B-type natriuretic peptides (BNP)
- Chest x-ray
  (enlarged heart & pulmonary infiltrate)
- Echocardiogram (measures ejection fraction)
- Cardiac stress test

BNP is a peptide released when the ventricle is filled with too much fluid and stretches. It's a marker for **CONGESTIVE HEART FAILURE (HF)**.



BNP	<100 pg/mL	Expected Range
BNP	100 - 300 pg/mL	HF is suspected
BNP	> 300 pg/mL	Mild HF
BNP	> 600 pg/mL	Moderate HF
BNP	> 900 pg/mL	Severe HF

RISK FACTORS

- Uncontrolled hypertension
- Congenital heart defect
- Arrhythmias
- Coronary artery disease
- Faulty heart valves
- Damage or inflammation of the heart muscle

PATIENT EDUCATION

- **REPORT** S&S of fluid retention (edema, weight gain)
- Elevate HOB
  (Semi or High-Fowler's position)
- Balance periods of activity & rest

DIET MODIFICATIONS:

- **Fluid restrictions**
- ↓ Sodium
- ↓ Fat
- ↓ Cholesterol

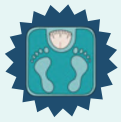
Spread fluids out during the day + suck on hard candy to ↓ thirst

NURSING CONSIDERATIONS

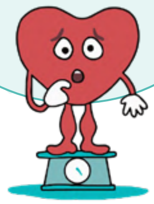
MONITOR:

- I&Os
- Daily weights
- For edema & pulmonary edema

Daily weights are the best way to monitor HF



Monitor for weight gain over a short period of time (2-3 lbs)



MEDICATIONS

- Diuretics
- Ace inhibitors
- Beta blockers
- Digoxin
- **MONITOR** potassium levels

POTASSIUM SPARING
(Spironolactone)

POTASSIUM WASTING
(Loop & thiazide)

NORMAL K+:
3.5 - 5.0 mEq/L

MEMORY TRICK

DIURESIS THE BODY

Diuretics = **D**iuresis = **D**ry inside

HEART FAILURE CONTINUED

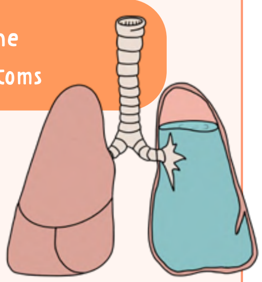
MOST COMMON — LEFT-SIDED HEART FAILURE

A PATIENT CAN HAVE BOTH!

Also called ***left ventricular (LV) heart failure***

	Description	Ejection fraction
SYSTOLIC HF	**Weakened heart muscle** The ventricle does not **EJECT (squeeze)** properly	**Ejection fraction reduced** Also called heart failure with reduced ejection fraction (HFrEF)
DIASTOLIC HF	**Stiff & non-compliant heart muscle** This is not an issue with the ejection fraction (the heart ejects properly). The issue is that the ventricles do not **FILL** properly	**Normal ejection fraction** Also called heart failure with preserved ejection fraction (HFpEF)

Fluid is backing up into the **LUNGS** = pulmonary symptoms



MEMORY TRICK

Left side think **L**ungs

RIGHT-SIDED HEART FAILURE

Also called ***right ventricular (RV) heart failure***

Typically occurs as a result of **LEFT-SIDED HF**

When the **LEFT** ventricle fails, pressure from fluid builds up and causes a back flow of fluids into the **RIGHT** side of the heart

This causes damage to the **RIGHT** side of the heart

Fluid is backing up into the **VENOUS SYSTEM**

MEMORY TRICK

Right = the **R**est of the body

SIGNS & SYMPTOMS

D Dyspnea
R Rales (crackles)
O Orthopnea
W Weakness/fatigue
N Nocturnal paroxysmal dyspnea
I Increased HR
N Nagging cough *(frothy, blood tinged sputum)*
G Gaining weight (2-3 lbs a day)

OTHER S&S
↑ UOP
Hypotension
S3 Gallop

CHRONIC HF may show **BOTH** of these signs & symptoms

S Swelling of the legs & hands
W Weight gain
E Edema (pitting)
L Large neck veins (JVD)
L Lethargy/fatigue
I Irregular heart rate
N Nocturia
G Girth (ascites)

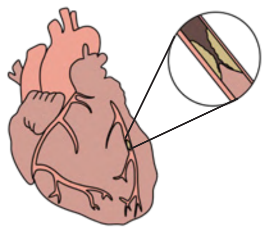
OTHER S&S
Hepatomegaly
Splenomegaly
Anorexia

CORONARY ARTERY DISEASE (CAD)

PATHOLOGY

Damage in the coronary arteries due to **atherosclerosis**.

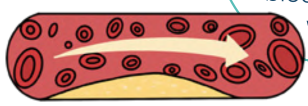
MOST COMMON TYPE OF CARDIOVASCULAR DISEASE



Atherosclerosis is plaque build-up that causes narrowing of the vessels and limits blood supply to the heart.

The plaque may rupture causing thrombi (clot) and may obstruct blood flow, leading to an acute MI.

Accumulation of fatty plaque that happens over time on the blood vessel walls



RISK FACTORS

NON-MODIFIABLE

- Age
- Gender
- Race
- Family history

MODIFIABLE

- Diabetes
- Hypertension
- Smoking
- Obesity
- Physical inactivity
- High cholesterol
- Metabolic syndrome

DIAGNOSTIC

- **Blood tests:**
  LDL, HDL, total cholesterol, triglycerides
- **EKG:** assess for changes in ST segments
- **Stress test**
- **Cardiac catheterization**

CHOLESTEROL

LDL → Want **L**OW levels (<100 mg/dL)
Low Density Lipoprotein
BAD cholesterol

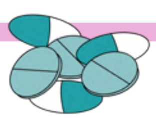
HDL → Want **H**IGH levels (>60 mg/dL)
High Density Lipoprotein
HAPPY cholesterol

SIGNS & SYMPTOMS

Usually **asymptomatic**

- Chest pain
  (stable angina which goes away with rest)
- Shortness of breath
- Epigastric distress (heartburn)
- Pain radiating to the jaw or left arm

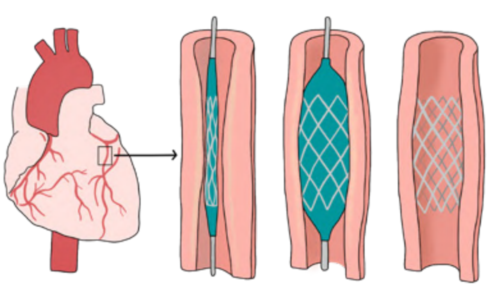
MEDICATIONS



- Antiplatelets
- Medications to normalize cholesterol levels
  (statins, bile acid sequestrants, fibric acids)

TREATMENT

- Percutaneous coronary intervention (PCI)



PATIENT EDUCATION

HEART HEALTHY DIET:

- ↓ in saturated fats
- ↑ in fiber

PREVENTATIVE MEASURES

- Check cholesterol levels
- Manage hypertension
- Control diabetes
- Smoking cessation
- Increase physical activity
- Weight loss if needed

ANGINA PECTORIS

Angina is chest pain associated with ischemia.
It's due to narrowing of at least one major coronary artery.

TYPES OF ANGINA

STABLE	"Predictable"	Occurs with **EXERTION** EXAMPLE Exercise or strenuous activity
UNSTABLE	"Preinfarction"	Occurs at **REST & MORE FREQUENTLY**
PRINZMETAL'S/ VARIANT	"Coronary artery vasospasm"	Pain at **REST** with reversible **ST-ELEVATION**

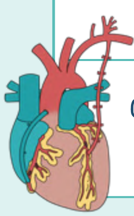
SIGNS & SYMPTOMS

- Chest pain (heavy sensation) may radiate to neck, jaw, or shoulders
- Unusual fatigue
- Weakness
- Shortness of breath
- Pallor
- Diaphoresis

INTERVENTIONS

GOAL: ↓ oxygen demand

- Reperfusion procedures
  - **CABG** Coronary Artery Bypass Graft
  - **PCI** Percutaneous Coronary Interventions



DRUG THERAPY

NITRATES	CALCIUM CHANNEL BLOCKERS	BETA BLOCKERS	ANTIPLATELET / ANTICOAGULANT
Vasodilators ↓ ischemia = ↓ pain Usually administered sublingual	Relaxes blood vessels ↑ oxygen supply to the heart ↓workload of heart	↓ myocardial oxygen consumption	Prevents platelet aggregation & thrombosis

PATIENT TEACHING

SUBLINGUAL NTG OR SPRAY

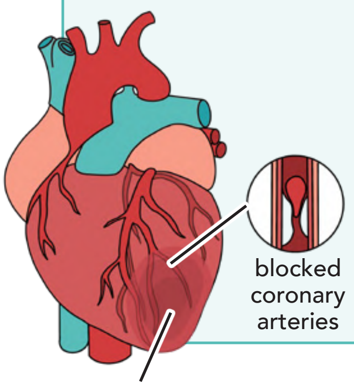
- **1** tab/spray sublingual every **5** minutes, up to **3** doses.
- If angina is not relieved or is worse 5 min after the first dose, call 911!

Keep in original container (dark, glass bottle) in a dry, cool place. Do not swallow or chew these tablets

MYOCARDIAL INFARCTION (MI)

PATHO

Complete blockage in one or more arteries of the heart **EMERGENCY!**



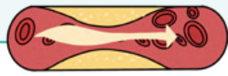
ATHEROSCLEROSIS

Coronary arteries become narrow due to plaque build-up

→

ANGINA

Due to ischemia (low O_2)



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MYOCARDIAL INFARCTION (MI)

Plaque rupture become a blood clot that blocks arteries of the heart

SIGNS & SYMPTOMS

Sudden, crushing, radiating chest pain that continues despite rest & medications

- Shortness of breath
- Nausea & vomiting
- Sweating
- Pale & dusty skin

WOMEN PRESENT WITH DIFFERENT SYMPTOMS

- Fatigue
- Shoulder blade discomfort
- Shortness of breath

PAIN FELT IN THE... Left arm • Mid back/shoulder • Heartburn

DIAGNOSIS

- ECG
  - ST-elevation (no O_2)
  - ST-depression (low O_2)
  - T-wave inversion
- Troponin
- Stress tests
  - Chemical & exercise

TREATMENT

IMMEDIATE

MNEMONIC

- **M** **MORPHINE** ↓workload of the heart & ↓ pain
- **O** **OXYGEN** ↑O_2 to the heart
- **N** **NITROGLYCERIN** opens up the vessels
- **A** **ASPIRIN** Prevents platelets from sticking together

→

CATH LAB OR CLOT BUSTER

MEDICATIONS

- Thrombolytics (clot busters)
- *Example: Streptokin*ase

Suffixes: -teplase -ase

SURGERY

- PCI "Percutaneous Coronary Intervention"
- CABG
- Endarterectomy
  - Cut out the blockage

Any time you give a thrombolytic, watch for signs of bleeding!

→

PREVENTION & REST

PREVENT / STABILIZE CLOT

- Heparin IV

REST THE HEART WITH...

- Nitro
- Beta-blockers
- Calcium channel blockers

TEST YOUR KNOWLEDGE! NO CHEATING :)

PERIPHERAL VASCULAR DISEASE WORKSHEET

is an umbrella term for...

PERIPHERAL VENOUS DISEASE (PVD)

Deoxygenated blood can't get back to the heart.

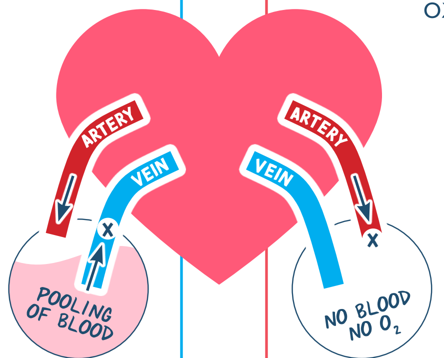
Pooling of oxygenated blood in the extremities.

PERIPHERAL ARTERIAL DISEASE (PAD)

Think "BAD"

Narrow artery (atherosclerosis) where oxygenated blood can't get to the distal extremities (hands & feet).

Ischemia & necrosis of the extremities



PVD			PAD		
PAIN ?			PAIN ?		
PULSE ?			PULSE ?		
EDEMA ?			EDEMA ?		
TEMP ?			TEMP ?		
COLOR ?			COLOR ?		
WOUNDS ?			WOUNDS ?		
GANGRENE ?			GANGRENE ?		
POSITIONING ?			POSITIONING ?		

CAUSES OF BOTH

________________ • ________ • ______________ • ____________________

DX: ____

TREAMENT

- **Position**

  ____

- **Medications**

  ____

  ____

- **Surgery**

  ____

  ____

  ____

TREAMENT

- **Position**

  ____

- **Perform** ________________________
- **Stop** ________________________
- **Avoid** ________________________
- **No** ________________________
- **Medications**

  ____

  ____

PERIPHERAL VASCULAR DISEASE

is an umbrella term for...

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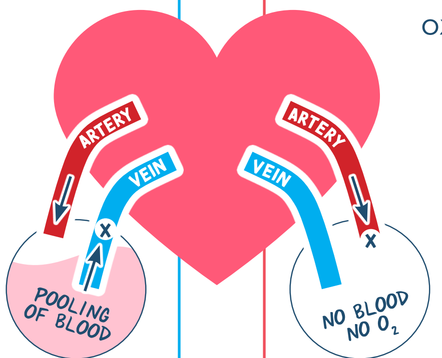


PVD

PAIN?	✔	Dull, constant, achy pain!
PULSE?	✔	May not be palpable due to edema
EDEMA?	✔	Blood is POOLING in the leg
TEMP?		Warm legs (Blood is warm)
COLOR?		Stasis dermatitis (Brown/yellow)
WOUNDS?		Venous STASIS ulcers, Irregular shaped wounds, shallow
GANGRENE?	✘	We have too much blood! Gangrene is caused by insufficient amounts of blood.
POSITIONING?		Elevate Veins. Positions that make it worse: dangling, sitting/standing for long periods of time

PAD

PAIN?	✔	Sharp pain: Gets worse at night "rest pain". Intermittent claudication
PULSE?	✔	Very poor or even absent
EDEMA?	✘	No blood in the extremities
TEMP?		Cool. No blood = cool leg (blood is warm)
COLOR?		Pale, hairless, dry, scaly, thin skin due to lack of nutrients (↓ O_2)
WOUNDS?		Regular in shape, red sores round appearance "punched out"
GANGRENE?	✔	Tissue death caused by a lack of blood supply
POSITIONING?		Dangle arteries

CAUSES OF BOTH

Smoking • Diabetes • High cholesterol • Hypertension

DX: Doppler Ultrasound or Ankle Brachial Index (ABI)

TREATMENT — KEEP VEIN OPEN!

- **Elevate Veins**
- **Medications**
  - Aspirin or Clopidogrel
  - Cholesterol lowering drugs "statin"
- **Surgery**
  - Angioplasty
  - Bypass (CABG)
  - Endarterectomy

TREATMENT — GET BLOOD MOVING!

- **DAngle Arteries** (Dependent position)
- **Perform daily skin care with moisturizer**
- **Stop smoking**
- **Avoid tight clothing** (vasoconstriction)
- **No heating pads!**
- **Medications**
  - Vasodilators
  - Antiplatelets

CARDIAC BIOMARKERS

TROPONIN

EXPECTED RANGE
TROPONIN I < 0.03 ng/mL
TROPONIN T < 0.1 ng/mL

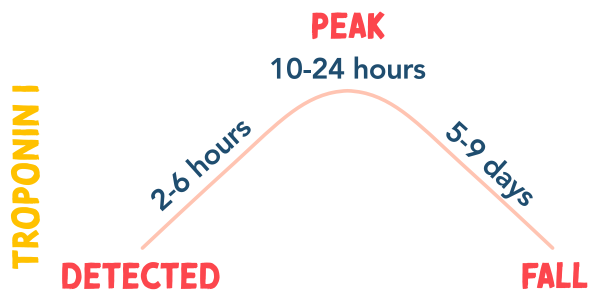
BEST indicator of an acute MI

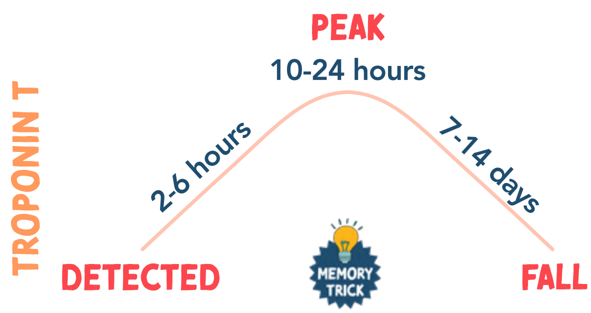
Protein released in the blood stream when the heart muscle is damaged.

There are 3 isomers of troponin:

TROPONIN C:
Binds calcium to activate muscle contraction

TROPONIN I & T:
Specific for cardiac muscle





MEMORY TRICK
Troponin T think Two weeks it can stay elevated

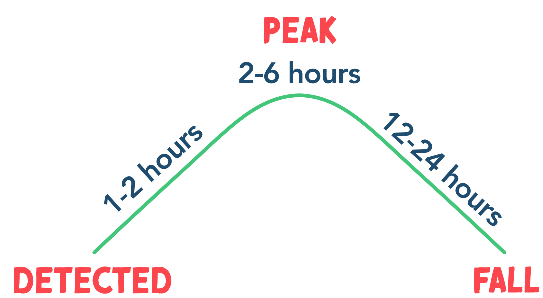
MYOGLOBIN

EXPECTED RANGE
5 – 70 ng/mL

Myoglobin is found in cardiac & skeletal muscle

NOT a specific indicator of an acute MI, but a (-) sign is good for ruling out an acute MI

MEMORY TRICK
Myoglobin think Muscle



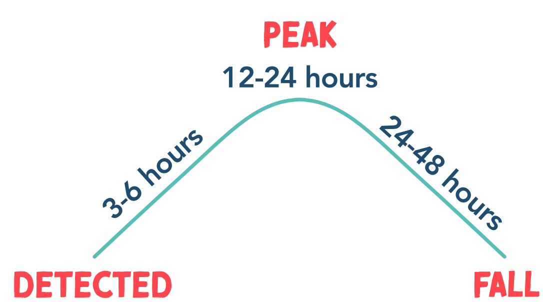
CK-MB

EXPECTED RANGE
0 – 5 ng/mL

CREATINE KINASE – MB

Cardiac-specific isoenzyme
BUT less reliable than Troponin

An enzyme released in the bloodstream when the heart, muscles or brains are damaged!



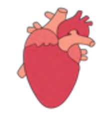
HYPERTENSION (HTN)

HYPERtension = **HIGH** BP

Most accurate diagnosis for HTN

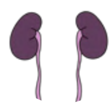
CATEGORIES	SYSTOLIC (SQUEEZE)	DIASTOLIC (DECOMPRESS)
HYPOTENSION	< 100	< 60
NORMAL	< 120	< 80
PRE-HTN	120 - 139	80 - 89
STAGE 1 HTN	140 - 159	90 - 99
STAGE 2 HTN	> 160	> 100
HTN CRISIS	> 180	> 120

AFFECTED ORGANS



CONGESTIVE HEART FAILURE (CHF)
Overworking of the heart muscle (ventricle enlarges)

STROKE
Weak & narrow vessels could lead to rupture of vessels



RENAL FAILURE
Too much blood flowing to the kidneys at a fast rate & high pressure

VISUAL CHANGES
Damages blood vessels in the retina (blurred vision, can't focus on objects)

RISK FACTORS

PRIMARY HTN (MOST COMMON)

Also called
ESSENTIAL or **IDIOPATHIC HTN**

- Cause is unknown
- Not curable, only controllable

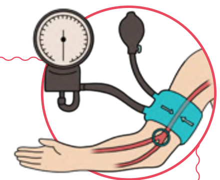
- **R** Race (African Americans)
- **I** Intake of Na/ETOH
- **S** Smoking
- **K** Low **K**+ & vitamin D levels
- **F** Family HX
- **A** Advanced age
- **C** ↑ Cholesterol
- **T** Too much caffeine
- **O** Obesity
- **R** Restricted activity
- **S** Sleep apnea

SECONDARY HTN

Has a direct cause / preexisting condition

- Chronic kidney disease
- Diabetes
- Hypo/Hyperthyroidism
- Cushing syndrome
- Pregnancy
- Certain drugs (oral contraceptives)

CHECKING BLOOD PRESSURE



→ Place stethoscope over brachial artery
→ Patients should not smoke, exercise, etc. within 30 minutes of having their BP checked *(could lead to inflated BP)*
→ Instruct the client to:
- Sit in a chair with legs uncrossed
- Arm at ❤ level
- Correct size cuff

Too small = false high BP
Too large = false low BP

→ No BPs should be auscultated in arms with:
- Mastectomy
  - HX of AV shunt
  - Blood clots
  - PICC lines/central lines

SIGNS & SYMPTOMS

Usually ***asymptomatic!***

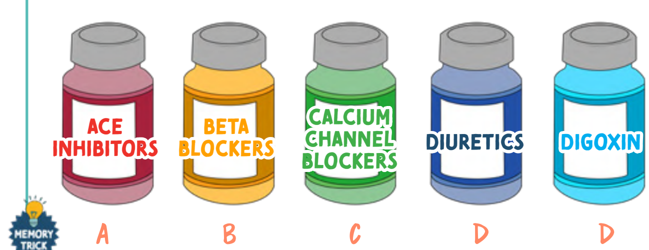
Commonly called the **"SILENT KILLER"**

Symptoms *(if seen)*:
- Blurred vision
- Headache
- Chest pain
- Nose bleeds

EDUCATION

- Limit sodium intake
- Limit alcohol intake
- Smoking cessation
- Teach how to measure BP & keep a record
- Exercise programs for weight loss if needed

ANTIHYPERTENSIVE MEDICATION OVERVIEW

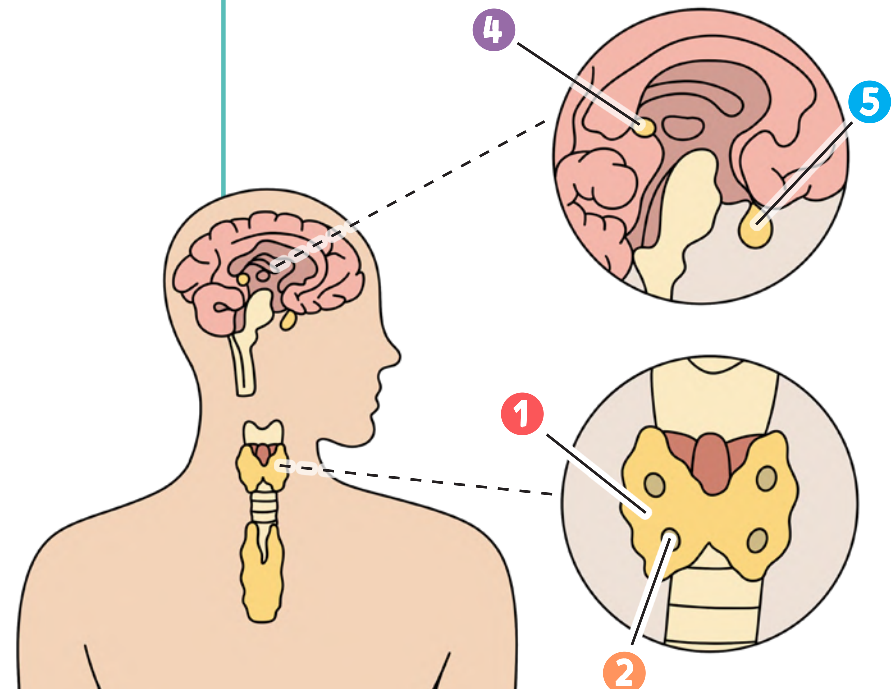


		SUFFIXES
A	ACE inhibitors	-pril
B	BETA Blockers	-olol
C	Calcium Channel Blockers	-pine -amil
D	Diuretics	
D	Digoxin	

ENDOCRINE SYSTEM OVERVIEW

FUNCTION OF THE ENDOCRINE SYSTEM: The endocrine system is made up of glands & organs that release hormones (chemical messengers). These chemical messengers carry information & instructions from one cell to another.

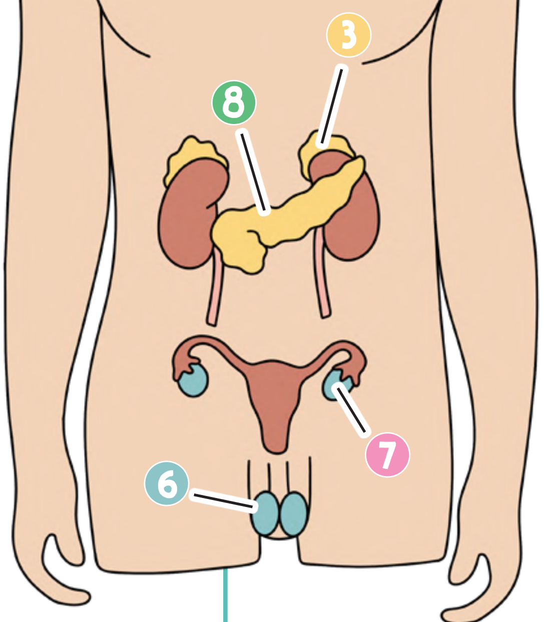
HORMONES RELEASED by the endocrine organs/glands



1 THYROID GLAND
- Thyroxine (T4)
- Triiodothyronine (T3)
- Calcitonin

2 PARATHYROID GLAND
- Parathyroid hormone (PTH)

3 ADRENAL GLAND
- **Adrenal cortex**
  - Aldosterone
  - Cortisol
- **Adrenal medulla**
  - Epinephrine
  - Norepinephrine



4 HYPOTHALAMUS
- Growth hormone-releasing hormone (GHRH)
- Thyrotropin-releasing hormone (TRH)
- Gonadotropin-releasing hormone (GnRH)
- Corticotropin-releasing hormone (CRH)

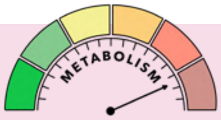
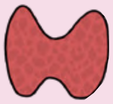
5 PITUITARY GLAND
- **Anterior**
  - Luteinizing hormone (LH)
  - Follicle-stimulating hormone (FSH)
  - Prolactin
  - Thyroid-stimulating hormone (TSH)
  - Growth hormone (GH)
  - Adrenocorticotropic hormone (ACTH)
- **Posterior**
  - Antidiuretic hormone (ADH) (Vasopressin)
  - Oxytocin

6 TESTES
- Testosterone

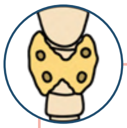
7 OVARIES
- Estrogen
- Progesterone

8 PANCREAS
- Insulin
- Glucagon

ENDOCRINE HORMONES

Thyroxine (T4) Triiodothyronine (T3)	These hormones are created and stored in the thyroid. Maintains body metabolism in a steady state. 
Calcitonin	Secreted by the thyroid gland. Regulates **calcium** in the body. MEMORY TRICK **CAL**Citonin think **CAL**Cium
Thyroid-Stimulating Hormone (TSH)	TSH stimulates the thyroid, causing T3 & T4 to be released 
Oxytocin	Muscle contractions to help expel the baby
Prolactin	Stimulates milk production after childbirth
Insulin	Works to **DECREASE** blood glucose levels. Insulin puts sugar & potassium into the cells to be used later as energy 
Glucagon	Works to **INCREASE** blood glucose levels. Breaks down stored glucose (glycogen) in the liver 
Epinephrine & Norepinephrine	Stress hormones. They are catecholamines that are released when blood pressure drops. Helps in times of **ACUTE** stress
Cortisol	Glucocorticoid. Helps regulate metabolism, ↑ blood glucose levels, and has anti-inflammatory properties. Helps in times of **CHRONIC** stress
Antidiuretic Hormone (ADH)	Helps regulate the amount of water in your body 
Aldosterone	Mineralocorticoid that helps in fluid balance
Parathyroid Hormone (PTH)	Helps to increase serum calcium in the blood
Estrogen	Helps to regulate the menstrual cycle, uterus growth during pregnancy, maintains the pregnancy, and supports the fetus as it grows
Progesterone	Helps to regulate the menstrual cycle, stimulates growth of maternal tissues and fetal organs during pregnancy MEMORY TRICK **P**rogesterone think **P**regnancy hormone
Testosterone	Helps in the development of male sex organs and reproductive tissue, plays a vital role in sperm production, promotes secondary sex characteristics (↑ bone mass, ↑ muscle mass, ↑ growth of body hair) MEMORY TRICK **TEST**osterone think **TEST**es

LAB VALUES RELATED TO THE ENDOCRINE SYSTEM



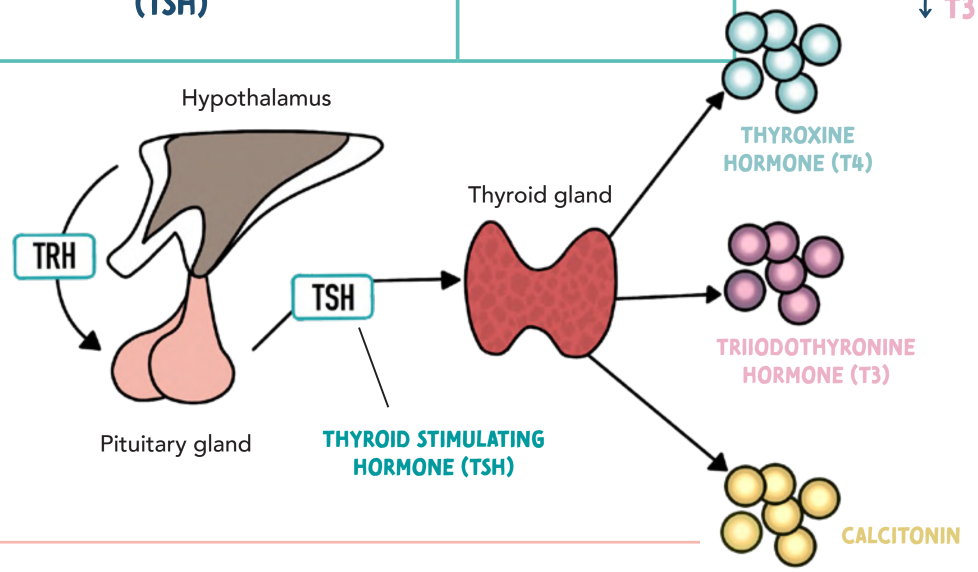
THYROID PANEL

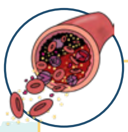
	EXPECTED RANGE
T3 (TRIIODOTHYRONINE)	80 - 220 ng/dL
T4 (THYROXINE)	4 - 12 mcg/dL
THYROID STIMULATING HORMONE (TSH)	0.5 - 5 mU/L

T3 & T4 are always opposite of TSH (negative feedback mechanism)

HYPERTHYROIDISM:
↑ T3 & T4 ↓ TSH

HYPOTHYROIDISM:
↓ T3 & T4 ↑ TSH





BLOOD GLUCOSE

	EXPECTED RANGE	DESCRIPTION
BLOOD GLUCOSE GOAL	70 - 110 mg/dL	Any time of the day (doesn't matter when the last meal was)
FASTING BLOOD SUGAR (FBS)	< 100 mg/dL	No caloric intake for at least 8 hours
2-HR ORAL GLUCOSE TOLERANCE TEST	< 140 mg/dL	Drink a glucose drink (75g of glucose dissolved in water)
HBA1C	< 5.7%	Blood test that measures the average blood glucose (sugar) levels for the last 2-3 months

A finger stick blood sugar test is the most common way people with diabetes check their blood glucose levels

DIABETES TYPE 1 & 2

PATHOLOGY

TYPE 1 DIABETES MELLITUS (T1DM)

NO INSULIN PRODUCTION

MEMORY TRICK: Type ONE we have nONE

- Caused by an autoimmune response
- The cells are starved of glucose since there is no insulin to bring glucose into the cells
- The cells break down protein and fat into energy, causing ketones to build up = **ACIDOSIS**!
- Usually diagnosed in **CHILDHOOD**

TYPE 2 DIABETES MELLITUS (T2DM)

DOES NOT PRODUCE ENOUGH INSULIN, OR PRODUCES BAD INSULIN THAT DOES NOT WORK PROPERLY

MEMORY TRICK: Terrible Twos are BAD

- Insulin resistance
- Insulin receptors are worn out & not working properly!
- Usually diagnosed in **ADULTHOOD** (due to a poor diet, sedentary lifestyle, and obesity)

MEMORY TRICK: Easy to remember because **CHILDHOOD** comes 1ST in life and **ADULTHOOD** comes 2ND

RISK FACTORS

Type 1:
- Genetics
- Family history

Type 2:
- High blood sugar
- Hypertension
- Obesity
- Inactivity
- High cholesterol
- Family history
- Smoking

S&S

Type 1 — ONSET: ABRUPT

3 P's
- **Polyuria**: excessive peeing
- **Polydipsia**: excessive thirst
- **Polyphagia**: excessive hunger

Type 2 — ONSET: GRADUAL

3 P's
- **Polyuria**: excessive peeing
- **Polydipsia**: excessive thirst
- **Polyphagia**: excessive hunger

TREATMENT

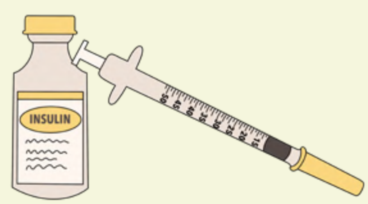
Type 1:

Only has 1 treatment:

INSULIN

Oral hypoglycemic agents will not work for this pt.

Insulin dependent for life!



Type 2:

Has 2+ treatments:

1. **DIET & EXERCISE**
2. **ORAL HYPOGLYCEMIC AGENTS**
   Example: Metformin
3. **POSSIBLY INSULIN**

Insulin is not administered routinely in a type 2 diabetic patient. Only in times of stress, surgery, or sickness will insulin need to be administered.

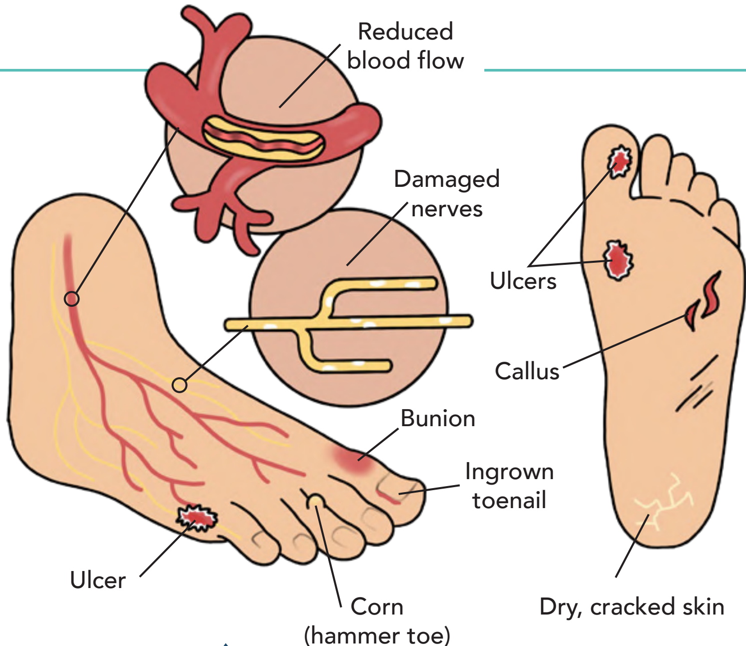
DIAGNOSTIC CRITERIA

CASUAL	FASTING BLOOD SUGAR (FBS)	GLUCOSE TOLERANCE TEST	HBA1C
Any time of the day (doesn't matter when the last meal was)	No caloric intake for at least 8 hours	Drink a glucose drink (75g of glucose dissolved in water)	Blood test that measures the average blood glucose (sugar) levels for the last 2-3 months
> 200 mg/dL	> 126 mg/dL	> 200 mg/dL	> 6.5%

DIABETES TYPE 1 & 2 CONTINUED

DIABETIC FOOT CARE

- Wash feet daily
- Use warm water (test temperature beforehand) & mild soap
- Gently pat feet completely dry
- Inspect feet daily with a mirror (check for any cuts, blisters, or sores)
- Avoid over-the-counter products (callus remover, alcohol, etc)
- Cut toe nails straight across
- Do not cross legs
- Report symptoms of infection to the HCP



Keep feet clean, dry, & avoid irritation!

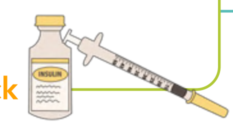
SICK DAY MANAGEMENT

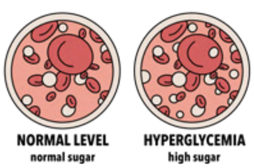
MONITOR
- Blood glucose often
- Temperature often
- Urine for ketones

REPORT TO THE HCP IF:
- Ketones are present in urine
- If blood sugar is > 250 mg/dL
- If temperature is > 101°F

- Stay hydrated (avoid dehydration)

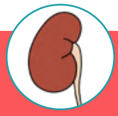
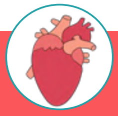
Do not skip insulin when you are feeling sick





DIABETES CAN NEGATIVELY AFFECT ALMOST EVERY ORGAN SYSTEM

This is because high levels of sugar in the blood damages the blood vessel walls and the nerves

ORGAN AFFECTED	KIDNEYS	NERVES	EYES	HEART	BRAIN
COMPLICATIONS	**NEPHROPATHY** Kidney damage Excessive blood glucose can damage the tiny blood vessels in the filtering system (glomeruli). **This causes kidney failure and even end-stage kidney disease.**	**PERIPHERAL NEUROPATHY** Damage to the nerves outside of the brain & spinal cord. Excessive blood glucose can injure the nerves. **This causes tingling, numbness, and eventually loss of sensation.** Nerve damage in the foot can cause serious complications such as major infections in cuts and blisters. **All this sugar in the blood also causes delayed wound healing = risk for infection**	**DIABETIC RETINOPATHY** Eye damage Excessive blood glucose damages the blood vessels of the retina. **This causes blindness, cataracts, glaucoma.**	**CARDIOVASCULAR DISEASE** Damage to the heart & major coronary arteries Excessive blood glucose damages the blood vessels and nerves controlling the heart. **This causes coronary artery disease, hypertension, atherosclerosis.**	**STROKE** Excessive blood glucose damages the blood vessels and makes them stiff. It also can cause a build up of fatty deposits. **This may cause a blood clot that travels to the brain causing a stroke.**

DKA VS. HHNS

DIABETIC KETOACIDOSIS (DKA)

PATHOLOGY

HAPPENS MOSTLY IN TYPE 1 DIABETIC PATIENTS

Not enough insulin
↓
Body can't allow blood sugar into the cells for energy
↓
Blood sugar becomes VERY high
↓
Cells break down protein & fat into energy
↓
KETONES build up = **ACIDOSIS!**

Ketones are a byproduct of metabolism

RISK FACTORS

4 S's

- **S**tress (surgery)
- **S**epsis (infection)
- **S**kipping insulin
- **S**tomach (stomach virus: nausea/vomiting)
- Undiagnosed diabetes

S&S

ONSET: ABRUPT

- **Hyperglycemia (300 - 500 mg/dL)**
- Ketosis & acidosis
- Dehydration
- Metabolic acidosis
- Kussmaul respirations (trying to blow off CO_2)
- Acid breath "fruity breath"

REMEMBER: CO_2 is an aCid

TREATMENT

- IV insulin with potassium (K+)
- Fluid replacement
- Correction of electrolyte imbalance
- Administer bicarbonate for metabolic acidosis

MEMORY TRICK: INsulin causes sugar & K+ to go IN the cells, causing hypokalemia unless we administer K+ with IV insulin

MEMORY TRICK: DKA remember to monitor K levels

NOTE FOR BOTH:

Regular insulin is the only insulin given IV

Regular goes Right into the vein

HYPERGLYCEMIC HYPEROSMOLAR NONKETOTIC SYNDROME (HHNS)

PATHOLOGY

HAPPENS MOSTLY IN TYPE 2 DIABETIC PATIENTS

NO acidosis present!

Simply high amounts of glucose in the blood

~~ACIDOSIS~~ ~~KETONES~~

RISK FACTORS

- Inadequate fluid intake
- ↓ kidney function
- Infection
- Stress
- Older adults

S&S

ONSET: GRADUAL

- **Hyperglycemia (>600 mg/dL)**
- 3 **P**'s (**P**olyuria, **P**olydipsia, **P**olyphagia)
- Dehydration **(hypovolemia)**
- Neurovascular changes (confusion, ↓ LOC, headache)

~~NO METABOLIC ACIDOSIS~~

TREATMENT

- Fluid replacement
- Correction of electrolyte imbalances
- Administer insulin
  - IV insulin with potassium (K+)
  - SubQ insulin

HYPERGLYCEMIA VS. HYPOGLYCEMIA

HYPERGLYCEMIA

↑ BLOOD SUGAR

>200 mg/dL
Gradual (hours to days)

BLOOD GLUCOSE GOAL: 70 - 110 mg/dL

SIGNS & SYMPTOMS

3 P's MOST COMMON SYMPTOMS

- **P**olyuria
- **P**olydipsia
- **P**olyphagia
- Hot & dry skin
- Dry mouth (dehydration)
- Fruity breath
- Deep, rapid breaths (air hunger)
- Numbness & tingling
- Slow wound healing
- Vision changes

MEMORY TRICK: Hot & dry = Sugars high

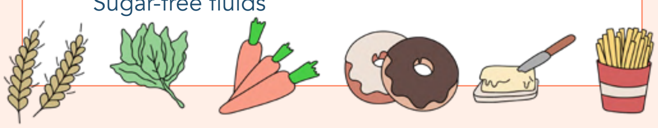
CAUSES

4 S's

- **S**epsis (infection)
- **S**tress
- **S**teroids
- **S**kipping insulin or oral diabetic medication
- Not eating a diabetic diet

DIABETIC DIET

✔	✘
Complex carbohydrates	Saturated fats
Fiber-rich foods	Trans fats
Heart-healthy fish	Cholesterol
"Good fats"	Sodium
Sugar-free fluids	



TREATMENT

- Administer insulin as needed
- Test urine for ketones

	RAPID	SHORT	INTERMEDIATE	LONG
GENERIC NAMES	Lispro Aspart Glulisine	regular	nph	Glargine Detemir
BRAND NAMES	Humalog Novolog Apidra	Humulin R Novolin R	Humulin N Novolin N	Lantus Levemir

HYPOGLYCEMIA

↓ BLOOD SUGAR

<70 mg/dL
Happens suddenly

THE BRAIN NEEDS GLUCOSE... NO GLUCOSE CAUSES BRAIN DEATH!

SIGNS & SYMPTOMS

- Cool & clammy skin
- Sweating (Diaphoresis)
- Palpitations
- Fatigue & weakness
- Confusion
- Headache
- Shakiness
- Inability to arouse from sleep
  - Can lead to coma ⚠

MEMORY TRICK: Cool & clammy needs some candy

CAUSES

- Exercise
  - Swimming, cycling, college athlete, etc.
- Alcohol
- Peak times of insulin

RAPID INSULIN has the highest risk for hypoglycemia

TREATMENT

CONSCIOUS PATIENTS

15 X 15 X 15

- Oral intake of **15 GRAMS** of carbohydrates
  Juices, soda, low fat milk. NOT peanut butter or high fat milk
- Recheck blood glucose in **15 MIN**
- Give another **15 GRAMS** of carbohydrates if needed

UNCONSCIOUS PATIENTS

Do not put anything in an unconscious client's mouth, they can **ASPIRATE**!

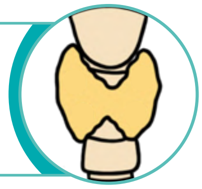
Administer IV 50% dextrose (D50) **OR** Glucagon (IM, IV, SubQ)

⚠ **EMERGENCY** call a rapid response

THYROID DISORDERS

FUNCTION

- The thyroid gland produces 3 hormone (T3, T4, & Calcitonin)
  - You need **IODINE** to make these hormones
- Thyroid gives you **ENERGY**!



HYPERTHYROIDISM

PATHOLOGY

Excessive production of thyroid hormone

TOO MUCH ENERGY!

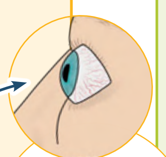
- Graves disease
- Too much iodine (helps makes T3 + T4)
- Toxic Nodular Goiter
- Thyroid replacement medication (Toxicity)

LAB VALUES

↑ T3 & T4 ↓ TSH

SIGNS & SYMPTOMS

- Hyper-excitable
- Nervous/tremors
- Irritable
- ↓ Attention span
- Increased appetite
- Weight loss
- Hair loss
- Goiter (enlarged thyroid)
- Hot
- **EXOPHTHALMOS**
- Increased:
  - Blood pressure
  - Pulse
  - GI function



Bulging eyes due to fluid accumulation behind the eyes

⚠ LIFE-THREATENING COMPLICATIONS ⚠

THYROID STORM!

⚠ **ACUTE / LIFE THREATENING EMERGENCY!**

TREATMENT

- Anti-Thyroid Medications
  - Methimazole or PTU
- Beta Blockers (↓ HR & BP)
- Iodine Compounds
- Radioactive Iodine Therapy
- Thyroidectomy

HYPOTHYROIDISM

PATHOLOGY

Low production of thyroid hormone

NOT ENOUGH ENERGY!

MOST COMMON

- Hashimoto's disease
- Not enough iodine
- Thyroidectomy
- Anti-thyroid medications
- Pituitary hormone
- Affects women more often then men

LAB VALUES

↓ T3 & T4 ↑ TSH

SIGNS & SYMPTOMS

- No energy
- Fatigue
- No expressions
- Weight gain
- Cold
- Amenorrhea
- Slurred speech
- Dry skin
- Coarse hair
- Decreased:
  - HR
  - GI function (constipation)
  - Blood sugar (Hypoglycemia)

⚠ LIFE-THREATENING COMPLICATIONS ⚠

MYXEDEMA COMA!

TREATMENT

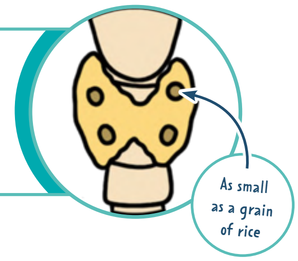
- Hormone replacement (replacing levothyroxine)
  - Synthetic levothyroxine
    - Synthroid or Levothroid
  - *Will be on this medication forever*

For more information about thyroid medications, see the Pharmacology Bundle

PARATHYROID GLAND DISORDERS

FUNCTION

The parathyroid gland produces and secretes PTH (parathyroid hormone) which controls the levels of **CALCIUM** in the blood



↑ PTH — HYPERPARATHYROIDISM

↑ CALCIUM ↓ PHOSPHORUS

CAUSES

PRIMARY CAUSE:
Tumor or hyperplasia of the parathyroid

SECONDARY CAUSE:
Chronic kidney failure

SIGNS & SYMPTOMS

- **STONES:** Kidney stones (↑ calcium)

- **BONES:**
  - Skeletal pain
  - Pathological fractures from bone deformities

- Abdominal **MOANS**
  - Nausea, vomiting, and abdominal pain
  - Weight loss / anorexia
  - Constipation

- Psychic **GROANS**
  - Mental irritability
  - Confusion

MEMORY TRICK: STONES, BONES, MOANS, & GROANS

TREATMENT

- Parathyroidectomy
- Removal of more than one gland
- Administer:
  - Phosphates, calcitonin, & IV or oral bisphosphonates
- **DIET:** ↑ fiber & moderate calcium

↓ PTH — HYPOPARATHYROIDISM

↓ CALCIUM ↑ PHOSPHORUS

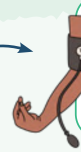
CAUSES

- Can occur due to accidental removal of the parathyroid
  - Thyroidectomy, parathyroidectomy, or radical neck dissection
- Genetic predisposition
- Exposure to radiation
- Magnesium depletion

SIGNS & SYMPTOMS

- Numbness & tingling
- Muscle cramps
- Tetany
- Hypotension
- Anxiety, irritability, & depression

Same S&S of hypocalcemia!



POSITIVE TROUSSEAU SIGN:
Carpal spasm caused by inflating a blood pressure cuff

CHVOSTEK'S SIGNS:
Contraction of facial muscles w/ light tap over the facial nerve

MEMORY TRICK: Think "C" for Cheesy smile

TREATMENT

- IV Calcium
- Phosphorus binding drugs
- **DIET:** ↑ Calcium ↓ Phosphorus

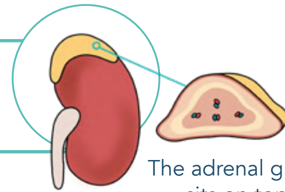
ADRENAL CORTEX DISORDERS

ADRENAL CORTEX HORMONES:

Glucocorticoids • Mineralocorticoids • Sex hormones

RETAINS: NA+ & H_2O

LOSES: K+



The adrenal gland sits on top of each kidney

CUSHING'S

Disorder of the adrenal cortex

TOO MANY STEROIDS

MEMORY TRICK: They "have a **CUSHION**"

CAUSES

- Females
- Overuse of cortisol medications
- Tumor in the adrenal gland that secretes cortisol

SIGNS & SYMPTOMS

- Muscle wasting
- Moon face
- Buffalo hump
- Truncal obesity w/ thin extremities
- Supraclavicular fat pads
- Weight gain
- Hirsutism (masculine characteristics)
- ↑ Glucose ↑ NA+
- ↓ K+ ↓ CA+
- Hypertension

TREATMENT

- Adrenalectomy
  - Requires lifelong glucocorticoid replacement
- Avoid infection
- Adm. chemotherapeutic agents if adrenal tumor is present

ADDISON'S

Disorder of the adrenal cortex

NOT ENOUGH STEROIDS

MEMORY TRICK: We need to "**ADD**" some

CAUSES

- Surgical removal of both adrenal glands
- Infection of the adrenal glands
- TB, cytomegalovirus, & bacterial infections

SIGNS & SYMPTOMS

- Fatigue
- Nausea / vomiting / diarrhea
- Anorexia
- Hypotension & Hypovolemia
- Confusion
- ↓ Blood sugar
- ↓ Na & H_2O ↑ K+
- Hyperpigmentation of the skin
- **Vitiligo**: white areas of depigmentation

⚠ ADDISONIAN CRISIS ⚠

SIGNS & SYMPTOMS

- Profound fatigue
- Dehydration
- Renal failure
- Rapid respiration
- Hyponatremia
- Hypokalemia
- Cyanosis
- Fever
- Nausea/vomiting

Think SHOCK!
- Hypotension
- Weak rapid pulse

TREATMENT: Fluid resuscitation & high-dose hydrocortisone

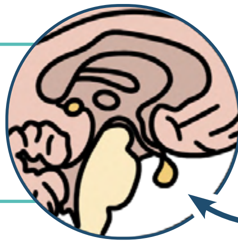
TREATMENT

- Adm. glucocorticoid and/or mineralocorticoid
- Diet: high in protein & carbs

PITUITARY GLAND DISORDERS

ANTIDIURETIC HORMONE (ADH):

ADH regulates & balances the amount of water in your blood



ADH is found in the PITUITARY GLAND!

SYNDROME OF INAPPROPRIATE ANTIDIURETIC HORMONE (SIADH)

SIADH think Soaked Inside

SIADH is often of non-endocrine origin

TOO MUCH ADH

RETAINS WATER

CAUSES

- Pulmonary disease
  - TB
  - Severe pneumonia
- Disorders of the CNS
  - Head injury
  - Brain surgery
  - Tumor
- HIV
- Medications
  - Vincristine
  - Phenothiazines
  - Antidepressants
  - Thiazide diuretics
  - Anticonvulsants
  - Antidiabetic drugs
  - Nicotine

SIGNS & SYMPTOMS

- Low urinary output of concentrated urine
- Fluid volume overload
- Weight gain without edema
- Hypertension
- Tachycardia
- Nausea & vomiting
- Hyponatremia

TREATMENT

- Implement seizure precautions
- Elevate HOB to promote venous return
- Restrict fluid intake
- Adm. loop diuretics
- Adm. vasopressin antagonists

INCREASED ICP can lead to an ADH problem

DIABETES INSIPIDUS (DI)

DI think Dry Inside

NOT ENOUGH ADH

LOSES WATER

CAUSES

- Head trauma, brain tumor
- Manipulation of the pituitary
  - Surgical ablation, craniotomy, sinus surgery, hypophysectomy
- Infections of the central nervous system (CNS)
  - Meningitis, encephalitis, or TB
- Failure of the renal tubules to respond to ADH

SIGNS & SYMPTOMS

- Excretes large amounts of diluted urine
- Polydipsia (increased thirst)
- Polyuria (increased urine output)
- Dehydration
- Decreased skin turgor
- Dry mucous membranes
- Muscle pain & weakness
- Headache
- Postural hypotension
- Tachycardia
- Low urinary specific gravity

Normal specific gravity **1.005 - 1.030**

TREATMENT

- Adequate fluids
- IV hypotonic saline
- ADH replacement (replace the missing hormone!)
  - Vasopressin or desmopressin
- Monitor
  - Intake & output
  - Weight

ADRENAL MEDULLA DISORDER

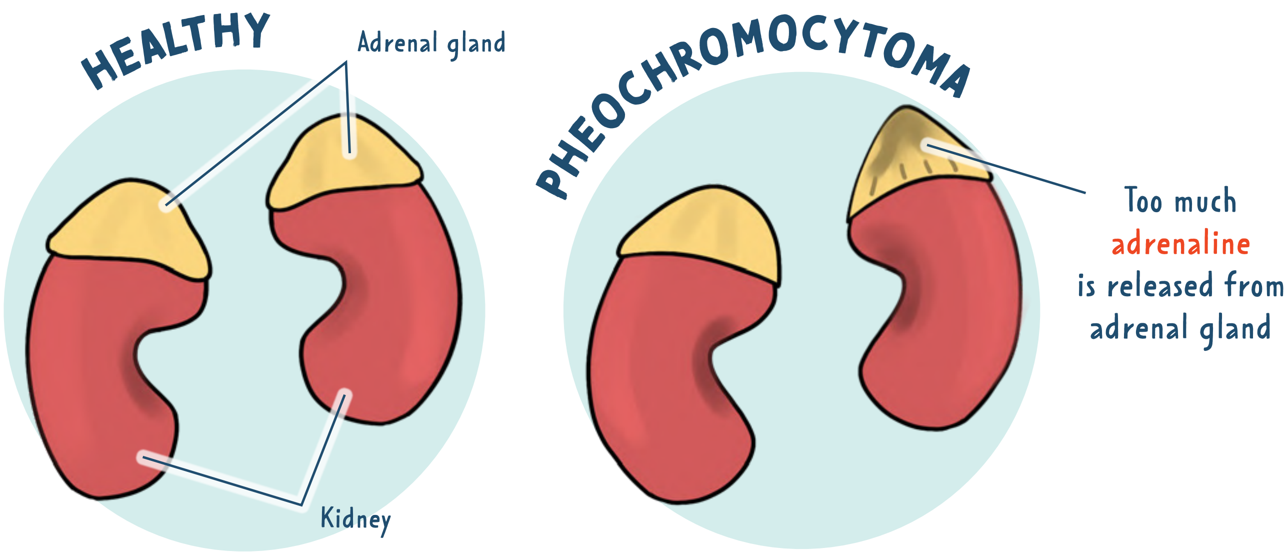
ADRENAL MEDULLA HORMONES:

Epinephrine ▪ Norepinephrine

↑

PHEOCHROMOCYTOMA

RARE tumor on the adrenal gland that secretes excessive amounts of epinephrine & norepinephrine



CAUSES

- Family history that makes them prone to developing the tumor

SIGNS & SYMPTOMS

H's

- **H**ypertension (severe)
- **H**eadache
- **H**eat (excessive sweating)
- **H**ypermetabolism
- **H**yperglycemia

AVOID STIMULI! It may cause a hypertensive crisis!

TREATMENT

- Adrenalectomy (if a tumor is present)
- Tell the client not to smoke, drink caffeine or change position suddenly
- Adm. anti-hypertensives
- Promote rest & calm environment
- **DIET:** high in calories, vitamins, & minerals

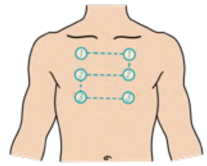
AUSCULTATING LUNG SOUNDS

TIPS FOR LISTENING

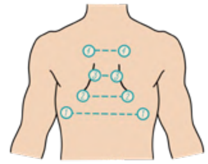
- Listen directly on the skin with the diaphragm
- Listening inside the **IN**tercostal spaces (**IN** *between the ribs)*
- Listen to the anterior & posterior chest
- Have the client sit upright (high fowler's), arms resting across the lap.
- Instruct client to take deep breaths
- Listen from top to bottom (comparing sides)

Listen for a FULL INHALATION TO EXPIRATION on each spot

ANTERIOR
Will hear **UPPER** lobes well



POSTERIOR
Will hear **LOWER** lobes well



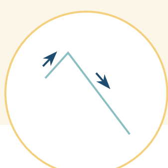
NORMAL SOUNDS

BRONCHIAL (TRACHEAL)

DESCRIPTION
High, loud & hollow tubular

LOCATION HEARD
Anteriorly only
(heard over trachea & larynx)

DURATION
Inspiration < expiration

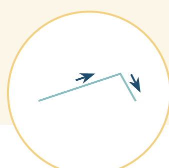


VESICULAR

DESCRIPTION
Soft, low pitched, breezy / rushing sound

LOCATION HEARD
Heard anterior & posteriorly

DURATION
Inspiration > expiration

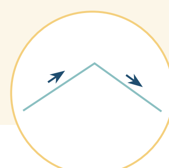


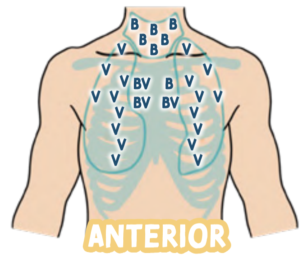
BRONCHOVESICULAR

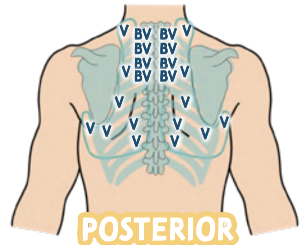
DESCRIPTION
Medium pitched, hollow

LOCATION HEARD
Heard anterior & posteriorly

DURATION
Inspiration = expiration







ABNORMAL (ADVENTITIOUS) SOUNDS

DISCONTINUOUS SOUNDS
DISCRETE CRACKLING SOUNDS

FINE CRACKLES (RALES)

DESCRIPTION: High pitched, crackling sounds (Sound like fire crackling, or velcro coming part)
DUE TO: Previously deflated airways that are popping back open
EXAMPLE: Pulmonary edema, asthma, obstructive diseases

COARSE CRACKLES (RALES)

DESCRIPTION: Low pitched, wet bubbling sound
DUE TO: Inhaled air collides with secretion in the trachea or large bronchi
EXAMPLE: Pulmonary edema, pneumonia, depressed cough reflex

PLEURAL FRICTION RUB

DESCRIPTION: Low pitched, harsh / grating sounds
DUE TO: Pleura is inflamed and loses it's lubricant fluid. It's literally the surfaces rubbing together during respirations
EXAMPLE: Pleuritis

CONTINUOUS SOUNDS
CONNECTED MUSICAL SOUNDS

WHEEZES

DESCRIPTION: High-pitched musical instrument with more than one type of sound quality (polyphonic)
DUE TO: Air moving through a narrow airway
EXAMPLE: Asthma, bronchitis, chronic emphysema

STRIDOR

DESCRIPTION: High pitched whistling or gasping with harsh sound quality
DUE TO: Disturbed airflow in larynx or trachea
EXAMPLE: Croup, epiglottis, any airway obstruction

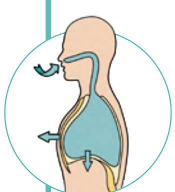
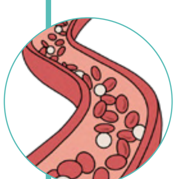
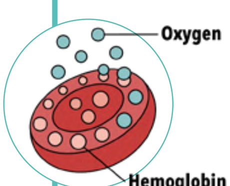
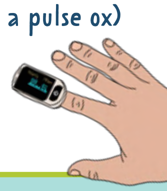
REQUIRES MEDICAL ATTENTION

LAB VALUES RELATED TO THE RESPIRATORY SYSTEM

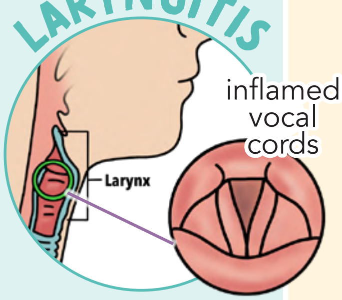
ABGs

	DEFINITION	EXPECTED RANGE	INTERPRETATION
PH	Measurement of how acidic or alkalotic your blood is	7.35 - 7.45	7.35 · 7.40 · 7.45 ACIDOSIS ← ABSOLUTE NORMAL → ALKALOSIS
$PaCO_2$	Measurement of carbon dioxide in the blood MEMORY TRICK: CO_2 think aCid	35 - 45	CO_2 **>45** = Acidosis CO_2 **<35** = Alkalosis
HCO_3	Measurement of bicarbonate in the blood MEMORY TRICK: Bicarbonate think Base	22 - 26	HCO_3 **>26** = Alkalosis HCO_3 **<22** = Acidosis
PaO_2	Measurement of oxygen in the blood	80 - 100	PaO_2 **<80** = Hypoxemia (the patient is not getting enough oxygen)
SaO_2	Percentage (%) of hemoglobin that is bound to oxygen	95-100%	SaO_2 **<95** = Hypoxemia (the patient is not getting enough oxygen) COPD pts are expected to have low O_2 levels (as low as 88%)

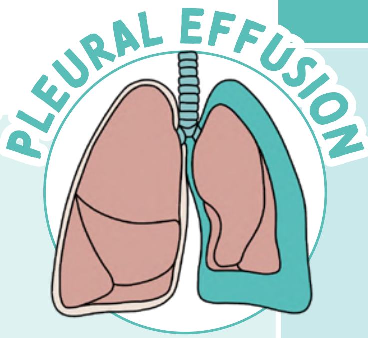
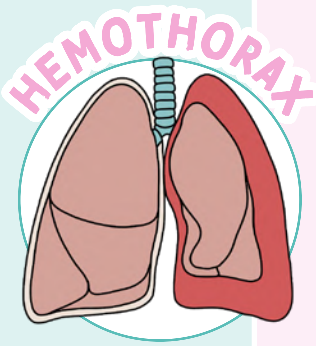
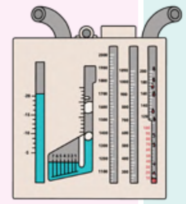
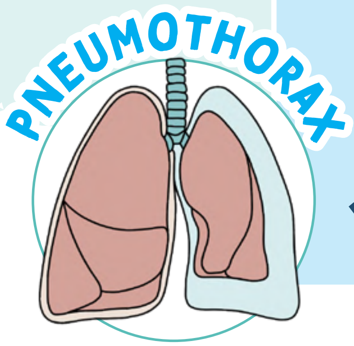
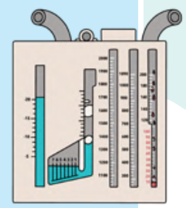
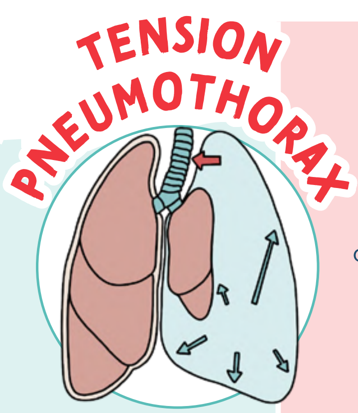
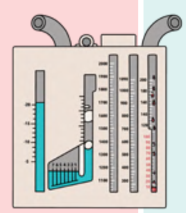
OXYGEN LEVELS EXPLAINED

	DEFINITION	EXPECTED RANGE	INTERPRETATION
FiO_2 	FiO_2 Fraction of inspired Oxygen (the air you breathe in)	Room air has **21%** oxygen	-
PaO_2 	The partial pressure of oxygen in the arterial blood MEMORY TRICK: PaO_2 = arterial	80 - 100 mmHg	**HYPOXEMIA** low · oxygen · in the blood Decreased oxygen in the **BLOOD** **HYPOXEMIA** usually leads to **HYPOXIA**
SaO_2 	Hemoglobin saturation percentage of hemoglobin that is bound to oxygen MEMORY TRICK: SaO_2 = Saturation (%)	95 - 100% (measured with a pulse ox) 	**HYPOXIA** low · oxygenation Decreased oxygen supply to the **TISSUES**

UPPER RESPIRATORY TRACT DISORDERS

	PATHOLOGY	SIGNS & SYMPTOMS	TREATMENT
RHINITIS 	Inflammation of the **mucous membrane in the nose** *Can be nonallergic or allergic*	▪ Runny nose ▪ Nasal congestion ▪ Nasal discharge ▪ Sneezing ▪ Headache	▪ Saline or steroid nasal sprays ▪ Antihistamines ▪ Decongestants
SINUSITIS 	Inflammation of the tissue lining the **sinuses** "sinus infection"	▪ Runny & stuffy nose ▪ Pressure & pain in the face ▪ Headache ▪ Post-nasal drip ▪ Mucus dripping down the throat ▪ Sore throat	▪ **VIRAL:** supportive measure ▪ **BACTERIAL:** antibiotics ▪ Nasal saline irrigation ▪ Corticosteroids ▪ Antihistamines
TONSILLITIS 	Inflammation of the **tonsils**	▪ Sore throat ▪ Fever ▪ Snoring ▪ Difficulty swallowing	▪ Fluids ▪ Salt water gargles ▪ Rest ▪ Humidified air ▪ Tonsillectomy (surgical removal of the tonsils)
LARYNGITIS 	Inflammation of the **larynx** (aka the "voice box")	▪ Hoarse voice ▪ Aphonia (loss of voice) ▪ Cough ▪ Dry sore throat ▪ Symptoms worsen with cold air or cold liquid	▪ Rest voice ▪ Avoid smoking & alcohol ▪ Avoid whispering and clearing throat (can irritate vocal cords) ▪ Humidified air & adequate hydration
PHARYNGITIS 	Inflammation of the **pharynx** (strep throat)	▪ Sore throat ▪ Red & swollen pharyngeal membrane & tonsils ▪ Lymph nodes ▪ White exudate ▪ Fever	▪ **VIRAL:** supportive measure ▪ **BACTERIAL:** antibiotics ▪ Rest ▪ Salt water gargles

HEMOTHORAX, PLEURAL EFFUSION, PNEUMOTHORAX, TENSION PNEUMOTHORAX

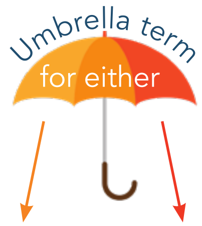
	PATHOLOGY	RISK FACTORS	TREATMENT
	Lung collapse due to collection of **FLUID** in the pleural space	▪ Trauma ▪ Infection (pneumonia)	▪ Thoracentesis 
	Lung collapse due to a collection of **BLOOD** in the pleural space "Hemo" means blood	▪ A pneumothorax is often followed by a hemothorax	▪ Chest tube 
	Lung collapse due to a collection of **AIR** in the pleural space	▪ Trauma (blunt or penetrating) ▪ Medical procedure (central line placement) ▪ Gun shot or stab wound	▪ Chest tube 
	**MEDICAL EMERGENCY** Complications of a **PNEUMOTHORAX**. Occurs when the opening to the pleural space creates a one-way valve, then air collects in the lungs and can't escape (pressure builds up)	**Signs & symptoms:** ▪ Jugular vein distention (JVD) ▪ Compression on the heart (tachycardia, hypotension, chest pain) ▪ Compression on other lung (tachypnea, hypoxia) ▪ Tracheal shift	▪ Needle decompression (aspirate the air) ▪ Chest tube 

CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

PATHOLOGY

$\downarrow O_2$ & $\uparrow CO_2$

Progressive pulmonary disease that causes chronic airflow obstruction. COPD causes the alveoli sacs to lose their elasticity (inability to fully exhale) leading to **AIR TRAPPING.**



EMPHYSEMA or **CHRONIC BRONCHITIS**

RISK FACTORS

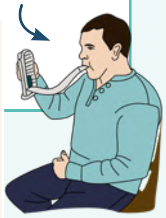
MOST COMMON

- Smoking
  - Breathing in harmful irritants
- Occupation exposure
- Infection
- Air pollution
- Genetic abnormalities
- Asthma
- Severe respiratory infection in childhood

Deficiency of Alpha1- antitrypsin (Protects the lining of the lungs)

DIAGNOSTIC

- Arterial blood gases (ABGs)
- Chest X-ray
- Pulmonary function test: **Spirometry**



Obstructive lung disease **FEV1** / **FVC** ratio of less than 70%

FEV1	**FVC**
=	=
FORCED EXPIRATORY VOLUME	**FORCED VITAL CAPACITY**

PATIENT EDUCATION

- Smoking cessation
- Regular exercise
- Avoid inhaling irritations
  (*Examples:* smoke, mold, pollen, dust)
- Stay up to date on vaccines
- Influenza & pneumococcal vaccine to $\downarrow$ the incidence of pneumonia
- Teach proper breathing techniques:

PURSED LIPS Promotes carbon dioxide elimination

DIAPHRAGMATIC BREATHING We want to use the DIAPHRAGM rather than the accessory muscles to breathe

DIET MODIFICATIONS:

- $\uparrow$ calories
- Small frequent meals
- $\uparrow$ protein
- Stay hydrated
  - Thins mucous secretion

Patients with COPD (especially emphysema) are using a lot of their energy to breathe, therefore burning a lot of calories

MEDICATIONS

- **Bronchodilators**
- Corticosteroids

End in suffixes: **-asone**, **-inide**, **-olone**

ORDER OF EVENTS:

1. **Bronchodilator:** Dilated airways
2. **Corticosteroids**: Now that airways are open, the steroids can do its job

NURSING CONSIDERATIONS

MONITOR RESPIRATORY SYSTEM:

- Lung sounds
- Sputum production
- Oxygen status

OXYGEN THERAPY

THOSE WITHOUT COPD	Healthy patients are stimulated to breath due to $\uparrow CO_2$
COPD PATIENTS	COPD patients are stimulated to breathe due to $\downarrow O_2$ (if you give too much O_2 ...they lose their "drive to breathe") Give oxygen with caution

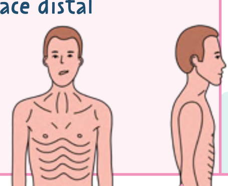
EMPHYSEMA VS CHRONIC BRONCHITIS

LIMITED AIR FLOW

$\downarrow O_2$ & $\uparrow CO_2$

EMPHYSEMA

- Abnormal distention of airspaces
- Enlargement & destruction of airspace distal to the terminal bronchiole
- Hyperventilation (breathing fast)
- Trying to blow off CO_2



PINK PUFFERS

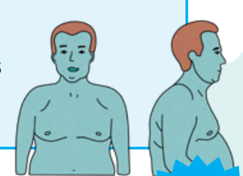
SIGNS & SYMPTOMS

- Hyperinflation of the lungs (barrel chest)
- Thin - weight loss
- Shortness of breath
- Severe dyspnea

Burning a lot of calories from trying to breathe off the excess CO_2

CHRONIC BRONCHITIS

- Mucus secretion
- Airway obstruction (inflammation)
- Chronic productive cough & sputum production for >3 months (within 2 consecutive years)



BLUE BLOATERS

SIGNS & SYMPTOMS

- Overweight
- Cyanotic (blue) - Hypoxemia
  - $\downarrow O_2$ & $\uparrow CO_2$
- Peripheral edema
- Rhonchi & wheezing
- Chronic cough

CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

NURSING MANAGEMENT & EDUCATION

MONITOR RESPIRATORY SYSTEM

- Lung sounds
- Sputum production
- Oxygen status

LIFESTYLE MODIFICATIONS

- Smoking cessation
  - Determine readiness
  - Develop a plan
  - Discuss nicotine replacement

DIET MODIFICATIONS

- Promote nutrition
- Increase calories
- Small frequent meals
- Stay hydrated
  - Thins mucous secretions

TEACH PROPER BREATHING TECHNIQUES

- Pursed lips
- Diaphragmatic breathing

SURGERY

- Bullectomy
- LVRS: lung volume reduction surgery
- Lung transplant

STAY UP TO DATE ON VACCINES

- Influenza & pneumococcal vaccine ↓ the incidence of pneumonia

OXYGEN THERAPY

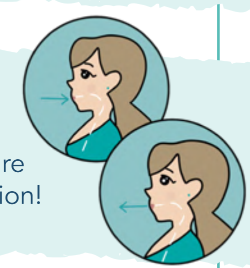
- COPD clients are stimulated to breathe due to ↓ O_2 (if you give too much O_2...they lose their "drive to breathe")
- Healthy clients are stimulated to breath due to ↑ CO_2

Adm. O_2 during exacerbations or showing signs of respiratory distress

Adm. oxygen with caution to clients with **CHRONIC HYPERCAPNIA** (elevated $PaCO_2$ levels)
1 - 2 liters max

Clients with COPD (especially emphysema) are using a lot of their energy to breathe, therefore burning a lot of calories

small, frequent meals that are rich in **PROTEIN**



PROMOTES CARBON DIOXIDE ELIMINATION

Allows better expiration by ↑ airway pressure that keeps air passages open during exhalation!

We want to use the **DIAPHRAGM** rather than the *accessory muscles* to breathe!

➥ This strengthens the diaphragm and slows down breathing rate

MEDICATIONS

BRONCHODILATORS

- Relaxes smooth muscle of lung airways = better airflow
- Symbicort (steroid + long-acting bronchodilator)

CORTICOSTEROIDS

- ↓ inflammation (oral, IV, inhaled)
- *Example:* Prednisone, Solu-Medrol, Budesonide

SUFFIX:
"-asone"
"-inide"
"-olone"

BUPROPION (ANTI-DEPRESSANT)

*For more information about respiratory medications, see the Pharmacology Bundle

ORDER OF EVENTS

1. **Bronchodilator**
   Dilated airways
2. **Corticosteroids**
   Airways are open; now the steroids can do their job

PNEUMONIA

PATHOLOGY

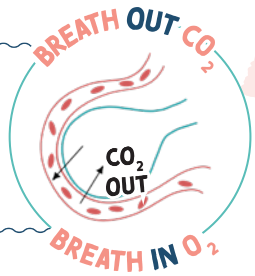
Lower respiratory tract infection that causes inflammation of **ALVEOLI SACS**!

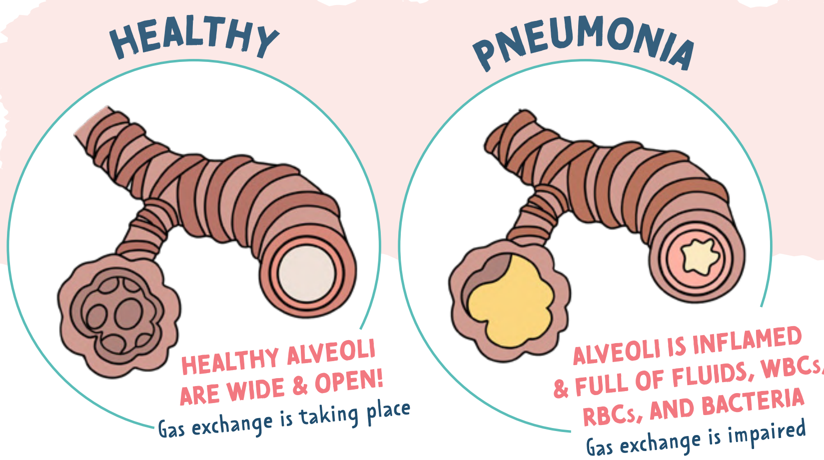
TYPES

- Community acquired pneumonia (CAP)
- Hospital acquired pneumonia (HAP)
- Healthcare associated pneumonia (HCAP)
- Ventilator-associated pneumonia (VAP)
- Aspiration pneumonia

REMEMBER

gas exchange takes place in the alveoli... so pneumonia causes **impaired gas exchange.**

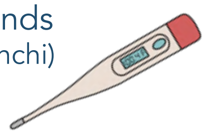




SYMPTOMS

MNEMONIC

- **P** Productive cough (purulent sputum)
- **N** Neuro changes (especially in the elderly)
- **E** Elevated Lab (↑ PCO_2 & ↑ WBCs)
- **U** Unusual breath sounds (course crackles & rhonchi)
- **M** Mild to high fever
- **O** Oxygen saturation low
- **N** Nausea & vomiting
- **I** Increased HR & BP
- **A** Achy (chills, fatigue)



RISK FACTORS

Can be **COMMUNITY-ACQUIRED** or **HOSPITAL-ACQUIRED**!

- Prior infection
- Immunocompromised
  - HIV, young/old, auto immune infections
- Postoperative
- Lung diseases
  - COPD
- Immobility
- Aspiration risk

INTERVENTIONS

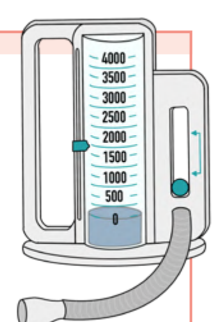
- Monitor...
  - Respiratory status
  - Vital signs: HR, temp, & pulse oximetry
  - Color, consistency & amount of sputum
- Diet
  - ↑ Calories
  - ↑ Protein
  - ↑ Fluids (oral or IV) → Thins secretions & compensates dehydration from fever
  - Small frequent meals
- Medications
  - Antipyretics
  - **Antibiotics** *(only for bacteria)* → Always take blood cultures BEFORE administering **antibiotics**
  - Antivirals
  - Bronchodilators
  - Cough suppressants
  - Mucolytic agents
- Semi Fowler's position → Helps lung expansion

DIAGNOSTIC

- Chest X-ray → shows pulmonary infiltrates or pleural effusions
- ↑ White blood cells
- Sputum culture → can be **BACTERIAL, VIRAL, or FUNGAL**

EDUCATE

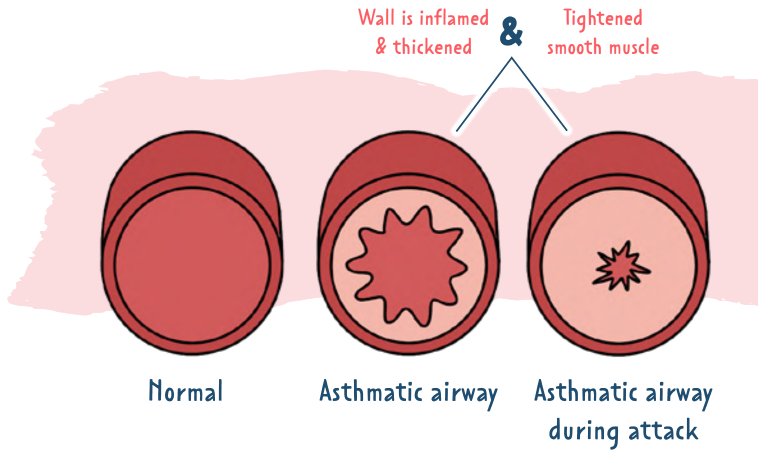
- Use of Incentive Spirometer
  - Helps to pop open the alveoli sacs & get the air moving
- Up-to-date vaccines
  - Annual flu shot
  - Pneumococcal vaccine
- Smoking cessation
- Hand washing & avoiding sick people



ASTHMA

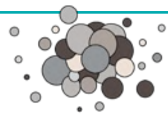
PATHOLOGY

Chronic lung disease that causes an inflamed, narrow, & swollen airway (bronchi & bronchioles)



CAUSES

NOT COMPLETELY KNOWN!



- Genetic
- Environmental
  - Smoke, pollen, perfumes, dust mites, pet dander, cold or dry air, etc.
- GERD
- Exercise-induced asthma
- Certain drugs
  - NSAIDS, aspirin

CLASSIFICATIONS:

Based on Symptoms

MILD INTERMITTENT	MILD PERSISTENT	MODERATE PERSISTENT	SEVERE PERSISTENT
< 2 a week	> 2 a week Not daily	Daily symptoms & exacerbations that happen 2x a week	Continually showing symptoms with frequent exacerbations

SIGNS & SYMPTOMS

CHARACTERIZED BY FLARE-UPS
(meaning: it comes & goes)

- Dyspnea (shortness of breath)
- Tachypnea (fast respiratory rate)
- Chest tightness
- Anxiety
- Wheezing
- Coughing
- Mucus production
- Use of accessory muscles
- **AIR TRAPPING**

Air trapping causes the client to retain CO_2 which is **ACIDIC** = **RESPIRATORY ACIDOSIS**

⚠ STATUS ASTHMATICUS ⚠

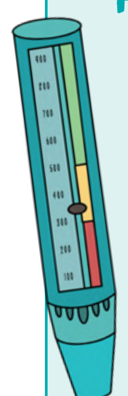
Medical emergency
Life-threatening asthma episode

oxygen
↓
hydration
↓
nebulization
↓
systemic corticosteroid

NURSING CARE

- Assess client's airway
- High Fowler's position
- Provide frequent rest periods
- Adm. oxygen therapy
  - Goal: keep the O_2 at 95 - 100%
- Maintain a calm environment to ↓ stress
- Asses **PEAK FLOW METER** reading
- Asses for cyanosis & retractions

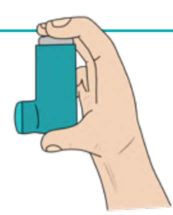
PEAK FLOW METER



- Shows how controlled the asthma is & if it's getting worse
- Establish a baseline by performing a "personal best" reading
  - Client will exhale as hard as they can & get a reading

GREEN = GOOD
YELLOW = NOT TOO GOOD
RED = BAD

MEDICATIONS



- BronchoDILATORS
  - Short-acting (Albuterol) *Rapid relief*
  - Long-acting (Salmeterol) *Prevents asthma attack*
  - Methylxanthines (Theophylline)
- Corticosteroids *Anti-inflammatory Agents*
  - Suffix **-ASONE** & **-IDE**
  - *Ex: Beclometh**asone***
- Leukotriene Modifiers
- Anticholinergics

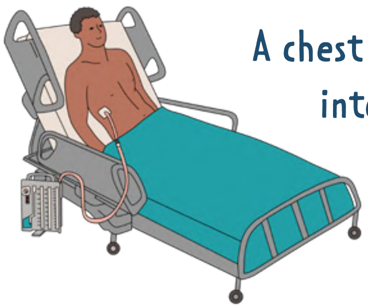
MNEMONIC

Certain medications are known to cause bronchospasms in patients with asthma. We want to "**BAN**" these medications from asthma patients.

- **B** Beta blockers
- **A** Aspirin
- **N** NSAIDs

**For more information about respiratory medications, see the Pharmacology Bundle*

CHEST TUBES



A chest tube is a tube that is inserted into the pleural space to remove excess air, blood, or fluid. This helps re-expand the lungs.

WHY IS IT USED?

- After thoracic surgery
- During cardiac surgery (drain fluid from around the heart)
- Spontaneous pneumothorax
- Pneumothorax
- Hemothorax
- Pleural effusion
- Empyema (infection)

3 CHAMBERS:

DRAINAGE CHAMBER

This is where the fluid is collected from the patient

WATER-SEAL CHAMBER

Allows air to be removed from the pleural space WITHOUT outside air entering the lungs

Tidaling (rise & fall with each breath) = **GOOD**

Excessive continuous bubbling in the water seal chamber = **BAD**

If the water stops fluctuating, this could mean:
1. The lung has re-expand
2. The tubing is kinked

SUCTION-CONTROL CHAMBER (Two types)

WET SUCTION & **DRY SUCTION**

If the tube becomes DISLODGED: Cover the insertion site with a sterile dressing

If the chamber becomes DAMAGED: Place the tubing in sterile water while waiting for a new system

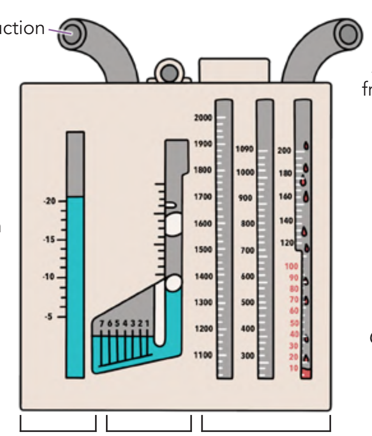
NURSING CONSIDERATIONS

- Always keep the drainage system BELOW the patient's chest
- **NEVER** strip the tubing
- **NEVER** clamp the tubing
- **EDUCATE** the patient to do **Valsalva maneuver** when the HCP is removing the chest tube
  - Deep breath, exhale, and bear down
- **MONITOR:**
  - Color & quantity of the drainage in the drainage collection chamber every hour
  - Lung sounds
  - Insertion site
- **REPORT** bright red blood (dark red is expected)

WET-SUCTION

Uses **WATER** to control the level of suction (actually filling the suction control chamber with water)

WILL HAVE GENTLE BUBBLING

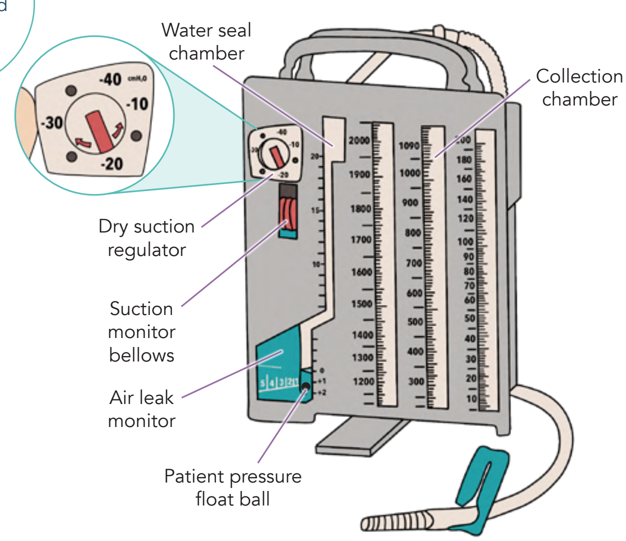


Both have a **collection chamber** and an **air leak monitor**

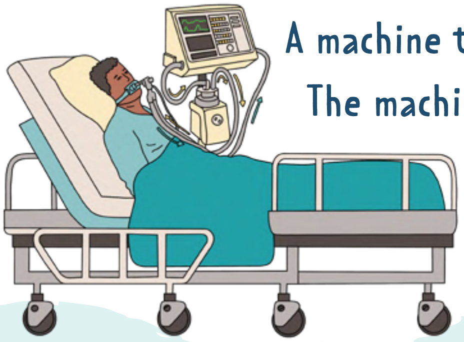
DRY-SUCTION

There is no water column (it's DRY). The suction is controlled by a suction monitor bellows that balances wall suction

THERE WILL BE NO BUBBLING



MECHANICAL VENTILATION



A machine that helps a person breathe. The machine pumps air into the lungs unlike normal breathing.

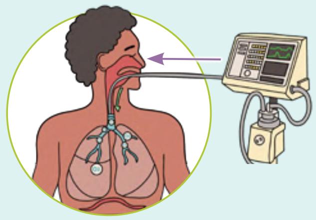
WHY IS IT USED?

- Control breathing during surgery
- Rest the respiratory muscles
- When a patient is unable to breathe on their own (respiratory failure such as ARDS)

POSITIVE PRESSURE VENTILATION*

THE AIR IS PUSHED INTO THE LUNGS

This forceful air entering into the lungs can cause **barotrauma**

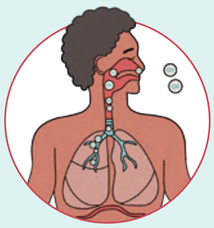


NEGATIVE PRESSURE VENTILATION*

NORMAL BREATHING

The diaphragm uses negative pressure to bring in oxygen

MEMORY TRICK: **N**egative think **N**ormal breathing



VENTILATOR SETTINGS

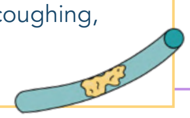
TIDAL VOLUME (VT)	Volume of gas delivered with each breath **500 - 800** mL
RESPIRATORY RATE	# of breaths delivered to the patient **12 - 20** breaths per min
FiO_2	Fraction of inspired oxygen (O_2 concentration of the air being delivered to the pt.) **21% - 100%**
POSITIVE END EXPIRATORY PRESSURE (PEEP)	The amount of pressure in the lungs after expiration (prevents collapse of the alveoli)

UNDERSTANDING ALARMS

HIGH PRESSURE ALARMS

MEMORY TRICK: **H**igh think **H**igh blockage of airflow

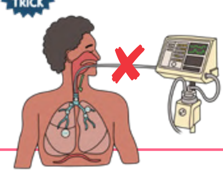
CAUSES:
Excessive mucous or secretions, kinks, coughing, pulmonary edema, or pneumothorax, a patient "fighting" the ventilator



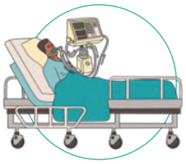
LOW PRESSURE ALARMS

MEMORY TRICK: **L**ow think **L**eaks

CAUSES:
Disconnection, cuff leak, tube displacement

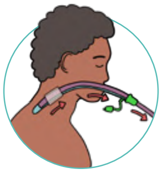


NURSING CONSIDERATIONS



MONITOR:

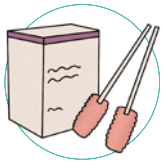
- Level of consciousness
- Vital signs
- Lung sounds
- Arterial blood gases
- Symptoms of ventilator associated pneumonia
- The gastrointestinal system
- Nutritional status



SUCTIONING

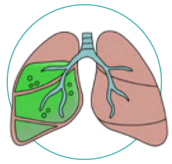
Suction secretions only when needed:

- Never suction when inserting a catheter into the airway
- Never suction for longer than 10 seconds
- Administered 100% oxygen before suctioning



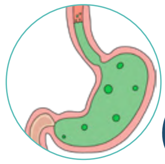
ORAL CARE

- Clean the mouth with chlorhexidine every 2 hours



MOBILIZE SECRETIONS

- Turn/reposition the patient every 2 hours
- Keep the head of the bed >30°



GASTROINTESTINAL SYSTEM

- Administer PPIs & H2 blockers to prevent stress ulcers and decrease acid

Proton pump inhibitors (PPIs) end in -prazole

Histamine H2 antagonists (H2-blockers) end in -dine

LAB VALUES RELATED TO THE HEMATOLOGICAL SYSTEM

		EXPECTED RANGE	DESCRIPTION	↓	↑
COMPLETE BLOOD COUNT (CBC)	RED BLOOD CELLS (RBCs)	F 4.2 – 5.2 X 10^6 /uL M 4.7 – 6.1 X 10^6 /uL	Red blood cells transport oxygen to the body's cells.	Fluid volume overload (More volume dilutes the RBCs) Hemorrhage Anemia Renal disease (lack of erythropoietin production)	Dehydration/ fluid volume deficit (Less volume concentrates the RBCs) Hyperactivity of the bone marrow
	WHITE BLOOD CELLS (WBCs)	4,500 – 11,000 / uL	The white blood cells are a part of the immune system and help to fight infections and diseases.	**LEUKOPENIA** WBCs < 4,500 /uL Immunosuppression	**LEUKOCYTOSIS** WBCs > 11,000 /uL Current or recent INFECTION & inflammation
	PLATELETS (PLT)	150,000 – 450,000 /uL	Platelets help clot the blood. Platelet aggregation is the clumping together of platelets that form a plug at the site of the injury.	**THROMBOCYTOPENIA** PLTs < 150,000 /uL ↓ Platelets think BLEEDING	**THROMBOCYTOSIS** PLTs > 450,000 /uL Certain cancers Infection
	HEMOGLOBIN (HGB)	F 12 - 16 g/dL M 13 - 18 g/dL	Hemoglobin is an iron containing protein found in red blood cells. It transports oxygen from the lungs to the tissues. It also returns CO_2 from the tissues back to the lungs.	Fluid retention (hemodilution) Anemia Hemorrhage	Dehydration (hemoconcentration)
	HEMATOCRIT (HCT)	F 36% - 48% M 39% - 54%	The percent of blood that is made up of red blood cells (expressed as a %).	Fluid retention (hemodilution) Anemia Hemorrhage	Dehydration (hemoconcentration) Low oxygen availability (smoking, pulmonary diseases (COPD), high altitudes)
TYPES OF COAGULATION TESTS	ACTIVATED PARTIAL THROMBOPLASTIN TIME (aPTT)	NORMAL (not on anticoagulants) 30 - 40 seconds ON HEPARIN THERAPY 1.5 - 2.0 x the normal value	aPTT measures how long it takes for a blood clot to form. It's also used to monitor the effectiveness of the anticoagulant: **HEPARIN**.	MEMORY TRICK Numbers are LOW = Clots will GROW	Heparin therapy MEMORY TRICK Numbers are too HIGH = Patient will DIE (from increased bleeding)
	PROTHROMBIN TIME (PT)	NORMAL (not on anticoagulants) 10 - 12 seconds ON HEPARIN THERAPY 1.5 - 2.0 x the normal value	Prothrombin time measures the amount of time needed to form a clot. It's also used to monitor the effectiveness of the anticoagulant: **WARFARIN**.		Deficiency in vitamin K Deficiency in clotting factor Liver disease
	INTERNATIONAL NORMALIZED RATIO (INR)	NORMAL (not on anticoagulants) < 1 ON HEPARIN THERAPY INR 2.0 - 3.0 INR 2.5 - 3.5 (heart valve replacement)	INR is calculated from the prothrombin time and is used to monitor oral anticoagulants such as **WARFARIN**.		Warfarin therapy MEMORY TRICK Numbers are too HIGH = Patient will DIE (from increased bleeding)
	D-DIMER	< 0.5 mcg/mL	D-dimers are fragments of fibrin that are in the blood when a clot dissolves or is broken down. D-dimer helps to determine if a clot is present somewhere in the body	Blood clot is ruled out	Additional tests are needed to confirm and determine a specific diagnosis Blood clot may be present in the body

IRON DEFICIENCY ANEMIA

PATHOLOGY

TYPE OF ANEMIA CAUSED BY ↓ IRON LEVELS

There are many types of anemias (**iron deficiency**, vitamin B12 deficiency, folate deficiency, etc).

MOST COMMON TYPE OF ANEMIA

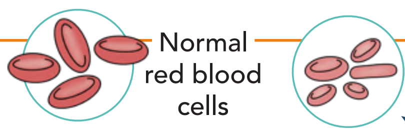
ANEMIA: the body doesn't have enough RBCs to carry oxygen to the tissues and the body.

RED BLOOD CELLS ROLE

- Transports O_2 & removes CO_2 from the body with the help of hemoglobin (Hgb)

HEMOGLOBIN (HGB)

- Found in the RBCs
- It's a protein that contains IRON



Normal red blood cells

Iron deficiency anemia

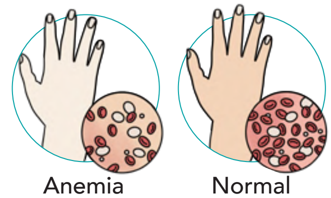
Smaller in size & more pale

(because hemoglobin makes the blood bright red)

SIGNS & SYMPTOMS

SAME SYMPTOMS AS ANEMIA

- Pallor
- Weakness & fatigue
- Shortness of breath (from lack of oxygen)
- Tachycardia
- Microcytic (small) red blood cells



specific to iron deficiency anemia

SEVERE SYMPTOMS

- Smooth, red tongue
- Brittle & ridged nails

TREATMENT/MEDICATIONS

TREAT THE CAUSE: D/C any drugs causing the anemia.

IRON SUPPLEMENTS (oral or liquid)

MOST COMMON TREATMENT

Examples: ferrous sulfate, ferrous gluconate, ferrous fumarate

IV ADMINISTRATION OF IRON

If oral iron is poorly absorbed or poorly tolerated

RISK FACTORS

- Lack of iron (vegetarian diet)
- Blood loss (excessive menstruation, surgery or trauma)
- Pregnancy
- Iron malabsorption (due to bariatric surgery or Celiac disease)

DIAGNOSTIC

- Complete blood count (CBC)
  - ↓ hemoglobin & ↓ hematocrit
- Bone marrow aspiration
- Stool sample, colonoscopy, endoscopy (checking for blood)

NORMAL VALUES:

Hemoglobin (Hgb)
Female: 12 - 16 g/dL *Male:* 13 - 18 g/dL

Hematocrit (HCT)
Female: 36% - 48% *Male:* 39% - 54%

PATIENT EDUCATION

EDUCATE on administering iron supplements:

Side effects of iron supplements:
Black stool
Constipation
Foul aftertaste

↑ ABSORPTION

Vitamin C:
Take iron with fruit juice & multivitamin.
Take on an empty stomach

↓ ABSORPTION

Calcium:
Do not take iron with milk or antacids

liquid iron stains the teeth!

1. Take with a straw
2. Brush teeth after

EDUCATE on foods high in IRON: MNEMONIC **"EAT LOTS OF IRON"**

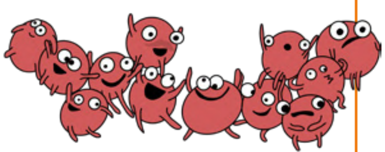
- **E**gg yolks
- **A**pricots
- **T**ofu
- **L**egumes
- **O**ysters
- **T**una
- **S**eeds
- P**O**tatoes
- **F**ish
- **I**ron-fortified cereals
- **R**ed meats
- P**O**ultry
- **N**uts

THROMBOCYTOPENIA

PATHOLOGY

↓ PLATELETS

- Platelets help clot the blood
- **Platelet aggregation:** the clumping together of platelets that form a plug at the site of injury

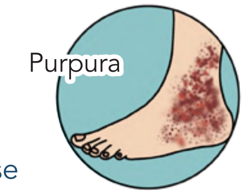


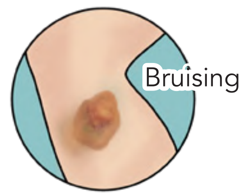
- ↓ Platelets = think bleeding

NORMAL PLATELET COUNT	THROMBOCYTOPENIA
150,000 - 450,000 /μL	< 150,000 /μL

SIGNS & SYMPTOMS

- Weakness, dizziness, tachycardia, hypotension
- Prolonged bleeding time
- Petechiae (pinpoint bleeding)
- Purpura
- Bruising
- Bleeding from the gums & nose
- Heavy menstrual cycles
- Blood in stool or urine





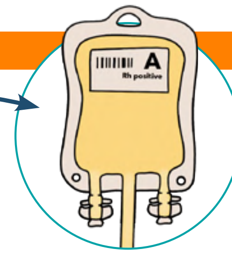
RISK FACTORS

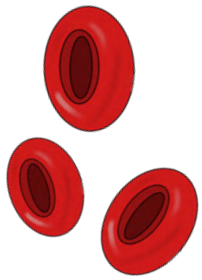
MNEMONIC

- **P** Platelet disorders
- **L** Leukemia
- **A** Anemia
- **T** Trauma
- **E** Enlarged spleen
- **L** Liver disease
- **E** Ethanol (alcohol-induced)
- **T** Toxins (drug-induced)
- **S** Sepsis

TREATMENT

- Platelet transfusion
- Bone marrow transplant
  - Platelets are made in the bone marrow
- Splenectomy
  - For those unresponsive to medical therapy





DIAGNOSTIC

- ↑ Bleeding time
- ↑ INR & ↑ PT/PTT
- ↓ Hgb & Hct
- Bone marrow aspiration & biopsy

PATIENT EDUCATION

EDUCATE they will need to follow **BLEEDING PRECAUTIONS**:

- Use electric razors
- Use small needle gauges
- NO aspirin
- Decrease needle sticks
- Protect from injury

IMMUNE THROMBOCYTOPENIC PURPURA (ITP)

Type of thrombocytopenia, formerly called "**idiopathic thrombocytopenia purpura**"

"Purpura" is in the name because the body bruises easily & petechiae may occur in the trunk & extremities

PATHOLOGY

Autoimmune disease where the body produces antibodies against its own thrombocytes (platelets)

ITP
< 100,000 /μL

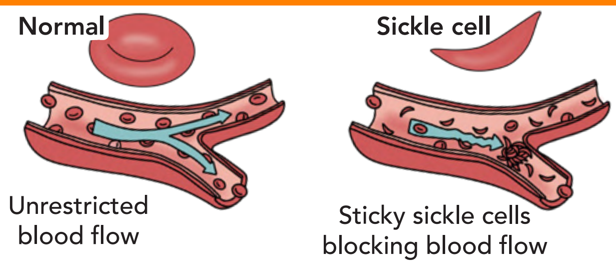
RISK FACTORS

- Children after viral illness
- Females (ages 20-40)
- Pregnancy

SICKLE CELL ANEMIA

PATHOLOGY

An inherited disease that causes the hemoglobin molecule to be defective. RBCs are weak and die earlier than healthy RBCs.



Hemoglobin S is sensitive to low amounts of oxygen in the body!

Low oxygen
↓
Causes RBCs to change their shape
↓
Sickled shaped, sticky, and stiff
↓
Causes clumping which blocks blood flow to the tissues
↓
⚠ **SICKLE CELL CRISIS** ⚠

3 CELL TYPES OF SICKLE CELLS CRISIS:

Acute vaso-occlusive crisis	RBCs sticking in vessels = hypoxia *(this is very painful!)* MOST COMMON
Aplastic crisis	The body stops producing enough RBCs (bone marrow can't keep up)
Sequestration crisis	The spleen stops working & becomes flooded with the sickle cells

RISK FACTORS

A patient is born with this genetic blood disorder. It's an autosomal recessive disorder (the sickle hemoglobin (HbS) gene is inherited).

It's commonly recognized early in life after maternal iron stores have been depleted.

DIAGNOSTIC

- Blood sample
- Test before birth (testing the amniotic fluid)

AUTOSOMAL RECESSIVE

Either parent can have the sickle cell trait, but this doesn't mean they have sickle cell anemia. BOTH parents have to pass down the sickle hemoglobin (HbS) gene.

MEDICATIONS

- Analgesics & opioids
  To help with the pain

TREATMENT/ NURSING CONSIDERATIONS

- IV fluids (stops the clumping of RBCs)
- Oxygen therapy
- RBC transfusions
- Stem cell transplant

SIGNS & SYMPTOMS

- Anemia symptoms (fatigued, tachycardia, pallor)
- ★ Pain
- Dactylitis (swelling of the hands & feet)
- Stroke
- Acute chest syndrome (tachypnea, wheezing, fever, cough)

PATIENT EDUCATION

EDUCATE on how to **prevent** sickle cell crisis:

- ✔ Vaccines up to date
- ✔ Prevent infection (hand hygiene, avoiding big crowds)
- ✔ Limit stress
- ✔ Avoid high altitudes
- ✔ Drink lots of water (stay hydrated)
- ✔ Smoking cessation
- ✔ Avoid over-exertion

DISSEMINATED INTRAVASCULAR COAGULATION (DIC)

PATHOLOGY

Causative factor (underlying disease)

↓

Inflammatory response causes inflammation & coagulation in the vasculature

↓

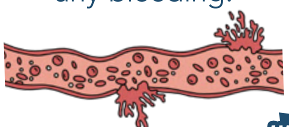
The fibrinolytic system is halted

↓

Causes lots of small clots & platelets to clump

↙ ↘

Lots of small clots are using all the blood's clotting factors. This leaves other parts of the body with no means of stopping any bleeding.



TOO LITTLE CLOTTING
(BLEEDING)

+

HAPPENING AT THE SAME TIME

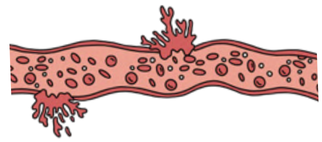
Excessive clotting causes blockage of blood vessel

TOO MUCH CLOTTING

↓

Can lead to organ ischemia
(because organs are not getting blood supply)

SIGNS & SYMPTOMS



BLEEDING

Bleeding can be minimal all the way up to hemorrhaging everywhere

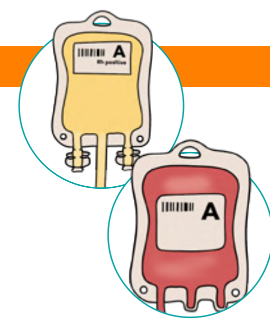
- Petechiae & purpura
- Hematuria
- Melena (black tarry stools)
- Nose bleeds

&

BLOOD CLOTS

Stroke, heart attack, deep vein thrombosis, or a pulmonary embolism

TREATMENT



- **Treat the underlying cause!**
- Transfusion
  - Packed RBCs
  - Fresh frozen plasma (FFP)
  - Platelets

MEDICATIONS

- **Vasopressors**
  Cause vasoconstriction which ↑ blood flow & increases perfusion to the organ
- **Heparin infusion**
  Stops the clotting which increases blood flow to the organs
- **Cryoprecipitate**
  Replaces fibrinogen, factors V & VII

RISK FACTORS

DIC is not a disease. Rather, DIC occurs due to an underlying condition or disease:

- Infection/sepsis
- Malignancy
- Allergic reactions
- Obstetric complications
- Trauma
- Shock
- Toxins

DIAGNOSTIC

- Lab tests
  - ↓ platelet & ↓ fibrinogen levels
  - Prolonged clotting time (↑ PT, aPTT)
  - ↑ D-dimer (indicates there is a clot somewhere in the body)

NURSING CONSIDERATIONS

- Administer oxygen
- IV fluids
- Correct electrolyte imbalances

MONITOR:

- For signs of bleeding
- Vital signs
- Lab values

GASTROINTESTINAL SYSTEM OVERVIEW

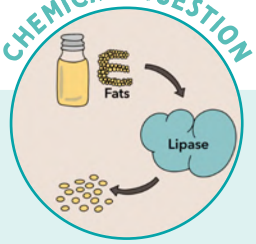
ORAL CAVITY COMPONENTS

MECHANICAL DIGESTION

Physical movement of food
(when food is broken down into smaller pieces)

EXAMPLES:
- Chewing
- Churning of the stomach

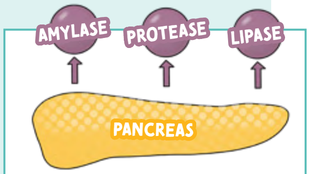
CHEMICAL DIGESTION



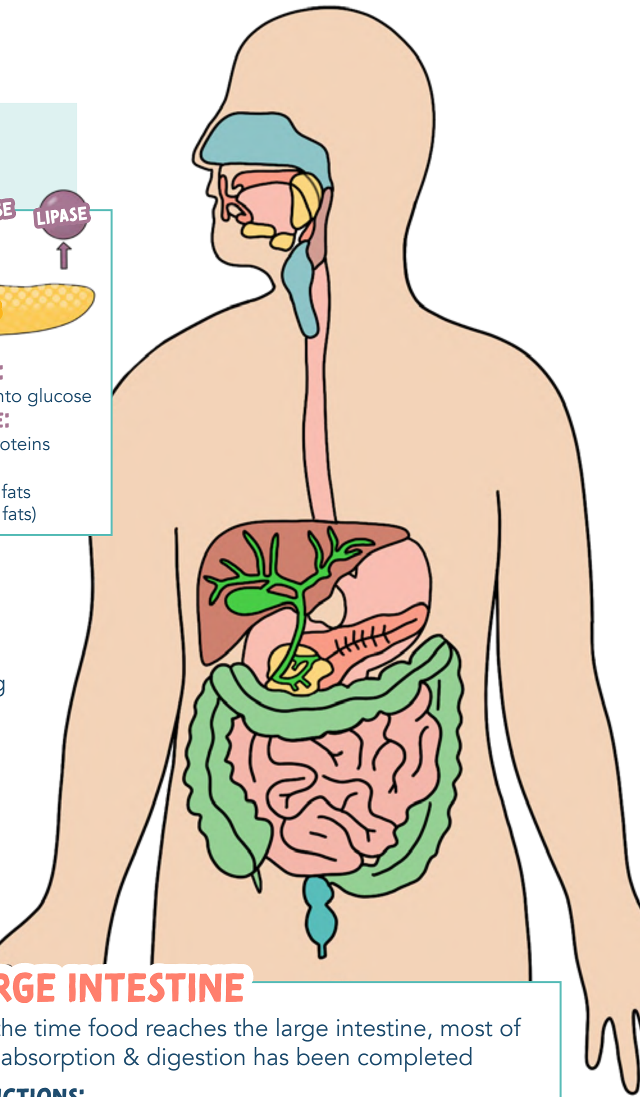
When food is broken down by ENZYMES and digestive juices

TIP: Enzymes end in "**-ASE**"

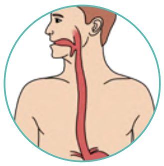
MEMORY TRICK: **P**rotease think **P**roteins
Lipase think **L**ipids (fat)



AMYLASE:
breaks down carbs into glucose
PROTEASE:
breaks down proteins
LIPASE:
breaks down fats
(think lipids = fats)



ESOPHAGUS



Is a hollow muscular tube that carries food & liquid from the mouth to the stomach. It does this by peristalsis.

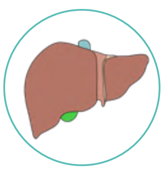
STOMACH

A hollow muscular organ

FUNCTIONS:
- Stores food during eating
- Secretes digestive fluids
- Moves partially digested food (chyme) into the small intestine

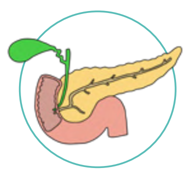
LIVER

FUNCTIONS:
- Filters the blood
- Metabolism of sugar, protein, and fat
- Synthesize lipoproteins (VLDL & HDL)
- Makes vitamin D
- Detoxifications
  (excretion of bilirubin and other toxins)



- Bile formation
- Drug metabolism
- Helps in blood clotting
- Synthesize proteins such as albumin and coagulation factors

PANCREAS

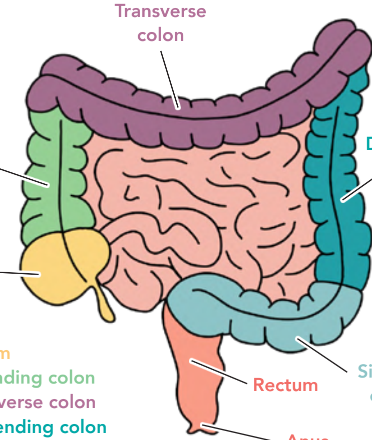


Helps make pancreatic juice (enzymes). This pancreatic juice break down sugar, fat, and starch. The pancreas has both exocrine and endocrine functions.

LARGE INTESTINE

By the time food reaches the large intestine, most of the absorption & digestion has been completed

FUNCTIONS:
- **ABSORBS** water and electrolytes from food that has not been digested yet
- **DEFECATION** rids the body of any waste leftover from food and removes it through the rectum and anus

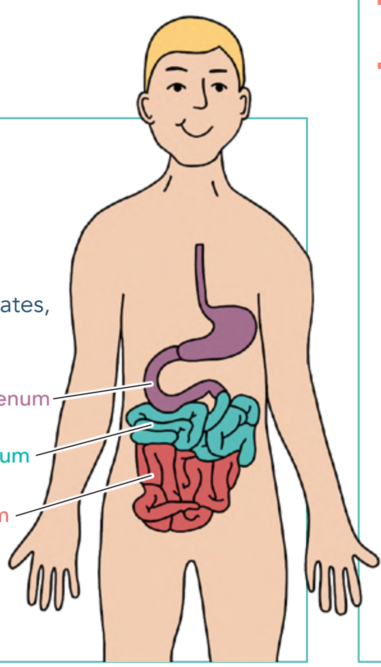


PROXIMAL → DISTAL
- Cecum
- Ascending colon
- Transverse colon
- Descending colon
- Sigmoid colon
- Rectum
- Anus

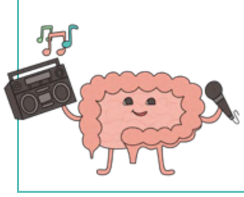
SMALL INTESTINE

The longest portion of the GI tract (longer than the large intestine)

FUNCTIONS:
- **DIGESTION** of food from the stomach
- **ABSORPTION** of nutrients, fats, carbohydrates, vitamins, minerals, etc.) and water from food into the bloodstream to be used by the body

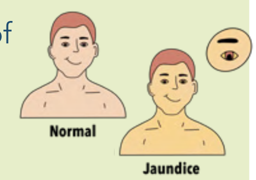


PROXIMAL → DISTAL: Duodenum, Jejunum, Ileum



MEMORY TRICK: To remember the order (PROXIMAL - DISTAL)
DJ Ileum in the club!

LAB VALUES RELATED TO THE GASTROINTESTINAL SYSTEM

	EXPECTED RANGE		
AMYLASE Pancreatic enzyme	**30 - 110** U/L	↑ levels could indicate pancreatitis	**LIPASE** is a better indicator of pancreatitis than **AMYLASE** because serum lipase remains elevated for a longer period of time. MEMORY TRICK: Lipase think Longer
LIPASE Pancreatic enzyme	**< 200** U/L		
BILIRUBIN Produced by the liver	Total **0.2 – 1.2** mg/dL	↑ levels could indicate liver dysfunction	**JAUNDICE:** Yellow discoloration of the skin due to high levels of bilirubin. Visible when serum bilirubin is > 2 mg/dL 
ALBUMIN	**3.5 - 5.5** g/dL	↑ levels could indicate dehydration	Albumin helps keep fluid in your bloodstream
PREALBUMIN	**15 - 36** mg/dL	↓ levels could indicate malnutrition	Prealbumin is great for assessing **NUTRITIONAL STATUS**
AST Liver enzyme	**0 - 35** U/L	↑ levels could indicate liver dysfunction	**AST** must be taken with **ALT** If ALT is normal, this means there is a problem other than liver disease, such as damage to another organ (heart, brain, muscle, kidneys)
ALT Liver enzyme	**0 - 48** U/L		
AMMONIA	**10 - 80** mcg/dL	↑ levels could indicate liver dysfunction 	Ammonia (NH3) is produced by cells throughout the body and is used by the **LIVER** to make urea. If the liver stops working, ammonia increases in the body. Too much ammonia is very toxic (especially to the brain) 

Part of the liver function test (LFT)

ACUTE & CHRONIC PANCREATITIS

PATHO

The islets of Langerhans secrete **INSULIN & GLUCAGON INTO THE BLOOD STREAM**

Pancreatic tissue: secrete digestive enzymes that break down **CARBOHYDRATES, PROTEINS & FATS**

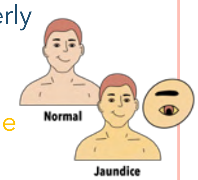
PANCREATITIS is an **AUTO-DIGESTION** of the pancreas by its own digestive enzymes released too early in the pancreas

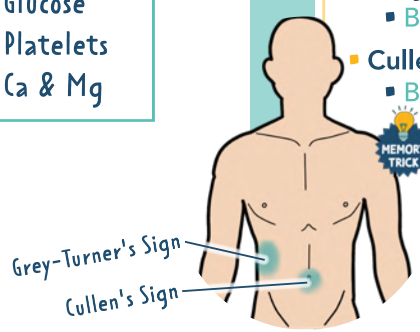
LABS

- ↑ Amylase
- ↑ Lipase
- ↑ WBCs
- ↑ Bilirubin
- ↑ Glucose
- ↓ Platelets
- ↓ Ca & Mg

ACUTE vs CHRONIC

	ACUTE	CHRONIC
	Sudden inflammation that is **REVERSIBLE** if prompt recognition and treatment is done	Chronic inflammation that is **IRREVERSIBLE**
CAUSES	**Gallstones** ▪ Blocks the bile duct **Alcohol (ETOH)** ▪ Damages the cells of the pancreas **Infection** **Medications** **Tumor** **Trauma**	**Repeated episodes of acute pancreatitis** **Excessive & prolonged consumption of alcohol (ETOH)** ▪ Recurrent damage to the cells of the pancreas **Cystic Fibrosis**
SIGNS & SYMPTOMS	In **ACUTE**, there will still be working functions of the pancreas. **Sudden sever PAIN!** ▪ Mid-epigastric pain LUQ **Nausea & vomiting** **Fever** **↑ HR & ↓ BP** **↑ Glucose** **Mental confusion & agitation** **Abdominal guarding** **Rigid/board-like abdomen** **Grey-Turner's Sign** ▪ Bluish discoloration at the flanks **Cullen's Sign** ▪ Bluish discoloration of the umbilicus MEMORY TRICK: **C**ullen's = **C**ircle belly button	In **CHRONIC**, you will see different S&S due to the prolonged damage & loss of function **Chronic epigastric pain or no pain** **Pain ↑ after drinking ETOH or after a fatty meal** **Steatorrhea "fatty stools"** ▪ Oily/greasy frothy stool **Weight loss** ▪ Can't digest food properly **Jaundice** ▪ Yellowish color of the skin from build up of bile **Diabetes Mellitus** ▪ Damage to the islet of Langerhans **Dark urine** ▪ From excess bile in the body





MEDICATIONS

- Opioid analgesics
- Antibiotics
- Pancreatic enzymes
- Insulin
- Proton Pump Inhibitors (PPI's), H2 antagonists, antacids

DIGESTIVE ENZYMES (EXOCRINE)

AMYLASE: Breaks down carbs to **glucose**

PROTEASE: Breaks down **proteins**

LIPASE: Breaks down **fats**

INTERVENTIONS

- Rest the pancreas!
  - NPO
    (we don't want stimulation of the enzymes)
- IV fluids
- Pain management
- Positioning
  - Side lying ➜ fetal position, NOT supine!
- Insert NG tube
  - Remove stomach contents

MONITOR:
- Glucose
- Blood pressure
- Intake & output
- Laboratory values
- Stools

DIET

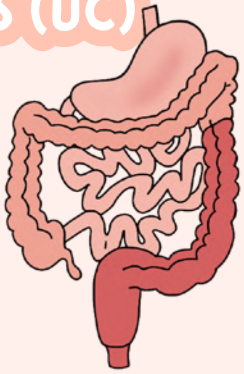
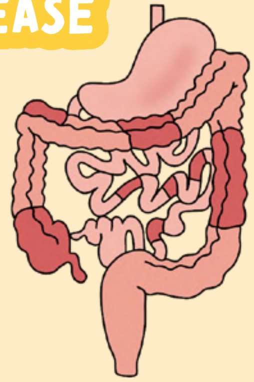
- NO ETOH!
- ↑ protein
- Limit sugars

- ↓ fat
  (no greasy, fatty foods)
- Complex carbohydrate
  (fruits, vegetables, grains)

ULCERATIVE COLITIS VS. CROHN'S DISEASE

TYPES OF Inflammatory Bowel Disease (IBD)

This is not the same thing as irritable bowel syndrome (IBS)

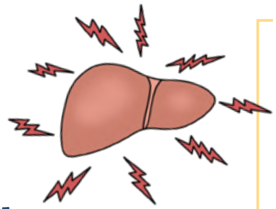
	MOST COMMON ULCERATIVE COLITIS (UC)	CROHN'S DISEASE
DESCRIPTION	Chronic ulceration & inflammation of the rectum & colon 	Inflammation of the gastrointestinal tract wall at ANY point through ALL layers 
LOCATION	Affects the large intestine & rectum only	Can affect anywhere in the GI tract (mouth to the anus)
THICKNESS	Inflammation affects the submucosa or mucosa	Inflammation is transmural (occurring across the entire wall)
APPEARANCE	Inflamed areas are continuous with no patches	Patches of inflammation throughout the bowel. This makes a cobblestone appearance! MEMORY TRICK: **C**rohn's think **C**obblestone
CURE	✔ **YES!** Colectomy	✘ **NO** cure, but surgery can help with symptoms
COMPLICATIONS	Toxic mega colon, rupture of bowel, dehydration. ⚠ Increased risk for hemorrhage/ shock	Abscess, fistulas. ⚠ Increased risk for infection (sepsis)
SIMILARITIES	• Both a form of inflammatory bowel disease (IBD) • Causes of both are not completely known • Both increase the risk for colon cancer • Both cause inflammation & ulcers • Both should consume the following diet: ↓ fiber, ↑ protein diet, & ↑ fluids	

TYPES OF HEPATITIS

HEPATITIS

HEPAT → LIVER ITIS → INFLAMMATION

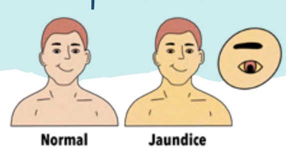
"INFLAMMATION OF THE LIVER"



CAUSED BY:

- Viral (A, B, C, D, E) MOST COMMON
- Excessive use of alcohol
- Hepatotoxic medications

	TRANSMISSION	SIGNS & SYMPTOMS	DIAGNOSTIC	TREATMENT	VACCINE
HAV ACUTE ONLY	**Fecal & oral** Food & water	GI symptoms (N&V, stomach pain, anorexia) Dark-colored urine Clay-colored stool Vomiting Flu-like symptoms Jaundice YELLOW DISCOLORATION of the skin from the buildup of bilirubin	Anti-HAV IgM = Active infection **IgG** = Recovered (**I**t's **G**one)	Supportive therapy... REST!	✔
HBV B IS BOTH ACUTE & CHRONIC	MEMORY TRICK **B** think **B**ody **F**luids *(Semen, saliva)* • Birth & blood • Childbirth, sex, & IV drugs		HBsAG = Active infection Anti-HBs = Immune / recovery	**ACUTE** Supportive therapy & rest **CHRONIC** Antivirals	✔
HCV ACUTE & CHRONIC	**Body fluids** Most common: IV drug users		Anti-HCV No post exposure immunoglobulin	Antivirals Interferon	✘
HDV ACUTE & CHRONIC	MEMORY TRICK **D**epends on **B** **B** & **D** = **B**u**D**s Hep D occurs with Hep B		HDAg Anti-HDV	Antivirals Interferon	✘
HEV ACUTE ONLY	**Fecal & oral** Food & water *uncooked meats, 3rd world countries*		Anti-HEV	Supportive therapy... REST!	✘



EDUCATION for ALL types of Hepatitis!

- Rest
- Diet
- Small frequent meals
- ↑ Carbohydrates
- ↑ Calories
- ↓ Protein & fat
- Proper hand hygiene
- Do not share personal hygiene products
- Avoid sex until hepatitis antibodies are negative
- Educate on toxic substances to avoid
- Alcohol, acetaminophen, aspirin, sedatives

LABS:

All will be elevated in Hepatitis

Liver enzymes:
ALT: 0 - 48 U/L
AST: 0 - 35 U/L

Bilirubin: 0.2 - 1.2 mg/dL

Ammonia: 10 - 80 mcg/dL

CIRRHOSIS

FUNCTIONS of a healthy Liver

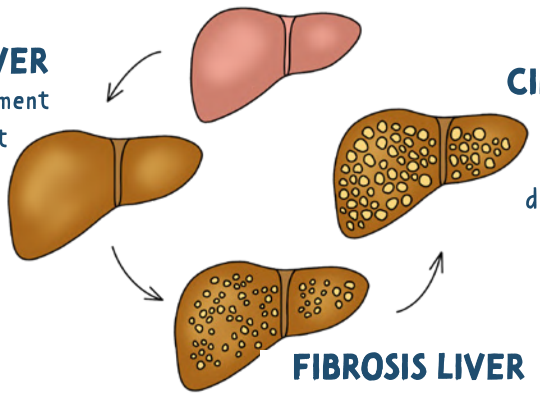
1. **DETOX** the body
2. Helps to **CLOT** the blood
3. Helps to **METABOLIZE** (break down) drugs
4. **SYNTHESIS** (makes) **ALBUMIN**

If the function of the liver is disrupted, then all these functions are not working properly

STAGES OF LIVER DAMAGE

HEALTHY LIVER

FATTY LIVER
liver enlargement due to fat deposits



CIRRHOSIS LIVER
liver cell destruction

FIBROSIS LIVER
healthy liver tissue is replaced with scar tissue

PATHOLOGY

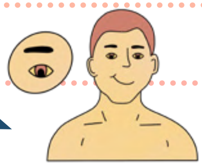
- Liver cells are **DESTROYED** and replaced with fibrotic (scar) tissue.
- Loss of normal function of the liver.

CAUSES

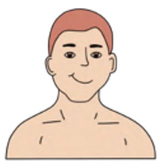
- **Alcoholic cirrhosis** MOST COMMON
  Caused by excessive alcohol intake
- Nonalcoholic fatty liver disease (NAFLD)
  - Viral hepatitis B & C
  - Autoimmune
  - Hepatotoxic drugs
  - Toxins & parasites
  - Fat collection in the liver (obesity, diabetes, ↑ cholesterol)

SIGNS & SYMPTOMS

- Asterixis
  - *Liver flap*
- Ascites
- Edema
- Abdominal pain
- Chronic dyspepsia (GI upset)
- Itchy skin
- **Jaundice**
  - *Yellow discoloration in the eyes & skin*
- ↑ Bilirubin & ammonia
- ↓ Platelets
  - *Risk for bleeding*
- ↓ WBC's
  - *Risk for infection*



Jaundice



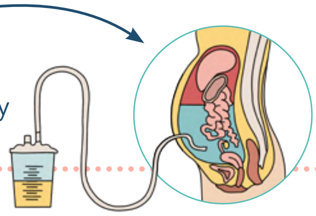
Normal

COMPLICATIONS

- Portal HTN
  - Portal veins become narrow due to scar tissue
- GI bleeding (esophageal varices)
- Splenomegaly
- Anemia
- Hepatic encephalopathy/coma
  - Due to ↑ ammonia levels (ammonia is a sedative)
- Gynecomastia
  - Breast development in men
- Hepatorenal syndrome
  - Acute kidney injury *in clients with liver failure*

TREATMENT

- Stop alcohol consumption
- Rest
- Measure abdominal girth
- Paracentesis
  - Removal of fluid from the peritoneal cavity (ascites)
- Daily weights & I&O's
- Liver transplant
- Prevent **BLEEDING**



REMEMBER: the liver normally helps clot the blood

BLEEDING PRECAUTIONS
- Use electric razor
- Use soft-bristled toothbrush
- Hold pressure on scrapes/cuts to minimize bleeding

MEDICATIONS

- Antacids
- Diuretics
- Vitamins
- Lactulose
  - ↓ serum ammonia through the stool

AVOID NARCOTICS The liver can't metabolize drugs when it's sick

MEMORY TRICK: Lactulose think LactuLOOSE because it LOOSENS the bowels

⚠ Do not give acetaminophen to people with liver issues!

NEUROLOGICAL ASSESSMENTS

LEVEL OF CONSCIOUSNESS (LOC)

LEVEL OF CONSCIOUSNESS (LOC)
is always **#1** with neurological assessment

A change in LOC may be the only sign that there is a PROBLEM!

PUPILLARY CHANGES

PERRLA

Pupils, **E**qual, **R**ound, **R**eactive to **L**ight & **A**ccommodation

NORMAL PUPIL SIZE: 2 - 6 mm

MENTAL STATUS

- Are they aware of their surroundings?
- Are they oriented to person, place, time, & situation?
- Do they have their short term & long term memory?

ASK these types of questions to assess mental status:

- What is your name?
- Do you know where you are?
- Do you know what month it is?
- Who is the current U.S. president?
- What are you doing here?

GLASGOW COMA SCALE

TOOL FOR ASSESSING A CLIENT'S RESPONSE TO STIMULI

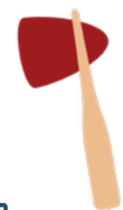


EYE OPENING RESPONSE	Spontaneous	4
	To speech	3
	To pain	2
	No response	1
VERBAL RESPONSE	Oriented	5
	Confused	4
	Inappropriate words	3
	Unclear sounds	2
	No response	1
MOTOR RESPONSE	Obeys command	6
	Moves to localized pain	5
	Flex to withdraw from pain	4
	Abnormal flexion	3
	Abnormal extension	2
	No response	1
TOTAL	**3 - 15**	

INTERPRETATION

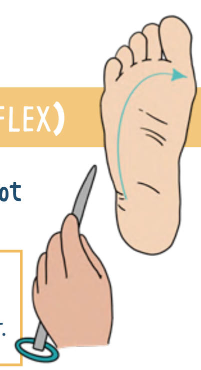
- WORST **3** Severe impairment of neurological function, coma, or brain death
- **<8** Unconscious patient
- BEST **15** Fully alert & oriented

DEEP TENDON REFLEX (DTR) RESPONSES



- **0** = No response **ABSENT**
- **1+** = Present, but sluggish or diminished
- **2+** = Active or expected response **NORMAL**
- **3+** = More brisk than excited; hyperactive
- **4+** = Brisk, hyperactive, with intermittent, or transient clonus

BABINSKI REFLEX (PLANTAR REFLEX)



Elicited by stroking the lateral side of the foot

INTACT CNS

The lateral sole of the foot is stroked and the toes contract & draw together.

BRAIN DYSFUNCTION

Toes fan out when stroked.

Remember this is **only normal in newborns & infants up to 2 years of age**, but **abnormal in adults!**

MEMORY TRICK

Babinski think normal in **B**abies & the **B**ig toe fans out

SEIZURES

WHAT IS A SEIZURE? Abnormal & sudden electrical activity of the brain

WHAT IS EPILEPSY? Chronic seizure activity due to a chronic condition

CAUSES

- ↑ fever (Febrile seizure in child)
- CNS infection
- Drug or alcohol withdrawal
- ABG imbalance
- Hypoxia
- Brain tumor
- Hypoglycemia
- Head injury
- Hypertension

STAGES OF A SEIZURE

PRODROMAL
When symptoms start before the actual seizure

(can be days before the seizure happens)

→

AURA
Warning sign right before the seizure happens:
- Weird smell or taste
- Altered vision
- Dizzy

Not all patients experience an aura

→

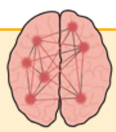
⚠ ICTUS ⚠
SEIZURE!

Status Epilepticus: a seizure that lasts >5 minutes without any consciousness during the seizure

→

POST-ICTUS
Recovery after the seizure
- Headache
- Possible injury
- Confusion
- Very tired

GENERALIZED SEIZURES



THE ENTIRE BRAIN IS AFFECTED

TONIC-CLONIC "Used to be called grand-mal" May begin with an aura. Stiffening (tonic) and/or rigidity (clonic) of the muscles.

MYOCLONIC Sudden jerking or stiffening of the extremities (arms or legs).

ABSENCE Usually looks like a blank stare that lasts seconds. Often goes unnoticed

ATONIC Sudden loss of muscle tone. May lead to sudden falls or dropping things.

PARTIAL (FOCAL) SEIZURES

ONE AREA OF THE BRAIN IS AFFECTED

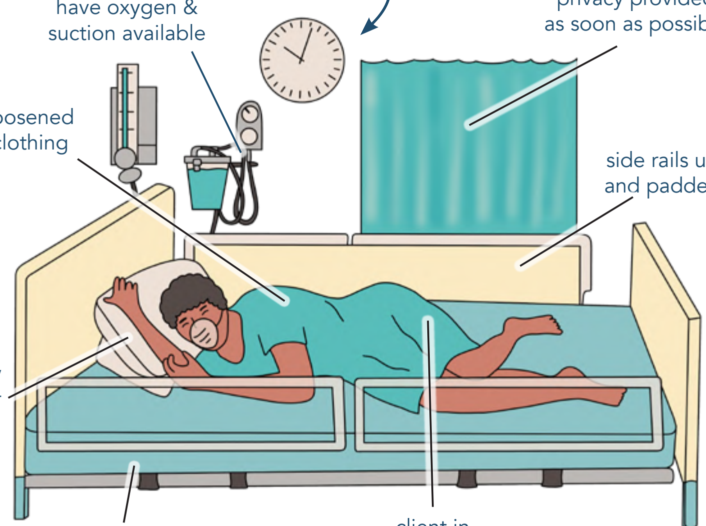
SIMPLE PARTIAL Sensory symptoms with motor symptoms and stays aware. They may report an aura.

COMPLEX PARTIAL Altered behavior/awareness and loses consciousness for a few seconds.

CARE DURING THE SEIZURE

SEIZURE PRECAUTIONS

Remember: if the seizure lasts > 5 minutes is status epilepticus. This needs IMMEDIATE attention



DON'T
- Restrain the client
- Force the jaw open
- Place anything in their mouths
- Leave the client

CEREBROVASCULAR ACCIDENT (CVA) "STROKE"

PATHOLOGY

Sudden interruption of blood supply to the brain.

The pathology of a stroke depends on the type of stroke.

RISK FACTORS

MODIFIABLE

- ★ Hypertension
- Atherosclerosis
- Anticoagulation therapy
- Diabetes mellitus
- Obesity
- Stress
- Oral contraceptives

NON-MODIFIABLE

- Family history of strokes
- Older age
- Male gender
- Black
- Hispanic

SIGNS & SYMPTOMS

MNEMONIC

- **F** **Face drooping**
  - Uneven smile
- **A** **Arm weakness**
  - Arm numbness; can't lift arm
- **S** **Speech difficulty**
  - Slurred speech
- **T** **Time to call 911**

RIGHT BRAIN

- Behavioral changes
- Lack of impulse control
- LEFT-sided hemiparesis (1-sided weakness)

MEMORY TRICK: Right think Reckless

Remember: If the stroke occurs in the left side of the brain, the right side of the body will be affected

LEFT BRAIN

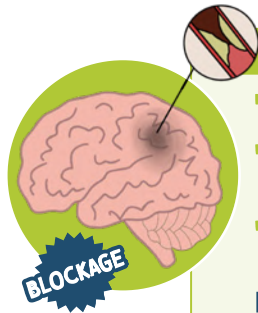
- Issues with language (aphasia)
- RIGHT-sided hemiparesis (1-sided weakness)

MEMORY TRICK: Left think Languages

TYPES OF APHASIA:

RECEPTIVE: Unable to comprehend speech **(Wernicke's area)**

EXPRESSIVE: Can comprehend speech, but can't respond back with speech **(Broca's area)**



ISCHEMIC STROKE

"THROMBOTIC OR EMBOLIC"

- Thrombosis:
  A blood clot that formed on the artery wall
- Embolism:
  A blood clot that has left part of the body

Blood flow is cut off which leads to **ischemia**

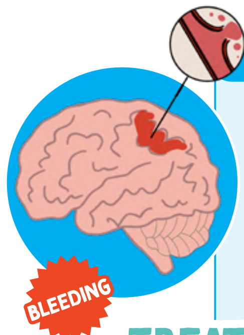
TRANSIENT ISCHEMIC ATTACKS (TIAs)

"Mini strokes" • No cerebral infarction occurs

MEDICATIONS:

- **Fibrinolytic therapy** ("clot buster")
  Suffix: **-ase**
  Examples: alteplase, streptokinase

Must be given within 4.5 hours from onset of symptoms



HEMORRHAGIC STROKE

- Ruptured artery
- Aneurysm (weakening of the vessel)
- Uncontrolled hypertension

The collection of blood in the brain leads to **ischemia & increased ICP**

TREATMENT:

- **Stop the bleeding**
- **Prevent ↑ ICP**

NURSING CONSIDERATIONS

POSITIONING OF THE CLIENT

- Elevate head of bed to ↓ ICP
- Place a pillow under the affected arm in a neutral position

PREVENTATIVE DVT MEASURES

- Compression stockings
- Frequent position change
- Mobilization
- Encourage passive range of motion every 2 hours

ASSIST WITH COMMUNICATION SKILLS

- Be patient
- Make clear statements
- Ask simple questions
- Don't rush!

ASSIST WITH SAFE FEEDING

- Do not feed until gag reflex has come back
- ↓ chances of aspiration
- Keep suction at the bedside
- Crush medications

DIET MODIFICATIONS

- After a stroke, a patient will start on a liquid diet and progress slowly to a regular diet.

LIQUID

- Thin
- Nectar-like
- Honey-like
- Spoon-thick

FOOD

- Pureed
- Mechanically altered
- Mechanically softened
- Regular

CRANIAL NERVES

WHAT ARE CRANIAL NERVES?

Nerves that originate from the brain stem.
They send information to & from various parts of the body.

SE SENSORY
M MOTOR
B BOTH

MNEMONICS

Ooh,	Olfactory	Some	Sensory
Ooh,	Optic	Say	Sensory
Ooh	Oculomotor	Marry	Motor
To	Trochlear	Money	Motor
Touch	Trigeminal	But	Both
And	Abducens	My	Motor
Feel	Facial	Brother	Both
Very	Vestibulocochlear / Acoustic	Says	Sensory
Good	Glossopharyngeal	Big	Both
Velvet.	Vagus	Brains	Both
Such	Spinal Accessory	Matter	Motor
Heaven!	Hypoglossal	More	Motor

START

I: OLFACTORY — SE

FUNCTION:
Sense of smell

TEST:
Smell substance with eyes closed (test each nostril separately)

II: OPTIC — SE

FUNCTION:
Vision

TEST:
- Snellen chart
- Ophthalmoscopic exam
- Confrontation to check peripheral vision

III: OCULOMOTOR — M

FUNCTION:
Ocular (eye) motor (movement)
Controls most eye movements, pupil constriction, & upper-eyelid rise

TEST:
- Look up, down, & inward
- Ask the client to follow your finger as you move it towards their face

IV: TROCHLEAR — M

FUNCTION:
Controls downward & inward eye movement

TEST:
- Look up, down, & inward
- Ask the client to follow your finger as you move it towards their face

V: TRIGEMINAL — B

FUNCTION:
MOTOR - Mastication (biting & chewing)
SENSORY - Facial sensation

TEST:
- Pressure on the forehead cheek & jaw with a cotton swab to check sensation
- Ask client to open mouth & then bite down

VI: ABDUCENS — M

FUNCTION:
Controls parallel eye movement
Abduction - moving laterally
AKA away from midline

TEST:
- Look up, down, & inward
- Ask the client to follow your finger as you move it towards their face

VII: FACIAL — B

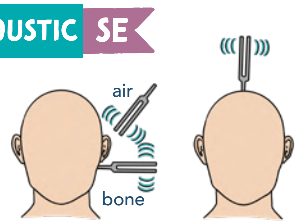
FUNCTION:
MOTOR - Facial expression
SENSORY - Taste (sweet & salty)

TEST:
- Ask client to do different facial expression
  (Frown, smile, raise eyebrows, close eyes, blow etc)
- Test tongue by giving client sour, sweet, bitter, and salty substances.

VIII: VESTIBULOCOCHLEAR / ACOUSTIC — SE

FUNCTION:
Balance & hearing

TEST:
- Stand with eyes closed
- Otoscopic exam
- Rinne & Weber Tests



IX: GLOSSOPHARYNGEAL — B

GLOSSO MEANS TONGUE!

FUNCTION:
MOTOR - Tongue movement & swallowing
SENSORY - Taste (sour & bitter)

TEST:
Test tongue by giving client sour, bitter, & salty substance.

X: VAGUS — B

FUNCTION:
MOTOR - Swallowing, speaking, & cough
SENSORY - Facial sensation

TEST:
Sensation coming from skin around the ear

XI: SPINAL ACCESSORY — M

FUNCTION:
Controls strength of neck & shoulder muscles

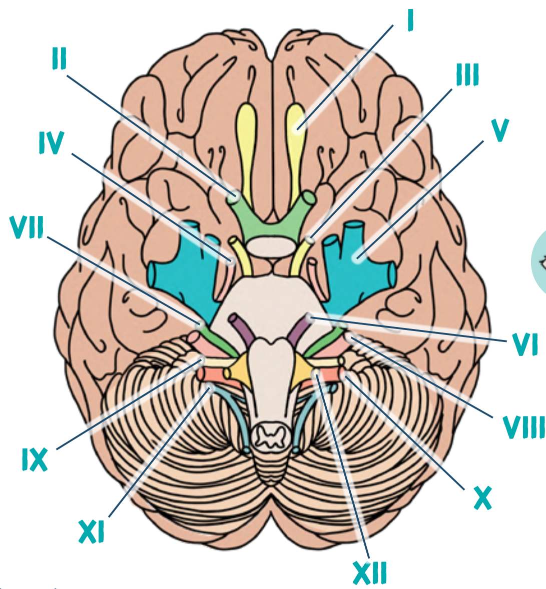
TEST:
Ask the client to rotate their head & shrug their shoulders

XII: HYPOGLOSSAL — M

GLOSSO MEANS TONGUE!

FUNCTION:
Tongue movement (swallowing & speech)

TEST:
Inspect tongue & ask to stick tongue out



CRANIAL NERVES

WHAT ARE CRANIAL NERVES?

Nerves that originate from the brain stem.
They send information to & from various parts of the body.

LABEL THE FLAGS:

- SE — SENSORY
- M — MOTOR
- B — BOTH

MNEMONICS

Ooh,	O_______	Some	S_______
Ooh,	O_______	Say	S_______
Ooh	O_______	Marry	M_______
To	T_______	Money	M_______
Touch	T_______	But	B_______
And	A_______	My	M_______
Feel	F_______	Brother	B_______
Very	V___________ / A_______	Says	S_______
Good	G___________	Big	B_______
Velvet.	V_______	Brains	B_______
Such	S_______	Matter	M_______
Heaven!	H_______	More	M_______

START

I:

FUNCTION:

TEST:

II:

FUNCTION:

TEST:

- _______ chart

E
F P
T O Z
L P E D
P E C F D

III:

FUNCTION:

TEST:

IV:

FUNCTION:

TEST:

V:

FUNCTION:

TEST:

VI:

FUNCTION:

TEST:

VII:

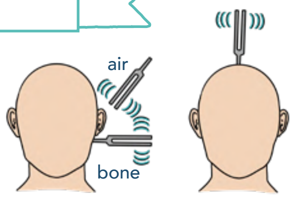
FUNCTION:

TEST:

VIII:

FUNCTION:

TEST:



_______ TEST _______ TEST

IX:

FUNCTION:

TEST:

X:

FUNCTION:

TEST:

XI:

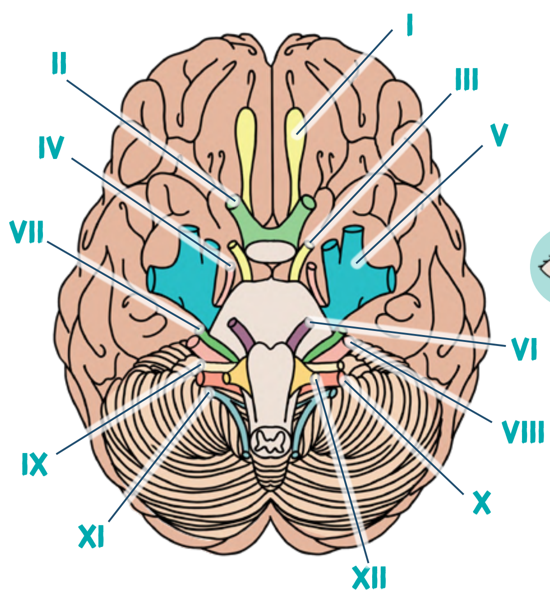
FUNCTION:

TEST:

XII:

FUNCTION:

TEST:



WANT MORE WORKSHEETS?
Check out The Complete Laminated Study Templates!

BURNS

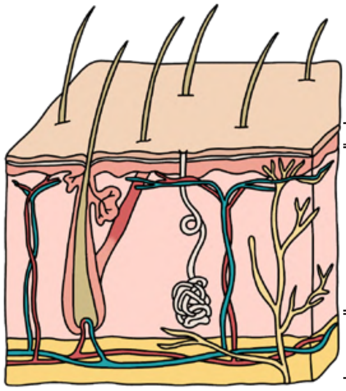
WHAT IS A BURN?

Damage to skin integrity

BURNS INJURY DEPTH

1ST DEGREE	Superficial	▪ Epidermis ▪ Pink & painful (still has nerves) ▪ No scarring	**BLANCHING**: present **HEALS**: a few days
2ND DEGREE	Superficial Partial Thickness	▪ Epidermis & dermis ▪ Blisters, shiny, & moist ▪ Painful	**BLANCHING**: present **HEALS**: 2 - 6 weeks
3RD DEGREE	Full Thickness	▪ Epidermis, dermis, & hypodermis ▪ May look black, yellow, red & wet ▪ No pain/ limited pain (nerve fibers are destroyed) ▪ Skin will not heal (need skin grafting) ▪ **Eschar**: dead tissue, leathery; must be removed!	

LAYERS OF THE SKIN



TYPES OF BURNS

MOST COMMON Thermal	Superficial heat *Examples:* *liquid, steam, fire*
Chemical	Burn caused by a toxic substance. Can be alkalotic or acidic *Examples: bleach, gasoline, paint thinner*
Radiation	Sunburns (UV radiation) & cancer treatment (radiation therapy)
Inhalation	Caused by inhaling smoke which can cause flame injury or carbon monoxide poisoning
Friction	Burn caused when an object rubs off the skin *Examples: road rash, scrapes, carpet burn*
Cold	Skin has been overexposed to cold *Examples: frostbite*
Electric	Electrical current that passes through the body, causing damage within

INHALATION INJURY

Damage to the respiratory system. Happens mostly in a **closed area**

SIGNS OF INHALATION INJURY:

- Hair singed around the face, neck, or torso
- Trouble talking
- Soot in the nose or mouth
- Confusion or anxiety

CARBON MONOXIDE (CO) POISONING

Carbon monoxide travels faster than oxygen, allowing it to bind to hgb first. Now oxygen cannot bind to hgb = **HYPOXIA**

Classic symptom: cherry red skin

Treatment: 100% O_2

POTENTIAL COMPLICATIONS

Dysrhythmias, fracture of bones. Release of **myoglobin & hemoglobin** into the blood which can clog the kidneys.

BURN LOCATION

RESPIRATORY
- Face
- Neck
- Chest
- Torso

DISABILITY
- Hands
- Feet
- Joints
- Eyes

TROUBLE HEALING
- Poor blood supply
- Diabetes
- Infection

INFECTION
Any open area where bacteria can easily enter
- Perineum
- Ears
- Eyes

COMPARTMENT SYNDROME
- In the extremities
  Tight skin such as eschar acting like a band around the skin cutting off blood circulation

PHASES OF BURN MANAGEMENT

MNEMONIC **"EAR" = EMERGENT, ACUTE, REHABILITATIVE**

EMERGENT PHASE — 24 - 48 HOURS after burn

Onset of injury to the restoration of capillary permeability

PATHO

↑ Capillary permeability (leaky vessels) causing:

- Plasma leaves the intravascular space
  - Albumin & sodium follows
- Fluids shift to the interstitial tissue

LEADS TO EDEMA

Leads to fluids volume deficit (FVD) in the intravascular space

VITAL SIGNS

↑ Pulse
↓ Blood pressure
↓ Cardiac output

Think HYPOVOLEMIC SHOCK!

↓ Urine output
(from ↓ perfusion to the kidneys)

LABS

↑ Potassium (K+)
↑ Hematocrit (HCT)
↓ White blood cells (WBCs)
↑ BUN/ceatinine

NURSING CONSIDERATIONS

Think ABCs

- Establish IV access *(preferably 2)*
- Fluids (Lactated Ringer's, crystalloids)
- Parkland formula
- Foley catheter to monitor urinary output (UOP)
  GOAL: > 30 mL/hr of UOP
- Decrease edema
  - Elevate extremities above heart level

ACUTE PHASE — 48 - 72 HOURS after burn & until wounds have healed

Capillary permeability stabilized - to wound closure

PATHO

Capillary permeability is restored which leads to the body diuresing (increased urine production). All the excess fluid that shifted from the interstitial tissue shifts back into the intravascular space.

GOALS

- **PREVENT INFECTION**
  - Systemic antibiotic therapy
- **ENSURE PROPER NUTRITION**
  - Needs ↑ calories
  - Protein & Vit C to promote healing
- **ALLEVIATE PAIN**
- **WOUND CARE**
  - Premedicate before wound care
  - Debridement or grafting

NURSING CONSIDERATIONS

- **RENAL**
  - Diuresis is happening
  - Foley catheter to monitor UOP
- **RESPIRATORY**
  - Possible intubation if respiratory complications occurred
- **GASTROINTESTINAL**
  - Since the client is in FVD, there is ↓ perfusion to the stomach
    - Paralytic ileus
    - Curlings ulcer
  - Medication to decrease chance of ulcers
    - H2 histamine blocking agent (↓HCl)
  - Monitor bowel sounds
  - May need NG tube for suctioning

REHABILITATIVE PHASE — Could be weeks - years

Burn healed and the patient is functioning mentally & physically

GOALS

- Psychosocial
- Activities of daily living (ADLs)
- Physical therapy (PT)
- Occupational theory (OT)
- Cosmetic corrections

FLUID RESUSCITATION FOR BURNS

THE PARKLAND FORMULA

Used to calculate the total volume of fluids (mL) that a patient needs **24 hours** after experiencing a burn

Apply only in 2nd & 3rd degree burns.

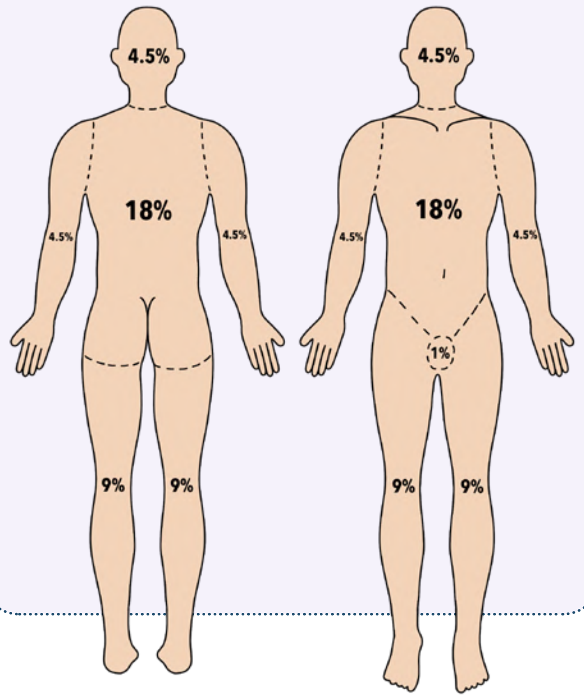
4 ML X TBSA (%) X BODY WEIGHT (KG) = TOTAL ML OF FLUID NEEDED

Give the first ½ of the solution in the **FIRST 8 HOURS**

Over the **NEXT 16 HOURS,** give the second ½ of the solution

RULE OF NINES

Quick estimate of the % of the **total body surface area (TBSA)** has been effected by a partial & full-thickness burn in an adult client.



PRACTICE QUESTION

PART 1: CALCULATING TBSA (%)

A 25 year old male patient who weighs **79 kg** has sustained burns to the back of the right arm, posterior trunk, front of the left leg, and their anterior head and neck. Using the **Rule of Nines**, calculate the total body surface area percentage that is burned.

Back of right arm - 4.5%
Posterior trunk - 18%
Front of left leg - 9%
Anterior head & neck- 4.5%

ANSWER: 36%

PART 2: THE PARKLAND FORMULA

Use the Parkland formula to calculate the total amount of Lactated Ringer's solution that will be given over the next 24 hours.

ANSWER: 11,376 mL

NOTE: The formula uses TBSA (%). However, you must calculate using 36. Not 0.36 (also written as 36%).

4 mL X 36% X 79 kg = 11,376 mL

11,376 / 2 = 5,688 mL
FIRST 8 HOURS

11,376 / 2 = 5,688 mL
NEXT 16 HOURS

Keep in mind: the question could ask you for mL given in the first 24 hours, the first 8 hours, etc., so read the question carefully.

SHOCK

WHAT IS SHOCK?

A life-threatening condition resulting from **INADEQUATE TISSUE PERFUSION.** This leads to possible cell dysfunction, cell death, and even organ failure.

HYPOVOLEMIC SHOCK

MOST COMMON TYPE OF SHOCK

ETIOLOGY

HYPOVOLEMIC

"LOW" "VOLUME" "IN THE BLOOD"

Decreased intravascular volume

CAUSES

NON-HEMORRHAGIC (not from bleeding)

- **FLUID SHIFT** (edema or ascites)
- **SEVERE DEHYDRATION** (vomiting, diarrhea, burns)

HEMORRHAGIC (from bleeding)

- **TRAUMA**
- **GI BLEED**
- **POSTPARTUM**

SIGNS & SYMPTOMS

Pulse	CO	HR	BP
Weak, thready pulse	↓ Not a lot of blood being pumped by the heart	↑ **Tachycardia** Compensating to increase blood flow	↓ **Hypotension**

Skin	CVP	SVR	O_2 Sat
Cyanosis *(Bluish tint of the lips, tongue, and fingertips)* Cool, pale skin ↓ capillary refill (>3 seconds)	↓	↑ **Vasoconstriction**	↓ ↓ blood being perfused to the body = low O_2

Other Signs & Symptoms

LABS CAN BE:

- **↑ HCT** hemoconcentration
- **↓ HCT** actually hemorrhaging the RBCs
- Oliguria (urine output of <30 mL/hr)
- Confused, agitated *due to decreased blood flow to the brain*

TREATMENT

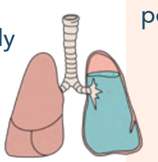
- → **Large gauge IVs** *(at least 2)*
- → **Fluids & blood replacement**
  - Crystalloids *(example: normal saline or Lactated Ringers)*
  - Colloids *(albumin)*
  - Blood products *(plasma, PRBCs, & PLTs)*

CARDIOGENIC SHOCK

ETIOLOGY

The heart can't pump enough blood to meet the perfusion needs of the body

NOTE: *There is enough blood,* the heart just can't pump it to the body which causes **fluid accumulation** in the lungs!



CAUSES

- Damage from an acute MI
- Severe hypoxemia
- Acidosis
- Hypoglycemia
- Cardiomyopathy
- Cardiac tamponade
- Dysrhythmias

SIGNS & SYMPTOMS

Pulse	CO	HR	BP
Weak peripheral pulses	↓ Not a lot of blood being pumped by the heart	↑ **Tachycardia** Compensating to increase blood flow	↓ **Hypotension**

Skin	CVP	SVR	O_2 Sat
Cool, clammy skin ↓ capillary refill (>3 seconds)	↑	↑ **Vasoconstriction**	↓ ↓ blood being perfused to the body = low O_2

Other Signs & Symptoms

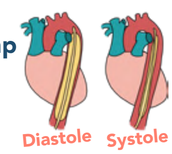
- Jugular vein distention (JVD)
- Chest pain
- Oliguria (urine output of <30 mL/hr)
- Confused, agitation *due to decreased blood flow to the brain*

From fluid accumulation in the lungs:

- Dyspnea
- Pulmonary edema

TREATMENT

- → **For an MI:** Angioplasty, Thrombolytics
- → **Oxygen**
- → **Vasopressors** *(example: epinephrine, dobutamine, dopamine)* — Vasopressors cause vasoconstriction which ↑ blood flow and increases perfusion to the organs
- → **Diuretics**
  - ↓ the workload of the heart
  - ↓ extra blood volume
- → **Intra-aortic balloon pump** *(helps to improve coronary artery blood flow & ↑ CO)*



BP = Blood pressure **HR** = Heart rate **CO** = Cardiac output **SVR** = Systemic vascular resistance **CVP** = Central venous pressure

DISTRIBUTIVE SHOCK
(Septic, Neurogenic, Anaphylactic)

DISTRIBUTIVE: Leaky blood vessels

Excessive **vasodilation** (widening of vessels) → Intravascular volume pools in the peripheral blood vessels → Since the blood is in the peripherals, it is NOT perfusing the vital organs which causes relative hypovolemia

SEPTIC SHOCK (SEPSIS)

MOST COMMON TYPE OF DISTRIBUTIVE SHOCK

ETIOLOGY

Caused by widespread infection or sepsis

CAUSES

- Pneumonia
- Urosepsis
- Bacteria
- Intra-abdominal infections
- Wound infection
- Invasive procedures
- Indwelling medical devices (catheters)

TREATMENT

CORRECT THE UNDERLYING CAUSE

- → Fluid replacement
- → **Broad-spectrum antibiotics**
- → Vasopressors (norepinephrine & dopamine)
- → Neuromuscular blockade agents & sedation
  - ↓ metabolic demands & provides comfort
- → Medications to prevent stress ulcers
  - H2 blocking agents
  - Proton pump inhibitors (PPIs)

Broad spectrum antibiotics are used when the organism is not yet known/determined. Once the organism is known, the client can be put on more specific antibiotics.

SIGNS & SYMPTOMS

Pulse	CO	HR	BP
Bounding pulses	↑	↑ Tachycardia	↓ Hypotension

Skin	CVP	SVR	O_2 Sat
Initially warm & flushed, but as the BP drops, the skin becomes cool, pale & mottled	↓	↓ Vasodilation	↑

Other Signs & Symptoms

- → Hyperthermia & fever
- → Increased respiratory rate
- → GI upset: Nausea, vomiting, diarrhea, decrease gastric motility
- → ↑ Inflammatory markers
  - ↑ WBCs
  - ↑ C-reactive protein (CRP)

NEUROGENIC SHOCK (VASOGENIC)

ETIOLOGY

Vasodilation due to a loss of balance between

In neurogenic shock, the client mainly experiences parasympathetic stimulation which causes VASODILATION for an extended period

MEMORY TRICK: P think Peaceful

PARASYMPATHETIC STIMULATION (Rest & digest) → Causes dilation (relaxing) of the smooth muscles

SYMPATHETIC STIMULATION (Fight or flight) → Causes constriction (tightening) of the smooth muscles

CAUSES

- Spinal cord injury (above T6, cervical)
- Spinal anesthesia
- Nervous system damage
- Insulin reaction

MEMORY TRICK: NEURogenic = Issue with NERvous system

SIGNS & SYMPTOMS

EVERYTHING IS DECREASED ↓

Remember parasympathetic means **RELAXED EVERYTHING**

CO	HR	BP
↓	↓	↓ Hypotension

the sympathetic NS is not working to compensate & ↑ the HR

RELATIVE HYPOVOLEMIA: There is enough blood volume. However, the vascular space is **dilated,** so blood volume is displaced causing hypovolemia.

Skin	CVP	SVR	O_2 Sat
Dry, warm extremities (venous blood pooling) Hypothermia: warm/dry extremities, cold body	↓	↓ Vasodilation	↓

TREATMENT

DEPENDS ON THE CAUSE OF THE SHOCK

- → Spinal cord injury
- → Assess & manage airway
  May need intubation or mechanical ventilation
- → Elevate the head of the bed
- → IV fluids ⚠ Watch for fluid volume overload
- → Increased risk for clots due to pooling of blood
  - Watch for signs of a clot
  - Compression devices
  - Antithrombotic agents (heparin)
- → Vasopressors (*example*: epinephrine, dobutamine, dopamine)

PROTECT THE SPINE: Keep spine immobilized (cervical collar, backboards, log-rolling)

S&S OF BLOOD CLOTS:
- Pain in the extremities
- Redness
- Tenderness
- Warmth

BP = Blood pressure **HR** = Heart rate **CO** = Cardiac output **SVR** = Systemic vascular resistance **CVP** = Central venous pressure

DISTRIBUTIVE SHOCK
(Septic, Neurogenic, Anaphylactic)

DISTRIBUTIVE: Leaky blood vessels / Excessive **vasodilation** (widening of vessels) → Intravascular volume pools in the peripheral blood vessels → Since the blood is in the peripherals, it is NOT perfusing the vital organs which causes relative hypovolemia

ANAPHYLACTIC SHOCK

ETIOLOGY

SEVERE ALLERGIC REACTION

Foreign substance (antigen)
↓
Systemic antigen-antibody reaction (IgE)
↓
Mast cells release potent vasoactive substance (histamine/bradykinin)
↓
Activates inflammatory cytokines
↓
Causes **VASODILATION** & capillary permeability

CAUSES/TRIGGERS

Often unknown (idiopathic)

- Foods (example: peanuts)
- Medications
- Insects (example: bee sting)
- Latex
- Exercise-induced anaphylaxis (EIA)

Signs & symptoms usually occur within 2 - 30 minutes of exposure to antigen

TREATMENT

REMOVE THE ALLERGEN!

- → High-flow oxygen
- → First-line drug: **EPINEPHRINE** ★
  - Causes vasoconstriction & bronchodilation
- → Other possible medications
  - Antihistamines
    - Diphenhydramine (Benadryl)
  - Albuterol (Proventil)
  - Corticosteroids
- → Fluids
- → Stay with the client & monitor

BIPHASIC ANAPHYLAXIS: A recurrence of anaphylaxis after appropriate treatment

SIGNS & SYMPTOMS

Pulse	CO	HR	BP
Rapid, weak pulse	↓	↑ Tachycardia	↓ Hypotension

CAPILLARY PERMEABILITY: Fluid is leaving the intravascular space

Skin	CVP	SVR	O_2 Sat
Generalized flushing	↓	↓ Vasodilation	↓

Other Signs & Symptoms

CARDIAC
- Cardiac dysrhythmias or cardiac arrest

GI
- Nausea/vomiting
- Acute abdominal pain

FEELING OF IMPENDING DOOM

RESPIRATORY
- Bronchoconstriction
  - Difficulty breathing
  - Wheezing
  - Coughing
  - Unable to speak

SKIN
- Itching, generalized flushing, redness, hives, or a rash may be present

HOW TO USE AN EPINEPHRINE AUTO-INJECTOR (EAI)

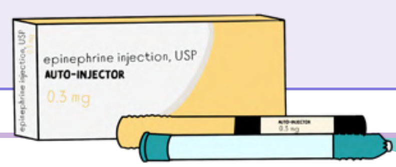
EDUCATION POINTS:

- → Store in dark room
- → Administer EAI immediately after the first sign of an allergic reaction

INJECT IN THE OUTER THIGH AT A 90° ANGLE

EXPECTED SYMPTOMS AFTER ADMINISTRATION:

- → Tachycardia
- → Palpitations
- → Dizziness



BP = Blood pressure **HR** = Heart rate **CO** = Cardiac output **SVR** = Systemic vascular resistance **CVP** = Central venous pressure

ABGs

ABG
ARTERIAL BLOOD GAS

ABGs MEASURE HOW ACIDIC OR ALKALOTIC THE BLOOD IS IN THE ARTERIAL CIRCULATION.

*also a measure of gases such as O_2 & Co_2

4 MUST-KNOW COMPONENTS

PH	Measurement of how acidic or alkalotic your blood is	regulated by both lungs & kidneys	7.35 - 7.45
$PACO_2$	Measurement of carbon dioxide in the blood CO_2 think aCid	Regulated by the lungs	35 - 45
HCO_3	Measurement of bicarbonate in the blood Bicarbonate think Base	Regulated by the kidneys	22 - 26
PAO_2	Measurement of oxygen in the blood	Regulated by the lungs	80 - 100

Value not needed to interpret alkalosis or acidosis. It just tells you if the patient is hypoxic or not.

ABG INTERPRETATION

1 KNOW YOUR LAB VALUES!

	ACIDOSIS	NORMAL	ALKALOSIS
PH	< 7.35 ↓	7.35 - 7.45	> 7.45 ↑
CO_2	> 45 ↑	35 - 45	< 35 ↓
HCO_3	< 22 ↓	22 - 26	> 26 ↑

2 RESPIRATORY OR A METABOLIC PROBLEM?

there are 2 ways to analyze the information

ROME METHOD

Respiratory	PH ↑	CO_2 ↓	Alkalosis
Opposite	PH ↓	CO_2 ↑	Acidosis
Metabolic	PH ↑	HCO_3 ↑	Alkalosis
Equal	PH ↓	HCO_3 ↓	Acidosis

TIC-TAC-TOE METHOD

ACID	NORMAL	BASE

3 UNCOMPENSATED, PARTIALLY COMPENSATED, OR FULLY COMPENSATED?

If the **PH** is out of range & **CO_2** or **HCO_3** is in range = **UNCOMPENSATED**

If **CO_2, HCO_3 & PH** are **ALL** out of range = **PARTIALLY COMPENSATED**

If **PH** is in range (7.35 - 7.45) = **FULLY COMPENSATED**

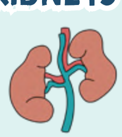
7.35 — 7.40 — 7.45

Acidosis ← Absolute Normal → Alkalosis

PH IN RANGE? Just because the PH is "normal", it can still fall on a acidotic side or alkalotic side

How do the organs Compensate?

KIDNEYS



BICARB
HYDROGEN

MEMORY TRICK: B think BASE

Excreting excess **ACID** & **BICARB (HCO_3)**

OR

Retaining **HYDROGEN** & **BICARB (HCO_3)**

hours - days to compensate

LUNGS

CO_2

MEMORY TRICK: CO_2 think ACID

HYPERventilation = ↓ CO_2 = **ALKALOSIS**

HYPOventilation = ↑ CO_2 = **ACIDOSIS**

compensates FAST!

ABG PRACTICE QUESTION EXAMPLE

QUESTION

A client with a bowel obstruction has been treated with gastric suctioning for 4 days. The nurse notices an increase in nasogastric drainage. Which Acid-base imbalance does that nurse correctly identify?

The patient labs are the following →

Ph **7.50**
$PaCO_2$ **50** mm Hg
PaO_2 **90** mm Hg
HCO_3 **32** mEq/L

Value not needed to interpret alkalosis or acidosis. It just tells you if the patient is hypoxic or not.

TIC-TAC-TOE METHOD

1 What does the problem give you?

PH	7.50	○ ACIDIC	✓ ALKALOTIC	○ NORMAL
CO_2	50	✓ ACIDIC	○ ALKALOTIC	○ NORMAL
HCO_3	32	○ ACIDIC	✓ ALKALOTIC	○ NORMAL

2

ACID	NORMAL	BASE
CO_2		PH
		HCO_3

- ○ RESPIRATORY ACIDOSIS
- ○ RESPIRATORY ALKALOSIS
- ○ METABOLIC ACIDOSIS
- ✓ METABOLIC ALKALOSIS

3 UNCOMPENSATED, PARTIALLY COMPENSATED, or FULLY COMPENSATED?

Is the pH in range?	○ YES	✓ NO
Is the CO_2 in range?	○ YES	✓ NO
Is the HCO_3 in range?	○ YES	✓ NO

- ○ UNCOMPENSATED
- ✓ PARTIALLY COMPENSATED
- ○ FULLY COMPENSATED

If CO_2, HCO_3 & PH are ALL out of range

FINAL ANSWER:

METABOLIC ALKALOSIS, PARTIALLY COMPENSATED

ROME METHOD

1 What does the problem give you?

PH	7.50	○ ACIDIC	✓ ALKALOTIC	○ NORMAL
CO_2	50	✓ ACIDIC	○ ALKALOTIC	○ NORMAL
HCO_3	32	○ ACIDIC	✓ ALKALOTIC	○ NORMAL

2 Which of the four scenarios from the ROME method matches the information given in your problem?

Respiratory	PH ↑	CO_2 ↓	Alkalosis
Opposite	PH ↓	CO_2 ↑	Acidosis
Metabolic	PH ↑	HCO_3 ↑	Alkalosis
Equal	PH ↓	HCO_3 ↓	Acidosis

- ○ RESPIRATORY ACIDOSIS
- ○ RESPIRATORY ALKALOSIS
- ○ METABOLIC ACIDOSIS
- ✓ METABOLIC ALKALOSIS

3 UNCOMPENSATED, PARTIALLY COMPENSATED, or FULLY COMPENSATED?

Is the pH in range?	○ YES	✓ NO
Is the CO_2 in range?	○ YES	✓ NO
Is the HCO_3 in range?	○ YES	✓ NO

- ○ UNCOMPENSATED
- ✓ PARTIALLY COMPENSATED
- ○ FULLY COMPENSATED

If CO_2, HCO_3 & PH are ALL out of range

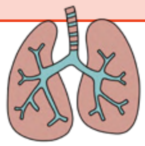
FINAL ANSWER:

METABOLIC ALKALOSIS, PARTIALLY COMPENSATED

RESPIRATORY ACIDOSIS VS. RESPIRATORY ALKALOSIS

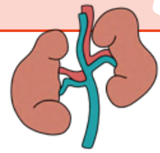
RESPIRATORY ACIDOSIS

PATHOPHYSIOLOGY



LUNG PROBLEM

The lungs are **RETAINING** too much **CO_2**



KIDNEYS COMPENSATE

The kidneys excrete excess **HYDROGEN** & retain **BICARB** (HCO_3)

PH	CO_2
< 7.35	> 45

CAUSES

RETAINING CO_2: "Depress" breathing

MNEMONIC

- **D**rugs (opioids & sedatives)
- **E**dema (fluid in the lungs)
- **P**neumonia (excess mucus in the lungs)
- **R**espiratory center of the brain is damaged
- **E**mboli (pulmonary emboli)
- **S**pasms of the bronchial (asthma)
- **S**ac elasticity damage (COPD & emphysema)

All these things cause impaired gas exchange

SIGNS & SYMPTOMS

- ↑ Blood pressure
- ↑ Respiration rate
- ↑ Heart rate
- Restlessness
- Confusion
- Headache
- Sleepy / coma

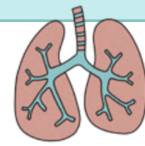
INTERVENTIONS

- Administer O_2
- Semi-Fowler's position
- Turn, cough, & deep-breathe (TCDB)
- Pneumonia: ↑ fluids to thin secretions & administer antibiotics
- If CO_2 >50, they may need an endotracheal tube
- Monitor potassium levels

NORMAL K+
3.5 - 5.0 mmol/L

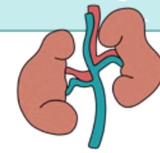
RESPIRATORY ALKALOSIS

PATHOPHYSIOLOGY



LUNG PROBLEM

The lungs are **LOSING** too much **CO_2**



KIDNEYS COMPENSATE

The kidneys excrete excess **BICARB** (HCO_3) & retain **HYDROGEN**

PH	CO_2
> 7.45	< 35

CAUSES

LOSING CO_2: "Tachypnea"

↑ Temperature

Aspirin toxicity

Hyperventilation

SIGNS & SYMPTOMS

- ↑ Heart rate
- Confused & tired
- Tetany
- EKG changes
- (+) Chvostek's sign

Twitching of the facial muscles when tapping the facial nerve in response to **HYPOCALCEMIA**

INTERVENTIONS

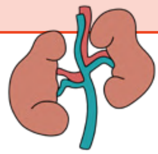
- Provide emotional support
- Fix the breathing problem!
- Encourage good breathing patterns
- Rebreathing into a paper bag
- Give anti-anxiety medications or sedatives to ↓ breathing rate
- Monitor K+ & Ca- levels

NORMAL CA-
9 - 11 mg/dL

METABOLIC ACIDOSIS VS. METABOLIC ALKALOSIS

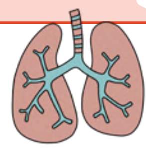
METABOLIC ACIDOSIS

PATHOPHYSIOLOGY



KIDNEY PROBLEM

Too much **HYDROGEN**
Too little **BICARB** (HCO_3)



LUNGS COMPENSATE

The lungs will blow off CO_2

PH	HCO_3
< 7.35	< 22

CAUSES

- Diabetic ketoacidosis → Not enough insulin = ↑ fat metabolism = excess ketones (acid)
- Acute/chronic kidney injury
- Malnutrition → Breaking down of fats = excess ketones (acid)
- Severe diarrhea → MEMORY TRICK: Remember Bicarb comes out of your Base

SIGNS & SYMPTOMS

- ↑ Respiratory rate → **KUSSMAUL'S BREATHING** Deep rapid breathing >20 breaths per minute
- Hyperkalemia
  - Muscle twitching
  - Weakness
  - Arrhythmias
- ↓ Blood pressure
- Confusion

Metabolic **ACIDOSIS** = ↑ serum potassium
Metabolic **ALKALOSIS** = ↓ serum potassium

INTERVENTIONS

NORMAL K+
3.5 - 5.0 mmol/L

- Monitor intake & output
- Administer IV solution of sodium bicarb to ↑ bases & ↓ acids
- Initiate seizure precaution
- Monitor K+ levels

DIABETIC KETOACIDOSIS (DKA)

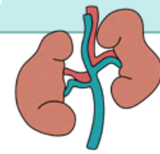
- **Give insulin** (this stops the breakdown of fats which stops ketones from being produced)
- **Monitor for hypovolemia** due to polyuria

KIDNEY DISEASE

- Dialysis to remove toxins
- Diet
  - ↑ Calories
  - ↓ Protein

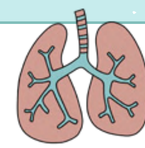
METABOLIC ALKALOSIS

PATHOPHYSIOLOGY



KIDNEY PROBLEM

Too much **BICARB** (HCO_3)
Too little **HYDROGEN**



LUNGS COMPENSATE

The lungs will retain CO_2

PH	HCO_3
> 7.45	> 26

CAUSES

- Too many antacids → Too much sodium bicarbonate (BASE)
- Diuretics
- Excess vomiting → Excess loss of hydrochloric acid (HCL) from the stomach
- Hyperaldosteronism

SIGNS & SYMPTOMS

- ↓ Respiratory rate → **HYPOVENTILATION** <12 breaths per minute
- ↓ Potassium (K+)
  - Dysrhythmias
  - Muscle cramps/weakness
  - Vomiting
  - Tetany
  - Tremors
  - EKG changes

INTERVENTIONS

NORMAL CA-
9 - 11 mg/dL

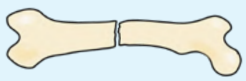
- Monitor K+ and Ca- levels
- Administer IV fluids to help the kidneys get rid of bicarbonate
- Replace K+
- Give antiemetics for vomiting (Zofran or Phenergan)
- Watch for signs of respiratory distress

FRACTURES

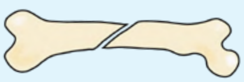
WHAT IS A FRACTURE? A fracture is a complete or incomplete disturbance in the progression of bone structure

TYPES OF FRACTURES

COMMINUTED
The bone is crushed causing lots of little fragments



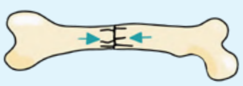
TRANSVERSE
The bone is fractured straight across



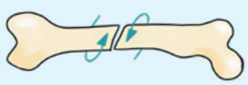
OBLIQUE
The fracture runs at an angle across the bone

commonly seen in children

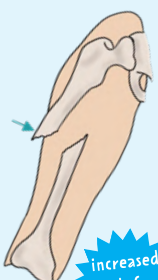
GREENSTICK
One side of the bone is bent, the other is broken



IMPACTED
The fractured bone is driven into another bone



SPIRAL
The break partially encircles the bone

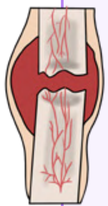


OPEN/COMPOUND
A fracture where the bone breaks through the skin

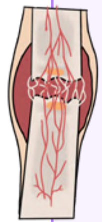
increased risk for INFECTION

STAGES OF BONE HEALING

MNEMONIC
He
Fell
Because
he was
Running



STAGE I
HEMATOMA FORMATION
- First 1-2 days of fracture
- Bleeding into the injured site occurs



STAGE II
FIBROCARTILAGINOUS CALLUS FORMATION
- Formation of granulation tissue
- Reconstruction of bone begins
- Still not strong enough to bear weight

STAGE III
BONY CALLUS FORMATION (OSSIFICATION)
- 3rd - 4th week of fracture healing
- Mature bone is replacing the callus

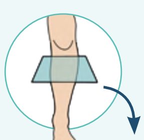
STAGE IV
REMODELING
- This may take months to years!
- Compact bone replaces spongy bone
- X-rays are used to monitor the progress of bone healing

NURSING ASSESSMENT POST-FRACTURE

Neurovascular assessments

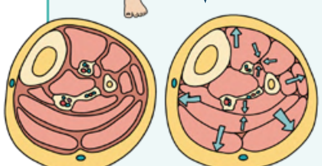
5 P's
- P Pain
- P Pallor
- P Pulselessness
- P Paresthesia — burning or tingling sensation
- P Paralysis

COMPARTMENT SYNDROME



Increased pressure and build-up, causes tissue impairment leading to cell death!

Pressure ↑
↓
Blood flow cut off
↓
Tissue damage due to HYPOXIA (lack of oxygen)



Normal

Compartment Syndrome
Muscle swelling causing compression of nerves and vessels

SIGNS & SYMPTOMS
- Deep, throbbing, unrelenting pain
- Pain unrelieved by medications
- Disproportional to the injury
- Intensifies with passive ROM

TREATMENT
IMMEDIATE
- Place extremity at the heart level (not above heart level)
- Open the cast or splint

FASCIOTOMY
Fascia is cut to relieve tension & pressure

GOUT

PATHOLOGY

Gout is a form of arthritis characterized by increased uric acid levels.

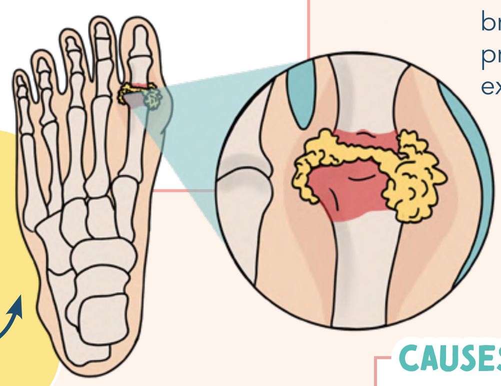
HYPERURICEMIA
"high" "uric acid" "in the blood"

This causes deposits of uric acid crystals in the joints.

TOPHI

Accumulation of sodium urate crystals in joints such as the big toe and hands, or other areas such as the ears.

MEMORY TRICK: **T**ophi think **T**oe



WHAT IS URIC ACID?

Uric acid is created from purine breakdown during digestion. It's produced by the liver and is mostly excreted by the kidneys.

EXPECTED RANGE:
F: **2.5 - 8** mg/dL
M: **1.9 - 7.5** mg/dL

SIGNS & SYMPTOMS

Can be **ACUTE** or **CHRONIC**

- Acute gouty arthritis
- Pain (severe)
- Swelling
- Warmth at the site
- Bone deformity
- Joint damage
- Tophi
- Renal calculi

CAUSES

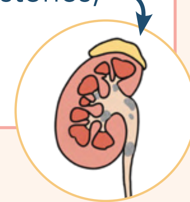
- Diet high in purines
- Certain medications
  - Diuretics (causes dehydration)
  - Aspirin
  - Cyclosporine
- Disorder of purine metabolism
- Kidney problems
  - Inadequate excretion of uric acid by the kidneys

EDUCATION

- Educate on avoiding:
  - Foods high in purines
  - Medications (aspirin)
  - Alcohol
  - Dehydration

foods high in purines:
ORGAN MEATS (liver, kidney)
RED MEATS
SEAFOOD
ALCOHOL (beer)

- Stay hydrated: 2-3 liters per day
- Uric acid deposits can cause kidney stones, fluids help prevent this!
- Weight loss program if overweight



MEDICATIONS

GENERIC	TRADE NAME
allopurinol	Aloprim, Zyloprim, Lopurin

MEMORY TRICK: Allo**P**urinol → **P**revents gout

GENERIC	TRADE NAME
colchicine	Mitigare, Colcrys

MEMORY TRICK: **C**olchicine → for a**C**ute gout attacks

For more information about gout medications, see the musculoskeletal section in the Pharmacology Bundle

OSTEOPOROSIS

PATHOLOGY

OSTEOPOROSIS

OSTEO → "relating to bone"
POROSIS → "porous"

Osteoporosis essentially means: **HAVING POROUS BONES**

Normal bone marrow has small holes in it, but osteoporosis causes much larger holes

- The rate of **BONE RESORPTION** (osteoCLASTS) is greater than the rate of **BONE FORMATION** (osteoBLASTS) = ↓ **DECREASED TOTAL BONE MASS**

DIAGNOSTIC

- **Bone density test:** Dual-energy x-ray absorptiometry **(DEXA)**

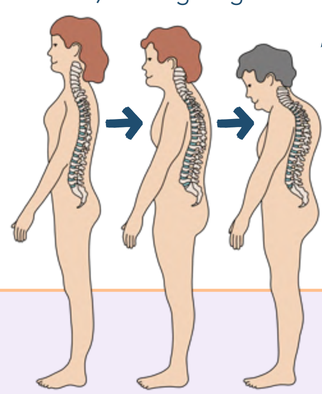
This process takes X-ray images measuring calcium and other minerals in the bones

RISK FACTORS

- **C**alcium & vitamin intake is LOW
- **A**ge: **women after menopause** (the decrease in estrogen at menopause causes increase bone resorption)
- **L**ifestyle (smoking, excessive alcohol intake, sedentary lifestyle, immobility)
- **C**aucasian or Asian women
- **I**nherited (family history)
- **U**nderweight/malabsorption disorder (Celiac disease, bariatric surgery, eating disorders)
- **M**edications: **long-term use of corticosteroids**, anticonvulsants, levothyroxine, long-term use of proton pump inhibitors, etc.

SIGNS & SYMPTOMS

- May be asymptomatic until a fracture occurs
- **FRACTURES** (hips, spine, wrist)
- Low back, neck, or hip pain
- The back will be rounded (hunch back) causing height loss



FRACTURES

Clients often think they fell and broke something, **BUT** bones may break first causing them to fall.

NURSING INTERVENTIONS

ASSESSING FOR RISK FACTORS

Educate on stopping smoking & limiting alcohol

EDUCATE ON WAYS TO PREVENT OSTEOPOROSIS

TEACHING ABOUT PREVENTING INJURY

AT HOSPITAL

- Use call light
- Non-slip socks
- Communicate falls risk
- Clutter-free environment

AT HOME

- No area rugs (risk for falling)
- Watch out for pets
- Keep glasses near by

PREVENTION

- Weight-bearing exercises (weights, hiking, etc).
- Consume foods rich in calcium & vitamin D

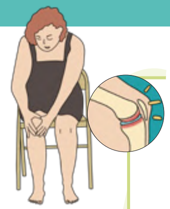
MEDICATIONS

- Calcium supplements with Vitamin D
- Bisphosphonates (ends in "**DRONATE**")

**For more information about bisphosphonates, see the Pharmacology Bundle*

OSTEOARTHRITIS (OA) & RHEUMATOID ARTHRITIS (RA)

OSTEOARTHRITIS (OA)



PATHOLOGY

OA is a noninflammatory degenerative disorder of the joints. It's caused by the breakdown of cartilage between the joints.

The articular cartilage breaks down, which leads to damage to the bone.

Eroded cartilage

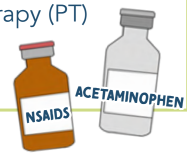
Bone ends rub together

RISK FACTORS

- **OBESITY**
- Older age
- Female gender
- Certain occupations (heavy labor)
- Genetics

TREATMENT

- Orthotic devices (splints, braces, knee braces)
- Walking aids (canes)
- Exercise
- Weight loss
- Occupational therapy (OT) & physical therapy (PT)
- Analgesics



DISTAL
Distal interphalangeal (DIP) called
HEBERDEN'S NODES

PROXIMAL
Proximal interphalangeal (PIP) called
BOUCHARD'S NODES

SIGNS & SYMPTOMS

- Pain
- Stiffness after activity (subsiding within 30 min)
- Functional impairment
- Bony enlargements

Occurring mostly at the weight-bearing joints (hips, knees)

MOVEMENT / EXERCISE → Aggravated / symptoms worsen

REST → Symptoms are relieved

RHEUMATOID ARTHRITIS (RA)

PATHOLOGY

Exact mechanism is unknown

RA is a chronic, inflammatory type of arthritis. It's classified as an autoimmune disease.

swollen, inflamed synovial membrane

SIGNS & SYMPTOMS

- **S**ymmetric joint pain
- **S**ymptoms are typically **BILATERAL** & symmetric
- **S**tiffness in the morning (lasting >1 hour)
- **S**welling, warmth, and redness
- Deformity of the fingers
- Can effect all joints (fingers, wrists, neck, shoulders, etc).
- Systemic effects: heart, lungs, skin, etc.

DIAGNOSIS

- Hard to diagnose because symptoms are very similar to other diseases
  - (+) Rheumatoid factor
  - Increase erythrocyte sedimentation
  - C-reactive protein (indicates inflammation in the body)
  - X-ray shows joint deterioration

RISK FACTORS

May cause an inflammatory response & destructive synovial fluid

- Environmental factors (smoking, pollution)
- Bacterial or viral illness
- Cigarette smoking
- Family history

STAGES OF RHEUMATOID ARTHRITIS

1. **SYNOVITIS**
   - Inflammation of the synovium
   - Synovial membrane thickens
2. **PANNUS FORMATION**
   - Pannus is a layer of vascular fibrous tissue
3. **FIBROUS ANKYLOSIS**
   - Joint invaded by fibrous connective tissue
4. **BONY ANKYLOSIS**
   - When the bones are fused together

This causes loss of...

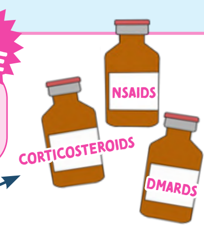
- Articular surfaces
- Joint motion
- Ligament elasticity

TREATMENT

NO CURE

GOAL: Decrease joint pain & swelling. Decrease changes of joint deformity & minimize disability.

- Medications

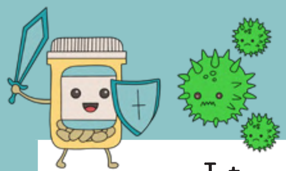


- Surgery
  - **SYNOVECTOMY:** removal of synovium
  - **JOINT REPLACEMENT**
  - **ARTHRODESIS:** "joint fusion"
- Joint support
  - Splints & assistive devices

- Range of motion (ROM) exercise
- Low impact exercise (walking, water aerobics, etc).
- Occupational therapy (OT) & physical therapy (PT)
- Heat or cold? **HEAT** → For stiffness
  COLD → For pain/inflammation

PHARMACOLOGY

BROUGHT TO YOU BY NURSE IN THE MAKING



ANTIBIOTICS / ANTIBACTERIALS

	PREFIXES / SUFFIXES	EXAMPLES
Tetracyclines	-CYCLINE	doxycycline, tetracycline
Sulfonamides	SULF-	sulfasalazine
Cephalosporins	-CEF, CEPH-	cefazolin, cephalexin
Penicillins	-CILLIN	ampicillin, oxacillin
Aminoglycosides & macrolides	-MICIN, -MYCIN	gentamicin, erythromycin
Fluoroquinolones	-FLOXACIN	ciprofloxacin, levofloxacin

ANTIVIRALS

	PREFIXES / SUFFIXES	EXAMPLES
Antiviral (undefined group)	VIR-, -VIR-, -VIR	oseltamivir, zanamivir
Antiviral (anti-herpes virus agents)	-CLOVIR	acyclovir, famciclovir
Antiretrovirals (protease inhibitors)	-NAVIR	atazanavir, nelfinavir
HIV / AIDS	-VUDINE	zidovudine, stavudine

ANTIFUNGAL

	PREFIXES / SUFFIXES	EXAMPLES
Antifungal	-AZOLE	fluconazole, voriconazole

ANESTHETICS / ANTIANXIETY

	PREFIXES / SUFFIXES	EXAMPLES
Local anesthetics	-CAINE	lidocaine, bupivacaine
Barbiturates (CNS depressant)	-BARBITAL	amobarbital, secobarbital
Benzodiazepines (for anxiety/sedation)	-ZOLAM, -ZEPAM	alprazolam, lorazepam

ANTIDEPRESSANTS

SCAN FOR ANTIDEPRESSANTS VIDEO

	PREFIXES / SUFFIXES	EXAMPLES
Selective serotonin reuptake inhibitors (SSRIs)	-OXETINE, -TALOPRAM -ZODONE	fluoxetine, escitalopram, vilazodone
Serotonin-norepinephrine reuptake inhibitors (SNRI/DNRI)	-FAXINE, -ZODONE - NACIPRAM	venlafaxine, nefazodone, milnacipran
Tricyclic antidepressants (TCAs)	-TRIPTYLINE, -PRAMINE	amitriptyline, clomipramine

ANALGESICS / OPIOIDS

	PREFIXES / SUFFIXES	EXAMPLES
Opioids	-DONE, - ONE	oxycodone, hydromorphone
NSAIDs (anti-inflammatory)	-PROFEN	ibuprofen, fenoprofen
Salicylates	aspirin (ASA)	
Nonsalicylates	acetaminophen	

UPPER RESPIRATORY

	PREFIXES / SUFFIXES	EXAMPLES
H1 Antagonists (second-generation antihistamines)	-TADINE, -TIRIZINE	loratadine, desloratadine, cetirizine, levocetirizine
Nasal decongestants	-EPHRINE, -ZOLINE	phenylephrine, naphazoline, oxymetazoline

LOWER RESPIRATORY

	PREFIXES / SUFFIXES	EXAMPLES
Beta2-agonists (Bronchodilator)	-TEROL	albuterol, levalbuterol
Xanthine derivatives (Bronchodilator)	-PHYLLINE	aminophylline, dyphylline
Cholinergic blockers	-TROPIUM	tiotropium
Immunomodulators & leukotriene modifiers	-ZUMAB, -LUKAST	reslizumab, montelukast

GASTROINTESTINAL

	PREFIXES / SUFFIXES	EXAMPLES
Histamine H2 antagonists (H2-blockers)	-TIDINE	cimetidine, famotidine
Proton pump inhibitor (PPIs)	-PRAZOLE	omeprazole, pantoprazole

ANTIDIABETIC

	PREFIXES / SUFFIXES	EXAMPLES
Thiazolidinedione	-GLITAZONE	rosiglitazone, pioglitazone
Inhibitor of the DPP-4 enzyme	-GLIPTIN	sitagliptin, linagliptin

CARDIAC: ANTIHYPERTENSIVES

	PREFIXES / SUFFIXES	EXAMPLES
ACE inhibitors	-PRIL	enalapril, captopril
Beta-blockers	-OLOL	metoprolol, nadolol
Angiotensin II receptor antagonists	-SARTAN	losartan, olmesartan
Calcium channel blockers	-PINE, -AMIL	amlodipine, verapamil
Vasopressin receptor antagonists	-VAPTAN	conivaptan, tolvaptan
Alpha-1 blockers	-OSIN	prazosin, doxazosin
Loop diuretics	-IDE, -SEMIDE	furosemide, bumetanide
Thiazide diuretics	-THIAZIDE	hydrochlorothiazide, chlorothiazide
Potassium-sparing diuretics	-ACTONE	spironolactone

CARDIAC: ANTIHYPERLIPIDEMICS

	PREFIXES / SUFFIXES	EXAMPLES
HMG-CoA reductase inhibitor	-STATIN	simvastatin, rosuvastatin

CARDIAC: OTHER

	PREFIXES / SUFFIXES	EXAMPLES
Anticoagulant (Factor Xa inhibitor)	-XABAN	apixaban
Low-molecular-weight heparin (LMWH)	-PARIN	enoxaparin, dalteparin
Thrombolytics (clot-buster)	-TEPLASE	alteplase
Antiarrhythmics	-ARONE	amiodarone

MISCELLANEOUS

	PREFIXES / SUFFIXES	EXAMPLES
Corticosteroids	-ASONE, -OLONE, -NIDE	betamethasone, fluocinolone, amcinonide
Triptans (anti-migraine)	-TRIPTAN	almotriptan, sumatriptan
Ergotamines (anti-migraine)	-ERGOT-	dihydroergotamine, ergotamine
Antiseptics	-CHLOR MEMORY TRICK Chlor think Clean	chlorhexidine, chloroxylenol
Bisphosphonates	-DRONATE	risedronate, alendronate
Neuromuscular blockers	-NIUM	vecuronium, rocuronium
Retinoids (anti-acne)	TRETIN-	tretinoin
Phosphodiesterase 5 inhibitors	-AFIL	sildenafil, tadalafil
Carbonic anhydrase inhibitors	-LAMIDE, - AMIDE	acetazolamide, diclofenamide

COMMON THERAPEUTIC LEVELS

Digoxin	**0.5 - 2.0** ng/mL **(> 2 = TOXIC)**
Lithium	**0.6 - 1.2** mEq/L
Theophylline	**10 - 20** mcg/mL
Dilantin (Phenytoin)	**10 - 20** mcg/mL
Magnesium sulfate	**4 - 7** mg/dL
Acetaminophen (Tylenol)	**10 - 20** mcg/mL
Gentamicin	**5 - 10** mg/L
Salicylate (aspirin)	**100 - 300** mcg/mL
Vancomycin	Peak: **20 - 40** mcg/mL Trough: **5 - 15** mcg/mL
Valproic acid	**50-100** mcg/mL

ANTIDOTES

Antidotes work to reverse the toxicity of a certain medication

✗ **ANTI**-dote
✗ Think **ANTI** - Drug!

Opioids / Narcotics	**NALOXONE (NARCAN)**	NO more Opioids NARCAN → OPIOIDS
Warfarin (Coumadin)	**VITAMIN K**	During WAR, Vitamin K Kills WARfarin
Heparin	**PROTAMINE SULFATE**	You will need HELP from a PRO to stop bleeding out
Digoxin	**DIGIBIND OR DIGIFAB**	DIGIBIND DIGIFAB
Anticholinergics	**PHYSOSTIGMINE**	
Benzodiazepines	**FLUMAZENIL (ROMAZICON)**	I FLU fast in my mercedes BENZ
Cholinergic crisis	**ATROPINE (ATROPEN)**	We don't have time to CHAT, we have a toxic situation CHOLINERGIC → ATROPINE
Acetaminophen (Tylenol)	**ACETYLCYSTEINE (MUCOMYST)**	ACETAMINOPHEN → ACETYLCYSTEINE
Magnesium sulfate	**CALCIUM GLUCONATE**	MAGgie CALLs for help! MAGNESIUM → CALCIUM
Iron	**DEFEROXAMINE**	DEFEROXAMINE → FERrous means "containing iron"
Lead	**SUCCIMER OR CALCIUM DISODIUM EDETATE**	These are chelation agents
Alcohol withdrawal	**CHLORDIAZEPOXIDE (LIBRIUM)**	
Beta blockers	**GLUCAGON**	Beta blockers be GONe with GlucaGON
Calcium channel blockers	**GLUCAGON, INSULIN, OR CALCIUM**	
Aspirin	**SODIUM BICARBONATE**	You take ASPIRIN when you have a headache. You may also want a SALTY snack when you have a headache.
Insulin Reaction	**GLUCAGON**	If you want your insulin GONe, you give GlucaGON
Pyridoxine	**DEFEROXAMINE**	
Tricyclic antidepressants	**SODIUM BICARBONATE**	
Cyanide	**HYDROXOCOBALAMIN**	

SCAN FOR ANTIDOTES VIDEO

SALICYLATES & NONSALICYLATES

SALICYLATES

GENERIC	TRADE NAME
aspirin	-

ACTION

- Analgesic & antipyretic
- Anti-inflammatory
- Anticoagulant

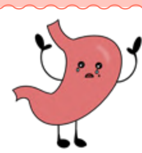
- Analgesic
  - Inhibits prostaglandins. Prostaglandins make pain receptors more sensitive to feel pain.
- Antipyretic
  - ↓ body temp by dilating the blood vessel & spreading the blood throughout the body.
- Aspirin
  - Prolongs bleeding times.
  - Inhibits the clumping of platelets.

USES

- Mild to moderate pain
- ↓ body temp
- Inflammatory conditions (RA, OA, & rheumatic fever)
- Aspirin is used to ↓ the risk of an MI & CVA

SIDE EFFECTS

- GI upset
  - Heartburn
  - Nausea / vomiting
  - Anorexia
  - GI bleeding



CONTRAINDICATIONS

PREGNANCY CATEGORY D

- Known sensitivity to Salicylates or NSAIDS
- Any **BLEEDING TENDENCIES**
  - GI bleeding (peptic ulcers)
  - Blood dyscrasia
  - Bleeding disorders
  - On anticoagulants
  - Vit K deficiency
- Children with recent viral infection
  - Risk for **REYE'S SYNDROME**!

NURSING CONSIDERATIONS

- Stop taking salicylates 1-week prior to major surgery (remember ↑ risk for bleeding)
- Monitor for GI bleeding

ANTIDOTE: activated charcoal

TOXICITY?
1) Gastric lavage
2) Activated charcoal (within 2 hours of ingestion)

NONSALICYLATES

GENERIC	TRADE NAME
acetaminophen	Tylenol

ACTION

- Analgesic & antipyretic

Action is not completely known. Does NOT have any anti-inflammatory or antiplatelet effects.

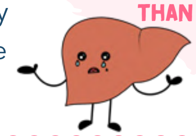
USES

- Mild to moderate pain
- Aspirin substitute for those with:
  - Allergy to aspirin
  - bleeding tendencies
- Children with fever / flu-like symptoms

SIDE EFFECTS

- Hives
- Hemolytic anemia
- Pancytopenia
- Hypoglycemia
- Liver damage
  - Hepatotoxicity
  - Hepatic failure
  - Jaundice

These side effects rarely occur when the medication is taken as directed. They occur due to **CHRONIC USE** or **HIGHER DOSAGE THAN RECOMMENDED**



CONTRAINDICATIONS

PREGNANCY CATEGORY B

- Known sensitivity to acetaminophen
- Those with liver dysfunction
  - Chronic alcohol use

NURSING CONSIDERATIONS

- Before adm. of acetaminophen, assess overall health & alcohol use
  - Malnourished clients & those with chronic alcohol use (>3 drinks /day) are at increased risk for liver damage
  - Limit dosage to 1000-2000 mg/day!

ANTIDOTE: Acetylcysteine (mucomyst)

This protects the liver cells & destroys acetaminophen metabolism

TOXICITY?
1) Gastric lavage (within 4 hours of ingestion)
2) Give antidote via nebulizer within 24 hours of ingestion

NSAIDS

NSAIDS Non-steroidal Anti-Inflammatory Drugs

Gets their name because they produce an anti-inflammatory effect but they are not steroids!

GENERIC	TRADE NAME
Ibu**profen**	Advil
feno**profen**	Nalfon
flurbi**profen**	–
diclofenac	–
celecoxib	Celebrex
ketor**olac**	Sprix (nasal spray)
naproxen	Aleve
indomethacin	Indocin

SUFFIXES: -PROFEN, -OLAC

Inhibits **COX 2** without inhibiting **COX 1**

ACTION

- Anti-inflammatory
- Analgesic
- Antipyretic

Inhibit prostaglandin synthesis by blocking cyclooxygenase (COX)

COX 1 Enzyme that maintains stomach lining

COX 2 Enzyme that triggers pain

This means they inhibit pain, but also inhibit the enzyme that maintains the lining of the stomach!

USES

- Mild to moderate pain
- Menstrual cramps
- ↓ fever
- Musculoskeletal disorders
  - OA & RA

SIDE EFFECTS

MOST COMMON

- GI upset
  - Nausea / diarrhea / vomiting
  - Anorexia
  - Abdominal pain / discomfort
- Heart
  - HTN & heart failure
- Kidney clogging
  - NSAIDs are nephrotoxic!
- Blood clots
  - Stroke

MEMORY TRICK

NSAIDs think **N**ephrotoxic!

CONTRAINDICATIONS

PREGNANCY CATEGORY **D**

- Known hypersensitivity to NSAIDs or aspirin
- Clients with clot history
  - MI, CVA, PE, DVT
- Clients with liver, kidney, or bleeding disorders

Certain medications are known to cause bronchospasms in clients with asthma.

We want to "**BAN**" these medications from asthma patients.

MEMORY TRICK

- **B** Beta blockers
- **A** Aspirin
- **N** NSAIDS

NURSING CONSIDERATIONS

- NSAIDs cause GI upset such as acid reflex
  - Administer proton pump inhibitors (PPIs)
    - Ompr**azole**
    - Pantopr**azole**
- **EDUCATE:** take with food to decrease stomach upset
  - Don't take on an empty stomach

PAIN MANAGEMENT MEDS

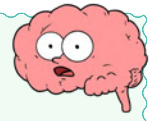
OPIOID ANALGESICS

GENERIC
hydromorph**one**
codeine
oxyco**done**
fentanyl
morphine sulfate
SUFFIXES: -DONE, -ONE

THE GOLD STANDARD

- Most commonly used opioid for chronic pain.
- Can be given in many forms:
  (PO, nasally, subcut, IM, IV, & suppository)

ACTION



CNS Depressant

Binds to opioid receptors in the brain which causes an analgesic sedative, & euphoric effect.

USES

- ↓ anxiety & sedate post-op
- ↓ anxiety in those with dyspnea
- Relieve pain (myocardial infarction)
- Manage opioid dependence
- Treat diarrhea & intestinal cramping

Opioids do **NOT** produce an anti-inflammatory effect or an antipyretic effect. So they are not used to reduce fevers or for gout / rheumatoid arthritis.

SIDE EFFECTS

- ↓ GI Function
  - Constipation

LONG TERM SIDE EFFECTS
Client will NOT build tolerance

- ↓ Vital signs
  - ↓ HR
  - ↓ BP (hypotension)
  - ↓ RR
- ↓ CNS function
  - Sedation, insomnia, weakness, dizziness
- Pruritus (itching)
- Nausea
- IV admin
- Burning sensation

SHORT TERM SIDE EFFECTS
Client WILL build tolerance

TOLERANCE VS. DEPENDENCE

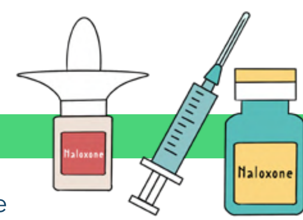
TOLERANCE

The body adapts to the drug (gets used to it)

Higher doses of medication are needed to achieve the same effect!

DEPENDENCE

The body goes through "withdrawals" & experiences negative effects when the medication is STOPPED!



ANTIDOTE: Naloxone (Narcan)

- → Reversal agent for opioid overdose
- → Opioids last longer than the effect of naloxone (Narcan)
- → Repeat doses may be needed

REMEMBER: This reverses the opioid's effects and the client's pain will come back!

NURSING CONSIDERATIONS

- Preventative measures for **CONSTIPATION**
  - Adm. stool softeners or laxatives
  - Daily exercise
  - Fluids, Fiber, & Fruits
  - Encourage client to defecate when they feel the urge (do not wait)
- Client respirations begin to drop
  - Coaching the client to breath may increase the respiratory rate
  - Administer naloxone (Narcan)
- Preventative measures for falls
  - Opioids causes orthostatic hypotension
  - Educate to rise slowly, assist the client with ambulatory activities
  - Keep the room well lit
- Take PO opioids with food to decrease GI upset
- Do not drink ETOH!

MEMORY TRICK: **F**luids, **F**iber, & **F**ruits — **F**ill up the toilet

TRANSDERMAL PATCH

- ☑ Remove old patch before placing a new one
- ☑ Dispose old patch in the sharps container
- ☑ Date & initial the patch
- ☑ Do not apply over hair
- ☑ Rotate sites
- ☑ Avoid the sun or heat (it increases absorption)

WHEN TO STOP THE MEDICATION:

- Respiratory depression
  - RR < 12
- If the client is unarousable

SULFONAMIDES & FLUOROQUINOLONES

ANTIBIOTICS

OVERVIEW OF ANTIBIOTICS

- Antibiotics and antibacterials are used interchangeably
- Antibiotics are only used for bacterial infection (not viral)
- Finish the entire prescription of antibiotics (even if you are feeling better)
- NO alcohol (antibiotics are hard on the liver)
- A culture & sensitivity test
  - **CULTURE** is a test to determine the type of bacteria
  - **SENSITIVITY TEST** is to determine what kind of medication will work best
  - Always obtain cultures *before* administering an antibiotic

SUPERINFECTION
Antibiotics disrupt the "normal flora" which can cause a super infection (secondary infection)

SULFONAMIDES "SULFA DRUGS"

GENERIC	TRADE NAME
sulfadiazine	–
sulfasalazine	Azulfidine
sulfamethoxazole	Bactrim
PREFIX: SULFA-	

ACTION

Bacteriostatic (slow-growing)

Inhibit folic acid metabolism.

It slows the growth of the bacteria enough for the body to take over with its own defense mechanic (WBCs)

NURSING CONSIDERATIONS

- Increase **FLUIDS** intake because sulfas dry out the body
- Since sulfas cause photosensitivity, we want to use sunblock and avoid the sun!
- Take folic acid daily
- Patient may bruise easily
  - Monitor skin & handle with care

MEMORY TRICK: **S**uflas think **S**unburn

SIDE EFFECTS

- **GI UPSET!**
  - Nausea, vomiting, anorexia, diarrhea, abdominal pain, stomatitis
- Chills / fever
- Crystalluria
  - Crystals in the urine
- Photosensitivity
  - Increased risk for sunburn!
- **HEMATOLOGIC CHANGES**
  - Leukopenia (↓ WBCs)
  - Thrombocytopenia (↓ platelets)
  - Aplastic anemia (↓ RBCs)

USES

- UTIs (commonly caused by E.coli)
- Acute otitis media
- Ulcerative colitis
- Topical: used for burn wounds

CONTRAINDICATIONS

PREGNANCY CATEGORY D

- Hypersensitive to sulfa drugs
- Allergy to sulfonylureas like Glyburide (antidiabetic medications)

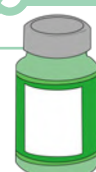
FLUOROQUINOLONES

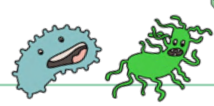
GENERIC	TRADE NAME
cipro**floxacin**	Cipro
gemi**floxacin**	Factive
o**floxacin**	Floxin
moxi**floxacin**	Avalox
levo**floxacin**	Levaquin
SUFFIX: -FLOXACIN	

ACTION

Interferes with the synthesis of bacterial DNA

(Causes death of the bacterial cell)





CONTRAINDICATIONS

PREGNANCY CATEGORY C

- Clients with a history of hypersensitivity to the fluoroquinolones
- Children <18 years old
- Give with caution to:
  - Diabetics, those with renal impairment, history of seizures, & the elderly.

SIDE EFFECTS

- **GI UPSET!**
  - Nausea, diarrhea, abdominal pain
- Dizziness
- Photosensitivity

↑ RISK FOR TENDONITIS & TENDON RUPTURE
(Especially the elderly taking corticosteroids)

MEMORY TRICK: Your tendon is near the FLOOR & can rupture due to FLOORoquinolones

USES

- Lower respiratory infections
- Bone & joint infections
- UTIs
- STIs
- Infections of the skin
- Ophthalmic solutions for eye infections

NURSING CONSIDERATIONS

- Fluoroquinolones cause photosensitivity. We want to use sunblock and avoid the sun!
- Take on an EMPTY stomach w/ full glass of water

ANTIBIOTICS

PENICILLIN & CEPHALOSPORINS

PENICILLIN

GENERIC
peni**cillin** G
peni**cillin** V
amoxi**cillin**
ampi**cillin**
pipera**cillin**
oxa**cillin**
SUFFIX: -CILLIN

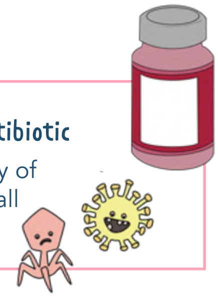
4 TYPES:

- Natural
- Penicillinase-resistant
- Aminopenicillins
- Extended-spectrum

ACTION

Broad Spectrum Antibiotic

Inhibits the integrity of the bacterial cell wall



CONTRAINDICATIONS

PREGNANCY CATEGORY C

- History of allergies to **CEPHALOSPORINS** or **PENICILLIN**
- Renal disease, asthma, bleeding disorders, GI disease

SIDE EFFECTS

- **GI UPSET!**
  - Stomatitis & dry mouth
  - Gastritis, nausea, vomiting, diarrhea, & abdominal pain
- **ORALLY** - Inflammation of the tongue (Glossitis)
- **IM INJECTION** - Pain at the site
- **IV INJECTION** - Irritation & inflammation (Phlebitis)

USES

Penicillins are commonly used as Prophylaxis (prevention) against secondary infections

- UTIs
- Septicemia
- Meningitis
- Intra-abdominal infections
- STIs (syphilis)
- Respiratory infections (pneumonia)

NURSING CONSIDERATIONS

- Pregnancy & breast-feeding safe
- Penicillin makes oral contraceptive ineffective (use additional contraceptive)

MEMORY TRICK: **P**enicillin Bumps the **P**ill

- **EDUCATE:** take with **FOOD** to ↓ GI upset
- Penicillin allergy is very common!

⚠ CROSS SENSITIVITY ⚠

Ask about allergy to **PENICILLIN** or **CEPHALOSPORINS** before administering the first dose!
A client who is allergic to penicillin also may be allergic to cephalosporins.

CEPHALOSPORINS

1ST GENERATION MEDICATIONS

GENERIC	TRADE NAME
cefadroxil	Duricef
cefazolin	An**cef**
cephalexin	Keflex

2ND GENERATION MEDICATIONS

GENERIC	TRADE NAME
cefaclor	Ceclor
cefoxitin	Mefoxin
cefotetan	–

3RD GENERATION MEDICATIONS

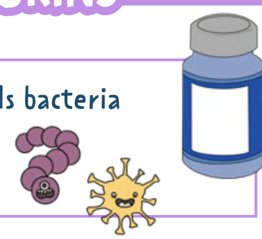
GENERIC	TRADE NAME
cefdinir	Omnicef
ceftriaxone	Ro**ceph**in
cefotaxime	Claforan

PREFIXES & SUFFIXES: -CEF- & -CEPH-

ACTION

Bactericidal - kills bacteria

(Causes death of the bacterial cell)



CONTRAINDICATIONS

PREGNANCY CATEGORY B

- History of allergies to **CEPHALOSPORINS** or **PENICILLIN**
- Administer with caution: clients with renal disease, hepatic impairment, bleeding disorder

USES

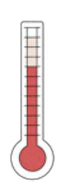
- Otitis media
- Respiratory infections
- Bone infections
- UTIs
- Use prophylactically pre-opt, intra-opt, and post-opt to prevent infection during surgery.

NURSING CONSIDERATIONS

- Cephalosporins make oral contraceptive ineffective (use additional contraceptive)
- Do NOT drink alcohol while on this medication

SIDE EFFECTS

- **GI UPSET!**
  - Nausea, vomiting, diarrhea
- Dizziness
- Malaise
- Heartburn
- Fever
- Nephrotoxicity



- Aplastic anemia (↓ RBCs)
- Stevens-Johnson syndrome (SJS)
- Toxic epidermal necrolysis
- **IV INJECTION** - Irritation & inflation (Phlebitis)
- **IM INJECTION** - Pain at the site

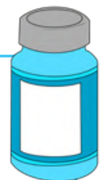
TETRACYCLINES & AMINOGLYCOSIDES

TETRACYCLINES

GENERIC	TRADE NAME
tetra**cycline**	–
doxy**cycline**	Atridox
mino**cycline**	Arestin
demeclo**cycline**	Declomycin

SUFFIX: -CYCLINE

ACTION



Bacteriostatic (slow-growing)

Inhibits bacterial protein synthesis

SIDE EFFECTS

- **GI DISTRESS!**
  - Nausea / vomiting / diarrhea
  - Stomatitis
- Skin rashes
- Photosensitivity reaction

USES

- **SKIN**
  - Skin & soft tissue infection
  - Severe acne
- Rocky mountain spotted fever
- Helicobacter Pylori (H. pylori)

MEMORY TRICK: **T**etra think **T**eeth

CONTRAINDICATIONS

PREGNANCY CATEGORY D

- Known allergy to tetracyclines
- Contraindicated in lactation

MEMORY TRICK: **T**etracyclines think **T**oxic to the developing fetus

NURSING CONSIDERATIONS

- Fluoroquinolones cause photosensitivity. We want to use sunblock and avoid the sun!
- Tetracyclines make oral contraceptives ineffective
  - Use additional contraception
- Take on an EMPTY stomach with a full glass of water

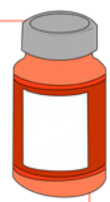
- Causes tooth discoloration
  - Do not give to children younger than 9
- Sit up for 30 min after taking medication
  - Do not lay down
  - Pill induced esophagus (HEARTBURN & scaring of the esophagus!)
- Avoid calcium/dairy products
  - These prevent the absorption of the drug

AMINOGLYCOSIDES

GENERIC	TRADE NAME
genta**micin**	–
kana**mycin**	–
neo**mycin**	–
strepto**mycin**	–

SUFFIXES: -MYCIN, -MICIN

ACTION



Bactericidal - kills bacteria

Blocks the ribosome from reading the mRNA. Then the bacterial can't multiply.

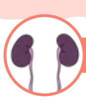
NURSING CONSIDERATIONS

- Monitor:
  - Renal status
  - Neuro status
  - Respiratory status
- Evaluate clients comments related to any hearing issues

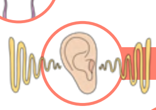
SIDE EFFECTS

- **GI DISTRESS!**
  - Nausea / vomiting / anorexia
- Rash & hives

MEMORY TRICK: **A**Minoglycosides are **A** mean antibiotic because they have very harmful side effects



NEPHROTOXICITY **Hurts the kidneys**: Proteinuria hematuria, & increase BUN & Creatinine.



OTOTOXICITY **Hurts the ears**: Tinnitus, vertigo, hearing loss, which may be permanent.

NEUROTOXICITY **Hurts the brain**: Numbness, tumors, convulsions, muscular paralysis.

CONTRAINDICATIONS

PREGNANCY CATEGORY D

- Known allergy to aminoglycosides
- Hearing loss
- Musculoskeletal disorders (Myasthenia gravis & Parkinson's disease)
- Contradicted for location

USES

- **BOWEL PREPARATION**: Decrease normal flora in the GI for those having abdominal surgery
- Management of hepatic coma
  - Decreasing the ammonia in the intestines

DIURETICS OVERVIEW

LOOP DIURETIC

GENERIC	TRADE NAME
furose**mide**	Lasix
bumeta**nide**	Bumex
torse**mide**	Demadex
SUFFIX: -NIDE, - MIDE	

⚠ Potent (strong) diuretic

ACTION
- Inhibit reabsorption of NA+ & Cl-

Acts on 3 sites
=
↑ reabsorption

PURPOSE
- Hypertension
- Heart failure
- Renal disease
- Edema
- Pulmonary edema

SIDE EFFECTS
- ↓ Hypokalemia ⚠
- ↓ Hypotension
- ↑ Hyperglycemia
- Photosensitivity
- ↓ Hyponatremia
- Dehydration

POTASSIUM WASTING!

REMEMBER NORMAL POTASSIUM **3.5 - 5.0**

NURSING CONSIDERATIONS
- Obtain baseline vital signs
- Replace K+ if < 3.5 mEq/L
- Adm. furosemide SLOWLY
  (rapid adm. can cause ototoxicity)

THIAZIDE DIURETIC

GENERIC	TRADE NAME
hydrochloro**thiazide**	Microzide
chloro**thiazide**	Diuril
methyclo**thiazide**	–
SUFFIX: -THIAZIDE	

ACTION
- Inhibit reabsorption of NA+ & Cl-
- Excretion of Na+, Cl-, & H2O

↑ UOP
=
↓ blood volume

PURPOSE
- Hypertension
- Heart failure
- Renal disease
- Cirrhosis
- Edema
- Corticosteroids
- Estrogen therapy

SIDE EFFECTS
- ↓ Hypokalemia
- ↓ Hypotension
- ↓ Hyponatremia
- ↓ Libido
- ↑ Hyperglycemia
- Photosensitivity
- Dehydration
- Azotemia

POTASSIUM WASTING!

NURSING CONSIDERATIONS
- Obtain baseline vital signs
- Monitor intake & output
- Give w/ meals to ↓ GI upset
- Replace K+ if < 3.5 mEq/L
  - **NEVER** give K+ IV push
- Avoid giving to pt.'s with gout
- Monitor renal function

- Daily weights
  - Same time, same scale!
- Clients with a **sulfa allergy** should avoid thiazide diuretics ⚠

DIURETICS OVERVIEW

MEMORY TRICK

DIURESIS THE BODY
Diuretics = Diuresis = Dry inside

- *WHERE SODIUM GOES...WATER FLOWS!*
- Sodium makes us retain water
  - Low sodium diet (**SODIUM SWELLS!**)
- Give diuretics in the morning, not at night
  - You don't want your client peeing all night long (Nocturia)
- Instruct the client to make slow position changes (diuretics cause orthostatic hypotension)
- Monitor:
  - Daily weights (report 2-3 lbs weight gain)
  - Intake & output
  - Vital signs
  - Potassium levels

OSMOTIC DIURETIC

GENERIC	TRADE NAME
mannitol	Osmitrol

ACTION
- ↑ the thickness of the filtrate so water can't be reabsorbed
- Excretion of Na+ & Cl-

PURPOSE
- Treatment of cerebral edema
- ↓ intraocular pressure (IOP)

SIDE EFFECTS
- Edema
- Blurred vision
- Nausea, vomiting, & diarrhea
- Urinary retention

NURSING CONSIDERATIONS
- Only administered IV
- May crystallize *(check solution before adm.)*
- Perform neuro assessment & LOC *(if using for cerebral edema)*

K+ SPARING DIURETIC

GENERIC	TRADE NAME
spironolactone	Aldactone

MEMORY TRICK **S** think **S**paring

ACTION
- Blocks aldosterone ("salt water" hormone)
- Lets fluid out of the body, into the potty!

- Excretion of Na+ & H_2O

NOT K+
(spares potassium)

PURPOSE
- Hypertension
- Edema
- Hypokalemia
- Hyperaldosteronism
- Cross-sex hormonal therapy

Spironolactone inhibits testosterone

SIDE EFFECTS
- Hyperkalemia (> 5.0)
- Diarrhea
- Gastritis
- Drowsiness
- Erectile dysfunction
- Gynecomastia *(enlargment of the breasts in men)*

EDUCATE: gynecomastia is usually reversible after therapy has stopped

NURSING CONSIDERATIONS
- Avoid eating foods high in potassium *(green leafy veggies, melons, bananas, avocado, etc.)*

- Avoid salt substitutes & potassium supplements
- Monitor K+ levels

⚠ Watch out for **HYPERKALEMIA** (K+ > 5.0 mEq/L)

CARDIAC MEDS

ANTIHYPERLIPIDEMIC DRUGS

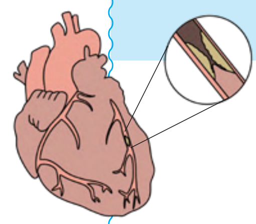
OVERVIEW

- Atherosclerosis is when lipids stick to the blood vessel walls which can obstruct blood flow
- The goal of all antihyperlipidemic drugs is to lower lipid levels in the blood

CHOLESTEROL

LDL – Low Density Lipoprotein → Want LOW Levels (<100 mg/dL) **BAD CHOLESTEROL**

HDL – High Density Lipoprotein → Want HIGH Levels (>60 mg/dL) **HAPPY CHOLESTEROL**



HMG-COA REDUCTASE INHIBITORS "STATINS"

GENERIC	TRADE NAME
atorva**statin**	Lipitor
fluva**statin**	Lescol
lova**statin**	Altoprev
pitava**statin**	Livalo
simva**statin**	Zocor
rosuva**statin**	Crestor

SUFFIX: -STATIN

USES

LOWERS CHOLESTEROL

- Hyperlipidemia
- PRIMARY PREVENTION: Preventable treatment for patients at risk for coronary artery disease (CAD)
- SECONDARY PREVENTION: Stabilizes fatty plaques in clients with current coronary artery disease (CAD)

ACTIONS

- Inhibits the enzyme HMG-CoA Reductase
- Statins are not a cure!

SIDE EFFECTS

NEURO
- Headache
- Nausea
- Dizziness

GI
- Constipation
- Cramping
- Abdominal pain
- Hyperglycemia

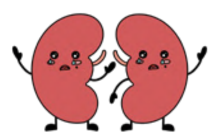
NURSING CONSIDERATIONS

- Monitor liver enzymes
  - ➥ ALT/AST
- Monitor therapeutic response
  - ➥ Statins should lower LDL, & increase HDL
- Avoid grapefruit consumption
  - ➥ Increases risk for toxicity of statins
- Statins are **pregnancy category X** & should not be taken while breastfeeding
- Monitor for signs of **rhabdomyolysis** because statins have been associated with this

RHABDOMYOLYSIS

- Rare condition where the muscles are damaged
- Myoglobin leaks into the blood which can cause **kidney damage**

Signs & Symptoms:
- Muscle pain, tenderness, or weakness
- Accompanied by malaise or fever
- ↑ creatine kinase levels
- Dark urine color (tea or cocoa like urine)



BILE ACID RESINS

GENERIC	TRADE NAME
cholestyramine	Prevalie
colestipol	Colestid
colesevelam	Welchol

USES

- Hyperlipidemia
- Gallstone dissolution
- Pruritus associated with partial biliary obstruction

ACTIONS

Bile is made & secreted by the **LIVER**

↓

Then, it's stored the **GALLBLADDER**

↓

Once emulsified, the fats & lipids are absorbed in the **INTESTINES**

↓

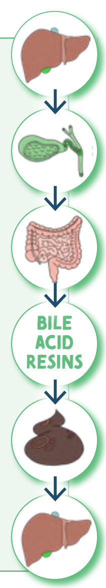
BILE ACID RESINS binds to the bile acid to form an insoluble substance (can not be absorbed by the intestine)

↓

So it's excreted with the **FECES**

↓

↓ bile acids = **LIVER** uses cholesterol to make more bile = ↓ cholesterol



SIDE EFFECTS

GI
- Constipation
- Increase risk for bleeding R/T Vit K malabsorption
- Vitamin A & D deficiencies

NURSING CONSIDERATIONS

- Bile acid resins may interfere with the digestion of fats, preventing the absorption of **fat-soluble vitamins**

MEMORY TRICK: **A**ll **K**ids **E**at **D**onuts
  - Vitamin A & D may be given in a water-soluble for long term therapy
- Bile acid resins may cause constipation, so educate to...
  - Increase fluids, fibers
  - Exercise regularly
  - Use stool softener

ANTIHYPERTENSIVES

ACE INHIBITORS

angiotensin-converting enzyme inhibitors

GENERIC	TRADE NAME
capto**pril**	–
enala**pril**	Vasotec
fosino**pril**	–
lisino**pril**	Prinivil
SUFFIX: -PRIL	

USES

- Hypertension
- Heart Failure

ACTION

Dilates blood vessels, which lowers blood pressure. They do not directly affect the heart rate.

- Inhibits RAAS Renin-Angiotensin-Aldosteron-System
- RAAS is the main hormonal mechanism involved in regulating the blood pressure
- ACE converts angiotensin I ➔ angiotensin II (a powerful vasoconstrictor)
- Inhibiting ACE will inhibit this vasoconstricting effect, decreasing blood pressure!

SIDE EFFECTS

A = ANGIOEDEMA
C = COUGH (DRY)
E = ELEVATED K+

Orthostatic Hypotension
Dizziness

NURSING CONSIDERATIONS

- Assess BP & pulse routinely
- Monitor for hypotension
  - Educate on changing positions slowly
- Monitor K+ levels
  - Normal 3.5 - 5.0
  - Educate to avoid foods high in potassium & avoid salt substitutes
- Assess for angioedema
  - Swelling of the area beneath the skin or mucosa *(deep edema)*
  - **DANGEROUS**: swelling of the face & mouth
- Educate to not suddenly stop the medication it can cause **rebound hypertension** *(needs to be tapered off)*
- Ace inhibitors are contraindicated in pregnancy due to the teratogenic effects on the fetus

BETA BLOCKERS

GENERIC	TRADE NAME
acebut**olol**	Sectral
metopr**olol**	Lopressor
propran**olol**	Inderal
nad**olol**	Corgard
SUFFIX: -OLOL	

USES

- Hypertension
- Stable angina
- Chronic / compensated heart failure (not acute heart failure)
- Dysrhythmias

ACTION

- Blocks norepinephrine & epinephrine (fight or flight hormones)
- Blocks the negative effects of the sympathetic nervous system
  - ➥ Beta blockers can be selective or non-selective
    - Meaning they can block different beta sites (beta 1 and/or beta 2)

↓ Resistance
↓ Workload
↓ Cardiac Output

BETA 1 (one heart)

BETA 2 (two lobes)

SIDE EFFECTS

THE B'S OF BETA BLOCKERS

- **B**radycardia & heart **B**locks
- **B**reathing problems
  - ➥ Bronchi spasms
- **B**ad for heart failure patients (in an acute setting)
- **B**lood sugar masking
  - ➥ Masks S&S of hypoglycemia (low blood sugar)
- **B**lood pressure lowered - Hypotension

NURSING CONSIDERATIONS

- Monitor for hypotension
- Educate on changing positions slowly
- Do not give non-selective beta blockers to asthma patients or COPD patients (*remember: non-selective works on Beta1 & **Beta2** = **Lung** constriction*)
- Educate to not suddenly stop the medication. It can cause rebound hypertension *(needs to be tapered off)*
- Monitor for S&S of heart failure
  - These medications produce inotropic effects (↑ contraction strength of the ♥)
  - S&S of ♥ failure: Wet lung sounds, weight gain, edema, etc

ANTIHYPERTENSIVES

CALCIUM CHANNEL BLOCKERS

MEMORY TRICK

VERY **N**ICE **D**RUGS

GENERIC	TRADE NAME
Verap**amil**	Calan
Nife**dipine**	Procardia
Diltiazem	Cardizem
amlo**dipine**	Norvasc
nicar**dipine**	Cardine
SUFFIXES: -DIPINE, -AMIL	

USES

LOWER HR & BP

- Hypertension
- Angina
- Dysrhythmias

SIDE EFFECTS

- Orthostatic hypotension
- Dizziness
- Flushing
- Headache
- Peripheral edema
- Constipation

ALL COMMON SIDE EFFECTS

ACTION

Blocks movement of calcium

(↓ calcium = ↓ available for transmission of nerve impulses)

- Relaxes blood vessels
- ↓ blood pressure
- ↑ supply of oxygen to the heart
- ↓ ❤'s workload

NURSING CONSIDERATIONS

- Antihypertensives cause **orthostatic hypotension**
  - Change positions slowly
  - Sit on the side of the bed for a few minutes before standing
- Educate to not suddenly stop the medication. It can cause **REBOUND HYPERTENSION** (needs to be tapered off).

- Do not drink grapefruit juice
- Leg elevation & compression to reduce edema
- To help with **CONSTIPATION**:

CAN CAUSE SERVERE HYPOTENSION!

Fluids, **F**iber, & **F**ruits
Fill up the toilet!

ANTICOAGULANTS

Prevents new clots or prevents current clots from getting bigger! Anticoagulants do not dissolve clots & do not thin the blood. Anticoagulants are use for clients who are at an increased risk for **CLOT FORMATION!**

Can a patient be on *both* at the same time?

YES! Commonly used together. Gives time for Warfarin to kick in!

WARFARIN

GENERIC	TRADE NAME
warfarin	Coumadin

ACTION

- Interferes with the production of **VITAMIN K**
  - ➥ **USED IN THE LIVER TO MAKE CLOTTING FACTORS**
- ↓ of clotting factors II (prothrombin), VII, IX, and X. (Prothrombin is required for the clotting)

USES

- **LONG-TERM THERAPY**
- Works slowly (a few days to take effect)

ROUTES

- Orally — MOST COMMON
- IV

SAFE DURING PREGNANCY? ☒ **NO!**

ANTIDOTE: VITAMIN K

NURSING CONSIDERATIONS

- Educate client to be consistent with their vitamin K food intake (green leafy vegetables, liver, etc.)
- Antibiotics increase the risk for **BLEEDING** (they increase INR)
- Will have freq. blood test to check therapeutic range
- Educate to take the pill at the same time every day

HEPARIN

ACTION

Heparin inhibits the formation of fibrin clots. Inhibits the conversion of fibrinogen to fibrin (inactivates factors needed for the clotting)

USES

- **SHORT-TERM THERAPY**
- Works quickly

ROUTES

- **NOT** given orally
- Given by injection (IV or subq)
- IV drip

Heparin is inactivated by gastric acid in the stomach

SAFE DURING PREGNANCY? ☑ **YES!**

ANTIDOTE: PROTAMINE SULFATE

LOW MOLECULAR WEIGHT HEPARIN (LMW)

GENERIC	TRADE NAME
enoxa**parin**	Lovenox
dalte**parin**	Fragmin

SUFFIX: -PARIN

LMW HEPARIN IS ADMINISTERED:

- Subq in the belly
- 2 inches from the umbilicus
- 90 degree angle!
- After subq injection, it's common to have bruising, irritation, & pain!
- Do not massage injection site after

For LMW heparins, we don't look at blood coags.
We want to monitor **PLATELET COUNT**!

Heparin Induced Thrombocytopenia **(HIT)**
Should check platelets while on LMW.
Normal PLT count: **150,000 - 450,000**

COAGULATION

NOT ON ANY ANTICOAGULANT:

PT: 10 - 12 seconds
INR: < 1
aPTT: 30 - 40 seconds

ABBBREVIATIONS:

PTT: Prothrombin Time
aPTT: Activated Partial Thromboplastin Time
INR: International Normalized Ratio

INTERPRETATION:

Numbers are TOO high = Patient will die (increased bleeding)
Numbers are **LOW** = Clots will **GROW**

WARFARIN

Measured with: INR

THERAPUTIC RANGE:
1.5 - 2 times the normal value
INR: **2 - 3**
INR: **2.5 - 3.5**
(Heart valve replacement)

HEPARIN

Measured with: aPTT

THERAPUTIC RANGE:
1.5 - 2 times the normal value
aPTT: **47 - 70 SECONDS**

DIGOXIN

MEDICATION CLASS: CARDIAC GLYCOSIDES

USES
- Heart failure
- Cardiogenic shock
- Antiarrhythmic
  - Atrial fibrillation

ACTION
- **(+) INOTROPIC ACTIVITY**
  - Increases the force of the contraction = increased cardiac output
- **(-) CHRONOTROPIC**: beats slower
- **(-) DROMOTROPIC**: slows impulses sent through AV node, able to squeeze more blood

TOXICITY

THERAPEUTIC RANGE: 0.8 - 2.0 NG/ML

> 2 = Think Toxic

GENERIC	TRADE NAME
digoxin	Lanoxin

SIGNS OF TOXICITY
Report these to the HCP

GI SYMPTOMS EARLY SIGN

Nausea, vomiting, diarrhea

VISUAL SYMPTOMS

Blurred vision, yellow or green vision, halo effect around dark object

NEUROLOGICAL SYMPTOMS

Headache, drowsiness, confusion, disorientation

ANTIDOTE: DIGIBIND

CAUSES OF TOXICITY?

D Decreased potassium (**HYPO**kalemia) **<3.5** mEq/L
- Potassium wasting diuretics (Loop)

I Injured kidneys

G GFR decreased (the elderly)

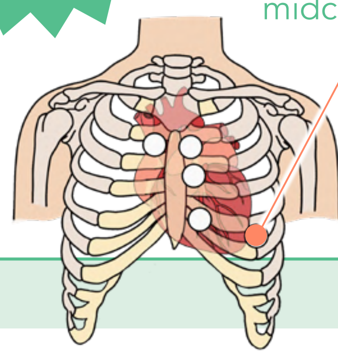
Digoxin is almost solely excreted by the kidneys

NURSING CONSIDERATIONS

- **HOLD THE MEDICATION IF**
  - Adults: <60 bpm
  - Children: <70 bpm
  - Infants: <90-110 bpm
- **KEEP ALL APPOINTMENTS**: drug levels & electrolytes will be monitored

THE APICAL PULSE MUST BE ASSESSED FOR 1 MINUTE **BEFORE** ADMINSTERING **DIGOXIN**

The **APICAL PULSE** is located at the fifth intercostal midclavicular space.



- The apex of the heart
- Point of maximal impulse (PMI)
- Mitral valve

NITROGLYCERIN (NTG)

CARDIAC MEDS

MEDICATION CLASS: ANTIANGINALS

USES

- Angina (chest pain)
- Prevent angina attacks
- Acute coronary syndrome

SIDE EFFECTS

THE H's

H = **H**eadache
H = **H**ypotension (orthostatic)
H = **H**ot flushing of the face

- Rash
- Sublingual
  - Tingling / burning sensation
- Transdermal
  - Contact dermatitis

ALARMING SIGNS

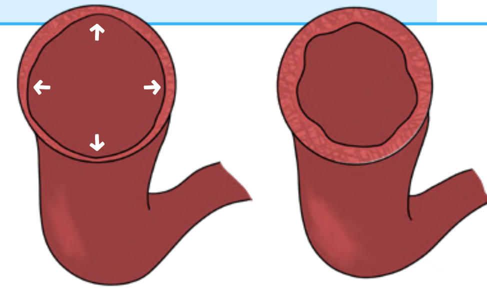
- Lack of coordination
- Lightheadedness
- Pallor
- Irritable

ACTION

- **VASODILATOR**

Dilators do the following:

D **D**ecrease blood pressure
D **D**ilates vessels
D ↓ Vascular resistance



VASODILATION NORMAL BLOOD VESSEL

D = decrease cardiac workload
D = decrease oxygen consumption

CONTRAINDICATIONS

PREGNANCY CATEGORY C

- Known hypersensitivity to nitroglycerin
- Allergy to adhesive (transdermal)
- Clients taking phosphodiesterase (PDE) inhibitors
- Head trauma, cerbral hemorrhage
- Severe anemia

QUICK VS. SLOW ONSET

QUICK	SLOW
IV	Nitro patch
Sublingual tabs	Nitro ointment
Transligual spray	Sustained-release tablets

NURSING CONSIDERATIONS

DO NOT TAKE with phosphodiesterase (PDE) inhibitors (erectile dysfunction (ED) drugs)

Ends in "**-afil**" Like silden**afil** (viagra)
Causes dangerously low blood pressure resulting in death

LONG-ACTING NITRATES DESIRABLE OUTCOME:
The client can perform activities without chest pain (shower, get dressed, etc)

- Monitor blood pressure
  - Stop the medication if systolic BP drops below 100 or the baseline drops below 30 mmHg

- Increased risk for falls due to orthostatic hypotension
  - Educate: rise slowly when getting up

PATIENT EDUCATION

TOPICAL & TRANSDERMAL PATCH

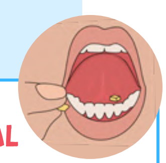
- Remove prior dose
- Rotate sites
- Place over a clean/hairless area
- Wear gloves
- Do not rub nitro ointment into the skin, it can cause rapid absorption!
- Patches can be worn in the shower

SUBLINGUAL NTG OR SPRAY

1 tab/spray sublingual every **5** minutes, up to **3** doses.

If angina is not relieved or is worse 5 min after the first dose, call 911!

SUBLINGUAL OR BUCCAL



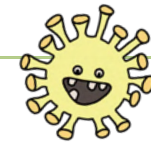
- Place **BUCCAL** tablet between the cheek and gum
- Place **SUBLINGUAL** under the tongue
- Rinse with water before placing the tablets in your cheek

Keep in original container (dark, glass bottle) in a dry, cool place.
Do not swallow or chew these tablets

CORTICOSTEROIDS

GENERIC	TRADE NAME
predni**sone**	Delt**asone**
hydrocorti**sone**	Hydrocort
dexameth**asone**	Ozurdex
flutic**asone**	Flovent HFA
beclometh**asone**	–
flunisol**ide**	Aerospan
cicleson**ide**	Zetonna
SUFFIXES: -SONE, -ASONE, -IDE	

ACTION



ANTI-INFLAMMATORY EFFECTS!

- They reduce the number of mast cells in the airway

THERAPUTIC USES

- COPD
- Rheumatoid arthritis
- Lupus

Can also be administered: IV, IM, PO, rectally, ocularly

TOPICAL CORTICOSTEROIDS

Dermatitis
Rashes
Eczema
Insect bites

INHALED CORTICOSTEROIDS (ICSS)

Chronic asthma

Nasal polyps & rhinitis

SIDE EFFECTS

S's of Steroids

Steroids cause...

- **Sugar**: Hyperglycemia
- **Soft Bones**: Causes osteoporosis
- **Sick**: Decreased immunity / sepsis
- **Sad**: Depression
- **Salt**: Water & salt retention (hypertension)
- **Sex**: Decreased libido
- **Swollen**: Water gain = weight gain
- **Sight**: Risk for cataracts

PATIENT EDUCATION

(Long-term corticosteroid replacement)

- Report signs of an **INFECTION**
  - Corticosteroids are immunosuppressing and can cause an infection
  - Since they are anti-inflammatory, it may hide the fact that the client has an infection
- Increase **CALCIUM** in the diet
  - Corticosteroids can cause osteoporosis & muscle weakness
- Yearly optometrist appointment
  - Corticosteroids may cause **CATARACTS**
- **STRESS OR SURGERY** causes a decrease in cortisol
  - You may need to increase your dose in times of stress
- Never stop steroids suddenly
  - Slowly taper off the medication!

IMPORTANT TEACHING!

After administration, rinse the mouth to decrease the risk of contracting a possible fungal infection from candidiasis

THRUSH: a type of yeast infection

TAKING Bronchodilators & Corticosteroids?

1. **B**ronchodilator first (to help open up the airways)
2. Wait 5 minutes
3. Administer the **C**orticosteroid

MEMORY TRICK: **B** comes before **C** in the alphabet — A B C

BRONCHODILATORS (SABA & LABA)

RESPIRATORY MEDS

SHORT-ACTING BETA2 AGONISTS (SABAs)

GENERIC	TRADE NAME
albu**terol**	Proventil
epinephrine	Adrenalin
levalbu**terol**	Xopenex
terbutaline	–

Think **A**lbuterol is for **A**cute **A**sthma **A**ttacks

LONG-ACTING BETA2 AGONISTS (LABAs)

GENERIC	TRADE NAME
salme**terol**	–
formo**terol**	Foradil
indaca**terol**	Arcapta
arformo**terol**	Brovana

SUFFIX: -TEROL

Think **S**almeterol is for **S**low and **S**teady working a LONG time

ACTION

BRONCHO-DILATORS

Dilates (opens up) the bronchi

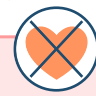
When an agonist binds to the beta-2 receptors the sympathetic nervous system "Fight or flight"takes effect. **THE AIRWAYS RELAX AND DILATE WHICH INCREASES OXYGEN FLOW** which makes it easier to breathe.

MEMORY TRICK

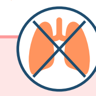
To remember that beta-**2** receptors are in the lungs: you have **TWO LUNGS**

To remember beta-**1** receptors found on the heart: you only have **ONE HEART.**

BETA 1 (one heart)



BETA 2 (two lobes)



USES

SHORT-ACTING BETA2 AGONISTS (SABAs)

Acute symptom relief

- Bronchospasms
- Asthma

LONG-ACTING BETA2 AGONISTS (LABAs)

Long-term management

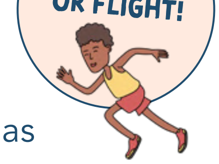
- COPD
- Chronic Bronchitis
- Prevention of bronchospasms

Airway Dysfunction

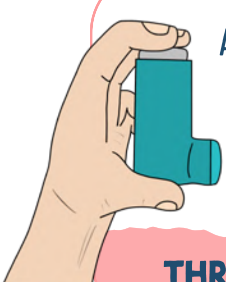
SIDE EFFECTS

- Tachycardia
- Palpitations
- Cardiac arrhythmias
- Hypertension
- Nervousness & anxiety
- Insomnia

REMEMBER Think **FIGHT OR FLIGHT!**



IMPORTANT TEACHING!



After administration, rinse the mouth to decrease the risk of contracting a possible fungal infection from candidiasis

THRUSH: a type of yeast infection

PATIENT EDUCATION

TAKING Bronchodilators & Corticosteroids?

1. **B**ronchodilator first (to help open up the airways)
2. Wait 5 minutes
3. Administer the **C**orticosteroid

MEMORY TRICK

B comes before **C** in the alphabet

A B C

BRONCHODILATOR (Xanthine derivatives) (Methylxanthines)

GENERIC	TRADE NAME
amino**phylline**	–
dy**phylline**	Lufyllin
oxtri**phylline**	Choledyl SA
theo**phylline**	Theochron
SUFFIX: -PHYLLINE	

THEOPHYLLINE

Therapeutic levels
10 - 20 mcg/dL

Toxic **>20** mcg/dL

SIGNS OF TOXICITY

- Tonic clonic seizures
- Tachycardia & dysrythmias

ACTION

BRONCHO-DILATORS
Dilates (opens up) the bronchi

Stimulate the central nervous system (CNS) to promote bronchodilation.

Relaxation of the smooth muscles of the bronchi.

USES

- Relief & prevention of bronchial asthma
- Tx of bronchospasms seen in COPD

SIDE EFFECTS

- Tachycardia
- Palpitations
- ECG changes
- Nervousness & anxiety
- Irritable

REMEMBER Think FIGHT OR FLIGHT!

CHOLINERGIC BLOCKING (Anticholinergic)

ALSO CALLED:

- **ANTICHOLINERGIC DRUGS**
- **CHOLINERGIC BLOCKERS**
- **PARASYMPATHOLYTIC DRUGS**

GENERIC	TRADE NAME
a**clidinium**	Tudorza
ume**clidinium**	Incruse
ipra**tropium**	Atrovent
tio**tropium**	Spiriva
SUFFIXES: -TROPIUM, -CLINDIDIUN	

Inhaled anticholinergic medications

ACTION

Cholinergic blocking drugs **BLOCK THE PARASYMPATHETIC NERVE** that causes the airway to constrict.

By blocking this, it allows the airways to remain open.

SIDE EFFECTS

BLOCKS SECRETIONS → Dry Inside

- Can't See - Blurred vision
- Can't Pee - Dysuria
- Can't Spit - Dry mouth
- Can't Poop - Constipation

RESPIRATORY USES

Prevention of bronchospasms associated with COPD

PATIENT EDUCATION

- Prevent constipation
  - Increase fluids & fiber
- To help with the dry mouth, increase fluids & suck on hard candies.

Fluids, Fiber, & Fruits
Fill up the toilet!

Anticholinergic drugs are used for many other purposes as well, such as:
PARKINSONISM, PEPTIC ULCER, VAGAL NERVE–INDUCED BRADYCARDIA & PREOPERATIVE REDUCTION OF ORAL SECRETIONS.

LITHIUM CARBONATE

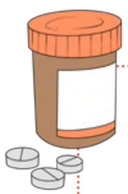
PSYCHIATRIC MEDS

MOOD STABILIZER:

Known for its side effects and narrow therapeutic range

THERAPEUTIC RANGE:
0.6 – 1.2 mEq/L

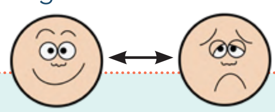
USES



MEMORY TRICK

Lithium is a **L**ong-term treatment

Bipolar disorder
Helps regulate the "mood swings" (depression & mania)



ADVERSE REACTIONS

- Nausea/drowsiness/fatigue
- Thirst
- Dry mouth
- Weight gain

TOXICITY!

- Confusion
- Blurred vision
- Diarrhea
- Tinnitus
  Ringing in ears
- Slurred speech
- Coma
- Convulsions
- Excessive urination
- Excessive thirst
- Tremors/ataxia

TOXICITY LEVELS

Mild: 1.5 - 2 mEq/L
Moderate: 2 - 3 mEq/L
Severe: > 3 mEq/L

HOW DOES TOXICITY HAPPEN?

- Dehydration
  causes ↑ lithium levels in blood
  - Excessive sweating such a high fever
  - Diarrhea
  - Diuretic therapy
- Hyponatremia
- Old age & kidney failure
  ↓ GFR = lithium builds up in the blood

CONTRAINDICATION

PREGNANCY D CATEGORY

- Contraindicted in pregnancy & breastfeeding
- Renal/cardiovascular disease
- Dehydrated patients
  Excessive diarrhea or vomiting
- Receiving diuretics
- Sodium depletion
- Hypersensitivity to tartrazine
- Avoid NSAIDs
  ↓ renal blood flow = ↑ risk for toxicity

Contraceptives may be prescribed

EDUCATION

- Carry ID that shows you are taking lithium
- Educate on signs & symptoms of toxicity
- Educate and stress importance of taking medication regularly
- Serum lithium levels should be checked every 1-2 months
- Do not operate heavy machinery or drive
- Educate on drinking plenty of water to avoid dehydration (therefore avoiding toxicity)
- Avoid starting a low salt diet
  Sudden ↓ in salt = ↑ in lithium

PSYCHIATRIC MEDS

ANTIDEPRESSANT DRUGS

SSRIs

Selective serotonin reuptake inhibitor

ACTION

Inhibits uptake of serotonin = ↑ serotonin

MEMORY TRICK: Think **S**miley **S**erotonin

USES

- Depression
- Anxiety
- OCD
- Eating disorders

SIDE EFFECTS

NEURO
- Headache
- Tremors
- Difficulty sleeping

GI
- Nausea
- Dry mouth / thirst
- Constipation
- Urinary retention
- Sexual dysfunction

3 S's OF SSRIs
- **S**erotonin syndrome
- **S**exual dysfunction
- **S**tomach issues

SEROTONIN SYNDROME
- Too much serotonin in the brain
- Mental changes
- Tachycardia
- Tightness in muscles
- Difficulty walking
- ↑BP & temp

NURSING CONSIDERATIONS

- May take 4-6 weeks to take effect
- Take medication in the morning
- **First line** drug for depression/anxiety

Educate on the importance of compliance

⚠ SUICIDE WARNING ⚠

A client who had suicidal plans may now have the **energy** due to the medication to carry out the plans!

DRUG TABLE

GENERIC	TRADE NAME
sertraline	Zoloft
ci**talopram**	Celexa
esci**talopram**	Lexapro
flu**oxetine**	Prozac
vila**zodone**	Viibryd
SUFFIXES: -TALOPRAM, -OXETINE, -ZODONE	

SNRIs / DNRIs

Serotonin / Norepinephrine & Dopamine / Norepinephrine reuptake inhibitor

ACTION

Affects serotonin, norepinephrine & dopamine

USES

- Depressive episodes
- Anxiety disorders
- Fibromyalgia
- Diabetic neuropathy pain

SIDE EFFECTS

NEURO
- Headache
- Dizziness
- Vertigo
- Photosensitivity
- Agitation/tremors
- Insomnia

GI
- Dry mouth/thirst
- Dehydration
- Constipation
- Nausea/diarrhea

NURSING CONSIDERATIONS

- May take 4-6 weeks to take effect
- Do not mix with TCAs or MAOIs
- **Zyban** is used for smoking cessation.

Do not use it while taking bupropion for depression – it could cause ***overdose***.

DRUG TABLE

GENERIC	TRADE NAME
bupropion	Zyban & Wellbutrin
duloxetine	Cymbalta
venla**faxine**	Effexor XR
mil**nacipran**	Savella
nefa**zodone**	–
SUFFIXES: -FAXINE, -ZODONE, -NACIPRAN	

ANTIDEPRESSANT DRUGS

TCAs
Tricyclic antidepressants

ACTION
Blocks reuptake of serotonin & norepinephrine in the brain

USES
- Depressive episodes
- Bipolar disorder
- OCD
- Neuropathy
- Enuresis

SIDE EFFECTS
- Constipation
- Dry mouth
- Drowsiness
- Blurred vision
- Orthostatic hypotension
- Urine retention
- Cardiotoxic

Causes heart problems in patients with pre-existing cardiac conditions or elderly clients...give with caution!

NURSING CONSIDERATIONS
- May take 2- 3 weeks to take effect
- **WAIT** 14 days after being off MAOIs to start taking TCAs
- **Amoxapine** is not an antipsychotic drug but similar to these drugs, it may cause TD & NMS (D/C the drug immediately if these symptoms occur)

Educate on the importance of compliance

DRUG TABLE

GENERIC	TRADE NAME
ami**triptyline**	—
amoxapine	—
clomi**pramine**	Anafranil
pro**triptyline**	Vivactil
nor**triptyline**	Pamelor
SUFFIXES: -TRIPTYLINE, -PRAMINE	

MAOIs
Monoamine oxidase inhibitor

ACTION
Blocks monoamine oxidase which causes ↑ in epinephrine, norepinephrine, dopamine, & serotonin, which causes stimulation of the CNS!

USES
Depression

SIDE EFFECTS
NEURO
- Orthostatic hypotension
- Dizziness
- Blurred vision

GI
- Constipation
- Dry mouth
- Nausea/ vomiting

HYPERTENSIVE CRISIS
- Headache
- Stiff neck
- Nausea / vomitting
- Fever
- Dilated pupils

Seek medical help to ↓ blood pressure

NURSING CONSIDERATIONS
- Can take up to 4 weeks to reach therapeutic levels
- Educate on the signs & symptoms of HTN crisis
- Avoid foods with Tyramine
  - Aged cheese
  - Fermented meats
  - Chocolate
  - Caffeinated beverages
  - Sour cream & yogurt

DRUG TABLE

GENERIC	TRADE NAME
phenelzine	Nardil
tranylcypromine	Parnate
isocarboxazid	Marplan

ANTIANXIETY DRUGS (ANXIOLYTICS)

BENZODIAZEPINES

USES

Bipolar disorder
Benzos are mainly prescribed for:

- acute anxiety
- sedation/muscle relaxant
- seizures
- alcohol withdrawal

***Not** a first-line drug for treating long-term psychiatric anxiety conditions*

ACTION

Binds to cell receptors enhancing the effects of

GABA

GABA
(inhibitory neurotransmitter)
slows/calms the activity of the nerves in the brain

GENERIC	TRADE NAME
alpra**zolam**	Xanax
lora**zepam**	Ativan
dia**zepam**	Valium
clona**zepam**	Klonopin
chlordiazepoxide	Librium

SUFFIXES: -ZOLAM, -ZEPAM

ANTIDOTE: FLUMAZENIL

MEMORY TRICK: I FLU fast in my Mercedes BENZ

ADVERSE DRUG REACTIONS (ADRs)

- Mild drowsiness, sedation
- Lightheadedness, dizziness, ataxia
- Visual disturbances
- Anger, restlessness
- Nausea, constipation, diarrhea
- Lethargy, apathy, fatigue
- Dry mouth

NURSING CONSIDERATIONS TO HELP WITH ADRs

Lightheadedness, dizziness, ataxia →
Take at night if it makes you dizzy/drowsy
Rise slowly from sitting or lying
Do not drive or operate heavy machinery

Nausea, constipation, diarrhea →
Fluids, fiber, & exercise!
Give with food to ↓ GI upset

Dry mouth →
Sips of water, suck on hard candy, chewing sugar-free gum

SYMPTOMS OF WITHDRAWAL

Withdrawals typically happen when the medication is stopped abruptly or taken for >3 months

- ↑ Anxiety
- ↑ HR
- ↑ BP
- ↑ Temp/sweating
- ↓ Memory
- Agitation
- Seizures/tremors
- Insomnia
- Vomiting
- Muscle aches

NURSING CONSIDERATIONS

- Not meant for long term therapy because ↑ risk for physical & psychological **DEPENDENCE**
- Use of long term therapy leads to **TOLERANCE**
  Larger doses of the drug are required to achieve the desired outcome
- Must be **TAPERED**
  ↓ the dosage gradually.
  NEVER stop the medication abruptly!

CONTRAINDICATIONS & PRECAUTIONS

- Pregnant, laboring & lactating women
- Elderly (↑ chance of dementia)
- Impaired liver or kidney function
- Debilitation

PREGNANCY CATEGORY D

NONBENZODIAZEPINES

ACTION

Depends on the drug

buspirone (Buspar)
acts on serotonin receptors

hydroxyzine (Vistaril)
acts on the hypothalamus & brainstem reticular formation

GENERIC	TRADE NAME
buspirone	Buspar
doxepin	Silenor
hydroxyzine	Vistaril
meprobamate	–

ANTIPSYCHOTICS

PSYCHIATRIC MEDS

Most commonly used for psychosis (schizophrenia)

REVIEW: WHY ARE SGAs BETTER THAN FGAS?
SGAs work on both positive & negative symptoms, and have a lower risk of developing tardive dyskinesia (TD)

FIRST GENERATION ANTIPSYCHOTICS (FGAs)

Also called **TYPICAL/CONVENTIONAL**

GENERIC	TRADE NAME
chlorpromazine	–
haloperidol	Haldol
loxapine	Adasuve

SECOND GENERATION ANTIPSYCHOTICS (SGAs)

Also called **ATYPICAL**

GENERIC	TRADE NAME
risperidone	Risperdal
clozapine	Clozaril
quetiapine	Seroquel
ziprasidone	Geodon
aripiprazole	Abilify

ACTIONS (FGAs)

- Blocks/inhibits dopamine from being released in the brain
- Helps diminish positive symptoms of schizophrenia

ACTIONS (SGAs)

- Acts on both serotonin & dopamine in the brain
- Helps diminish positive symptoms of schizophrenia & helps negative symptoms as well!

SIDE EFFECTS (FGAs)

- Higher risk of **TD**, **EPS**, & **NMS**
- Orthostatic hypotension

SIDE EFFECTS OF BOTH

- Anticholinergic effects
- Photophobia
- Photosensitivity
- Sedation/lethargy

SIDE EFFECTS (SGAs)

- Lower risk of TD, EPS & NMS
- ↑ Weight
- ↑ Cholesterol
- ↑ Triglyceride
- ↑ Blood sugar

TARDIVE DYSKINESIA (TD)
- Involuntary movements of the face, tongue, or limbs that may be irreversible.

EXTRAPYRAMIDAL SYNDROME (EPS)
- Parkinson's like symptoms
- Akathesia (restlessness)
- Dystonia (muscle twitching)

NEUROLEPTIC MALIGNANT SYNDROME (NMS)
- Combination of symptoms: EPS, high fever, & autonomic disturbance
- One can recover 7-10 days after DC of medication, but it can be fatal if not treated in time

CONTRAINDICATIONS

- Hypersensitivity
- Comatose client
- Depressed
- Bone marrow depression
- Blood dyscrasias
- Parkinson's disease
- Liver problems
- Coronary artery disease
- **Hyper** or **hypo**tension

NURSING CONSIDERATIONS

- Educate that it may take 6 - 10 weeks to take effect
- Tell client about adverse reactions and emphasize that adherence is very important

FGAs
- Teach S&S of **TD**, **EPDS**, & **NMS**!
- Advise the client to get up slowly

SGAs
- Check labs *(blood sugar, LDL, triglycerides)*
- To decrease the risk of gaining weight, advise the client about exercise, low-calorie diet, & monitor their weight.

LEVOTHYROXINE

GENERIC	TRADE NAME
levothyroxine	Synthroid

MEMORY TRICK: think THYroid — SYNthetic THYroid

ACTION

- The exact mechanisms are not fully known

Levothyroxine increases the metabolic rate of tissues

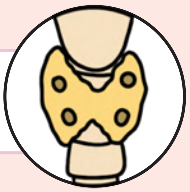
SIDE EFFECTS

- Anxiety
- GI upset
- Sweating
- Weight loss
- Heat intolerance

SAME AS HYPERTHYROIDISM!

MEDICATION CLASS

- Synthetic Hormone



THERAPEUTIC USES

- Treats hypothyroidism
- Thyroid-stimulating hormone suppression
- Thyroid diagnostic testing
- Hormone supplement after thyroidectomy

✗ Should not be used as a weight loss regimen

THERAPEUTIC RESPONSE

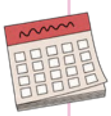
NO LONGER SHOWING SIGNS OF HYPOTHYROIDISM

- Normal heart rate (60 – 100 BPM)
- Improved energy levels (not fatigued)
- Normal skin (not cool or pale)

SAFE DURING PREGNANCY? ✓ YES!

Do not stop the medication if symptoms resolve. Thyroid hormone is needed for fetal brain development!

PATIENT EDUCATION



- It may take 8 weeks to see the full effect
- Report signs of **HYPERTHYROIDISM** ⚠
  - Tachycardia, heart palpitations, weight loss, insomnia, anxiety
- Monitor T4 & T3 levels
- Take once a day (in the morning before breakfast)
- Take at the same time everyday
- Take on an empty stomach

Educate on the importance of compliance

MEMORY TRICK: Levothyroxine is a Life Long therapy

ANTITHYROID DRUGS

METHIMAZOLE

GENERIC	TRADE NAME
methimazole	Tapazole

MEDICATION CLASS

- First-line antithyroid drugs

PROPYLTHIOURACIL (PTU)

GENERIC
propylthiouracil (PTU)

MEMORY TRICK

Prevents **T**hyroid from being **U**p

MEDICATION CLASS

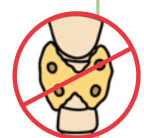
- First-line antithyroid drugs

ACTION

- Inhibits the manufacture of thyroid hormones
- Does not affect existing thyroid hormones circulating in the blood or stored in the thyroid gland

USES

- Treats hyperthyroidism
- Treats thyrotoxicosis
- Treats Graves' disease (autoimmune disease that causes hyperthyroidism)
- Used before thyroidectomy surgery (shrinking it before the surgery)



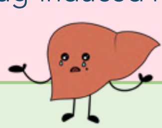
SIDE EFFECTS

- Hay fever
- Skin rash
- Headache
- Nausea & vomiting
- Paresthesias

SYSTEMIC ADVERSE REACTIONS

Risk for:

- Agranulocytosis → Increased risk for infection
- Drug-induced hepatitis → Monitor liver values



PATIENT EDUCATION

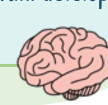
- It may take 1-2 weeks to see the full effect → Educate on the importance of compliance
- Report signs of **HYPOTHYROIDISM** (Bradycardia, weight gain, lethargy, cold intolerance, depression)
- Report signs & symptoms of an infection to the health care provider (Fever, sore throat, etc.)
- Do not abruptly stop the medication (could cause **THYROID STORM** ☠)

PREGNANCY CONSIDERATIONS

PREGNANCY CATEGORY D

- Use with extreme caution during pregnancy because they can cause **hypothyroidism** in the fetus
- If it's necessary, **PROPYLTHIOURACIL** is the preferred drug (does not cross the placenta)

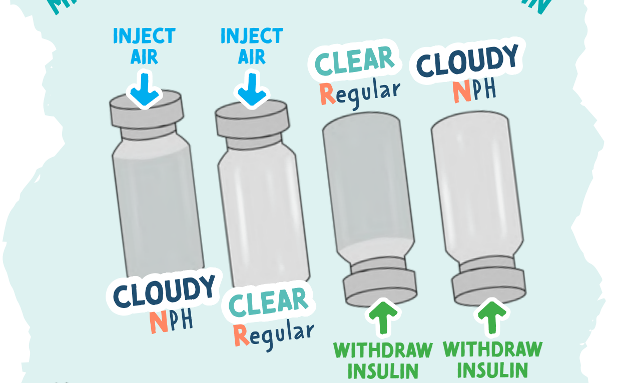
REMEMBER: The fetus needs thyroid hormone for proper brain development



INSULIN TYPES

	GENERIC	BRAND NAME	ONSET / PEAK / DURATION	
RAPID	Lispro Aspart Glulisine	Humalog Novolog Apidra	**ONSET:** 5 - 30 min **PEAK:** 30 - 90 min **DURATION:** 3 - 5 hrs	**HIGHEST RISK FOR HYPOGLYCEMIA**
SHORT (CLEAR)	Regular	Humulin R Novolin R	**ONSET:** 30 - 60 min **PEAK:** 2 - 4 hrs **DURATION:** 5 - 7 hrs	**ONLY INSULIN GIVEN IV** MEMORY TRICK: Regular goes Right into the vein
INTERMEDIATE (CLOUDY)	NPH	Humulin N Novolin N	**ONSET:** 1 - 2 hrs **PEAK:** 4 - 12 hrs **DURATION:** 18 - 24 hrs	**NEVER GIVE IV**
LONG	Glargine Detemir	Lantus Levemir	**ONSET:** 1 - 2 hrs **PEAK:** None **DURATION:** 24 hrs+	**LOWEST RISK FOR HYPOGLYCEMIA** **DO NOT MIX WITH ANY OTHER INSULIN** MEMORY TRICK: Long think Lonely

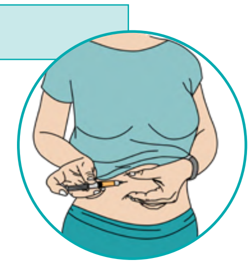
MIXING REGULAR INSULIN & NPH INSULIN



MEMORY TRICK: how to remember this order?

"You are Not Retired, you are an RN"

ADMINISTRATION



- Must be given subcut or IV
  - Insulin is destroyed by the GI tract so it can not be given PO
- Remove all air bubbles
- Rotate site 1 inch from previous site
- Common sites: back of arms, thighs & abdomen (at least 2 inches away from the belly button)

COMPLICATIONS

- Hypoglycemia (especially with rapid insulin)
- Weight gain
  - ➥ Insulin is a growth hormone
- Lipoatrophy (loss of subcut fat)

ALLOPURINOL VS. COLCHICINE

ALLOPURINOL

GENERIC	TRADE NAME
allopurinol	Aloprim, Zyloprim, Lopurin

MEDICATION CLASS

- Uric acid inhibitors

THERAPEUTIC USES

- **PREVENTS** gout attacks ★
- Does not help with acute attacks

Take NSAIDs for acute attacks, **NOT** aspirin

MEMORY TRICK: AlloPurinol → Prevents gout

SIDE EFFECTS

- **GI UPSET:** Nausea, vomiting, abdominal pain, diarrhea
- **SKIN RASH**

same GI side effects

EDUCATION

Stop the medication if a **RASH** occurs
- This may indicate a **HYPERSENSITIVITY REACTION** ☠ (Stevens-Johnson syndrome)

G Gulp a lot of fluid during the day (2-3 L/day) & take the medication with a glass of water

O No Organ meats

U Urine output up to 2 L/day

T Takes several months to take effect

- Uric acid deposits can cause kidney stones
  - Fluids help prevent this
- Allopurinol + aspirin = ↑ uric acid levels
  - Take acetaminophen instead

COLCHICINE

GENERIC	TRADE NAME
colchicine	Mitigare, Colcrys

MEDICATION CLASS

- Antigout agent

THERAPEUTIC USES

- **RELIEVES** acute gout attacks ★
- Also **PREVENTS** gout attacks as well
  - Does not help with pain relief, only helps decrease inflammation

Take NSAIDs for acute attacks, **NOT** aspirin

MEMORY TRICK: Colchicine → for aCute gout attacks

SIDE EFFECTS

- **GI UPSET:** Nausea, vomiting, abdominal pain, diarrhea
- **ADVERSE REACTION:** Risk for **BONE MARROW SUPPRESSION** ☠

EDUCATION

G Gulp a lot of fluid during the day (2-3 L/day) & take the medication with a glass of water

O No Organ meats

U Urine output up to 2 L/day

T Takes several months to take effect

- Uric acid deposits can cause kidney stones
  - Fluids help prevent this

BISPHOSPHONATES VS. CALCITONIN (SALMON)

BISPHOSPHONATES

GENERIC	TRADE NAME
alen**dronate**	Binosto
eti**dronate**	Didronel
iban**dronate**	Boniva
pami**dronate**	Aredia
rise**dronate**	Actonel

SUFFIX: -DRONATE

MEDICATION CLASS

- Bone resorption inhibitors

MODE OF ACTION

- Bisphosphonates inhibit normal & abnormal **BONE RESORPTION** which leads to increased bone mineral density!

THERAPEUTIC USES

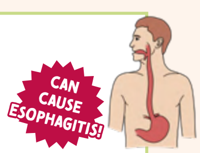
- **BUILDS BONE DENSITY & PREVENTS BONE FRACTURES**
- Treats & prevents osteoporosis (postmenopausal & long term use of steroids)
- Treats paget's disease
- Treats hypercalcemia

SIDE EFFECTS

- **GI UPSET:** Nausea, diarrhea, dyspepsia, acid reflux, abdominal pain

EDUCATION

- Take with a full glass of water on an empty stomach
- Stay upright for 30 minutes



- Separate iron, antacids, & multiple vitamins at least 30 minutes apart from taking bisphosphonates

These decrease absorption

- Encourage increased intake of calcium & vitamin D
- Encourage weight-bearing exercises to preserve bone mass

"If you don't use it, you lose it!"

NURSING CONSIDERATIONS

- Monitor serum calcium levels before, during, & after therapy

NORMAL CALCIUM: **9 - 11** mg/dL

CALCITONIN (SALMON)

GENERIC	TRADE NAME
calcitonin (salmon)	Miacalcin

MEDICATION CLASS

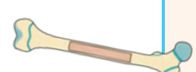
- Hormone
- Hypocalcemic agent

MEMORY TRICK: CalciTONin helps TONe down calcium levels in the blood!

MODE OF ACTION

- Inhibits **OSTEOCLASTS** (Cells that cause bone breakdown)
- ↓ the rate of bone breakdown

THERAPEUTIC USES



- Treats & prevents postmenopausal **OSTEOPOROSIS**
- Treats hypercalcemia
  - Too much calcium in the bloodstream (we want it in the bones, not in the bloodstream)

SIDE EFFECTS

- **GI UPSET**
- **INTRANASAL ROUTE** Nasal irritation & nasal dryness

EDUCATION

- Encourage increased intake of calcium & vitamin D
- Encourage weight-bearing exercises to preserve bone mass

"If you don't use it, you lose it!"

TEMPLATES & PLANNERS

NURSING DIAGNOSIS

SUPPORTING DATA

GOALS

NURSING DIAGNOSIS

SUPPORTING DATA

GOALS

PATIENT INFO

MEDICAL HISTORY

NURSING DIAGNOSIS

SUPPORTING DATA

GOALS

NURSING DIAGNOSIS

SUPPORTING DATA

GOALS

Course Tracker

COURSE:

SUBMITTED	ASSIGNMENT/PROJECT	DUE DATE	SCORE

Test / Quiz Tracker

COURSE:

TEST DATE	CHAPTERS/TOPICS COVERED	GRADE	PASSED?
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
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			YES NO
			YES NO

HOURLY Planner

MONTH: ____________________

☐ SUNDAY	☐ MONDAY	☐ TUESDAY	☐ WEDNESDAY	☐ THURSDAY	☐ FRIDAY	☐ SATURDAY
PRIORITIES	PRIORITIES	PRIORITIES	PRIORITIES	PRIORITIES	PRIORITIES	PRIORITIES
6 AM	6 AM	6 AM	6 AM	6 AM	6 AM	6 AM
7 AM	7 AM	7 AM	7 AM	7 AM	7 AM	7 AM
8 AM	8 AM	8 AM	8 AM	8 AM	8 AM	8 AM
9 AM	9 AM	9 AM	9 AM	9 AM	9 AM	9 AM
10 AM	10 AM	10 AM	10 AM	10 AM	10 AM	10 AM
11 AM	11 AM	11 AM	11 AM	11 AM	11 AM	11 AM
12 PM	12 PM	12 PM	12 PM	12 PM	12 PM	12 PM
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3 PM	3 PM	3 PM	3 PM	3 PM	3 PM	3 PM
4 PM	4 PM	4 PM	4 PM	4 PM	4 PM	4 PM
5 PM	5 PM	5 PM	5 PM	5 PM	5 PM	5 PM
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8 PM	8 PM	8 PM	8 PM	8 PM	8 PM	8 PM
9 PM	9 PM	9 PM	9 PM	9 PM	9 PM	9 PM
PRODUCTIVITY METER BAD GOOD GREAT!	PRODUCTIVITY METER BAD GOOD GREAT!	PRODUCTIVITY METER BAD GOOD GREAT!	PRODUCTIVITY METER BAD GOOD GREAT!	PRODUCTIVITY METER BAD GOOD GREAT!	PRODUCTIVITY METER BAD GOOD GREAT!	PRODUCTIVITY METER BAD GOOD GREAT!

MONTHLY Planner

MONTH: ______________________________ YEAR: __________

WEEKLY *Planner*

MONDAY:

TUESDAY:

WEDNESDAY:

THURSDAY:

FRIDAY:

SATURDAY:

SUNDAY:

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NOTES

TESTS / EXAMS

PROJECTS / ASSIGNMENTS

SELF-CARE ❤

DISEASE:

PATHOLOGY

SIGNS & SYMPTOMS

RISK FACTORS

COMPLICATIONS

DIAGNOSIS

TREATMENT

PHARMACOLOGY TEMPLATE

DRUG CLASS:

GENERIC NAME	TRADE NAME

SUFFIXES OR PREFIXES:

ANTIDOTE:

ACTION

THERAPEUTIC USES

SIDE EFFECTS

CONTRAINDICATIONS

NURSING CONSIDERATIONS

PATIENT EDUCATION

NCLEX Study Schedule

Month: ____________________

NCLEX Date: ____________________

SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:
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Dear future nurse,

You may be stressed, you may feel tired, and you may want to give up. Nursing school is hard, there's no doubt about it. Everyone cries, everyone has meltdowns, and there will be moments you don't feel qualified for the task at hand. But take heart, the challenge only makes you stronger. Put in the work, show up on time, and find an amazing study group. You got this! :)

— Kristine Tuttle, BSN, RN

You got this, future nurse!